

Common Physical Sources for a Constraint-Derived Standard Model

Physical Higgs responses, composite boundaries and quantum–Einstein action

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21 September 2026

Primitive-standing reference edition

Abstract

We construct a common physical realization of the constraint-derived Standard Model and its Einstein response. The quantum and gravitational descriptions share one premetric source architecture. The physical comparison preserves the same fields, state, source operations, continuations and joint parameter point, rather than combining separately occupied sectors.

On the occupied interacting Standard Model source, the invariant Higgs mass insertion yields a represented derivative current and scalar descendant. The physical Higgs channel gives a nonzero descendant preparation with a nonzero component in a bounded positive-invariant-mass band. Exact Higgs, flavor and metric-source transformations retain the full interaction prescription and its contact terms. The same preparation has smooth stress, finite positive excitation energy and a nonzero local curvature-contact response. The source-dependent closed-time-path action supplies the causal stress and Ward identities of the nonlinear minimal Einstein equation before its stationary member is selected.

For color, we prove a complete interpretation of the independently formed source inputs and their licensed operations. Actual physical readers, exact domains and compatible continuation families are preserved, including mixed Higgs responses and the source-qualified phase and heavy-hadron constructions. Neutral physical output maps factor through the invariant part of the joint input. The resulting comparison covers every physical use declared by the original target and retains the distinct authority of compact responses, persistence, spectral data, asymptotic attachments and complete continued germs. These observations coexist with the electroweak channels and Einstein action on one unchanged source. The conclusion retains the original quantum-action order and physical source qualifications; it does not assert a stronger all-state scattering or nonperturbative quantum-gravity theorem.

Complementary constructions give an explicit interacting gauge–Higgs–fermion Hamiltonian, physical domains, dressed observations, positive asymptotic states and a source-attached nonlinear Einstein family. Their relation to the direct physical realization is specified through source comparisons, not inferred from a common interaction grammar.

Keywords: AASC; Standard Model; physical Higgs response; positive source; composite operator; stress-energy; Einstein dynamics; source continuation.

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1 Premetric architecture and common physical sources

The Standard Model and gravity enter this construction through a common premetric source architecture. Admissibility, standing, reference and irreversibility govern the identity of every nondegenerate determinate target before metric or Hilbert-space representation. Gravity is a structural necessity at that level; the Einstein equation is its minimal first-response metric projection, with its full nonlinear curvature content [9, 39]. Quantum and gravitational representations are thus compared as projections of the same upstream construction, not joined as independently fundamental theories [10, 38].

The preceding carrier-sector-projection classification fixes matter, gauge and Higgs identities, their admissible response relations and the complete observational quotient at its stated source level [1, 3]. Its classical successor supplies an occupied Einstein-matter action, source and constrained development [2]. The symmetry closure supplies a nonempty compatible physical assignment and one lawful assignment orbit at its frozen target [37]. These positive source results are used here. We construct the joint physical observations and continuation maps that realize these conclusions on one source.

The handoff retains the rooted generation-response and four-kind matter correspondence of FCGS-I, the whole-doublet and paired-conjugation comparisons of FCGS-II, and the one-Higgs, kind-preserving naturality of FCGS-III [48, 28, 29]. On the bosonic side, BCRS-I supplies mixed source-law coverage with fixed spectator inputs [30]; its electroweak and color successors supply their source-dependent joint-kernel, domain, continuation and metric-response comparisons [4, 5]. These are mathematical identities to preserve, rather than particle names from which a joint realization is inferred. The primitive generation architecture is fixed by joint adequacy and minimal necessity before its neutral and matter occupation is formed [55, Theorems 5.4 and 6.2]. FCGS-I supplies the complete root receipt used here [48, Sections 3–4]; representation-algebra specialization supplies its separate response interpretation [49]. Rank one is a consequence of the derived primitive triangle, not an input selecting the generation count [55, Corollary 5.5]. The chiral, infrared and operator-classification results supply physical phase and heavy-hadron data and the original local operation grammar, each with its stated source assumptions [36, 51].

1.1 Target data and source identity

Definition 1.1 (Physical target). A physical target retains the field and representation inventory, global gauge form, spin and boundary data, source and regulator record, parameter point, allowed preparation and observation classes, and the required continuation and dynamical domains. A comparison of two realizations specifies maps on these data and on all source observations that it uses. A comparison preserving a marginal source need not preserve a coupled source or its dynamics.

The occupied Standard Model source includes the selected gauge, Higgs and chiral interaction inventory, its physical state and boundary convention, renormalized insertion prescription and authorized physical channel data. The invariant Higgs insertion, rather than an elementary gauge-dependent scalar renamed invariant, provides the direct physical starting point [34, 20, 33]. A source operation is received with its actual normalization, local contacts, common domain and continuation. In particular, a finite source derivative, a represented field and an entire analytically continued channel are different mathematical objects. Their comparisons are proved below where they are used.

The scalar and derivative constructions take place on that occupied physical source itself. The explicit gauge-Higgs-fermion Hamiltonian provides a second, constructive realization on a native regulated target. Matching its interaction grammar to the physical source is useful, but does not alone identify their states, operator domains or correlation functions. Conversely, receiving

a physical Higgs channel does not require that it first be obtained as a limit of this particular Hamiltonian.

Target	Source and observations used here
T_{ph}	The occupied interacting Standard Model source with its physical local-observable interface, original Higgs and current/source operations, electroweak continuation, and metric-response prescription. The direct constructions preserve its fixed normalization, state, domains and sibling inputs.
T_{reg}	A finite spatial graph, faithful gauge quotient, Higgs coordinates, finite CAR matter inventory, Gauss-invariant space and interacting Hamiltonian, with mass insertions, currents, dressed observations, exact feedback and metric histories.
T_{as}	The bounded asymptotic algebra of a declared regulator family, its cluster states, continuum equal-time canonical observations and physical time-continuous subalgebra.
T_{hom}	A declared homogeneous realization of the native quantum source and Einstein dynamics. The initial smoothed example and the stronger literal-Hamiltonian, reference-resolved construction retain their respective source and preparation hypotheses.
$T_{\text{YM,pair}}$	The continuum Yang–Mills vacuum source with a bounded auxiliary singlet-pair interaction, a new interacting ground state and a proved gap for that changed Hamiltonian.
T_{res}	A positive-energy continuum-coupled comparison channel with its exact spectral density and marked continuation sheet.
$T_{\text{WDW},B}$	The separately constructed tetrad–Standard Model quantum constraint source, its coefficient/probe responses and nonpolynomial graph limits.

The physical Higgs resonance is received and transported on T_{ph} ; the resonance on T_{res} is a separate constructive example. The native Einstein family has an actual interacting quantum source and is not merely an external background. A separate direct construction below gives the original physical source its own metric-dependent quantum–Einstein action and an occupied stationary member. It does not identify the native family with that physical source. The WDW route is available where its complete hypotheses are met, not mandatory for every gravitational realization.

1.2 One preparation across physical responses

The central direct construction begins with the represented mass insertion O_ρ . Its derivatives

$$K^\mu = \partial^\mu O_\rho, \quad D = \square O_\rho$$

are defined by compact-test pullback on the same field domain. The nonzero physical Higgs channel proves that a real test f gives $\phi = D(f)\Omega \neq 0$, without treating an unstable pole as a spectral eigenvalue. This one preparation admits three comparisons: a nonzero current-generated physical spectral boundary, an exact metric-source contact, and a smooth stress-energy profile.

The metric contact is obtained from the original source protocol $\rho \mapsto \rho + \sigma R(g)$, not appended as an independently chosen stress tensor. At the flat baseline it is

$$I^{\mu\nu} = 2(\partial^\mu \partial^\nu - \eta^{\mu\nu} \square) O_\rho, \quad \eta_{\mu\nu} I^{\mu\nu} = -6D.$$

Its complete source response and its nonzero expectation derivative are proved on the original physical preparation. Separately, the physical vacuum stress and inherited Einstein law give an exact all-test matching condition and a contact-complete equation for dynamical continuation. The mixed curvature contact is not equated with the entire baseline stress.

1.3 The common physical realization

The principal result is Theorem 11.5. It gives one occupied physical source carrying the electroweak observations, the complete originally declared color/composite and asymptotic operations, and

compatible Einstein dynamics. The comparison retains whole source mechanisms and their exact domains, not merely equal values at a selected vacuum. Its input class is independently formed before any terminal observation is evaluated.

At the complete physical target, the gauge-invariant electroweak channels, confined/composite and asymptotic color channels, interacting quantum fields, and dynamical gravity must retain one field inventory, branch, global form, boundary, source, parameter point and authority. Compatibility cannot be replaced by a list of separately occupied marginals [8, 7]. A direct source-preserving construction is sufficient when it provides these comparisons; regulator convergence is a method used by some results below, not an extra universal premise.

The color comparison covers every original input and licensed operation, including all terminally relevant subclasses. Actual witnesses are retained for the positively claimed physical realizations, and every invoked boundary retains its own source authority. This is different from demanding that every pre-boundary configuration produce a hadron. The whole family is compared without selecting only successful observations. Similarly, the direct Einstein action is constructed at the original quantum-action order and admits a stationary member, but does not assert that every preparation is stationary at the same metric.

Part I develops the physical source calculus, complete color-input interpretation, represented Higgs/current channels, actual boundaries and quantum–Einstein action, culminating in their common realization. Part II constructs the interacting native source, its physical inclusions, controlled observations and asymptotic states. Part III develops its all-component metric response and source-attached Einstein continuation. Part IV treats distinct spectral and quantum-constraint realizations and their exact comparison criteria. These complementary constructions retain their respective domains; none is made an additional universal premise of the principal result.

Part I

Common sources and physical observations

2 Existence of a common positive source

BCRS-3 reconstructs a represented source from its complete joint Gram kernel [3]. The existence problem is earlier: when do separately prescribed source values extend to such a positive kernel? We specialize the positive Hahn–Banach extension argument to the common observation space, retaining its precise hypotheses.

Definition 2.1 (Prescribed observation space). Let $\mathcal{A}$ be an independently constructed nonzero unital complex C^* -algebra. Each acquired source has a unital algebra $\mathcal{A}_i$, a state ω_i , and a specified unital $*$ -homomorphism $j_i : \mathcal{A}_i \rightarrow \mathcal{A}$. Let $E \subset \mathcal{A}_{\text{sa}}$ be the real linear span of the common identity, the self-adjoint images of these source algebras, and any independently prescribed bounded self-adjoint mixed observations. The candidate values determine $f : E \rightarrow \mathbb{R}$ when they respect every finite linear relation in $\mathcal{A}$.

The order on E is inherited from the actual common algebra. It is not the collection of orders on the separate source algebras.

Theorem 2.2 (Positive common-state extension). *There exists a state ω on $\mathcal{A}$ extending the prescribed values if and only if f is well-defined, $f(1) = 1$, and*

$$e \in E \cap \mathcal{A}_+ \implies f(e) \geq 0.$$

The theorem does not require an already supplied complete mixed kernel.

Proof. Necessity follows by restriction. For sufficiency, for $a \in \mathcal{A}_{\text{sa}}$, define

$$p(a) = \inf\{f(e) : e \in E, a \leq e\}.$$

The set is nonempty because $\|a\| 1 \in E$. If $a \leq e$, then $e + \|a\| 1 \geq 0$, so $f(e) \geq -\|a\|$. Thus p is finite. Order addition and positive scaling show that p is sublinear. If $e \in E$, positivity of f on $E \cap \mathcal{A}_+$ gives $p(e) = f(e)$.

Real Hahn–Banach extends f to a real linear functional $L : \mathcal{A}_{\text{sa}} \rightarrow \mathbb{R}$ dominated by p . If $a \geq 0$, then $p(-a) \leq 0$, hence $-L(a) = L(-a) \leq 0$. The complex-linear extension ω is therefore positive and has $\omega(1) = 1$. The usual positive-functional norm identity gives $\|\omega\| = 1$. Its restriction is the prescribed f . $\square$

The extension fills previously unspecified mixed values. It need not be unique. If the complex span of E is norm-dense in $\mathcal{A}$, continuity makes it unique. The actual source-completeness requirement must establish such determination, or its appropriate unbounded analogue.

Example 2.3 (An explicit incompatible pair). Let X, Z be Pauli matrices in $M_2(\mathbb{C})$. Prescribe the $X = 1$ pure state on $C^*(X)$ and the $Z = 1$ pure state on $C^*(Z)$. They agree on their scalar intersection. However,

$$b = \sqrt{2}I - X - Z \geq 0, \quad f(b) = \sqrt{2} - 2 < 0,$$

because $(X + Z)^2 = 2I$. No common positive state extends these values. Shared marginal identity therefore does not establish joint order consistency.

2.1 Finite compatibility and finite obstructions

Theorem 2.4 (Finite positive-obstruction criterion). *For bounded self-adjoint $a_\lambda \in \mathcal{A}$ and prescribed real b_λ , a state satisfying $\omega(a_\lambda) = b_\lambda$ for all λ exists if and only if, for every finite set F and every $\varepsilon > 0$, there is a state satisfying*

$$|\omega_{F,\varepsilon}(a_\lambda) - b_\lambda| \leq \varepsilon \quad (\lambda \in F).$$

If no common state exists, there are finitely many such observations, real $c_1, \dots, c_n$ and $t \in \mathbb{R}$ such that

$$t1 - \sum_{j=1}^n c_j a_j \geq 0, \quad t - \sum_{j=1}^n c_j b_j < 0.$$

Proof. The state space is weak-* compact. Each finite system of the displayed approximate equalities defines a weak-* closed subset. These subsets have the finite-intersection property by taking unions of finite test sets and minima of their tolerances. Compactness gives a state satisfying all exact equalities. The converse is immediate.

If no common state exists, a finite set already fails. Its prescribed moment vector lies outside the compact convex image

$$M = \{(\omega(a_1), \dots, \omega(a_n)) : \omega \in S(\mathcal{A})\}.$$

Finite-dimensional strict separation gives c with $\sum c_j b_j > \sup_{m \in M} c \cdot m$. Put $t = \sup_{\omega \in S(\mathcal{A})} \omega(\sum c_j a_j)$. The characterization of order by states gives $t1 - \sum c_j a_j \geq 0$, while strict separation gives its negative prescribed value. $\square$

This theorem does not assume a full mixed state, but it also does not assert that every finite physical test is compatible. Those finite occupants must be constructed. The pure Yang–Mills finite-cap source argument provides an example in which actual positive mixed families, replacement estimates and overlaps are proved [17]. Additional matter and Higgs insertions require their corresponding mixed source comparisons.

2.2 Constraint and covariance consequences

For a bounded constraint c , the equation $\omega(c^*c) = 0$ gives $\pi_\omega(c)\Omega = 0$. It does not give $\pi_\omega(c) = 0$ on the whole representation. The latter requires all the tests $\omega((ca)^*(ca)) = 0$. This distinction is essential when passing from a kinematical algebra to physical observables.

Proposition 2.5 (An occupied observable-preserving constraint space). *Suppose an actual positive source has a normalized cyclic vector Ω in a common closed constraint kernel. Let $\mathcal{B} \subset \mathcal{A}$ be a unital $*$ -subalgebra containing the common identity, whose represented elements preserve that kernel. Then $\pi(\mathcal{B})\Omega$ is a nonzero physical representation space. For unbounded constraints, the statement retains the complete specified operator domains and their invariance.*

Proof. Each $\pi(b)\Omega$ lies in the common kernel. Closedness retains its norm closure. The space contains the unit vector Ω and is invariant under $\mathcal{B}$; the $*$ -property makes the bounded restricted action a representation. $\square$

The proposition receives both the positive source and preservation of its kernel. Constraint commutator closure alone supplies neither. Nor may one replace physical reduction by an arbitrary two-sided quotient: in the canonical polynomial algebra, imposing $p = 0$ in a two-sided quotient together with $[q, p] = i\hbar 1$, $\hbar \neq 0$, forces $1 = 0$.

Suppose source states ω_x, ω_y have an actual $*$ -isomorphism $\alpha_{yx} : \mathcal{A}_x \rightarrow \mathcal{A}_y$ with $\omega_y \alpha_{yx} = \omega_x$. The GNS assignment

$$U_{yx}[a]_x = [\alpha_{yx}(a)]_y$$

is well-defined and isometric by equality on a^*a , is onto, preserves the cyclic vector and intertwines all bounded observations. Composition of the source maps gives literal composition of these unitaries. When a point-norm continuous one-parameter automorphism group preserves a state, the resulting unitaries are strongly continuous: continuity holds on cyclic vectors by the algebra norm, and unitarity extends it to their closure. Stone's theorem supplies the generator. Its spectrum and asymptotic properties remain properties to prove.

2.3 Regularity is an additional physical requirement

Even the Weyl relations admit positive nonregular realizations. On $\ell^2(\mathbb{R})$, with its discrete orthonormal basis e_x , set

$$W(q, p)e_x = e^{ip(x+q/2)}e_{x+q}.$$

These unitaries give a positive vector state and the Weyl phase relations. Nevertheless, $\|W(q, 0)e_0 - e_0\| = \sqrt{2}$ for every $q \neq 0$. There is no strongly continuous translation generator in that variable. Positive extension therefore does not by itself produce the regular unbounded fields or dynamical domain required by a continuum quantum field theory. The occupied construction in Section 12 provides those domains at its fixed spatial regulator; the direct physical constructions use the already represented local-field source at its stated domain.

3 The occupied physical source and its complete insertion calculus

The physical comparison starts directly from the occupied Standard Model source. Its shared premetric architecture, nonempty compatible physical interior and gravitational role are inherited [1, 34, 52, 9, 10]. This section constructs the joint insertion calculus of that source, and Section 4 identifies its represented scalar and current channels. The explicit spatial Hamiltonian developed later is a distinct constructive realization, not an antecedent of this source-first comparison.

Fix one physical source point, including the complete chiral matter and gauge inventory, one Higgs doublet, the charged Yukawa matrices, global gauge form, spin and boundary data, preparation, renormalization and improvement prescription, and parameter point. The electroweak restriction of this source carries the physical spectral packet of SYM-8, Definitions 2.1–2.2 and Theorem 9.3 [20]. Its local source and trace operations are those of SYM-10, Theorem 6.1 [33]. Color, weak, hypercharge, Higgs and charged-lepton siblings remain in that common source when a displayed calculation selects only two of its insertion directions.

Three kinds of object will be kept distinct. A renormalized insertion jet specifies finite source derivatives and their contacts. A represented local observable acts on the actual positive-metric field domain and has unordered positive kernels. A continued physical channel additionally has its actual marked analytic domain, physical ports, boundary procedures and failure data. The occupied source supplies all three for the particular scalar channel used below; its finite-jet calculus does not assign the second and third kinds to every other source direction. This distinction preserves the positive physical content without replacing an operator or an analytic map by its name.

The local metric, current, coupling and dimension-two source coordinates and coincident-counterterm calculus are also explicit in the named local-RG source construction [40]. Its anomaly-free flavor specialization does not remove the anomalous flavor terms retained below.

3.1 One relative source and all finite contacts

Use finitely many real compact smooth probe coordinates $j = (j^1, \dots, j^n)$ in the fixed source chart. Complex Yukawa variations mean their independent real and imaginary coordinates, with the conjugate terms varied together. Let $F(j)$, $F(0) = 0$, be the source's complete renormalized relative action in its admitted baseline comparison. Its derivatives

$$F_I = \partial_I F(0), \quad I \neq \emptyset,$$

include source coefficients, counterterms, composite mixing, metric densities and the dependent coframe and torsion-free spin connection. All the insertions used here are Grassmann even. The chiral interactions remain composite fermion bilinears, not unpaired odd probes.

Let ω_* and $T_{R,k}$ be the preparation and symmetric multilinear time-ordering maps of this same source. They have the unit and central-factorization properties, and $T_{R,1}(A) = A$ on already renormalized insertions. Write

$$Z(j) = \sum_{k \geq 0} \frac{i^k}{k!} \omega_*(T_{R,k}(F(j), \dots, F(j))), \quad W(j) = -i \log Z(j), \quad Z(0) = 1. \quad (1)$$

For a formal source this denotes its finite jet at each separately fixed order: only $k \leq |I|$ contributes to $\partial_I Z(0)$. When the actual relative functional is differentiable it denotes that functional, with the logarithm branch fixed at one. No convergence of an infinite perturbative series follows from the finite notation.

For even insertions put

$$\kappa_T(A_1, \dots, A_k) = \sum_{\pi \in \mathcal{P}(\{1, \dots, k\})} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{B \in \pi} \omega_*(T_R(A_b : b \in B)),$$

where $\mathcal{P}(I)$ is the set of partitions of the labelled set I . In particular $\kappa_T(A) = \omega_*(A)$.

Theorem 3.1 (Complete finite source jets). *At the actual finite-jet or differentiability level of the fixed source,*

$$\partial_I W(0) = \sum_{\pi \in \mathcal{P}(I)} i^{|\pi|-1} \kappa_T(F_B : B \in \pi). \quad (2)$$

Thus the joint source responses are computed on one preparation and prescription, without independently choosing their mixed pairings.

Proof. Since $F(0) = 0$, every surviving interaction factor in a derivative of Z receives a nonempty set of labels. Ordering its k blocks gives the factor $k!$ cancelling the denominator in (1). Consequently

$$Z_I = \sum_{\pi \in \mathcal{P}(I)} i^{|\pi|} \omega_*(T_R(F_B : B \in \pi)).$$

The finite partition formula for derivatives of $\log Z$ at $Z = 1$ groups these block insertions into their connected cumulant. Multiplication by $-i$ gives (2). Every sum is finite at the chosen order. In an actual differentiable realization these are ordinary product and chain rules. $\square$

In particular,

$$\begin{aligned} W_a &= \omega_*(F_a), \\ W_{ab} &= \omega_*(F_{ab}) + i\kappa_T(F_a, F_b), \\ W_{abc} &= \omega_*(F_{abc}) + i\{\kappa_T(F_{ab}, F_c) + \kappa_T(F_{ac}, F_b) + \kappa_T(F_{bc}, F_a)\} - \kappa_T(F_a, F_b, F_c). \end{aligned} \quad (3)$$

The correct connected two-insertion operation is therefore $(W_{ab} - \omega_*(F_{ab}))/i$. A source-only term $q(j)1$ multiplies Z by e^{iq} , so it adds q to W and its actual derivative to F_{ab} . It cancels in this connected operation. This does not delete the distinct local extension terms already present in $T_R(F_a, F_b)$.

If the preparation or multilinear prescription actually moves, use $M_k(j) = \omega_j \circ T_{R,k}(j)$. The complete formula is

$$\begin{aligned} Z_I &= \sum_{S \subseteq I} \sum_{\pi \in \mathcal{P}(I \setminus S)} i^{|\pi|} (\partial_S M_{|\pi|})(0)(F_B : B \in \pi), \\ W_I &= -i \sum_{\pi \in \mathcal{P}(I)} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{B \in \pi} Z_B. \end{aligned} \quad (4)$$

Here $M_0 = 1$ and the empty partition has its usual meaning. Assigning labels first to M_k , then to the interaction factors, proves the first line; the logarithm proves the second. A fixed baseline prescription can represent moving metric sources only through the source's actual action-principle transport. It is not an identification with an independently renormalized moving vacuum.

3.2 Original metric and trace operations

The scalar coordinate is the coefficient ρ in $+\int dV_g \rho [H^\dagger H]_R$. It has engineering dimension two; if the action uses $-m_H^2 H^\dagger H$, then $\rho = -m_H^2$. For definiteness in this paragraph use the inverse-metric Weyl coordinate

$$g^{\mu\nu}(\sigma) = e^{2\sigma} g_*^{\mu\nu}, \quad \Theta = \frac{2}{\sqrt{|g|}} g^{\mu\nu} \frac{\delta S_R}{\delta g^{\mu\nu}}.$$

An opposite stress convention transports this coordinate and every trace contact together. The explicit four-dimensional density gives

$$F_{\rho(f), \sigma(h)}^{\text{density}} = -4O_\rho(fh), \quad F_{\rho(f), \sigma(h), \sigma(k)}^{\text{density}} = 16O_\rho(fhk).$$

These are density terms, not the complete mixed insertion. Composite, curvature, coframe and source-prescription derivatives remain. The engineering scalar trace row is instead $-2\rho O_\rho$, or $+2m_H^2 O_\rho$, before its anomalous mixing and EOM/improvement completion. The factors four and two answer different questions.

The iterated Weyl vector field also differentiates its own coefficient:

$$\begin{aligned} D_\Theta(h)D_\Theta(k)W &= 4 \int dx dy h(x)k(y)g^{\mu\nu}(x)g^{\alpha\beta}(y)W_{g^{\mu\nu}(x),g^{\alpha\beta}(y)} \\ &\quad + 4 \int dx h(x)k(x)g^{\mu\nu}(x)W_{g^{\mu\nu}(x)}. \end{aligned}$$

This contact is already included when the finite nonlinear metric path is differentiated; it is not added twice. For example $q = c \int dV_g \rho^2$ contributes $q_{\rho(f)\rho(k)} = 2c \int f k$ and $q_{\rho(f)\rho(k)\sigma(h)} = -8c \int f k h$. A permitted $\rho R(g)1$ term, or $R(g)[H^\dagger H]_R$ improvement, can have a nonzero metric derivative at flat space. Setting curvature to zero before differentiating loses that response.

Write the complete local trace identity of the same source as

$$D_\Theta(h)W = B_C(h)W + V_R(h)W + \mathcal{A}_W(h; j), \quad B_C(h) = \int h(x)b_C(j; x)\frac{\delta}{\delta c(x)}dx. \quad (5)$$

The color coordinate c includes its actual source-normalization Jacobian. The remainder V_R retains weak and hypercharge kinetics, quartic and all charged Yukawa coordinates, the dimension-two scalar row, virial/improvement, EOM/contact, current and background operations. The local identity/anomaly functional $\mathcal{A}_W$ is not set to zero at a flat baseline. Equality has the operator, correlator or source authority actually supplied by the local trace theorem [33].

Proposition 3.2 (Trace adjunction with every coefficient contact). *At a constant-coupling point with the actual coefficient $b_* = b_C(j_*) \neq 0$, every nonempty labelled set I satisfies*

$$\begin{aligned} b_* \partial_I \partial_{c(h)} W &= \partial_I \{ D_\Theta(h)W - V_R(h)W - \mathcal{A}_W(h; j) \} \\ &\quad - \sum_{\emptyset \neq S \subseteq I} (\partial_S B_C(h)) \partial_{I \setminus S} W, \end{aligned} \quad (6)$$

with all quantities evaluated at j_* .

Proof. Differentiate (5). The subset product rule for $\partial_I(B_C W)$ has one term with no derivative on B_C , namely $b_* \partial_{c(h)} \partial_I W$. Move the others to the right. Only the stated nonzero coefficient is divided out. In particular nonzero gauge coupling is not itself a proof of $b_* \neq 0$. $\square$

If the flavor Ward identity is $G_\alpha W = \mathcal{A}_F(\alpha; j)$, a quotient-beta representative change has the exact form

$$V' = V - G_{hS(j)}, \quad \mathcal{A}'_W = \mathcal{A}_W + \mathcal{A}_F(hS(j); j), \quad V = B_C + V_R.$$

Thus $V'W + \mathcal{A}'_W = VW + \mathcal{A}_W$. Differentiation retains $(\partial_a V')W$ and $\partial_a \mathcal{A}'_W$, including the source dependence of S . Gauged Standard-Model anomaly cancellation does not remove an anomalous flavor/background current or its phase data.

At an actually represented common invariant domain, an anchored operator identity $\Theta = b_* C + R$ yields $C = (\Theta - R)/b_*$, including every ordered word obtained by substitution. Its Gram columns are therefore determined by the actual Θ, R and sibling columns and are positive by squared norms. This is a useful conditional ordered realization, not a new domain deduced from (6) alone. More generally, in a fixed finite independent coefficient space, a desired operator row ℓ is recoverable from known rows A exactly when ℓ lies in the row space of A . Indeed $\ell = uA$ constructs it; otherwise finite-dimensional duality gives $v \in \ker A$ with $\ell v \neq 0$, so the known rows do not determine the requested one. Quotient relations may be used only at their licensed equality level. This prevents a single trace row from being mistaken for separate access to every chiral or color field.

3.3 Genuine Higgs-dressed quark insertions

Use the right-handed-field convention of the physical flavor source [35, 20]; the conjugate-Weyl presentation has its corresponding contragredient generators and tensor conjugations. With $\tilde{H} = i\sigma_2 H^*$, define the original source words

$$B_{u,ij} = \bar{u}_{Rj} \tilde{H}^\dagger Q_{Li}, \quad B_{d,ij} = \bar{d}_{Rj} H^\dagger Q_{Li}, \quad \Phi(H) = (\tilde{H}, H).$$

Spin and color contractions are those of the original Lorentz-scalar Yukawa interaction; the indices displayed are generation indices.

Lemma 3.3 (Polynomial full-source Higgs dressing). *The words are gauge-invariant Lorentz scalars of engineering dimension four, defined also at $H = 0$. They obey*

$$\Phi^\dagger \Phi = \Phi \Phi^\dagger = (H^\dagger H) I_2, \quad B_u \mapsto V_Q B_u V_u^\dagger, \quad B_d \mapsto V_Q B_d V_d^\dagger.$$

Their Hermitian original complex-Yukawa tangent is

$$O_Z^0 = - \int dV_g \sum_{r=u,d} \text{Tr}(Z_r B_r^\dagger + Z_r^\dagger B_r). \quad (7)$$

Its physical insertion F_Z is the complete renormalized derivative in that coordinate, including all mixing and counterterm derivatives.

Proof. The hypercharges of $\tilde{H}^\dagger Q_L$ and $H^\dagger Q_L$ are $2/3$ and $-1/3$; the barred right fields cancel them. Each weak factor is a singlet, and each color triplet is contracted with its dual. Thus the whole word is invariant under the inherited gauge quotient. Its dimension is $1 + 3/2 + 3/2 = 4$. If $H = (h_1, h_2)^T$, then $\tilde{H} = (\bar{h}_2, -\bar{h}_1)^T$; multiplication proves the two matrix identities. Generation transformations act only on the two free indices, giving the displayed laws. Finally differentiate $-\int \sum_r \text{Tr}(Y_r B_r^\dagger + Y_r^\dagger B_r)$ along a complex tangent regarded as a real direction. This gives (7) and its actual source completion. $\square$

No division by $\sqrt{H^\dagger H}$ or globally unitary Higgs frame is used. The polynomial identity does not multiply independently renormalized composites without their normalization contacts. For a nonzero entry of Z , the corresponding independent Grassmann monomial and a nonzero Higgs component exhibit a nonzero source word. Invertible perturbative renormalization retains its leading direction. This does not make every expectation or pole contraction nonzero.

The actual mixed operation is

$$C_{\rho,Z}(f, Z) = \frac{W_{\rho(f),Z} - \omega_*(F_{\rho(f),Z})}{i} = \kappa_T(F_{\rho(f)}, F_Z). \quad (8)$$

The first slot is the physical scalar detector identified below; the second is the full Higgs-dressed quark derivative. A vanishing canonical $F_{\rho,Z}^0$ in independent coordinates neither sets this connected response to zero nor removes a renormalized contact. Additional weak, color, lepton, Higgs and metric siblings use Theorem 3.1. The dressing also obeys

$$D_\mu^u(\tilde{H}^\dagger Q_L) = (D_\mu \tilde{H})^\dagger Q_L + \tilde{H}^\dagger D_\mu Q_L,$$

and the analogous down formula. Weak connections cancel by the covariant product rule, rather than being turned off.

3.4 Broken-current and fixed-sibling source identities

For a Hermitian traceless generation matrix T , rotate

$$Q_L \mapsto e^{-i\alpha T} Q_L, \quad u_R, d_R \mapsto e^{i\alpha T} (u_R, d_R).$$

This commutes with the complete gauge representation. In the convention $\delta\mathcal{L}_{\text{kin}} = -J_T^\mu \partial_\mu \alpha$, its current word and Yukawa tangent are

$$J_T^\mu = -\bar{Q}_L \gamma^\mu T Q_L + \bar{u}_R \gamma^\mu T u_R + \bar{d}_R \gamma^\mu T d_R, \\ V_T Y_r = -i\{T, Y_r\}, \quad V_T Y_r^\dagger = +i\{T, Y_r^\dagger\}.$$

Direct field variation gives, on the classical equations, the canonical divergence row

$$\partial_\mu J_T^\mu = \mathcal{D}_T, \quad \mathcal{D}_T = i \sum_{r=u,d} \text{Tr}(\{T, Y_r\} B_r^\dagger - \{T, Y_r^\dagger\} B_r).$$

Its renormalized version is the actual derivative $F_{V_T Y}$, with Ward, EOM, mixing and anomaly contacts. It is not a conserved current at a generic fixed Yukawa point.

Introduce $+\int a_\mu J_T^\mu$. For one fixed generator at zero background, $\delta a = d\alpha$. Let $\mathcal{V}_\alpha$ include this source variation and $\alpha V_T Y$, retaining

$$\mathcal{V}_\alpha W = \mathcal{A}_\alpha. \tag{9}$$

Several axial triples $(-T, T, T)$ are not a non-Abelian Lie algebra: their bracket is a vector direction. Non-Abelian or nonzero-background extensions therefore use the full inherited left/right $U(3)^5$ source family or an actual closed subgroup, with its vector and anomaly rows [35, 36, 21].

Proposition 3.4 (Computed current–Yukawa contact). *For a source-independent Yukawa tangent Z , put $(V_\alpha Z)_r = -i\alpha\{T, Z_r\}$, with its conjugate variation. On the preceding restriction,*

$$\mathcal{V}_\alpha W_Z = (\mathcal{A}_\alpha)_Z - W_{V_\alpha Z}, \\ \mathcal{V}_\alpha W_{Z,s} = (\mathcal{A}_\alpha)_{Z,s} - W_{V_\alpha Z,s}, \\ \mathcal{V}_\alpha W_{Z,K} = (\mathcal{A}_\alpha)_{Z,K} - W_{V_\alpha Z,K} - W_{Z,V_\alpha K},$$

where s is an independent flavor-inert sibling and K is a second Yukawa direction.

Proof. The source coefficients are linear in Y , so $[\partial_Z, \mathcal{V}_\alpha] = \partial_{V_\alpha Z}$. Differentiate (9) and move this coefficient contact to the right. A derivative in s commutes with the vector field. One in K contributes its first coefficient contact; the second derivative of the vector-field coefficient is zero. All source insertions then use (2) or (4). $\square$

3.5 The mass-shaped map retains its weak and Higgs terms

The complete quark interaction admits the polynomial rewrite

$$M_H = \Phi(H) \text{diag}(Y_u, Y_d) = \begin{pmatrix} H^{0*} Y_u & H^+ Y_d \\ -H^{+*} Y_u & H^0 Y_d \end{pmatrix}, \tag{10} \\ \mathcal{L}_{Yq} = -\bar{q}_L M_H q_R - \bar{q}_R M_H^\dagger q_L.$$

This is an action-word map, not substitution of a quantum Higgs field into an independent QCD functional. In particular

$$d_Y M_H[Z] = \Phi(H)Z, \quad d_H M_H[k] = \Phi(k)Y, \quad d_H d_Y M_H[k, Z] = \Phi(k)Z,$$

where Y, Z denote the up/down block diagonal matrices. Along moving coordinates, $M_{H,ab} = \Phi(H_{ab})Y + \Phi(H_a)Y_b + \Phi(H_b)Y_a + \Phi(H)Y_{ab}$. These are real Higgs derivatives. An original ρ derivative in a fixed field frame has $H_a = 0$; it is not differentiation of a vacuum value inside a quantum insertion.

The original left/right electroweak connections remain

$$\ell_\mu = g_2 W_\mu^a t_a + \frac{g_1}{6} B_\mu I, \quad r_\mu = g_1 B_\mu \text{diag}(2/3, -1/3),$$

with the common color connection as well. A rotation $\delta q_L = i\alpha A_L q_L$, $\delta q_R = i\alpha A_R q_R$ with these connections fixed contributes

$$\Delta_{\text{weak}} = i\bar{q}_L \gamma^\mu [\ell_\mu, A_L] q_L + i\bar{q}_R \gamma^\mu [r_\mu, A_R] q_R$$

and $\Delta_Y = i\bar{q}_L (A_L M_H - M_H A_R) q_R + \text{h.c.}$. The divergence is $-\Delta_{\text{weak}} - \Delta_Y$, before its quantum contact completion. The weak commutators vanish for the generation action above, but not for an arbitrary rotation between inequivalent electroweak components.

In an admitted split $H = h + \eta$, $h = (0, v/\sqrt{2})^T$,

$$M_H = M_h + M_\eta, \quad M_h = \frac{v}{\sqrt{2}} \text{diag}(Y_u, Y_d).$$

Apply the same linear renormalized insertion prescription after this source-word decomposition. Both terms remain in each insertion slot; the fluctuation correlation is not an error discarded in a limit. Holding M_h fixed gives $\delta Y = -(Y/v)\delta v$, which still changes the fluctuation coupling. A mere re-split of the unchanged field instead has $\delta\eta = -\delta h$.

The mass-spurion and heavy-current constructions of SYM-6 remain positive inputs at their stated targets [36]. Its conserved massless nonsinglet QCD current, nonzero axial vacuum variation and Goldstone spectral channel retain the registered massless and weak/QED-off conditions. In the full source the actual breaking/current correlator above remains. Restricted-current conservation cannot be used to erase it. Likewise the physical charm/bottom hadron records and selected heavy-current amplitudes matched through their registered inverse-mass and coupling order retain their matching remainder; top is not in that same hadronic record. These statements neither undo those occupations nor identify all their domains or continuations with the full source by reversing decoupling. A confined/composite attachment must retain these weak, Higgs, breaking and source observations [5].

3.6 Channel variations and an actual lawful flavor family

For two fixed original mass-detector labels 1, 2, define $C_{12} = (W_{12} - \langle F_{12} \rangle)/i$ with the source's normalized relative insertion expectation. In a fixed relative frame, differentiating normalized cumulants gives

$$\partial_a \kappa(A_1, \dots, A_n) = \sum_j \kappa(A_1, \dots, (A_j)_a, \dots, A_n) + i\kappa(A_1, \dots, A_n, F_a).$$

This follows by adjoining independent linear probes for the A_j and differentiating their connected generating functional. Consequently the original first and mixed second channel variations are

$$C_{12;a} = \kappa(F_{1a}, F_2) + \kappa(F_1, F_{2a}) + i\kappa(F_1, F_2, F_a), \quad (11)$$

$$\begin{aligned} C_{12;ab} = & \kappa(F_{1ab}, F_2) + \kappa(F_{1a}, F_{2b}) + \kappa(F_{1b}, F_{2a}) + \kappa(F_1, F_{2ab}) \\ & + i\{\kappa(F_{1a}, F_2, F_b) + \kappa(F_1, F_{2a}, F_b) + \kappa(F_{1b}, F_2, F_a)\} \\ & + i\{\kappa(F_1, F_{2b}, F_a) + \kappa(F_1, F_2, F_{ab})\} \\ & - \kappa(F_1, F_2, F_a, F_b). \end{aligned} \quad (12)$$

The first formula requires the derivative of the whole subtracted expectation: $\partial_a \langle F_{12} \rangle = \langle F_{12a} \rangle + i\kappa(F_{12}, F_a)$. Subtracting only $\langle F_{12a} \rangle$ would leave an erroneous contact. A physical source path adds $C_{12;IJ} j_a^I j_b^J + C_{12;IJ} j_{ab}^I$; moving detector tests and normalizations also differentiate. For a moving preparation or prescription use (4). These are third/fourth original-source operations, not the two-point Higgs–Yukawa response (8).

Theorem 3.5 (Anomaly-free local flavor transport and fixed siblings). *Choose a permutation of $T = \text{diag}(1, -1, 0)/2$. On the admitted contractible local source patch, the finite map*

$$Y_{r,\alpha} = e^{-i\alpha T} Y_r e^{-i\alpha T}, \quad a_\alpha = a + d\alpha, \quad r = u, d \quad (13)$$

constructs a lawful presentation of the occupied source for real compact smooth α , with the fields, preparation, renormalization and boundary data co-transported. In its Ward-preserving representative the calibrated invariant Higgs channel is unchanged on its entire admitted marked sheet. At fixed current sibling instead,

$$W[Y_\alpha, 0] = W[Y, -d\alpha]. \quad (14)$$

Actual local anomaly-representative contacts accompany these equations if another prescription is used.

Proof. Substitute the local generation transformations in the kinetic and Yukawa action. The derivative of the quark transformation is cancelled by $a + d\alpha$; the interaction is exactly its bifundamental spurion transform. The inverse map is the $-\alpha$ map. Compact support keeps the state and boundary endpoints fixed. Pulling back the actual field/test operations and relative prescription along this invertible map constructs the source presentation; no new quark Fock-space implementer or summation of a formal Ward equation is required.

The anomaly check retains the full chiral inventory. In all-left notation the quark axial background charge is $-t$ on Q_L, u_R^c, d_R^c . Mixed triangles with one such insertion and two Standard-Model gauge fields, and its gravitational triangle, vanish by $\text{Tr } T = 0$. The cubic coefficient is proportional to $-12 \text{Tr } T^3 = 0$. The two-background/one-hypercharge coefficient is

$$\text{Tr } T^2 \{6(1/6) + 3(-2/3) + 3(1/3)\} = 0.$$

The other mixed non-Abelian single-generator traces vanish. Thus this restricted local Abelian background has no nontrivial triangle class. No statement about every large/global transformation or a general non-Abelian flavor background follows. Any actual coboundary/prescription term is transported. On its admitted path the general relation is

$$W[j_\alpha] - W[j] = \int_0^1 \mathcal{A}_\alpha(j_{t\alpha}) dt;$$

in the specified Ward-preserving representative the right side is zero.

The invariant Higgs detector is carried into itself, so its fully calibrated distribution is its original distribution. Its existing authorized continuation is therefore the same complete germ, not only the same pole. Applying the map to current background $-d\alpha$ gives (14). Neither side of that equation has been asserted equal to $W[Y, 0]$. $\square$

For $\alpha = sf + tg$, let $Z_f Y = -if\{T, Y\}$, let $L_f a = df$, and put $U_{fg} Y = -fg\{T, \{T, Y\}\}$, with all conjugate variations. The fully transported channel satisfies

$$C_{12;Z_f+L_f} = 0, \quad C_{12;(Z_f+L_f)(Z_g+L_g)} + C_{12;U_{fg}} = 0.$$

The fixed-sibling identity instead gives

$$C_{12;Z_f} = -C_{12;L_f}, \quad C_{12;Z_f Z_g} + C_{12;U_{fg}} = C_{12;L_f L_g}.$$

The full formulas (11)–(12) and their normalization contacts occur in each term. The zero combined response has an actual zero continuation. It does not independently continue its separate current or Yukawa summands. If one side of the fixed-sibling experiment has an actual parameter-regular continued family, the licensed source equality transports it to the other.

For two constant noncommuting generation directions T, S , the actual group path $Y(s, t) = e^{-isT} e^{-itS} Y e^{-itS} e^{-isT}$ has $Y_{st} = -\{T, \{S, Y\}\}$. Reversing the order changes this contact by $-[[T, S], Y]$, the required vector direction. This is why the complete left/right source family, not an imagined non-Abelian group of axial directions alone, is used.

3.7 The exact radial Higgs map and its total Euler insertion

The represented descendant constructed in Theorem 4.4 has a full interaction meaning. Use the physical signature $(+ - - -)$ and retain the signed kinetic coefficient k_H . At the canonical source-word level write

$$\mathcal{L}_{HY} = k_H K + \rho O - \lambda O^2 + \mathcal{L}_Y, \quad K = (D_\mu H)^\dagger D^\mu H, \quad O = H^\dagger H,$$

where $\mathcal{L}_Y$ contains all up, down and charged-lepton Yukawa contractions and their conjugates. Gauge, metric and all other source terms remain in the same action.

Proposition 3.6 (Finite radial source map). *For real compact smooth α , the field change $H \mapsto e^\alpha H$ has the exact coefficient map*

$$\begin{aligned} k_{H,\alpha} &= e^{2\alpha} k_H, & \lambda_\alpha &= e^{4\alpha} \lambda, & Y_{r,\alpha} &= e^\alpha Y_r \quad (r = u, d, e), \\ \rho_\alpha &= e^{2\alpha} \{\rho - \operatorname{div}(k_H d\alpha) - k_H (d\alpha)^2\}, \end{aligned} \tag{15}$$

and $S_0[e^\alpha H; j] = S_0[H; j_\alpha]$.

Proof. The up-conjugate doublet scales by the same real factor as H . Every Yukawa word therefore scales once, while O and O^2 scale twice and four times. Covariant differentiation gives

$$K(e^\alpha H) = e^{2\alpha} \{K + \partial_\mu \alpha \partial^\mu O + (d\alpha)^2 O\}.$$

Integrating the middle term by parts with its actual factor $e^{2\alpha} k_H$ gives the stated mass coefficient, including the negative gradient-square term. Compact support removes the boundary term. All identities use the original covariant derivative. $\square$

This is an invertible field/source reparametrization, not a scale symmetry with all coefficients fixed. At $\alpha = tf$, let V_f vary k_H, ρ, λ, Y_r by $2fk_H, 2f\rho, 4f\lambda, fY_r$, omitting only the derivative term in $\delta\rho$. If $E_f = \delta_{H=fH} S_0$, then

$$E_f = F_{V_f}^0 - O(\operatorname{div}(k_H df)).$$

At constant $k_H = 1$ this is the local off-shell Euler identity

$$\square(H^\dagger H) = 2(D_\mu H)^\dagger D^\mu H + 2\rho H^\dagger H - 4\lambda(H^\dagger H)^2 + \mathcal{L}_Y - E_H. \tag{16}$$

Equivalently expand $\square(H^\dagger H) = 2(DH)^\dagger DH + H^\dagger D^2 H + (D^2 H)^\dagger H$ and the full scalar Euler variation. The term E_H is not silently zero in quantum insertions.

At that flat $k_H = 1$ baseline, in the actual source frame $S_R = S_0 + S_{\text{ct}}$, put

$$E_{R,f} = \delta_{H=fH} S_R, \quad \mathcal{R}_f = \delta_{H=fH} S_{\text{ct}} - (S_{\text{ct}})_{V_f} + (S_{\text{ct}})_{\rho(\square f)}.$$

Adding these actual counterterm derivatives to the preceding identity proves the complete source equation

$$F_{\rho(\square f)} = F_{V_f} + \mathcal{R}_f - E_{R,f}. \tag{17}$$

Its left side is the represented operator $D(f)$ of the next section. Thus the total right-side source operation has that representation and physical channel. Equality of the sum does not assign separate domains or nonzero residues to each kinetic, quartic, quark, lepton, Euler or counterterm summand.

The actual change-of-variables prescription also retains its measure contact: schematically in a fixed-product frame,

$$0 = \omega(\delta_f A) + i\omega(AE_{R,f}) + \omega(A\mathcal{J}_f),$$

with the connected subtraction and any prescription derivatives. Here $\mathcal{J}_f$ is the source's logarithmic Jacobian/contact, not an adjustable term. The leading scalar variation $\delta_f O(h) = 2O(fh)$ receives its own actual mixing corrections. Dropping $E_{R,f}$ inside all products would erase these contacts. Writing $V_f = V_{q,f} + V_{\text{rest},f}$, with V_q the up/down Yukawa rows only, gives the exact quark/source comparison

$$F_{V_{q,f}} = F_{\rho(\square f)} - F_{V_{\text{rest},f}} - \mathcal{R}_f + E_{R,f}.$$

The retained remainder is interaction content, not a term proved small.

For $\alpha = sf + th$ at constant k_H , the nonlinear contacts are

$$\begin{aligned} \rho_{st} &= 4fh\rho - 2k_H(f\square h + h\square f) - 2k_H\partial_\mu f \partial^\mu h, \\ (k_H)_{st} &= 4fhk_H, \quad \lambda_{st} = 16fh\lambda, \quad (Y_r)_{st} = fhY_r. \end{aligned}$$

They follow by differentiating (15), not by repeating only its first tangent. Variable k_H and metric variations differentiate the displayed divergence, contractions and volume. Together with (4), these formulas retain the full source, including its chiral and geometric contacts.

These identities provide the original interaction-sensitive operations used throughout the physical construction. Their complete signatures, readers and domains are compared in Theorem 10.7; the whole compatible family enters the common realization of Theorem 11.5. The color, Higgs, current and metric directions are thereby retained as operations of this one source, with their respective represented, continued or coefficientwise authority.

4 Represented Higgs and current channels on the physical source

This section constructs physical channels directly on the occupied source of Section 3. The starting scalar is the invariant detector $H^\dagger H$, with its actual physical correlator and continuation, not an elementary radial field renamed invariant. The physical local-observable interface supplies its positive-metric representation, vacuum, translation spectrum and common field domain [34, 20]. The following maps introduce neither an independent state nor a required equality with the finite regulator. They also do not turn the unstable Higgs pole into a stable one-particle eigenvalue.

4.1 The original mass coefficient selects the physical detector

Definition 4.1 (The calibrated original scalar port). On the fixed flat constant-parameter minimal-SM fiber, let O_ρ be the complete original insertion for $+\int dV_g \rho H^\dagger H$. Let O_H be the certified renormalized invariant Higgs detector of the same source. Retain their actual invertible normalization and identity mixing

$$O_\rho = nO_H + b1, \quad n \in \mathbb{R} \setminus \{0\}.$$

If their time-ordering extension conventions differ, retain the actual local contact transport L . Their connected operational channels are consequently

$$G_\rho(z) = n^2 G_H(z) + L(z). \tag{18}$$

The numbers and contact map are fixed by the source prescription, not chosen to set a residue. Here z is a physical channel variable, not a source coordinate.

Lemma 4.2 (The flat dimension-two invariant line). *The nonconstant Hermitian Lorentz-scalar proper-gauge-invariant field-polynomial grammar of engineering dimension two in this one-doublet minimal source is the real line generated by $H^\dagger H$. The original mass derivative has the invertible normalization of Definition 4.1.*

Proof. A Higgs field has dimension one, a fermion dimension $3/2$, and a field strength dimension two. A single antisymmetric field strength cannot be contracted to a Lorentz scalar; a gauge potential is not a proper-gauge-invariant polynomial. Fermion bilinears are too large, and one derivative cannot turn a single doublet into an invariant scalar. Of two scalar factors, the hypercharge-neutral invariant endomorphisms of the doublet are scalar multiples of the identity. The candidate ϵHH also vanishes for identical commuting fields and has the wrong hypercharge. The only nonconstant Hermitian line is therefore the one stated. An invertible renormalization of the original mass derivative retains its nonzero leading coefficient on that line; a projection to zero would not be such a normalization. Its field-independent mixing is separately recorded. $\square$

The lemma concerns the stated flat polynomial grammar, not arbitrary nonpolynomial observables, BRST ghosts or curved/source-dependent counterterms. In particular $R(g)1$ is an allowed dimension-two background mixing away from this fiber. Its derivatives and those of a ρR source contact remain in the metric comparison. If the action uses $-m_H^2 H^\dagger H$, the original m_H^2 insertion is $-O_\rho$; odd mixed responses retain that sign.

For compact tests f, g , Theorem 3.1 constructs

$$C_\rho(f, g) = \frac{W_{\rho(f), \rho(g)} - \omega_*(F_{\rho(f), \rho(g)})}{i} = \kappa_T(O_\rho(f), O_\rho(g)). \quad (19)$$

This is the actual boundary distribution of G_ρ . The subtraction contains the complete source seagull, including nonlinear coordinates and counterterm derivatives. Identity mixing drops from connected cumulants, but the independent extension contact L in (18) remains. On this translation-invariant fiber a finite local contact is a momentum polynomial: it is regular at an isolated physical pole, yet matters for the full regular germ.

Theorem 4.3 (Physical mass-source channel and full germ). *The map*

$$(f, g) \mapsto C_\rho(f, g) \mapsto n^{-2}(C_\rho - C_{\text{loc}}) \mapsto \text{the detector's authorized continuation}$$

recovers the actual physical Higgs channel with its full marked germ, current/source normalization, principal part and boundary/failure data. At an isolated pole its nonzero leading principal coefficient cannot be lost under this source normalization.

Proof. The physical packet supplies the invariant detecting correlator and its actual continuation [20]; Lemma 4.2 identifies its nonconstant insertion with the original source line. Equation (19) computes that same distribution and (18) transports its complete convention. Following the continuation of this identified distribution therefore gives the source-owned channel, not an inferred continuation of an arbitrary response.

Near an isolated pole p_H , write

$$G_H(z) = \sum_{k=1}^r c_{-k}(z - p_H)^{-k} + \sum_{m \geq 0} c_m(z - p_H)^m, \quad c_{-r} \neq 0.$$

The leading mass-normalized coefficient is $n^2 c_{-r} \neq 0$. For a positively oriented small contour on that sheet,

$$c_{-k} = \frac{1}{2\pi i} \int_\Gamma (z - p_H)^{k-1} G_H(z) dz.$$

At a marked regular point its convergent Taylor germ retains every derivative. The meromorphic identity theorem determines the actual channel on the connected admitted domain. Thus it instantiates the analytic-germ interface of BCRS–IIEW/H [4], rather than substituting a pole list or a finite source jet for that interface. $\square$

This theorem neither derives pole existence from the radial Hessian nor assumes every pole simple. The physical source determines the actual local sheet, real/complex character and residue. If a supplied calculation is only formal through an order, the source-derivative identity has that accuracy; it does not sum the series. The full-germ claim refers to the actual continued detecting channel already received. No global sheet beyond its authority is added.

4.2 The physical charged and neutral current ports

Physical detector normalization is dual to its original source. For $O' = nO + b1$, use $\rho' = \rho/n$ and retain the resulting central action term. More generally $O' = ZO$ and the action pairing $j^T O$ require $j' = Z^{-T} j$; the analytic bilinear transforms as $C' = ZCZ^T$. An unordered positive Gram has its own adjoint convention. Parameter-dependent frame changes differentiate all these factors and the local contact transport.

The W/Z ports likewise come from the same original coupling map. With $Q = T_3 + Y$, $T_a = \sigma_a/2$,

$$s_W = \frac{g_1}{\sqrt{g_1^2 + g_2^2}}, \quad c_W = \frac{g_2}{\sqrt{g_1^2 + g_2^2}}, \quad e = \frac{g_1 g_2}{\sqrt{g_1^2 + g_2^2}},$$

substitution of $A = s_W W^3 + c_W B$, $Z = c_W W^3 - s_W B$ gives

$$g_2 T_3 W^3 + g_1 Y B = e Q A + \frac{g_2}{c_W} (T_3 - s_W^2 Q) Z. \quad (20)$$

For $W^\pm = (W^1 \mp iW^2)/\sqrt{2}$ and $T_\pm = T_1 \pm iT_2$, the same substitution gives

$$g_2 (T_1 W^1 + T_2 W^2) = \frac{g_2}{\sqrt{2}} (T_+ W^+ + T_- W^-).$$

Conjugate-Weyl fields use $-T^T$ and reversed hypercharge; integer hypercharge $6Y$ has the reciprocal coupling change. Flavor rotations transport the charged-current mixing matrix and its conjugate channel, rather than only moving Yukawa coordinates.

These are original current normalizations, not computations of the full loop vertex. The occupied physical amplitude retains its interacting vertices, dressing, boundary and infrared conditions [20, 4]. On an actually admitted finite mixing chart,

$$G = J_L D^{-1} J_R + B_{\text{reg}}.$$

If its Nielsen comparison is $D_\xi = AD + DB$, solve $L_\xi = AL$, $R_\xi = RB$ and use

$$D = LD_0 R, \quad J_L = J_{L0} R, \quad J_R = LJ_{R0}.$$

Direct multiplication cancels the endpoint factors in the inverse term; the regular physical background is transported as well. At an actual simple determinant zero the residue is

$$\frac{J_L(p) \text{adj } D(p) J_R(p)}{\partial_z \det D(p)}.$$

Nonzero detection concerns the actual physical port, not every contraction of that tensor. These statements require the chart's regularity and nonsingular transport coefficients. They identify the occupied W/Z physical amplitudes; they do not introduce naked local charged-field correlators or turn unstable channels into stable Hilbert vectors.

4.3 Derivative fields on the same physical domain

Use signature $\eta = \text{diag}(1, -1, -1, -1)$, $\square = \eta^{\mu\nu} \partial_\mu \partial_\nu$, and the existing common invariant field domain $\mathcal{D}_0$, containing the vacuum Ω , for O_ρ . Define on compact smooth tests

$$K(b) = -O_\rho(\partial_\mu b^\mu), \quad D(f) = O_\rho(\square f). \quad (21)$$

Thus $K^\mu = \partial^\mu O_\rho$ and $D = \partial_\mu K^\mu$ are derivative distributions, not products of new point fields. The current K is distinct from both the axial quark Noether current and the Higgs phase current; no conservation is asserted.

Theorem 4.4 (Represented current, descendant and translation domains). *The operators in (21) preserve $\mathcal{D}_0$, are symmetric for real tests and closable there. They retain local support, physical gauge invariance and adjoint and tensor covariance. Their vacuum vectors have positive joint Gram kernels on the same Hilbert space and are smooth vectors for translations. With the field-vector convention $e^{iP \cdot x}$,*

$$k^\mu(f) = iP^\mu o(f), \quad d(f) = -So(f), \quad o(f) = O_\rho(f)\Omega, \quad S = P_\mu P^\mu \geq 0. \quad (22)$$

Proof. Each right side in (21) is an existing scalar smearing. The adjoint contains the conjugate-test operator on the dense common domain, proving symmetry and closability. Test derivatives preserve support and commute with the appropriate source covariance. For finitely many current/descendant vectors v_a , positivity is exactly $\sum_{ab} \bar{c}_a \langle v_a, v_b \rangle c_b = \|\sum_a c_a v_a\|^2$.

Translation of a compact smooth test is smooth in its test-function topology. The actual positive scalar two-point distribution implies continuity of $f \mapsto O_\rho(f)\Omega$ in Hilbert norm. The Stone difference quotients and their iterations therefore give all translation-generator derivatives as the corresponding derivative smearings. Covariance yields (22) with the displayed sign convention. Forward spectrum gives $S \geq 0$. No point vector $O_\rho(0)\Omega$, essential self-adjointness of a field smearing, or independent tensor factor was assumed. $\square$

The original source experiment is the test pullback

$$\rho_{b,d} = \rho - \partial_\mu b^\mu + \square d.$$

It constructs every current/descendant mixed response from the same relative source, with the contact subtraction of (19). Its kernel includes $b = \partial d$; that does not make its separate directions zero. The constant identity component in O_ρ drops from compact derivative smearings, while its source-dependent or curvature derivatives remain in other operations.

A covariant metric variation $k = \delta g$ also differentiates this test map:

$$\delta_k \square f = -k^{\mu\nu} \partial_\mu \partial_\nu f - \left(\partial_\mu k^{\mu\nu} - \frac{1}{2} \partial^\nu \text{tr}_\eta k \right) \partial_\nu f.$$

The mixed source insertion contains both the original metric derivative and $F_{\rho(\delta_k \square f)}$. All original density, coframe and composite-prescription contacts stay in the former. This chain rule does not construct a new curved-state family.

4.4 The complete descendant germ and a positive current boundary

The operational derivative channel is exactly

$$G_{(O,D)}(z) = \begin{pmatrix} 1 & -z \\ -z & z^2 \end{pmatrix} G_\rho(z). \quad (23)$$

For current slots the first and second analytic factors are $-ip^\mu$ and $+ip^\nu$, with the actual source duals. The identity is first a test-pullback identity of the boundary distribution, and

then a polynomial times its authorized continuation. All regular terms in G_ρ are multiplied too. A separately normalized time product of local derivative densities can have additional equal-time/extension contacts; their actual local polynomial comparison is retained rather than set to zero.

At an actual nonzero simple Higgs pole p_H ,

$$\text{Res } G_{OD} = -p_H \text{Res } G_\rho, \quad \text{Res } G_{DD} = p_H^2 \text{Res } G_\rho \neq 0.$$

For a higher-order pole the leading principal coefficient is likewise multiplied by p_H^2 ; the complete lower and regular coefficients follow by multiplication. The full scalar germ is recovered from G_{DD}/z^2 on its admitted domain avoiding zero. A deliberately massless scalar branch requires a new strict-nonzero check; the derivative operation need not preserve a pole at zero.

Theorem 4.5 (Nonzero physical descendant and current-generated boundary). *At the preceding actual nonzero Higgs pole there is a compact smooth test f with $d(f) \neq 0$. Let E be the joint translation spectral measure and*

$$B_n = \{p \in \bar{V}_+ : 1/n \leq p^2 \leq n, \ p^0 \leq n\}.$$

For some n , $E(B_n)d(f) \neq 0$. The independently current-generated closed space

$$\mathcal{H}_K(B) = \overline{\text{span}}\{h(P)E(B)K^\mu(g)\Omega : h \text{ bounded Borel, } g \in C_c^\infty, \mu = 0, \dots, 3\}$$

contains $E(B_n)d(f)$ and reduces translations. For its isometric inclusion a_K and $j_D(c) = cD(f)\Omega$,

$$\|a_K^* e^{-ia \cdot P} j_D(1)\|^2 = \|E(B_n)D(f)\Omega\|^2 > 0.$$

This is an actual source-to-current-boundary attachment.

Proof. If every $D(f)\Omega$ vanished, positivity would make its unordered two-point distribution zero. Its time-ordered distribution would then vanish off the coincidence set; any local extension would be a finite delta-derivative contact, with polynomial Fourier transform. It could not have the nonzero principal part in (23). Hence some compact test has a nonzero image.

By (22), $d(f)$ has no component on $p^2 = 0$. The bands increase to the positive-mass part of the forward cone, so one band image is nonzero. On B_n , all momentum components and S are bounded and $p^0 \geq 1/\sqrt{n}$. For $q = E(B_n)o(f)$,

$$E(B_n)k^0(f) = iP^0 q, \quad q = \frac{-i}{P^0} E(B_n)k^0(f) \in \mathcal{H}_K(B_n), \quad E(B_n)d(f) = -Sq \in \mathcal{H}_K(B_n).$$

All multipliers used here are bounded on the band. The generating space is invariant under every bounded spectral function and its adjoint, hence reduces translations. Because it lies in $E(B_n)\mathcal{H}$ and contains the displayed band vector, its projection of $d(f)$ is exactly that vector. Translation preserves its norm, proving the last assertion. $\square$

The boundary is defined by the current before the descendant is chosen; it is neither the whole Hilbert space by declaration nor a renaming of the desired image. If $\nu_f(A) = \langle o(f), E(A)o(f) \rangle$, the full unordered spectral matrix is

$$d\nu_{ab}(p) = \overline{c_a(p)} c_b(p) d\nu_f(p), \quad (c_O, c_{K^\mu}, c_D) = (1, ip^\mu, -p^2).$$

Every finite quadratic form is nonnegative. Its band restriction has bounded moments. It is not the time-ordered analytic tensor (23), and a complex resonance is not an atom of ν_f .

The real and imaginary parts of a nonzero compact test show that f may be chosen real. Then $D(f)\Omega$ is orthogonal to the vacuum, translation-smooth and has strictly positive finite energy. Its normalized vacuum-plus-descendant preparation stays on the common polynomial domain. A band projection gives another actual bounded-energy vector, but is not asserted to preserve that polynomial domain or the resonance sheet. The unspectrally-projected preparation is the one used for the all-test stress construction of Theorem 8.1.

4.5 A represented kinetic-current contact with complete test access

There is another original-source route to the same physical scalar, without assuming a separately represented full Higgs phase current. Retain the signed nonzero kinetic coefficient k_H , and introduce the test connection $D_\mu^{(a)} = D_\mu - ia_\mu$. At the canonical source-word level define $J_0^\mu = iH^\dagger \overleftrightarrow{D}^\mu H$. Direct expansion gives

$$\mathcal{L}_{\text{kin}}(a) = k_H \{ (D_\mu H)^\dagger D^\mu H + a_\mu J_0^\mu + a_\mu a^\mu O \}. \quad (24)$$

The polarized quadratic coefficient, or equivalently the even finite difference before renormalization, is

$$S^0(b, c) = 2k_H O(g^{-1}(b, c)).$$

Under $H \mapsto e^{i\alpha} H$, $J_0^\mu \mapsto J_0^\mu - 2(\partial^\mu \alpha) O$. The finite difference of this original current coefficient is thus

$$R_\alpha^0(h) = -2k_H O(h^\mu \partial_\mu \alpha).$$

If the background a is also shifted by $d\alpha$, the covariant current $J_0 + 2aO$ is unchanged. The relative-current experiment instead holds that original detector convention fixed.

Theorem 4.6 (Represented kinetic and relative-current contact). *Extract the preceding polynomial coefficients or finite differences first, then apply the same linear renormalized insertion map that defines the original O_ρ . The resulting physical operations are*

$$S(b, c) = O_\rho(2k_H g^{-1}(b, c)), \quad R_\alpha(h) = O_\rho(-2k_H h^\mu \partial_\mu \alpha). \quad (25)$$

Both are represented on $\mathcal{D}_0$. Each family accesses every compact smooth scalar test on the flat source. Its positive kernels and full source-normalized physical scalar response are the exact test pullbacks of O_ρ , including its identity and extension conventions.

Proof. Multiplying $(DH)^\dagger + iaH^\dagger$ by $DH - iaH$ gives the linear current and positive $a^2 O$ term with the displayed signed k_H . Two source derivatives give $2k_H g^{-1}(b, c) O$. Alternatively $\mathcal{L}(a) + \mathcal{L}(-a) - 2\mathcal{L}(0) = 2k_H a^2 O$; polarization gives the mixed coefficient. The phase identity follows from $D(e^{i\alpha} H) = e^{i\alpha} (DH + i d\alpha H)$. The two opposite linear terms in $iH^\dagger \overleftrightarrow{D} H$ sum to $-2d\alpha O$. Applying the fixed linear insertion map after these exact cancellations proves (25). No separately renormalized product or separate phase-current operator was subtracted on an unproved common domain.

For test coverage, let f be compactly supported in K . Choose a smooth cutoff $\chi = 1$ near K and a covector e with $g^{-1}(e, e) \neq 0$ on a neighborhood. In flat Minkowski space one may use $e = dx^0$ throughout that neighborhood. Set

$$b = \chi e, \quad c = \frac{f}{2k_H g^{-1}(e, e)} e.$$

The second expression is extended by zero outside the neighborhood; its support stays inside K . Then $2k_H g^{-1}(b, c) = f$. For the relative-current access choose a compact α equal to the coordinate x^0 near K , so $d\alpha$ is non-null there, and put

$$h^\mu = -\frac{f \partial^\mu \alpha}{2k_H (d\alpha)^2}.$$

Again the expression is taken only near K and zero-extended. It gives $-2k_H h^\mu \partial_\mu \alpha = f$. These explicit right inverses use no inverse wave operator or noncompact source. The right sides of (25) are existing scalar smearings, proving their domain and kernel claims. $\square$

The theorem concerns the original kinetic coefficient and its calibrated insertion. It does not assert that the full independently renormalized background- a action has only the polynomial (24). Source counterterms such as derivative background terms have their own derivatives. If the raw nonlinear background functional is the experiment, all those derivatives must be included. Nor does subtraction of two formal current symbols establish separate physical domains for both of them; here the finite source-word difference is formed before the representation map.

For the fixed tensor coefficient $S^{\mu\nu} = 2k_H\eta^{\mu\nu}O_\rho$, the complete analytic responses are

$$G_{S^{\mu\nu}, O_\rho} = 2k_H\eta^{\mu\nu}G_\rho, \quad G_{S^{\mu\nu}, S^{\alpha\beta}} = 4k_H^2\eta^{\mu\nu}\eta^{\alpha\beta}G_\rho.$$

Choose an actual nonzero contracted coefficient to recover the whole scalar germ, not only its pole. Thus these contact channels have the occupied nonzero Higgs principal part. Their positivity is separately the Gram positivity of the represented vectors $O_\rho(f)\Omega$, not a sign assertion about a Feynman Hessian.

This distinction is visible in a raw four-current derivative. For two separated polarized pairs 12 and 34, with the canonical quadratic background source so that the cross-pair and higher action contacts vanish, the complete formula is

$$\begin{aligned} W_{1234} = & i\kappa(F_{12}, F_{34}) - \kappa(F_{12}, F_3, F_4) - \kappa(F_{34}, F_1, F_2) \\ & - i\kappa(F_1, F_2, F_3, F_4). \end{aligned}$$

Thus the isolated physical S/S response is not the entire raw four-current response. Additional source counterterms restore their actual higher contacts through Theorem 3.1. No standalone phase-current spectrum is claimed by the contact theorem.

4.6 Actual detector families and continued response calculus

The operations just constructed have useful proved parameter regularity without postulating a new family of physical vacua. On a fixed compact support, multiplication and differentiation of smooth test profiles are continuous multilinear maps of their test spaces. A smooth profile family therefore gives a smooth family of the tests in (25) and (21). Pairing with the existing field distribution gives smooth matrix elements on its admitted domain; the positive two-point distribution gives smooth vacuum vectors in Hilbert norm. This is a test/preparation statement, not operator-norm differentiability of unbounded fields.

For finite constant detector coefficients $v_i(\theta)$, the continued family is actually

$$G_{ij}(z; \theta) = v_i(\theta)v_j(\theta)G_\rho(z),$$

with its retained dual and contact convention. It is C^k in the uniform norm on any compact regular part of the existing sheet whenever v_i is C^k . Likewise the fully transported flavor family of Theorem 3.5 has its actual constant calibrated Higgs germ. These are discharged physical families; they do not imply regularity of an arbitrary transverse change of Yukawa invariants or metric background.

For reference, the exact continued-family calculation can be stated without imposing a rational model on an interacting amplitude.

Proposition 4.7 (Source-calibrated contour response). *Let an actual channel family on one marked sheet be C^2 in source coordinates with values in the uniform norm on a compact contour Γ , with its original source derivatives transported through a finite chain of locally uniform holomorphic overlap neighborhoods from the detecting germ. Then*

$$M_n = \frac{1}{2\pi i} \int_\Gamma z^n G(z) dz, \quad (M_n)_I = \frac{1}{2\pi i} \int_\Gamma z^n G_I(z) dz, \quad I = a, ab,$$

where G_I is the actual contact-complete channel derivative. If the family has exactly one noncancelled simple pole inside Γ , choose a fixed detecting contraction with $m_j = \ell M_{jr}$, $m_0 \neq 0$. Its pole satisfies

$$p = \frac{m_1}{m_0}, \quad p_a = \frac{m_{1,a} - p m_{0,a}}{m_0},$$

$$p_{ab} = \frac{m_{1,ab} - p m_{0,ab} - p_a m_{0,b} - p_b m_{0,a}}{m_0}.$$

The residue and entire regular background are differentiated as well.

Proof. The contour integral is a bounded linear map on $C(\Gamma)$; Banach-valued difference quotients pass through it. The differentiated identities on the locally uniform holomorphic overlaps identify its integrand with the original source derivative. In the one-pole case $M_0 = R_p$, $M_1 = pR_p$, giving the ratio and its two quotient derivatives. For $G = R_p/(z - p) + B(z)$,

$$G_a = \frac{R_{p,a}}{z - p} + \frac{R_p p_a}{(z - p)^2} + B_a,$$

$$G_{ab} = \frac{R_{p,ab}}{z - p} + \frac{R_{p,a} p_b + R_{p,b} p_a + R_p p_{ab}}{(z - p)^2} + \frac{2R_p p_a p_b}{(z - p)^3} + B_{ab}.$$

Inside the contour the regular part is $B(z) = (2\pi i)^{-1} \int_{\Gamma} G(w)/(w - z) dw$: the residues of the pole term at $w = p$ and $w = z$ cancel. Cauchy's formula and the same parameter bounds prove its derivatives. The actual detector and lawful-presentation families above meet these regularity conditions on their preserved domains. $\square$

For a genuine source-coordinate variation of (18), put $K = G_\rho - L$. Its calibration contacts are

$$(G_H)_a = n^{-2} K_a - 2n^{-3} n_a K,$$

$$(G_H)_{ab} = n^{-2} K_{ab} - 2n^{-3} (n_a K_b + n_b K_a) + (6n^{-4} n_a n_b - 2n^{-3} n_{ab}) K.$$

The numerators are (11) and (12), not a bare three-point insertion. For a current presentation $J_L D^{-1} J_R + B$, the derivatives distribute among both endpoints, inverse and background, with

$$(D^{-1})_a = -D^{-1} D_a D^{-1},$$

$$(D^{-1})_{ab} = D^{-1} D_a D^{-1} D_b D^{-1} + D^{-1} D_b D^{-1} D_a D^{-1} - D^{-1} D_{ab} D^{-1}.$$

At second order the three second-factor derivatives and the six assignments to two different factors are all retained. Moving external contractions have their own identical product rule.

An actual analytic matrix presentation with a single simple determinant zero and no zero on the contour gives a local one-pole neighborhood by the implicit-function and Rouché theorems; nonzero detecting residue persists by continuity. This is a sufficient test on a supplied physical family, not an inference from a quadratic mass matrix. Thresholds, cancelled detector contractions, singular Nielsen coefficients or lost uniform continuation regularity stop this particular comparison at their actual boundary.

Finally, metric variation of the kinetic test coefficient retains

$$\delta(2k_H g^{\mu\nu} b_\mu c_\nu) = 2(\delta k_H) g^{\mu\nu} b_\mu c_\nu - 2k_H k^{\mu\nu} b_\mu c_\nu$$

$$+ 2k_H g^{\mu\nu} \{(\delta b_\mu) c_\nu + b_\mu \delta c_\nu\}.$$

The separate covariant-metric density derivative is $\delta dV_g = \frac{1}{2} \text{tr}_g(k) dV_g$. Composite, spin/coframe, curvature and moving-prescription derivatives stay in the original source insertion. These formulas concern admitted source protocols; they do not infer a curved physical vacuum or a new continuation sheet.

4.7 The joint physical output

The constructions yield the original Higgs detector, its full source-normalized Higgs germ and actual W/Z current ports, the represented gradient current and Euler descendant, a nonzero current-generated positive boundary, and separate kinetic/contact access to the complete scalar response. Equation (17) identifies the represented descendant with its total full-SM kinetic/potential/Yukawa source combination. The fixed-sibling flavor identities remain available after replacing each detector test by $\square f$; their contacts do not disappear, and a zero continued sum does not separately continue its summands.

The curvature construction in Theorem 8.2 uses these same represented objects and source normalization, and Theorem 8.3 uses the same actual descendant preparation. This is a common physical-source chain, not a substitution of the native spectral state for the physical vacuum. It provides a represented total composite containing the genuine quark and lepton source words without assigning its nonzero Higgs pole separately to those words. The complete original color/composite and asymptotic interfaces retain their own source authority under Theorems 10.5 and 10.8; the causal Einstein action is constructed in Section 9. Their simultaneous comparison is Theorem 11.5. That result preserves the full channels and domains proved here without asserting a new transverse continuation family where none belongs to the original source.

5 Compact responses of the interacting physical source

The renormalized physical field prescription fixes products with uncontracted field legs, not only the scalar functional (1). It therefore supplies mixed chronological and ordinary products before the physical expectation is taken. We use these products to construct compact color and Yukawa responses on the original Higgs detector ports, with the full interaction, preparation and source normalization unchanged.

The distinction is substantive. A symmetric scalar third derivative does not determine every ordering of three operators. Nor does a list of physical poles determine a composite-field domain. The construction below uses the original algebra-valued prescription, and compares its scalar restriction with the already received source functional. Its authority is coefficientwise in the original perturbative regime, or through the retained order with that source's error convention. The physical Higgs port and its stronger baseline domain and continuation remain the objects of Section 4; they are not reconstructed by summing a formal series.

5.1 The original prescription and its graded carrier

Fix the same occupied full source as in Section 3. SYM-10, Definition 2.1, includes its renormalized field and interaction inventory, composite-operator basis, mixing, EOM and source conventions, subtraction scheme, physical quotient, state and Ward data [33]. The perturbative BRST formulation and physical scalar packet are received with the actual hypotheses of SYM-8, Definitions 2.1–2.2 and Section 9 [20]. The complete color, weak, hypercharge, Higgs, quark and charged-lepton interactions remain in the common source. An electroweak restriction is not used to invent its color or chiral siblings.

In a perturbative field formulation, the normalization rule applies to the numerical distribution coefficients of partially contracted graphs, not only to fully contracted vacuum graphs. Uncontracted fields carry their original matrix, spinor and Grassmann indices. Composite-source counterterms, ghost and gauge-fixing terms, and the furnished Ward normalization are part of the same rule. This is the field-valued content of causal Wick expansion [41, Section 4 and Appendix A.1]. The corresponding local-covariant construction and its graded extensions are described in [42, Sections 3.1–3.7]. Their model-specific gauge-existence claims are not substituted

for the occupied chiral Standard-Model prescription. In particular, the pure-Yang–Mills gauge theorem does not supply a missing chiral Ward identity here; the original source already specifies the Ward identity used by this construction.

Let Φ denote the original computational multiplet in a fixed perturbative chart, including its gauge-fixing, auxiliary and ghost variables. It contains one Higgs doublet and the original chiral matter with the conjugations of Section 3; no additional matter field is introduced. Let H be its original matrix-valued Hadamard contraction. This computational contraction is not a new choice of interacting physical vacuum. We use units $\hbar = 1$, retaining the original perturbative coefficient grading.

For polynomial Wick blocks, define the ordered product by cross contractions:

$$N_H(P) \star N_H(Q) = \sum_c \epsilon(c) H_c N_H(P_c Q_c). \quad (26)$$

Here c runs over pairings between the two blocks, H_c is the product of their original contractions, P_c, Q_c retain the uncontracted legs, and $\epsilon(c)$ is the original graded permutation sign. The sum is finite for fixed polynomial blocks. In the equivalent functional notation the product is $m \exp(\Gamma_H)$, with the graded contraction operator Γ_H .

Lemma 5.1 (Original Wick multiplication). *Equation (26) is associative and obeys $(A \star B)^* = B^* \star A^*$ on the original Wick carrier. It extends to the source’s renormalized compact microcausal blocks. The graded commutator of the computational generators is the original causal propagator and vanishes at spacelike separation.*

Proof. Both parenthesizations of a product of three polynomial blocks enumerate the same contractions among their ordered legs. Each pair occurs once, and the total graded permutation is independent of parenthesization. This proves associativity. Reversing the blocks and conjugating the original contraction gives the adjoint identity. The antisymmetric graded part of the contraction is the causal propagator of the original quadratic field equations.

The distributional extension uses the original microcausal domain. The functional derivative kernels of a block exclude collections of covectors lying entirely in the closed future cone or entirely in the closed past cone. In a cross contraction, an oriented Hadamard kernel could produce a forbidden zero covector sum only if such a collection were already present in an operand. Thus the contraction products obey the wavefront multiplication criterion. Finite matrix indices and Grassmann signs do not change this criterion. The original local counterterms retain the permitted diagonal wavefront structure, so a renormalized chronological block can be multiplied by a second compact block without choosing another extension across that block boundary. All statements use the exact domain and support envelope of the original prescription [42, Sections 3.1–3.3 and 3.5]. $\square$

If the Wick reference changes from H to $H + w$, with w smooth, self-contraction gives its invertible re-expansion α_w . The same counting proves

$$\alpha_w(A \star_H B) = \alpha_w(A) \star_{H+w} \alpha_w(B).$$

The source action, normalization, adjoint, BRST operation and expectation must be transported with this map. Such a change is a computational coordinate change, not a replacement of the prepared state. We may throughout keep the original H fixed. A compact Yukawa variation, including a quadratic term induced in that chart, then remains a relative interaction rather than silently changing the propagator.

5.2 Coefficient identity and the original chronological operation

Distinguish the numerical normalization rule t_n^S , the algebra-valued chronological map T_0^S before inserting the full interaction, and the interacting maps to be defined below. The scalar functional extracting a numerical Wick coefficient is not the physical expectation E_S .

In a field-independent Wick presentation the original coefficient lift has the form

$$T_0^S(P_1, \dots, P_n) = \sum_{\gamma} \frac{\epsilon(\gamma)}{\gamma_1! \dots \gamma_n!} t_n^S(P_1^{(\gamma_1)}, \dots, P_n^{(\gamma_n)}) N_H(\Phi^{\gamma_1} \dots \Phi^{\gamma_n}). \quad (27)$$

The graded submonomial derivatives, spacetime arguments, tensor indices and signs are those of the original expansion. The sum is finite at each fixed list of polynomial vertices. If the source uses additional finite composite mixing, include its complete local Wick-polynomial coefficients in this expansion. In particular, no EOM, central or source-contact term is removed to force a simpler normalization.

Theorem 5.2 (Identity with the original field prescription). *Construct (27) from the original normalization of every partially contracted coefficient in its admitted insertion domain. The result is the original renormalized field operation, with the same local extensions, finite mixing, causal factorization and adjoint identities, at every furnished perturbative order.*

Proof. If the original field maps are given algebra-valued, expand those maps in their Wick coordinates. If they are given by a diagram and subtraction rule, apply that very rule to each partially contracted graph, retaining the indicated external field legs. These are two descriptions of the coefficients in (27).

Order the comparison first by perturbative order and then by Wick degree. Away from the relevant diagonals, both expressions have the same original contraction, propagators, vertex labels and graded signs. At each extension step they use the same numerical distribution and the same local normalization. Every coefficient of every uncontracted Wick monomial therefore agrees. Equivalently, in the field-independent presentation, commuting their difference with a free generator reduces it to lower Wick degree. After those coefficients agree, the remaining numerical coefficient agrees as well. There is no free central term left to choose. Induction proves the asserted equality, or the equality modulo the source's retained remainder ideal.

On causally ordered blocks the original off-diagonal operation is the ordered product. Equation (26) enumerates exactly its cross contractions, so

$$T_0^S(P_I, P_J) = T_0^S(P_I) \star T_0^S(P_J) \quad \text{if block } I \text{ is later than block } J. \quad (28)$$

The original diagonal normalization is retained off this ordered region. The Hermitian/unitary normalization is retained coefficient by coefficient too. It determines the anti-chronological coefficients by the formal $\star$ -inverse, not by an independent contact choice. $\square$

The coefficient identity uses the actual field prescription, rather than inferring it from selected scalar values $E_S(T(A_1, \dots, A_n))$. Those values would forget uncontracted-leg coefficients. Conversely, the fixed normalization rule is mathematical data even when its infinitely many coefficient values are not printed. Only its supplied polynomial, source, support and order domain is used.

5.3 The full interaction and exact restriction to the physical source

Let V be the original full interaction, including its gauge, Higgs, quark and lepton Yukawa terms and all original counterterm and source vertices. Define

$$\begin{aligned} S_0^S(F) &= T_0^S(e^{iF}), \\ S_V^S(F) &= S_0^S(V)^{-1} \star S_0^S(V + F), \\ R_V^S(f) &= -i \frac{d}{dt} S_V^S(tf) \Big|_{t=0}, \end{aligned} \tag{29}$$

$$T_V^S(f_1, \dots, f_n) = i^{-n} \partial_{t_1} \cdots \partial_{t_n} S_V^S \left(\sum_j t_j f_j \right) \Big|_{t=0}. \tag{30}$$

All inverses and exponentials here are formal operations with unit constant coefficient, or their furnished finite-order restrictions. They are not finite exponentials of unbounded fields. Compact interaction cutoffs, local comparison maps and the infrared/preparation envelope are exactly those of the original source; no new adiabatic vacuum is chosen.

The lower-case f 's retain raw source labels in the original computational presentation. This distinguishes a label from its realized insertion, not an unrenormalized action from the renormalized one. The label includes its original coefficient and counterterm convention. When F_I denotes an insertion of Section 3, its corresponding raw label will be written f_I . The source normalization is applied once, through the original maps, not again to a previously normalized field.

Theorem 5.3 (Same-source interacting realization). *Equations (29)–(30), formed with the same V and the same original expectation E_S , realize the original interacting field prescription. In particular,*

$$E_S(T_V^S(f_1, \dots, f_n)) = \omega_*(T_{R,n}(F_1, \dots, F_n)). \tag{31}$$

For the full nonlinear relative action $F(j)$, with its raw source presentation $f(j)$,

$$Z_S(j) = E_S(S_V^S(f(j))), \quad W_S(j) = -i \log Z_S(j) \tag{32}$$

has exactly the source jets and all action contacts of Theorem 3.1. No change of physical preparation is made.

Proof. Insert the same baseline V vertices into Theorem 5.2, sum the same original orders, and form the same formal inverse. Every coefficient in (30) is then the original interacting coefficient, with its local terms and uncontracted legs. Applying the original interacting reconstruction gives (31); applying it after the original nonlinear source substitution gives (32). The partition derivatives in (2) therefore retain every f_I contact. This argument never identifies the free coefficient extractor with E_S .

For later F and earlier G , causal factorization of the original chronological maps gives

$$S_0^S(V + F + G) = S_0^S(V + F) \star S_0^S(V)^{-1} \star S_0^S(V + G).$$

Expanding baseline vertices verifies this identity coefficientwise: the inverse removes the duplicated baseline block. Left multiplication by $S_0^S(V)^{-1}$ proves $S_V^S(F + G) = S_V^S(F) \star S_V^S(G)$. Thus the interacting maps retain causal factorization without deleting any sibling interaction. Original Hermitian normalization likewise makes $S_V^S(F)$ $\star$ -unitary for real V, F . Its differentiated adjoint and anti-chronological identities are the original ones. $\square$

For even Hermitian normalized inputs, the two-entry inverse identity is

$$\begin{aligned}\bar{T}_V(f, g) &= R_V(f) \star R_V(g) + R_V(g) \star R_V(f) - T_V(f, g), \\ T_V(f, g)^* &= \bar{T}_V(f, g).\end{aligned}\tag{33}$$

The computational adjoint may be the gauge chart's Krein adjoint. Physical positivity is asserted only where the original physical quotient and representation supply it.

5.4 Causal source differentiation with the original contacts

For a real source direction z , let $b = f_z$ be its complete raw action derivative and let f_i be a scalar detector label. Put

$$B = R_V(b), \quad A_i = R_V(f_i), \quad C_{iz} = R_V(f_{iz}).$$

All these maps have source S ; that superscript is suppressed in the following formulas. The source variation keeps the original full interaction in the baseline:

$$A_i(\epsilon) = R_{V+\epsilon b}(f_i + \epsilon f_{iz}) + O(\epsilon^2).$$

The contact f_{iz} includes the nonlinear coefficient, source, counterterm and insertion-normalization derivatives. It is not set to zero merely because the canonical polynomial has no cross term.

Proposition 5.4 (Original retarded coefficient). *In a fixed product and normalization-rule frame,*

$$D_{i,z} := \partial_z A_i = C_{iz} + i\{T_V(b, f_i) - B \star A_i\}.\tag{34}$$

The formula holds on the full jointly admitted compact test domain, including overlapping supports and their original local contacts. For real Hermitian source and detector directions, $D_{i,z}$ is Hermitian. Away from local contacts its response vanishes if the input support cannot causally precede the output.

Proof. Write $C = S_0(V)$. Differentiation gives $\partial_z C = iC \star B$, hence $\partial_z C^{-1} = -iB \star C^{-1}$. Differentiate $-iC^{-1} \star \partial_t S_0(V + tf_i)|_0$. The inverse term is $-iB \star A_i$; the numerator term is $iT_V(b, f_i)$; the moving insertion gives C_{iz} . This proves (34), including its sign and order. It is the first chronological-to-retarded coefficient, obtained here in the original full-source algebra [43, Appendix E].

Equation (33) gives

$$[i\{T_V(b, f_i) - B \star A_i\}]^* = -i\{\bar{T}_V(b, f_i) - A_i \star B\} = i\{T_V(b, f_i) - B \star A_i\}.$$

The contact is Hermitian in the original real direction as well. On a region where the input cannot precede the output, causal factorization identifies $T_V(b, f_i)$ with $B \star A_i$. Their difference vanishes there; overlap contacts retain their original prescription. These are distributional statements, not bounds for a new unbounded operator product. $\square$

5.5 Moving normalization, reconstruction and physical quotient

If the actual comparison changes the chronological rule independently of its representation in the relative action, that derivative must also be retained. At fixed product define

$$P_z(G) = \partial_z S_0^{(z)}(G) \Big|_{z=0, G \text{ fixed}}, \quad L_i = \partial_t S_0(V + tf_i)|_{t=0}.$$

Differentiating the inverse and numerator gives the extra term

$$K_{z,i} = iC^{-1} \star P_z(V) \star C^{-1} \star L_i - iC^{-1} \star \partial_t P_z(V + tf_i)|_{t=0}.\tag{35}$$

Here P_z is the derivative of the original coefficient rule, not a counterterm chosen to force a comparison. Add this term, and any actual field-coordinate derivative, to (34). A normalization derivative encoded in the relative action is not counted again as an independent P_z .

If the product is μ_z , its fixed-operand derivative $\dot{\mu}$ occurs at each product vertex. If the preparation is E_z , its derivative occurs at each expectation. For example, with $K_{12} = (A_1 \star A_2 + A_2 \star A_1)/2$ and $C_{12} = E(K_{12}) - E(A_1)E(A_2)$,

$$\begin{aligned} \dot{C}_{12} = & \frac{1}{2}E(\dot{A}_1 \star A_2 + A_2 \star \dot{A}_1 + A_1 \star \dot{A}_2 + \dot{A}_2 \star A_1) \\ & + \frac{1}{2}E\{\dot{\mu}(A_1, A_2) + \dot{\mu}(A_2, A_1)\} + \dot{E}(K_{12}) \\ & - \{E(\dot{A}_1) + \dot{E}(A_1)\}E(A_2) - E(A_1)\{E(\dot{A}_2) + \dot{E}(A_2)\}. \end{aligned} \quad (36)$$

This is the ordinary product rule on the actual source comparison. It constructs neither a new moving vacuum nor a finite curved-state family. Source-dependent identity mixing and curvature contacts obey the same rule: a constant central term drops from a centered covariance, but its coefficient derivatives and source seagulls need not.

Proposition 5.5 (Physical representative independence). *The compact responses of physical source insertions descend through the original physical quotient, with the complete moving Ward, prescription and reconstruction contacts. This does not require every off-shell summand in (34) to be separately closed under the frozen BRST differential.*

Proof. Let d_S be the original interacting nilpotent differential. The coefficient identity preserves its furnished normalization and Ward identities. It is a graded derivation of the interacting field algebra, so the product of closed representatives is closed. If $A = d_S X$ and C is closed, then $A \star C = d_S(X \star C)$, with the corresponding graded formula in the reverse order. The original physical expectation annihilates exact insertions. Physical product readouts are consequently representative-independent at the furnished source authority.

For an actual moving source the identities are

$$d_z A_i(z) = 0, \quad E_z(d_z X_z) = 0.$$

Differentiation gives

$$d_0 \dot{A}_i = -(\dot{d})A_i,$$

and the derivative of the second identity vanishes with its $\dot{d}$, $\dot{E}$ and product terms included. Applying the same argument to products proves representative independence of the full response. In a furnished BRST-compatible relative frame these contacts are already absorbed into that frame. It is not legitimate to impose such a frame by deleting a genuine source derivative. All equations retain the original on-shell/EOM and contact convention. $\square$

In particular, the Higgs mass and full Hermitian Yukawa directions are even physical source words. The latter are (7), including their Higgs fluctuation, weak, color and conjugate contractions. Holding sibling parameters fixed leaves their interactions in V ; it does not set them to zero. Gauge anomaly cancellation is not used to discard an anomalous flavor/background Ward term.

5.6 The full compact mixed Higgs response

Let f_1, f_2 be arbitrary jointly admitted compact real scalar tests and let b be any original even Hermitian compact insertion. The connected Hermitian detector covariance is

$$C_{12}(\epsilon) = \frac{1}{2}E_S\{A_1(\epsilon) \star A_2(\epsilon) + A_2(\epsilon) \star A_1(\epsilon)\} - E_S(A_1(\epsilon))E_S(A_2(\epsilon)). \quad (37)$$

The detector maps, rather than just their numerical expectation values, are part of this operation. On the original physical scalar interface they have the calibration of Definition 4.1.

Theorem 5.6 (All-compact physical source response). *In the fixed relative-source frame, put $m_i = E_S(A_i)$, $d_i = E_S(D_{i;z})$, with $D_{i;z}$ from (34). Then the original response of (37) is*

$$\begin{aligned} \mathcal{R}_{12;z} = & \frac{1}{2} E_S \{ D_{1;z} \star A_2 + A_2 \star D_{1;z} + A_1 \star D_{2;z} + D_{2;z} \star A_1 \} \\ & - d_1 m_2 - m_1 d_2. \end{aligned} \quad (38)$$

It is an original compact physical source-response operation on the calibrated Higgs ports, retaining the intrinsic field-valued result, exact domain, source and normalization. For the full Hermitian quark-Yukawa direction it is the mixed Higgs–Yukawa response with all original interactions present. Moving source frames use (35)–(36).

Proof. Theorem 5.3 supplies each original interacting field and each mixed chronological/ordinary block in (34). Differentiate (37). The two field derivatives give its four product terms; differentiating its two means gives the two subtractions. This proves (38) without an unordered-product inference from a scalar third jet.

Each fixed-order coefficient is a distribution on the original compact test space. The chronological extensions and Wick products used in the formula have the original wavefront and support authority, and the reconstruction has its original distributional continuity. Consequently the operation is defined on all jointly admitted compact tests, not only a separated chronological chamber or a finite set of smearings. The reality statement follows from Proposition 5.4 and Hermitian symmetrization; the physical quotient follows from Proposition 5.5.

The endpoint attachment precedes numerical evaluation: the actual maps $f_i \mapsto A_i$ are the original scalar ports, and $f_i \mapsto (A_i, D_{i;z})$ is their source response in the same field prescription. Thus this is not an arbitrary scalar decorated with a Higgs label. Its expectation is the readout of that intrinsic operation. The genuine quark-Yukawa source label is supplied by Lemma 3.3; substituting it changes none of the preceding arguments or sibling data. $\square$

The covariance of two individual detectors is distinct from an original protocol which also reads a joint detector seagull f_{12} . That protocol retains $A_{12} = R_V(f_{12})$ and the additional response

$$\partial_z E_S(A_{12}) = E_S(R_V(f_{12z}) + i\{T_V(b, f_{12}) - B \star A_{12}\}) \quad (39)$$

in the fixed frame, with the preceding moving terms otherwise. Its original source-functional prefactor is retained. It is not replaced by the prefactor of a Hermitian covariance. In particular, f_{12} and f_{12z} remain in the complete chronological jets even when (37) selects the individual-detector experiment.

At a fixed normalization point $O_\rho = nO_H + b_0 1$, the connected detector covariance and response carry the factor n^{-2} on conversion to O_H . For a moving n , differentiate that factor:

$$\partial_z(n^{-2}\mathcal{C}_\rho) = n^{-2}\partial_z\mathcal{C}_\rho - 2n^{-3}n_z\mathcal{C}_\rho.$$

Moving test duals, products and identity coefficients have their original contacts. If the compared chronological extensions differ, retain the independent local map L in (18) as well. On a curved source, the permitted $R1$, improvement and density derivatives are not erased by setting the baseline curvature to zero before differentiation.

No positive spectral atom or nonzero cross response is asserted for every insertion b . A coefficientwise compact response is not a new positive-Hilbert graph for every quark composite and does not continue a mixed amplitude by itself. The original physical Higgs germ remains unchanged as baseline boundary data; its existence does not promote (38) to an independently continued mixed germ.

5.7 Exact relation to the separated chronological response

There is a useful exact comparison with the source numerator in (11). Suppose the compact input is strictly earlier than both compact output supports, and the outputs are mutually spacelike. In the fixed relative-source experiment all cross-source local contacts vanish on this disjoint chamber. Causal factorization and physical locality give

$$T_V(b, f_i) = A_i \star B, \quad D_{i;z} = i[A_i, B]_\star, \quad A_1 \star A_2 = A_2 \star A_1.$$

The last equality follows inside the original causal field algebra; it is not inferred from symmetry of a scalar chronological coefficient.

Proposition 5.7 (Connected chronological restriction). *On this chamber, let $a = \kappa_{\text{ord}}(A_1, A_2, B)$, using the original ordered three-cumulant. Then*

$$\mathcal{R}_{12;z} = i(a - \bar{a}) = 2 \operatorname{Re}(ia) = 2 \operatorname{Re} C_{12;z}. \quad (40)$$

At fixed detector normalization its physical-Higgs version has the factor n^{-2} .

Proof. Original adjoint reversal and output commutation identify $\bar{a} = \kappa_{\text{ord}}(B, A_1, A_2)$. Expanding the two three-cumulants cancels the terms containing a separate mean of B and gives

$$a - \bar{a} = E_S([A_1 \star A_2, B]_\star) - m_2 E_S([A_1, B]_\star) - m_1 E_S([A_2, B]_\star).$$

Substitution in (38), and the product rule for the commutator, gives i times this expression. On the same chamber the original chronological restriction (31) and the vanishing cross contacts give $C_{12;z} = ia$ in (11). This proves (40) with its connected subtractions. $\square$

Outside this chamber the answer is (38), not a guessed extension of $2 \operatorname{Re} C_{12;z}$. In particular no singular time-ordering distribution is multiplied by a discontinuous step function to manufacture the full compact operation. The original algebra-valued extensions already supply it.

5.8 Admitted neutral words and color-response attachment

The original colored input records retain the ordered U/D, conjugate and gauge-support constituents, their tensor-product color representation and every weak, Higgs, charge, anomaly, global and source incidence [28, Definition of the joint colored configuration category]. They are pre-boundary records, not isolated-quark to hadron maps.

Consider the actual covariant finite Wick-word branch. A fermion-parity-homogeneous word P_C here has its source-specified constituent realization, ordering, arguments and internal representation R_C . This branch also retains the completed $P_C^* \otimes P_C$ insertion with its original composite normalization, product and domain authority. Admission of P_C alone is not permission to choose an unprovided higher-degree normalization or multiply singular fields on an unlicensed diagonal. Covariant derivatives and field strengths retain their original transformation laws; a connection coefficient is not falsely treated as a homogeneous tensor.

Proposition 5.8 (Source-admitted neutral completion). *On this branch, the original internally paired Hermitian completion*

$$O_C = \operatorname{Herm}[\operatorname{ev}_{R_C}(P_C^* \otimes P_C)]_S \quad (41)$$

is an even neutral source insertion. It retains its actual tensor tests, quantum Ward normalization and full source provenance. Its original raw source label $f_C(h)$ gives $B_C(h) = R_V(f_C(h))$ and the compact physical Higgs-port response of Theorem 5.6 on every jointly admitted input.

Proof. Each constituent has its actual source transformation law. Finite tensor formation, conjugation, covariant differentiation and admitted intertwiners give the transformation law of the complete word. The original internal evaluation cancels $R_C^* \otimes R_C$. Fermion parity doubles, while the internal graded ordering and Pauli relations remain. Hermitian conjugation preserves neutrality. Any exact zero remains zero; no nonzero operator is selected in its place.

The brackets in (41) mean the actual normalized composite insertion, not a naive pointwise product. Its assumed original completion authority supplies its raw insertion label and the original mixing. Theorem 5.2 applies that same normalization, and Theorem 5.3 restores the same V and E_S . Its furnished physical Ward prescription, with Proposition 5.5 when it moves, gives the physical readout. Theorem 5.6 then applies directly. $\square$

Remaining spacetime and spin indices in (41) are retained and paired with the original dual tensor test. No positive Lorentz-invariant norm on a finite spinor representation is introduced. For an admitted multilocal word, dual pairing occurs at the original endpoints and the original joint smearing remains. There is no relocation of constituents, unsupported restriction to a diagonal, or creation of a quantum parallel transporter from a classical path.

The construction is $C \mapsto (C, O_C)$, with C retained in full. It neither replaces the physical input by additional conjugate matter nor identifies C with a neutral particle. It is not a theorem that every abstract support record is a Wick word. More generally, Theorem 5.6 applies to every original admissible even Hermitian color insertion, whether or not it was constructed by (41).

For complex detector tests, the unsymmetrized response kernel is

$$D_{f;C}^* \star A_g + A_f^* \star D_{g;C}.$$

It uses the original antilinear/linear source-port convention. The Hermitian symmetrization

$$\frac{1}{2}\{D_{f;C}^* \star A_g + A_f^* \star D_{g;C} + D_{g;C}^* \star A_f + A_g^* \star D_{f;C}\}$$

is instead real-bilinear on the realified test spaces. The two conventions must not be conflated. The original algebra also supplies the ordered insertion $E_S(A_f^* \star B_C \star A_g)$; it exhibits the color operation between actual physical detector ports, but is not substituted for the causal response and its mean/contact subtractions.

5.9 Finite source networks and physical boundary authority

The constructor comparison retains all already supplied operations of the same source: constituent and transport maps; local field and relative-source maps; global or extended-support maps with their actual authority; physical boundary, spectral and continuation maps; proper-color comparisons; and admissibility and failure readouts. An original primitive may have an infinite-dimensional argument or an entire analytic germ. Retaining its full graph does not truncate that object to a finite table. The finite object is a network of uses, not necessarily the domain of a primitive operation [5, 1].

Theorem 5.9 (Exact finite-use reception). *Every finite network of these original supplied operations and the compact source maps just constructed has a same-source receiving network with the same intrinsic output, exact operation domain, first-failure diagnostic, physical boundary grade and licensed continuation. Shared operands remain shared. This is reception of the furnished operation grammar, not an assertion that every independently required physical output is already furnished.*

Proof. A primitive leaf retains its complete original graph and source record. The new compact field and response leaves have those graphs by Theorems 5.3 and 5.6; the completed-word leaves

retain the domain of Proposition 5.8. Every other supplied physical or global leaf is its actual operation, with its own boundary authority, not a new evaluator inferred from a label.

For successive maps f and g , the domain is exactly

$$\{x \in \text{dom } f : f(x) \in \text{dom } g\}.$$

Inductive equality of both values and domains preserves this set, the result and the first failed vertex. Finite joint formation uses the original joint operation and domain; it is not a tensor product of independently selected physical states. Grafting on a shared input retains that input once, including every mixed Higgs, weak, metric and source incidence. Restrictions, comparisons and continuations are further actual vertices, so the same induction preserves their domains, boundary types and failure behavior. Finally, original source relations among paths are equalities of these same operations on the same domains and are therefore preserved. Reading the network in its original evaluation order gives the converse identification. $\square$

The resulting endpoint is an actual compact current/source-response interface of the physical Higgs port. The original neutral-boundary category permits such authorized singlet channels; it does not require a new resonance for every compact response. A claim whose output is itself a continued resonance, persistent composite or asymptotic sector must nevertheless retain that stronger boundary map. An unchanged baseline Higgs pole is not the pole of an uncontinued mixed response [5, 3].

Defined zero response, proved exclusion, inapplicability, inactive request, undefined operation and unresolved required authority are different outcomes. A well-typed zero response remains a defined channel; productive or nonzero use needs its actual additional witness. The finite-use theorem preserves an unresolved authority test if one is present; it does not discharge it by reception. Conversely, no unactivated charged sector, all-state scattering completeness or universal Wilson-loop expectation is imposed merely because the source has color.

The conclusion of this section is the complete original compact mixed-source operation, its intrinsic physical-port attachment and its exact composition with the supplied source graphs. This local result retains the distinction between operation reception and physical existence. The independent configuration comparison is proved in Theorem 6.11; the original physical declarations and their actual boundary witnesses are matched in Theorem 11.3. Their simultaneous realization at the same full source is Theorem 11.5. Those results use the compact operation at its stated authority, without changing the premetric architecture, field inventory, preparation or stronger baseline physical channels.

6 Original source incidences and finite configurations

A colored configuration, a presentation of its source operations, a renormalized insertion and a physical boundary are different objects. The first records the original joint constituents and their incidences; the second provides the actual maps acting on those records; the third uses the source's normalization and product domain; the fourth retains its independently supplied physical authority. We construct the passage from the first two objects to their original source realization, without choosing configurations by the success of the last passage.

The source is the occupied interacting Standard Model of Section 3. Its algebra-valued prescription is constructed in Section 5. We first specialize the independent input domain, then exhibit its polynomial and source-block operations. The final construction uses the actual rooted current-source incidence to interpret every original finite configuration and licensed operation. It does not force a support graph or physical history to be a local field polynomial. The physical-interface application is completed in Theorem 11.5.

6.1 The independent source fibre

Let C_T be the original pre-attachment configuration category [28, Sec. 3]. An object is a finite ordered, or otherwise incidence-complete, family of up-type, down-type, conjugate and gauge-support entries. It retains the complete tensor color representation, charge, weak and Higgs data, anomaly and global-form incidences, processes and source provenance. Its source and constituent prechecks precede physical boundary activation. This is a source-admitted generating category, not an exhaustive universe of QCD states.

Let A_T consist of its full source anchors, with their original licensed source arrows, identities and composition. A configuration arrow retains its source arrow, so provenance defines a functor

$$p : C_T \longrightarrow A_T. \quad (42)$$

An anchor is not merely a coupling tuple or the global-form label Γ . Retain the distinguished incidence (C, a, ι) if the notation for C suppresses it. Relative constituent charts and their actual joining maps remain present; no universal amalgam of independent marginal sources is assumed [30, Sec. 17][1, Secs. 1–2].

Fix the already occupied full source anchor a_* and set

$$C_* = p^{-1}(a_*). \quad (43)$$

This is the strict fibre on source-preserving arrows. For presentation independence, use the groupoid of actual invertible full-source comparisons and retain (C, σ) with $\sigma : p(C) \rightarrow a_*$; an arrow u obeys $\sigma'p(u) = \sigma$. Source automorphisms can act on marks and are not discarded. When the source supplies only comparison spans or common refinements, keep those witnesses instead of inverting a raw quotient map. Source-changing continuations are treated below.

Theorem 6.1 (Full-record source specialization). *The original configurations formed at a_* are exactly (43), with every original record and precheck. Terminal-relevance predicates and required request relations specialize by this same independent pullback, without selecting successful outputs.*

Proof. A configuration at a_* has precisely the finite joint entries, full incidences and source/constituent prechecks of C_T there. Retaining its provenance gives an object of the fibre. Conversely an object of the fibre already has those same records and predicates at a_* . Forgetting only the redundant anchor equality reverses the construction on full records and source-preserving arrows. Under an invertible presentation change the original source comparison transports and reflects the prechecks; retaining σ prevents erasure of its marks.

Thus, for an original relevance predicate Rel_T and request relation Req_T , one has

$$\text{Rel}_*(C) = \text{Rel}_T(C), \quad \text{Req}_*(C, r) = \text{Req}_T(C, r) \quad (C \in C_*).$$

The request r retains its original output type, authority and continuation data. This is restriction of an independently defined relation, not definition by an available response. $\square$

An input remains in this fibre when a response is zero, a process fails or a physical request is not activated. An input only at a different source is not an input of this fixed-source application. If the original application specifies a family of starting anchors, take the union over that family; the theorem does not replace it by a successful point.

6.2 Exact domains, source-changing paths and presentation witnesses

For actual source-preserving partial graphs $f : D_f \subset X \rightarrow Y$ and $g : D_g \subset Y \rightarrow Z$, restrict the full input records. Then

$$\begin{aligned} D_{g_*f_*} &= \{x \in D_f \cap X_* : f(x) \in D_g \cap Y_*\} \\ &= D_{gf} \cap X_* = D_{(gf)_*}, \quad g_*f_* = (gf)_*. \end{aligned} \quad (44)$$

The first failed predicate and its diagnostic agree as well. At a many-input vertex pull back the whole joint domain and common-interface equations, not independently selected marginal fibres.

Fixing the initial source does not freeze subsequent sources or preparations. Keep each licensed path $(a_*, u_1, a_1, \dots, u_n, a_n)$ and all its actual operation records. Its complete domain is

$$\left\{ x_0 : \begin{array}{l} p(x_0) = a_*, \quad x_0 \in D_{f_1}, \quad f_1 x_0 \in D_{f_2}, \dots, \\ f_{n-1} \cdots f_1 x_0 \in D_{f_n} \end{array} \right\}. \quad (45)$$

A licensed path may leave a chart and return. Restricting every intermediate vertex to one static fibre would incorrectly remove such a path. All incoming charts and joining maps at joint vertices remain those supplied by the source.

Theorem 6.2 (Specialization of source-network reception). *The original whole-record source-network reception restricts to rooted diagrams at a_* and commutes with initial-source pullback. If N_* is the restricted normal-form construction, then*

$$\text{ev}_* N_* d = \text{ev}_* d. \quad (46)$$

The equality includes full operands, intrinsic outputs, normalization, authority, domains, boundary/continuation and failures. Network drawings need not be unique.

Proof. The original source-law and mixed-source reception theorems supply a normal form for every actual source use [30, Sec. 17][1, Sec. 2]. A primitive vertex retains its actual graph, full operands, intrinsic parameter and chart incidences. Its starting anchor, value and domain are therefore unchanged. Grafting compositions uses the same shared edges; (44)–(45) give the same domain, value and first failing predicate. Restrictions and comparisons use their original domain-reflecting witnesses. Induction proves (46) without approximating a whole operator or other infinite-dimensional intrinsic parameter.

Every original use rooted at a_* has the original normal form. Because its full source record is retained, that normal form lies in the same rooted pullback. Conversely every network in this pullback uses original operations on their supplied domains. Thus coverage of source uses, not only equality on a successful subset, commutes with specialization. $\square$

Let R_T retain configurations with their actual response realizations and let $E_T : R_T \rightarrow C_T$ be the erasure functor over the same source-anchor category A_T . It erases only realization data. Its objects include the actual product domains, normalization, preparation and comparison records. A functorial section of E_T would be sufficient for input realization, but the original profunctor target does not require a canonical word or such a global selection. The appropriate choice-free comparison is

$$\begin{aligned} P &\xrightarrow{J} R_T \xrightarrow{E_T} C_T, \\ E : P &\rightarrow C_T, \quad E_T J = E, \end{aligned} \quad (47)$$

where P is an independently formed category of all complete original presentation witnesses. A specified lawful natural isomorphism may replace equality; it must then be retained as a comparison.

Corollary 6.3 (Choice-free specialization and finite composition). *Equation (47) pulls back to the selected source fibre, retaining every presentation witness and licensed comparison. If original input interpretations, actual operation graphs and the physical introductions or source-proved alternatives used by a request are supplied at their original grades, their comparisons extend to every finite request generated by the original context rules.*

Proof. Form $P_* = P \times_{C_T} C_*$. For each witness w the equality $E_T J(w) = E(w)$ places $J(w)$ in the corresponding realization fibre. Hence J restricts to J_* and $E_* J_* = E|_{P_*}$. The same equation restricts original arrows, domains and compositions. With a natural isomorphism use the marked pullback and retain that isomorphism. No representative is chosen.

An input leaf uses its actual witness and this commuting square; an operation leaf uses its original graph; a physical introduction uses its independently admitted object and witness. Grafting uses (45) with the actual compatible joining maps. Original source equations identify only equal complete records. Induction preserves intrinsic outputs, authority and first failures. A required unresolved leaf is not a successful base case, and separate inhabitants of fibres are not presumed jointly compatible. A supplied section is the special case $E = I$; none was needed by the argument. $\square$

6.3 Graded words, endpoints and original composite normalization

Fix the original perturbative source chart of Section 5. Its graded jet alphabet contains the original quark, lepton and Higgs fields, their conjugates, gauge computational fields and admitted derivatives. Covariant words use the actual field strengths, covariant derivatives and intertwiners. A raw connection retains its inhomogeneous transformation law.

Let $\mathcal{P}(\mathcal{J})$ be the finite off-shell graded symbol algebra: bosonic symbols commute and fermionic symbols anticommute in its raw product $\odot$. This is not the quantum $\star$ product. EOM, total-derivative and physical-equivalence relations are used only with their original insertion/contact policy.

Proposition 6.4 (Raw word and neutral dual completion). *Every original parity-homogeneous finite expression built from this alphabet by ordered graded tensors, conjugation, finite differentiation and admitted contractions has its evaluated raw polynomial. If its covariant internal representation is R , then*

$$q_C = \text{Herm ev}_R(w_C^\dagger \odot w_C) \quad (48)$$

is a finite even internal-neutral polynomial of degree at most $2 \deg w_C$, with all remaining spacetime/spin indices retained. This holds uniformly at every finite degree, including exact zeros.

Proof. A leaf is its original generator, derivative or coefficient tensor at the supplied incidence. At an ordered tensor vertex multiply the previously constructed symbols in their original graded order and retain every index. At a contraction apply its actual intertwiner; at a derivative use the supplied derivative and product rule. Finite induction terminates in a polynomial. Representation actions commute with these contractions and act by the full tensor representation, so the same induction proves covariance. Conjugation reverses order and takes R to its original dual. The evaluation $R^* \otimes R \rightarrow 1$ cancels the internal action, parity doubles and Hermitian symmetrization is the original real operation. All original graded relations survive. $\square$

For an expression with different parity components keep those components separately. Internal neutrality is not a positive Lorentz norm: a spinor word may leave tensor indices, paired with the original compact dual test. A marked component of Q_L is not an independent weak representation. The common doublet, all-left/right-handed conjugations and the different $\tilde{H}/H$ incidences remain together [28, Sec. 7]. Their anomaly and global data are those of the full chiral multiplet, not separately inferred U/D theories.

For multilocal words retain the original arguments and endpoint representations. Pair dual factors at the same original endpoint; a contraction between endpoints requires its actual transport/incidence map. Diagonal global-color invariance is not invariance under independent local endpoint transformations. A general compact joint tensor test need not be separable. Where

the original multilinear distribution kernel is continuous, pair it with every test in that supplied envelope, without new diagonal restriction or support enlargement.

Three maps must remain distinct:

$$\begin{aligned} q &\longmapsto N_H(q), \\ f_q &\longmapsto [q]_S = T_1^S(f_q), \\ f &\longmapsto R_V^S(f), \quad \text{with original } E_S. \end{aligned} \tag{49}$$

The first is the raw Wick realization. The second uses the original composite basis, finite mixing and normalization; the third uses its original raw source label and full interacting prescription. No normalization is applied twice. Forming a new doubled raw polynomial is not restriction of the product of two renormalized distributions to a diagonal, and is not automatically their separately smeared Wick product.

If the full original composite rule is supplied on the polynomial sector of q_C , then (48) gives its actual argument and the same rule gives its normalized insertion. A composite probe at zero probe strength need not add a nonzero higher-dimensional coupling to the baseline interaction. Admission of a factor P alone, however, does not supply an unprovided normalization of a new local P^*P vertex. The original composite-domain condition of Proposition 5.8 remains necessary. On that domain the construction retains

$$C \longmapsto (C, q_C, f_C, [q_C]_S, R_V^S(f_C)),$$

not a replacement of C by extra conjugate matter or a neutral particle.

6.4 Source contacts and faithful change of operator basis

For a furnished real-multilinear vertex $M_s(u_1, \dots, u_r)$, finite labelled even source derivatives satisfy

$$\partial_I M_s(u_1, \dots, u_r) = \sum_{I_0 \sqcup \dots \sqcup I_r = I} (\partial_{I_0} M_s)(\partial_{I_1} u_1, \dots, \partial_{I_r} u_r). \tag{50}$$

The ordered blocks may be empty. Derivatives of the coframe, connection, test, intertwiner, adjoint, normalization and product occur at the vertex, not silently in a frozen sibling. For $q_C(s) = B_s(w_C(s), w_C(s))$ this is the three-block rule; B_s includes the original adjoint, raw product, internal evaluation and Hermitian pairing and is real-bilinear, not complex-bilinear. If N_s is the original linear normalization map, then

$$\partial_I [q_C]_s = \sum_{J \subset I} (\partial_J N_s)(\partial_{I \setminus J} q_C). \tag{51}$$

Lemma 6.5 (Complete finite contact rule). *Equations (50)–(51) hold on every original finite source tree at each furnished derivative order, with all repeated-direction multiplicities and original contact terms.*

Proof. One derivative acts either on the vertex or on one of its r operands. For a labelled set I , assigning each label to one of these $r+1$ positions enumerates every ordered block decomposition exactly once. This proves (50) by induction and already includes multiplicities when several labels denote the same direction. Apply it recursively to the finite tree. For the linear normalization vertex there are two positions, which gives (51). Its coefficient derivatives retain the original finite mixing, central, EOM and prescription terms. A relation valid throughout the compared source family differentiates as an identity; an accidental zero at one parameter value does not. $\square$

Let O be an actual finite operator column and j its source row. A furnished invertible parity-compatible basis comparison is

$$O' = MO, \quad j' = jM^{-1}, \quad \rho'(g) = M\rho(g)M^{-1}, \quad j'O' = jO. \quad (52)$$

Keep identity, EOM and contact components in that block or in its original accompanying source relations. The action is the full source-admitted action, including its global and overlap kernel.

Proposition 6.6 (Source-dual basis covariance). *Equation (52) preserves the full relative action, faithful operator action and every furnished differentiated source readout, with its coefficient-support, scheme, topology and order fixed.*

Proof. Direct substitution gives $j'O' = jO$ before reconstruction and conjugates each original action/intertwiner relation. For a moving lawful M ,

$$\partial_s j' = (\partial_s j)M^{-1} - jM^{-1}(\partial_s M)M^{-1}, \quad \partial_s O' = (\partial_s M)O + M\partial_s O.$$

The two basis-motion terms cancel in $\partial_s(j'O')$, leaving $\partial_s(jO)$. Repeated differentiation uses Lemma 6.5 and derivatives of $M^{-1}M = I$. The complete source normalization and reconstruction are transported with the same comparison, so their original readouts agree. This is not a change of physical coefficient support, topology or field content. $\square$

SYM-7 supplies concrete original operands for this construction. Its minimal renormalizable grammar exhausts the gauge/Higgs/fermion kinetic, Higgs-potential and up/down/lepton Yukawa classes relevant to the accidental-charge span: the two-Weyl/one-Higgs terms follow from their Lorentz, hypercharge and nonabelian contractions, and dimension excludes an omitted renormalizable baryon-violating singlet [51, Theorems 3.1–3.2]. These are the actual vertices of the shared full interaction, not merely equal representation labels. Its separately declared dimension-five, six, seven and nine sectors retain their exact field, contraction, flavor, coefficient-support and EFT domains [51, Sec. 6]; they are used only when activated in the original source or probe domain. No right-handed neutrino or unselected Wilson coefficient is thereby added, and the dimension-eighteen topological effect is not retyped as a local Wilson vertex.

For example, on the specified faithful-remnant target the $QQQL$ word has raw $\mathbb{Z}_{18}$ exponent 12, trivial color-center exponent and even parity. It annihilates the order-six overlap kernel and descends to exponent $12/6 = 2$ modulo 3 [51, Eqs. (19)–(22)]. This is a character of the original operator on that target, not a nonzero Wilson coefficient or expectation. A nontrivial character forces a vanishing expectation only when the same original state and Ward/topology data are invariant; applying that actual invariance then proves the selection rule. The character alone does not supply the missing state premise.

6.5 Interpretation of polynomial presentations and source networks

An original polynomial/source presentation is a finite typed diagram with its complete configuration record. It contains original field, adjoint, covariant-jet or field-strength leaves and their arguments; finite linear, ordered tensor, intertwiner, conjugation, derivative and raw-polynomial vertices; and only the licensed Wick, chronological, relative-source or differentiation vertices on their exact domains. Coefficients, general joint tensor tests, normalization and every actual root, weak, Higgs, charge, anomaly and global incidence remain attached. An original nonpolynomial support/process vertex, where present, keeps its own graph and grade.

Let $\mathbf{P}_*^{\text{fld}}$ be the independently admitted presentations in which every vertex asserted to realize a field has its actual source-field incidence. Structural-only legs remain structural. Retain all triples

$$\mathbf{I}_*^{\text{poly}} = \{(C, p, \vartheta) : C \in \mathbf{C}_*, p \in \mathbf{P}_*^{\text{fld}}, \vartheta : C(p) \simeq C \text{ an original full-record comparison}\}. \quad (53)$$

Arrows are licensed whole-presentation comparisons with their original operation domains. Different contractions, source expressions and arguments are not identified unless the complete source comparison identifies them. A noninvertible continuation retains its forward graph, not an invented inverse.

The target SDiag_* consists of the original source interfaces, independently of C . Its field part contains graded jet modules, tests, multilinear labels and the actual raw, Wick and renormalized operations. Its structural part contains the root/response, gauge, Higgs, support and continuation graphs. Thus structural graph, raw field label, normalized interacting insertion and physical boundary operation retain four different grades. The object image of (53) does not by itself make erasure full on every arrow between those objects.

Theorem 6.7 (Complete polynomial-presentation interpretation). *There is an interpretation*

$$\begin{aligned} \mathcal{J}_{\text{poly}} : \mathbb{P}_*^{\text{poly}} &\longrightarrow \mathbb{C}_* \times \text{SDiag}_*, \\ \text{pr}_1 \mathcal{J}_{\text{poly}}(C, p, \vartheta) &= C. \end{aligned} \tag{54}$$

At every furnished field vertex it is the actual source operation. It preserves all original constituents and shared incidences, certified relations, licensed comparisons, exact partial domains and failures. No canonical presentation over a coarse configuration is required.

Proof. Interpret a leaf by its original field/test map. A Q_L leaf retains its common weak doublet, color fundamental, hypercharge and chirality; its marked U/D components do not acquire independent weak actions. Conjugate leaves use their original dual representation and field map. Every supplied source-slot and rooted incidence remains attached.

At an ordered tensor vertex form the finite tensor of coefficient/test modules with all arguments and the full tensor action. Graded permutation acts on these labels; it does not make arbitrary quantum products supercommutative. At a contraction use its actual intertwiner, at a linear vertex its source coefficients, and at a derivative its original connection and derivative action. Use the raw graded algebra at a polynomial vertex and the actual domain at its subsequent Wick/renormalized vertex. Finite induction constructs the expression. Changing a contraction, coefficient, argument, support or source map therefore changes an actual operand; the construction is not C placed beside a constant probe.

Each formation equation is preserved separately: coefficient linearity, tensor associativity, connection Leibniz and transformation laws, conjugation with order reversal, and the actual intertwiner equations. Raw Pauli identities give their defined zeros. Quantum CAR, chronological and Ward identities hold at their furnished operation vertices with their contacts, not by the raw graded quotient. Theorems 5.2 and 5.3 identify every partially contracted coefficient with the same original prescription, full V and interacting expectation E_* . Thus normalization, causal, adjoint and Ward equations are preserved at each original order, or modulo its retained remainder.

A nonpolynomial supplied vertex is its exact original graph and incident maps. The source-law chart comparison acts at its proved structural grade; an independently supplied physical graph keeps its physical grade. No structural graph becomes an operator graph. A licensed comparison transports all coefficients, tests, roots, connections, Higgs, anomaly, global and provenance data. Its actual equations commute on the original domains. An invertible comparison reflects domains by its inverse; a general continuation makes only its original forward-domain assertion. These comparisons preserve identities and composition.

Finally quotient only by the original equation congruence. A certified relation is equality of these source operations on their common domain; a conditional relation retains its predicate, and an EOM or integration-by-parts relation retains its contact remainder. Interpretation therefore descends. Keeping the complete C, ϑ gives (54) exactly, with every shared operand still shared. No detector inversion or output-first selection occurs. $\square$

This theorem interprets every original presentation in its stated domain; it does not assert that every support or history record is a polynomial. The all-input result below uses actual heterogeneous source incidences in addition to this explicit field branch.

Let $f_1, \dots, f_n$ be any finite list of original admitted compact source labels with their complete incidences. At the furnished orders,

$$T_V(f_1, \dots, f_n) = i^{-n} \partial_{t_1} \cdots \partial_{t_n} \left\{ S_0(V)^{-1} \star S_0 \left(V + \sum_j t_j f_j \right) \right\} \Big|_0. \quad (55)$$

For physical even labels the t_j are even parameters. Odd computational labels use their original Grassmann-dual parameters and ordered graded derivatives, or are read directly from the multilinear coefficient. A commuting exponential is not used to impose a false symmetry on odd labels.

Theorem 6.8 (All finite original source-block networks). *Every finite network of (55), original Wick products, relative derivatives and licensed source comparisons has its actual source-diagram interpretation, retaining all legs, contractions, contacts and support domains. Whenever it carries a presentation (53), its interpretation is exactly (54). A complete original nonlinear physical source coefficient retains its full Ward and reconstruction combination; its separate summands need not be independently physical fields.*

Proof. At a fixed order the original partially contracted coefficient has finitely many original vertices and external field legs. Theorem 5.2 identifies that coefficient before applying E_* . Apply the constructor proof of Theorem 6.7 to the actual leaves, tensors, contractions, coefficients and test arguments, then their original linear sum. Baseline interaction vertices retain all gauge, quark, weak, Higgs and lepton siblings; they are not reclassified as extra external constituents of C . When configuration incidences are furnished, the same interpretation lies over that complete C .

Each subsequent product uses the original microcausal Wick operation and its cross contractions. Each relative derivative uses the same unit-constant-coefficient inverse and chronological maps. Their exact vertex domains give induction on every finite network, not just the single action monomials of the renormalizable grammar.

For an original nonlinear even protocol $F(t)$ with $F(0) = 0$, write $f_B = \partial_B F(0)$. In the fixed relative-rule and product frame, the ordinary exponential substitution gives

$$\partial_I S_V(F(t))|_0 = \sum_{\pi \in \text{Part}(I)} i^{|\pi|} T_V(f_{B_1}, \dots, f_{B_{|\pi|}}). \quad (56)$$

To see every contact, assign the labelled derivatives to nonempty blocks acting on the same occurrence of F . The exponential factorial cancels the ordering of these occurrences, leaving each set partition once. Thus nonsingleton block contacts remain. Independently moving rules or products contribute their actual fixed-operand derivatives in addition to this substitution rule; they are not absorbed into an f_B unless the original source encodes them there. The original Ward/reconstruction comparison applies to the complete coefficient, with moving differential and preparation contacts. This proves the assertion at the original grade. $\square$

For a concrete coupled family, the original interaction is

$$\begin{aligned} \mathcal{L}_Y = & -\bar{Q}_L \tilde{H} Y_u u_R - \bar{Q}_L H Y_d d_R \\ & - \bar{L}_L H Y_e e_R + \text{h.c.} \end{aligned} \quad (57)$$

Arbitrary original compact complex matrix directions $Z = (Z_u, Z_d)$, with conjugates co-varied to give a Hermitian direction, have canonical insertion

$$b_Z = -\bar{Q}_L \tilde{H} Z_u u_R - \bar{Q}_L H Z_d d_R + \text{h.c.},$$

together with their original renormalized contacts. Every furnished finite list is covered by (55)–(56). The full flavor tensors, common weak doublet, shared quantum Higgs and color contractions survive. If an original comparison also admits a Higgs displacement h , the canonical mixed coefficient replaces $\tilde{H}, H$ by $\tilde{h}, h$ with all coordinate and prescription contacts. This does not introduce an absent displacement or replace H by a vacuum number.

6.6 The exact mixed physical source readout

For a furnished moving source label $f(s)$, in a fixed product/rule frame, differentiating the original relative inverse gives

$$\begin{aligned}\partial_s R_{V(s)}(f(s)) &= R_V(f_s) + i\{T_V(b, f) - R_V(b) \star R_V(f)\}, \\ b &= \partial_s V.\end{aligned}\tag{58}$$

Indeed $\partial_s S_0(V) = iS_0(V) \star R_V(b)$, so its inverse contributes $-iR_V(b) \star S_0(V)^{-1}$; the numerator derivative gives the chronological term and the label derivative gives $R_V(f_s)$. No inverse of the possibly noninjective Moller map is taken. Independent normalization-rule, field-coordinate, product and reconstruction motion is added by (35)–(36). An action-encoded normalization derivative is not counted again.

If an interpreted presentation supplies an original admitted even Hermitian physical insertion f_C , set

$$\begin{aligned}A_i &= R_V(f_i), & B_C &= R_V(f_C), \\ D_{i;C} &= R_V(f_{iC}) + i\{T_V(f_C, f_i) - B_C \star A_i\}.\end{aligned}$$

For fixed original E_* , product, field-coordinate and normalization frame, the connected Hermitian response is

$$\begin{aligned}\mathcal{R}_{12;C} &= \frac{1}{2}E_*(D_{1;C} \star A_2 + A_2 \star D_{1;C} + A_1 \star D_{2;C} + D_{2;C} \star A_1) \\ &\quad - E_*(D_{1;C})E_*(A_2) - E_*(A_1)E_*(D_{2;C}).\end{aligned}\tag{59}$$

This follows by differentiating the original centered covariance, not by equating an unordered product with a symmetric scalar third jet. The actual detector/source maps exist before readout by Theorem 5.6; hence it is a source response on the existing physical ports, not a scalar with a detector label.

For the general original moving comparison replace $D_{i;C}$ by the complete $\dot{A}_i$ and put $S_{12} = (A_1 \star A_2 + A_2 \star A_1)/2$, $m_i = E_*(A_i)$. The additional terms are

$$\begin{aligned}(\dot{E})(S_{12}) &+ \frac{1}{2}E_*((\dot{\star})(A_1, A_2) + (\dot{\star})(A_2, A_1)) \\ &- (\dot{E})(A_1)m_2 - m_1(\dot{E})(A_2).\end{aligned}\tag{60}$$

Here $\dot{E}$ and $\dot{\star}$ are the actual fixed-operand derivatives. The product rule on $E_s(S_{12}(s)) - E_s(A_1(s))E_s(A_2(s))$ proves the formula and separates these terms from the moving-rule terms inside $\dot{A}_i$. At $O_\rho = nO_H + b_01$ the fixed conversion is n^{-2} ; a moving n also contributes $-2n^{-3}n_s\mathcal{C}_\rho$. Joint detector seagulls f_{12}, f_{12C} retain their separate terms and original prefactors.

The differentiated quantum Ward relation is $d_0\dot{A} = -(\dot{d})A$, not necessarily $d_0\dot{A} = 0$ term by term. The complete source, reconstruction and product derivative descends as proved in Proposition 5.5. These formulas construct every furnished finite coefficient, not an unproved finite deformation, positive-Hilbert domain for every summand, or continued mixed pole. A source/Pauli zero remains zero; an absent composite normalization is not reclassified as zero.

6.7 The actual field-role and Higgs-response incidences

The passage to the full original configuration domain uses actual incidences, not equality of the number of roots and the number of field components. The established structural Standard-Model tuple contains

$$\begin{aligned} Q_L &= (\mathbf{3}, \mathbf{2})_{1/6}, & u_R &= (\mathbf{3}, \mathbf{1})_{2/3}, & d_R &= (\mathbf{3}, \mathbf{1})_{-1/3}, \\ L_L &= (\mathbf{1}, \mathbf{2})_{-1/2}, & e_R &= (\mathbf{1}, \mathbf{1})_{-1}, \end{aligned}$$

and the one-doublet Higgs/bridge, together with the Yukawa-realizability, anomaly-neutrality, vacuum and response/mixing records [53, Definition 1.7]. Its generation-erased projection $\pi_{\text{cf}} : \Sigma_{\text{SM}} \rightarrow \Sigma_{\text{SM}}^{\text{cf}}$ deletes family cardinality, names, indices, indexed matrix dimensions and numerical family coordinates, while retaining representation, color, weak component, relative charge, chirality, bridge identity, operator status and boundary/continuation types [48, Sec. 5.1]. In particular it gives the actual roles

$$r_U = \rho_{\text{mat}}(Q_L^u, u_R, O_u), \quad r_D = \rho_{\text{mat}}(Q_L^d, d_R, O_d). \quad (61)$$

Here ρ_{mat} removes presentation choices, not physical sibling information. Their weak member, relative charge and $\tilde{H}/H$ operator incidences prohibit identifying these two roles by a lawful full-target redescription [48, Proposition 5.1].

The fully occupied primitive root object I_3 received from FCGS-I positions these count-free roles:

$$E_f = I_3 \times \{r_f\}, \quad u_f : E_f \longrightarrow \text{Occ}_f(R_f). \quad (62)$$

The positioned family-cell and rooted-realization results retain full coverage, duplicate-reduced occupancy, weak pairing, eligibility, charge/anomaly and Higgs/operator compatibility [48, Secs. 6–7]. A complete rooted realization has an active presentation, occurrence and certified incidence, with its consumer comparisons. Its canonical rooted class is not a choice of one such complete representative.

We use the literal accepted chain

$$\text{Disc}(E_f) \xrightarrow{\eta_f} \mathcal{S}_f(C_f) \xrightarrow{\chi_f} Q_f^Y \xrightarrow{\beta_{F,f}} \mathcal{H}_{F,f}(C_H), \quad (63)$$

with its one-Higgs, kind-preserving naturality on the maximal lawful same-target groupoid [29, Secs. 1 and 5]. The root-normalized $\text{Disc}(I_3)$ presentation is a different displayed form; it does not change the source of η_f .

The representation-response realization has a different domain:

$$\alpha_{\pi,U} : E_{\pi,U} \longrightarrow V_{\pi,U}. \quad (64)$$

$E_{\pi,U}$ is the original pre-linear response carrier, not the three-point role index E_f . Its realization preserves every admitted response witness, including the actual transition, transport and loop squares and the furnished tensor, dual, quotient, chiral, field, persistence and boundary comparisons [49, Definitions 4.1–4.3, Eqs. (3)–(5), Theorem 13.1]. The quark fields are received through precisely this algebraic representation of matter response intersected with color transport [47, Secs. 8.1–8.2 and 9.2]; it is not an additional free-field model inferred from a basis label.

The actual Yukawa presentation also has the original canonical coupling carrier

$$\mathcal{Y}_D = (\mathcal{W}_D^Y / \sim_D^Y, \text{Sig}^Y, \gamma_D^{Y,\text{red}}, \rho_Y). \quad (65)$$

It retains standing-entry and zero/nonzero statuses, ratios and obstructions through the admitted coupling witnesses and canonical Higgs bridge, together with representation, charge, anomaly, residual, order/deformation and continuation data [50, Definitions 3.3–3.4 and 5.1–6.2,

Theorems 6.3–7.1]. These incidences license the actual full two-index Higgs-dressed Yukawa source, not an inference from a fitted matrix. A zero baseline coefficient does not remove its operator-availability record or an admitted source variation; neither this carrier nor the root theorem asserts that every Yukawa entry is nonzero.

Primitive generation receipt. *For a source with the FCGS–I complete primitive generation receipt, the root base in (62) has three positions. The four count-free matter kinds, the local representation carrier $E_{\pi,U}$, the coupling carrier $\mathcal{Y}_D$, and the terminal response inventories retain their own identities. Additional physical source data do not add primitive positions for the unchanged generation work.*

Proof. The original primitive duties have the exact architectural classification and complete occupation theorem of [55, Theorems 6.2 and 5.4]. The received FCGS–I occupation identifies that architecture with its complete neutral occurrence base, and its positioned-fibre maps supply each E_f [48, Sections 3–4]. These are the maps used in (62), so their base has the same three positions. No count of matter kinds, response directions or terminal bearers enters this inference. The inherited source conditions and the full coupling-witness data in (65) remain required. Nonpromotion for the unchanged primitive task follows from [55, Theorem 6.4]; it does not erase the additional physical answers of the present source. $\square$

6.8 Reindexing the complete physical prescription

Let J be the family-copy alphabet in an actual presentation of the source’s repeated typed profile. It includes the field-role, Yukawa, mixing and boundary data of that profile [54, Definition 9.1 and Proposition 9.2]. This citation supplies the typing of the copies, not the independent primitive root-count theorem. On the received complete-root target, J is the family-copy presentation of that full occupation. Retain the nonempty family of all compatible bijections $m : J \rightarrow I_3$ and every original lawful full-source comparison. A source without this complete occupation is not assigned such a marking merely from its matrix dimension. A marking is not a physical ordering. In one displayed presentation set

$$\begin{aligned} Q_{m(a)}^m &= Q_a, & u_{m(a)}^m &= u_a, & d_{m(a)}^m &= d_a, \\ L_{m(a)}^m &= L_a, & e_{m(a)}^m &= e_a, & (Y_f^m)_{m(a),m(b)} &= (Y_f)_{a,b}. \end{aligned} \tag{66}$$

Transport all other indexed tensors, source labels, probes, counterterms and response records by this same bijection. Unindexed siblings remain unchanged as objects. Both weak components of Q use the same marking; all cross-family Yukawa entries remain. The all-left presentation conjugates fields and representation data consistently and commutes with this relabeling. Spin, color, weak, conjugation and spacetime-support slots are not altered.

Theorem 6.9 (Full-prescription reindexing). *Equation (66) extends to an isomorphism r_m of the original full coefficient/source presentation at its furnished quantum grade. It preserves the action, renormalized products, interacting preparation, all original source derivatives and contacts, and every licensed response comparison. Its U/D role projection is (61), and its coupling carrier is (65) with the same reindexed legs.*

Proof. Reindex every finite family sum in the full interaction by $(a, b) \mapsto (m(a), m(b))$. The Yukawa word and its adjoint have identical readback. Gauge and kinetic terms commute with this relabeling, and the two weak components keep their common action. Scalar, ghost, anomaly/global, source-normalization and preparation records remain present. This is simultaneous re-presentation of the whole source, not an active anomalous transformation with its phase record held fixed.

Reindex every coefficient, contraction and distributional argument before applying its original renormalized map. Thus

$$\begin{aligned}
T_0^m(r_m A_1, \dots, r_m A_n) &= r_m T_0^*(A_1, \dots, A_n), \\
r_m(A \star_* B) &= (r_m A) \star_m (r_m B), \\
V_m &= r_m V_*, \quad E_m(r_m A) = E_*(A), \\
S_{V_m}^m(r_m F) &= r_m S_{V_*}^*(F).
\end{aligned} \tag{67}$$

The first two identities include the same diagonal extensions and finite mixing. The expectation is transported by precomposition, not replaced by the free coefficient extractor or another vacuum. The last identity follows by applying r_m to the unit-constant inverse and product defining the original relative S operation. These are the actual algebra-valued maps of Theorems 5.2 and 5.3, rather than a reconstruction of products from scalar moments.

Differentiate this full identity in any original source direction. It reindexes that direction and every nonlinear, counterterm, prescription, preparation and mixed contact at the supplied order. The PSM3 realization preserves the response witnesses themselves; the MGN4 comparison preserves the coupling-witness quotient and signature. Their readouts therefore commute with the reindexing, not merely their names. Every supplied partial domain and failure is transported with its original map. This proves the assertion. $\square$

A general flavor-coordinate change is not a root permutation. If v_i is a marked vector of a computational frame and U is a lawful field-coordinate comparison, transport both

$$\Phi' = \Phi U^{-1}, \quad v'_i = U v_i, \quad \Phi'(v'_i) = \Phi(v_i).$$

Transport the Yukawa/source tensors and original anomalous/global phase data with it. The root belongs to the full marked response record, not to whichever raw component has subscript i afterward. An arbitrary $U(3)$ matrix is not thereby an external S_3 action on roots. Rooted naturality applies only to the actual complete same-target comparison. Fixed ancestral, apparatus or boundary marks restrict the comparison to their original lawful stabilizers.

6.9 The rooted current-source square and its physical refinement

Let s_* be the structural restriction of S_* . It is the same joint Σ_{SM} source, not a separately chosen family of compatible marginals. The canonical source construction supplies, for $n = m^{-1} : I_3 \rightarrow J$, the actual current restriction

$$\begin{aligned}
\rho_f^{s_*,n} : \text{Disc}(E_f) &\longrightarrow \mathsf{I}_f(s_*[n]), \\
(i, r_f) &\longmapsto \text{the full kind-}f \text{ source restriction at } n(i).
\end{aligned} \tag{68}$$

It uses the original U/D field roles and Yukawa source at that position, not a mass eigenstate or the partial physical-particle dictionary. In particular it retains the shared Q_L doublet and the different U/D charge, Higgs and operator incidences.

The same source provides the restriction maps to the rooted count-free record groupoid $\mathbf{R}_f^{\text{cf}}$ and the specified natural isomorphism

$$\begin{aligned}
d_f^{s_*,n} \rho_f^{s_*,n} &\xrightarrow{\theta_f^{s_*,n}} d_f^0 \beta_{F,f} \chi_f \eta_f \\
&: \text{Disc}(E_f) \longrightarrow \mathbf{R}_f^{\text{cf}}.
\end{aligned} \tag{69}$$

These are the rooted-source restriction and common-reduct square of the canonical apex [1, Canonical-apex construction, rooted source restriction and comparison]. Their components

are restrictions of the actual inherited comparison of the same generation-erased role and full Higgs bridge. They are not arbitrary bijections between equally sized sets. The common reduct retains root, kind, representation/color/weak/relative-charge/chirality, full bridge identity, operator-status laws and boundary/continuation types, not only values at one coupling.

At each constituent retain

$$(A_{i,f}, \rho_f^{s*,n}(i, r_f), \theta_f^{s*,n}(i, r_f)). \quad (70)$$

$A_{i,f}$ is the full supplied ancestral record, including its intermediate η, χ, β incidences, complete target, response, continuation and failure data. If ancestry is given by its canonical rooted class, retain all furnished representatives and witnesses; the zero-truncated class does not select a canonical complete realization. The current record keeps the full current source. Equality is asserted only after the specified d maps. An ancestral loop is not made a new quark holonomy, nor an ancestral history identified with a different physical history. All old consumers still act on their original records.

The original full ancestral realization target and its eligibility conditions remain intact. Equation (69) does not require the current physical source to be a new member of that entire ancestral target. It is the already supplied comparison of the common reduct, with both full records retained separately.

Let $\mathbf{l}_f^{\text{phys}}(S_*[n])$ consist of actual kind- f restrictions of the original physical source, retaining the response/field realization, algebra-valued coefficients, coupled action, interacting preparation, normalization, tests, partial domains and current comparisons. These are restrictions of supplied objects, not new products or physical boundaries. Write

$$h_f : \mathbf{l}_f^{\text{phys}}(S_*[n]) \longrightarrow \mathbf{l}_f(s_*[n])$$

for forgetting those physical refinements while retaining the current structural source record.

Theorem 6.10 (Actual rooted current-leg refinement). *The original occupied source constructs*

$$\begin{aligned} \widehat{\rho}_f^{S_*,n} : \text{Disc}(E_f) &\longrightarrow \mathbf{l}_f^{\text{phys}}(S_*[n]), \\ h_f \widehat{\rho}_f^{S_*,n} &= \rho_f^{s_*,n}, \\ d_f^{s_*,n} h_f \widehat{\rho}_f^{S_*,n} &\xrightarrow{\theta_f^{s_*,n}} d_f^0 \beta_{F,f} \chi_f \eta_f. \end{aligned} \quad (71)$$

For $f = U, D$ these are genuine nonterminal quark/Higgs field and original source-operation incidences. Every lawful whole-source marking and comparison is retained, without selecting a rooted representative or an asymptotic quark.

Proof. Use $n = m^{-1}$ in Theorem 6.9. At (i, r_U) , restrict the actual Q_L^u, u_R, O_u response/field record at $n(i)$; for D use Q_L^d, d_R, O_d . Every other leg of its Yukawa and source operations remains in the same full $S_*[n]$. Selecting an incident slot is not diagonalizing the coupling matrix or replacing the quantum Higgs field by its expectation.

The original SM3 field reception uses (64), with its actual response, transition, transport, loop and source-domain witnesses. MGN4 supplies the Higgs coupling carrier and operator-status comparisons. The original interacting coefficients are the maps of Section 5, not scalar-moment reconstructions. Thus the restriction is an actual object of $\mathbf{l}_f^{\text{phys}}$ at precisely that supplied grade. Forgetting its physical refinements returns (68) literally in the same presentation. Composition with (69) proves the final line of (71). No equality of the distinct full ancestral and current targets has been inserted.

Every original source comparison acts on the physical record by (67) and on its current structural restriction by that same restriction. On the common reduct it is the specified θ ; on

the ancestral record only its own licensed comparison acts. This proves compatibility. The source $\text{Disc}(E_f)$ is discrete, so these object assignments give the stated functor; nontrivial source comparisons remain in the surrounding presentation category. Keeping all supplied representatives and markings makes no global choice. $\square$

6.10 Every original finite configuration and licensed operation

The domain remains the independent C_* of Theorem 6.1, not the image of the polynomial presentations or the configurations already carrying a hadron. Its original source-admission record gives the full operands and their source incidences. At a rooted constituent retain (70), refining only the current leg by (71). A support or process record at a structural grade uses its original graph and evaluation/continuation maps at that grade. A local-field operation uses its actual algebra-valued map and normalization; a field name alone is not that map.

Theorem 6.11 (Complete original input and operation interpretation). *Every object of the independently formed C_* has an interpretation retaining its full original record, and every licensed original arrow retains its operation, exact domain, source relations and failures. Field constituents use Theorems 6.9 and 6.10; other original constituents and operations retain their actual source grade.*

More precisely, let F_ be the category of all original full source presentations, constituent-admission witnesses and compatible whole-source markings, formed before interpretation or boundary evaluation, and let $e : F_* \rightarrow C_*$ read their configurations. There is a category P_* of these interpretations and functors*

$$F_* \xrightarrow{\mathcal{J}} P_* \xrightarrow{E} C_*, \quad E\mathcal{J} = e. \quad (72)$$

Every original object has interpretation witnesses. A specified lawful natural isomorphism replaces equality when original objects are compared by re-presentation. All witnesses and licensed comparisons are retained; no global selection over coarse objects or new full ancestral-target membership is required.

Proof. Take an arbitrary original finite presentation of $C \in C_*$. Its primitive constituents and constructors are exhausted as follows.

Rooted U/D entries. Insert (70) and refine its current member using $\hat{\rho}_f$. Readback returns the actual constituent slot, not only a root index or representation dimension. The ancestral root, neutral anchor, full response, failure and deletion data remain on the ancestral record. The current field, shared weak doublet, $\tilde{H}/H$ word and every coefficient leg remain on S_* . The shared fields commute by the actual θ square.

Conjugate and gauge/support entries. Retain the original dual/conjugation witness, representation and source action, with the consistent all-left/right-handed dictionary. The furnished dual compatibility of (64) and the graded maps of (67) interpret the conjugate entry. A gauge/support entry retains its actual object, action, support and continuation graph. If the source supplies a local gauge field or nonlocal physical operation, use that very operation and domain. If it supplies only structural support, retain that structural object. No pure-Yang–Mills sector is transplanted to another representation from a support label.

Finite joint families. Apply one whole-source marking to the entire ordered or incidence-complete family. Retain its full tensor color representation, weak/Higgs and global/anomaly attachments, preparation and every cross-constituent response. At the representation grade this is the original tensor in the configuration definition. At the physical grade a joint operation is the actual furnished joint operation. The tensor comparison of PSM3 applies to the supplied joint-response witness; it does not create an unprovided joint Hilbert product domain from marginal realizations.

Contractions and local-field operations. A certified invariant contraction remains its actual tensor map, with the full intertwining equation and all mixed legs. At the coefficient grade every admitted completed Wick vertex, ordinary or chronological product, relative operation and source derivative is the map of Section 5, with the same V_* and E_* . If μ has input/output transports $r_{\text{in}}, r_{\text{out}}$, then

$$\mu_n(r_{\text{in}}x) = r_{\text{out}}\mu_*(x), \quad \text{Dom } \mu_n = r_{\text{in}}(\text{Dom } \mu_*). \quad (73)$$

The diagonal extensions and source, prescription, preparation and BRST contacts are part of this map. A completed insertion retains its original completed normalization/product/domain authority; admission of a factor alone does not adjoin a new P^*P operation. An unsupported proposed operation remains a request with its original status, not an edge.

Support, process and history attachment. Attach the original source graph and actual partial evaluation/continuation maps. Structural, algebraic, operational-history and physical maps retain their different types. A source graph is not replaced by a polynomial; a physical history is not replaced by identity continuation. Class changes, weak/Higgs response, decay, failure and inference remain their actual outputs. Any originally invoked boundary map remains that map, not its label or the observation of a different boundary.

These assignments respect formation equations because they restrict and simultaneously re-present the same operands and actual maps. Ancestor/current equations use the specified θ and the common $S_*[n]$, not equality of incompatible full fingerprints. A comparison wholly on one leg is evaluated there; a mixed comparison uses its actual original source warrant, not equality of the reduct alone.

Induct on the number of nodes of the finite constituent/support/ process presentation. The first two cases supply the leaves. Adjoining any other constructor uses its actual map on the interpreted incident records. Its source-admission guard has unchanged arguments and comparisons, so the construction exists at exactly the original grade. A primitive graph is interpreted as that actual graph, not as an arbitrary function of the same type. This includes structural and history records which are not erased polynomial words.

For a licensed arrow use its original source operation and warrants, or its original composition of licensed steps. For f and g with transports r_X, r_Y, r_Z , their re-presented graphs satisfy

$$\begin{aligned} f_n &= r_Y f r_X^{-1}, & g_n &= r_Z g r_Y^{-1}, \\ g_n f_n &= r_Z (g f) r_X^{-1}, \\ \text{Dom}(g_n f_n) &= r_X \{x \in \text{Dom } f : f(x) \in \text{Dom } g\}. \end{aligned} \quad (74)$$

Cancellation proves identities and associativity as well. The first failure and its source witness read back identically, including an empty composite domain. Every licensed equality of operations, domains, failures and authority records remains an equality, so interpretation descends through the original relation congruence. This is the exact partial-operation law, not a theorem that a product of closed unbounded operators is closed.

Erasure now returns the entire original configuration and each arrow's original operation, domain and warrant. The same induction proves (72). Every original configuration has its source-admission presentation, and the accepted whole-source marking torsor is nonempty after the independent I_3 theorem. Retain all presentations, compatible markings and ancestral witnesses to obtain P_* ; no boundary outcome selected them. Fixed marks or apparatus attachments keep their original stabilizer comparisons. This proves totality on the original domain without adjoining an unauthorized symmetry. $\square$

The physical-use assertion in this theorem is an operation identity, not metadata. For an originally furnished $M : D_M \subset X \rightarrow Y$, keep the carrier/response algebra, preparation, ports,

exact domain, continuation and source warrant. Its interpreted graph obeys $r_Y^{-1}M_n r_X = M$, or is literally unchanged on unindexed coordinates. Equality is not inferred by observing only scalar outputs. It is instantiated by (67) for the full original field prescription and by the actual PSM3/SM3 current-leg realization. Existing Higgs-response or persistence readers are identified as their original maps, not re-proved physically from a tensor.

The input theorem does not convert an unactivated request into a fulfilled boundary claim: if a requested physical operation is absent, its original missing-contract/status record remains, not a defined zero or a source-proved exclusion. This does not shrink C_* . Theorem 11.3 uses the independently fixed physical declaration family and its actual source-qualified witnesses in addition to the all-input result above. Theorem 11.5 then receives that complete family at the same quantum–Einstein source. No further root selection, particle naming or full ancestral-target test is inserted between these comparisons.

7 Physical color ports and composite persistence boundaries

Two complementary constructions connect color response with physical boundaries. The proper-gauge-invariant observable sector gives an exact quotient of every correctly attached color port. The occupied localized composites give actual configuration-to-boundary operations with their licensed histories and classification readouts. The second construction is not inferred from the first: a singlet projector does not create a physical composite. Conversely, a physical persistence boundary need not first be replaced by a stable Hilbert-space mass eigenvector.

We work on the full source of Section 3, retaining the three-generation chiral inventory, color, weak and hypercharge couplings, the fluctuating Higgs field and charged Yukawa tensors, global form, anomaly data, preparation, normalization and boundary conventions. The physical heavy-hadron instance used below is the actual point at which SYM–6 receives charm and bottom occupancy [36, Theorem 19.2]. It is not every arbitrary Yukawa point, a separate QCD vacuum, or the native finite Hamiltonian of Section 21. The same physical source can admit different preparations and operational histories without being a product of independently chosen sector states.

7.1 The actual color action on the observable sector

Let $\mathcal{H}$ and Ω be the positive physical representation and vacuum of the occupied local-observable interface [34, Sec. 8.2]. Let $\mathcal{D}_{\text{obs}}$ be the span of the admitted finite gauge-invariant local field words applied to Ω , on their actual successive-product domains, and set

$$\mathcal{H}_{\text{obs}} = \overline{\mathcal{D}_{\text{obs}}}.$$

Only already represented words enter this definition. It does not assign a Hilbert domain to every renormalized insertion jet or include every charged superselection sector. Quark and gluon composites, when actually represented as gauge-invariant observables, are included; this is not a Higgs-only or pure-Yang–Mills subtheory.

Lemma 7.1 (Induced proper-color action). *For the inherited group $G_\Gamma = (\text{SU}(3) \times \text{SU}(2) \times \text{U}(1))/\Gamma$, with Γ contained in its common $\mathbb{Z}_6$ matter kernel, the map $g \mapsto [(g, 1, 1)]$ embeds $\text{SU}(3)$. The proper-color comparison induced on $\mathcal{H}_{\text{obs}}$ is $V_c(g) = I$. This space reduces the actual translation representation.*

Proof. The kernel generator is $(\omega_3, -1, e^{i\pi/3})$. If one of its powers has the form $(g, 1, 1)$, the last entry forces that power to be divisible by six, so $g = 1$. This proves the subgroup assertion without claiming that a separate fundamental-color bundle exists on every topology of the full quotient.

For a finite compactly supported observable word in a Minkowski trivialization, choose $X \in \mathfrak{su}(3)$ with $e^X = g$, and a smooth compact cutoff χ equal to one near all its supports. The proper transformation $e^{\chi X}$ equals g there and the identity outside a larger neighborhood. The connection transformation in the intervening region is retained. The path $e^{t\chi X}$ connects this comparison to the identity, so no extra large-gauge or boundary action is introduced.

Each observable and the physical vacuum representative define the same gauge-quotient vector after this operation. Thus the induced map is the identity on $\mathcal{D}_{\text{obs}}$, including its null relations, and extends uniquely to I on its closure. This constructs the action on the stated source sector; it does not postulate a new global color charge. If a larger representation has a boundary color action, its restriction agrees with this action only when it implements these same comparisons and fixes the vacuum.

Translation covariance and vacuum invariance take every admitted finite word to another such word. Positive and negative translations therefore preserve $\mathcal{H}_{\text{obs}}$, so its orthogonal projection commutes with the translation group and the space reduces its generators. $\square$

The gauge and global premises are the anomaly-neutral full-source ones [20, 21]. In all-left notation the elementary color checks are $2-1-1=0$ for the cubic anomaly and $2(1/6)-2/3+1/3=0$ for the color-squared hypercharge coefficient per generation. These checks do not replace the complete anomaly theorem or discard an axial anomaly. The represented descendant $D(f)\Omega = O_\rho(\square f)\Omega$ from Section 4 gives a nonzero vector orthogonal to Ω for a real compact f . Hence the sector just constructed is not the vacuum ray alone. Its nontriviality does not make this vector a hadron or turn the unstable Higgs pole into a spectral atom.

Theorem 7.2 (Full-joint-input neutral-port factorization). *Let $\mathcal{K}$ carry the actual strongly continuous unitary color representation R of an admitted input, including every constituent, dual, support and dressing factor. The strong Haar average*

$$P_c = \int_{\text{SU}(3)} R(g) dg$$

is its orthogonal invariant projection. Every bounded source port $J : \mathcal{K} \rightarrow \mathcal{H}_{\text{obs}}$ with the actual comparison equation $JR(g) = V_c(g)J$ satisfies

$$J = JP_c. \tag{75}$$

In particular a nonzero isometric color-intertwining in/out port from a representation with no invariant vector cannot have its range in this sector. The same factorization holds for an actually admitted finite multilinear neutral-boundary response after linearization in its complete joint colored tensor input.

Proof. Strong continuity on the compact group makes the vector-valued integral well defined. Haar inversion gives $P_c^* = P_c$, Haar invariance gives $P_c^2 = P_c$, and $R(h)P_c = P_c$ for every h . Conversely an invariant vector is fixed by the integral. Thus its range is exactly the fixed space. Lemma 7.1 gives $JR(g) = J$; boundedness of J permits integration and proves (75). If $P_c = 0$, then $J = 0$, which cannot be an isometry on a nonzero space. Adding a time-intertwining requirement does not evade this identity.

For the operational statement let A_B be the existing linearized value map of the full multilinear response. Its source comparison, with the neutral boundary and all siblings retained, is $A_B R(g) = A_B$. Finite-dimensional integration gives $A_B = A_B P_c$. When the source admits only an invariant linear subspace, carry out this calculation on that subspace. No unadmitted input or failed operation is extended by zero, and no nonlinear history is averaged as though it were a linear amplitude. $\square$

The intertwining equation is the required source comparison for the port, not a consequence of matching representation names. In particular the theorem is an exclusion on the indicated observable-sector interface, not a classification of all charged or extended-support sectors. An isolated $\mathbf{3}$, $\bar{\mathbf{3}}$ or $\mathbf{8}$ has no fixed vector, whereas complete joint tensors can have one. When the columns of E form a basis of the full invariant subspace, using the original source-admitted invariant contraction tensors,

$$P_c = E(E^*E)^{-1}E^*, \quad A_B x = A_B E(E^*E)^{-1}E^*x. \quad (76)$$

Indeed the displayed matrix is self-adjoint, fixes every column and annihilates its orthogonal complement. The inherited inner products give the unit tensors $\delta_{ab}/\sqrt{3}$ in $\mathbf{3} \otimes \bar{\mathbf{3}}$, $\delta_{AB}/\sqrt{8}$ in $\mathbf{8} \otimes \mathbf{8}$, and $\epsilon_{abc}/\sqrt{6}$ in $\mathbf{3}^{\otimes 3}$ [5, Sec. 13]. Their nonzero invariant spaces preserve neutral quark, gluonic and baryonic composite possibilities. Zero triality alone would not prove this result: the adjoint has zero triality but no fixed vector. Color averaging commutes with the retained weak and hypercharge actions on honest full-group inputs; it does not neutralize those charges or remove their physical boundary requirements.

Proposition 7.3 (Continuation of the exact quotient). *An actual bounded source comparison $J'S = TJ$, with S a color intertwiner on the Hilbert input spaces, obeys $SP_c = P'_c S$, and hence preserves (75). For an already admitted holomorphic or meromorphic component amplitude, the factorization holds on its entire connected continuation chain, including both principal and regular Laurent terms.*

Proof. Integrate $SR(g) = R'(g)S$. Substitution in the original port comparison then retains all its input and output factors. In an analytic detecting germ, $A_B(z)(I - P_c) = 0$; uniqueness propagates that zero across every overlap of its actual continuation chain. It annihilates every Laurent coefficient of this complementary component. This uses the supplied germ and sheet, not a continuation inferred from finite jets. $\square$

If the color frame moves, differentiation is of the complete identity:

$$(A_B)_{ab} = (A_B)_{ab}P_c + (A_B)_a(P_c)_b + (A_B)_b(P_c)_a + A_B(P_c)_{ab}.$$

All original source, product, preparation and endpoint contacts remain inside these derivatives. A Hilbert expression $a_B^* U j_C P_c$ likewise retains its actual readout comparison. Unitary source and boundary comparisons supply the adjoint identity. Merely isometric V, V_B with $Va_B = a_B' V_B$ do not imply $a_B'^* V = V_B a_B^*$; the latter must then be supplied separately. No physical boundary is created by an adjoint shortcut.

7.2 Occupied physical composites and their persistence boundary

Let X_S denote the full record interface of the independently fixed joint colored configuration category on the physical source S . Its finite configurations retain U-type, D-type, conjugate and gauge-support incidences, their complete tensor representation and the charge, weak, Higgs, anomaly, global-form, process and source data [28, Sec. 3]. The superscript “pre” used for this interface means before attachment to an output boundary, not before hadron formation. Overall color neutrality does not erase an occupied composite’s colored constituent and support incidences.

The particle-role source supplies a localized carrier configuration

$$\kappa_U = (U, \Phi_S(U), \sigma_U, \chi_U, \beta_U). \quad (77)$$

Here U is the actual local continuation regime, $\Phi_S(U)$ its field-carrier profile, σ_U the standing signature, χ_U the boundary/classification witness, and β_U the bounded transport/restoration

profile. Its equivalence retains the complete standing, witness, transport, quotient-comparison, identity-memory, boundary-trace and admitted-transport tuple

$$(\sigma, \chi, \beta, Q, I, \partial, T).$$

This is the source's same-preserved-class relation, not equality of a name, a single readout, directed reachability or reversible dynamics [44, Definitions 3.2–3.6, Proposition 5.2].

Definition 7.4 (The independently typed persistence branch). Let $\mathcal{P}_S$ consist of the full source records with the actual field mediation, local continuation and bounded transport/restoration data of (77), stable localization and identity retention in the declared regime, the complete comparison tuple above, and independently admitted color-neutral composite status. Every required weak, Higgs, charge, global, anomaly, preparation and boundary incidence remains attached. These conditions precede the boundary map; membership is not defined by success of that map or by overlap with a selected Hilbert subspace.

The particle-role theorem derives persistence identity from these antecedents; it does not make an arbitrary localized particle color neutral. The additional composite typing is the actual physical matter/color reception [47, Secs. 10–11]. A member is not a simultaneously sharp classical assignment to quantum gauge and Higgs fields. Joint multiparticle records require their actual joint admission; separate particle memberships do not construct a tensor product state.

Retaining the entire existing record gives a typed re-representation

$$\begin{aligned} i_S : \mathcal{P}_S &\longrightarrow X_S, & D_{\text{pers}}(S) &= i_S(\mathcal{P}_S), \\ \pi_S : D_{\text{pers}}(S) &\longrightarrow \mathcal{P}_S, \\ \pi_S i_S &= I, & i_S \pi_S &= I \quad \text{on } D_{\text{pers}}(S). \end{aligned} \tag{78}$$

To check membership, the local profile and matter/color response give the constituent and support incidences; the full S record gives the remaining siblings and authority. The existing witness and identity memory are retained before the output is evaluated. Thus π_S reads the same full record; it is not a section of a particle-class quotient or an inverse to a lossy forgetting map. The whole independent X_S and its other operations are unchanged.

The selected physical source has actual members. SYM-6's heavy-hadron theorem receives charm and bottom at the stated physical-SM point, retaining the heavy-mass, low-residual-momentum and lifetime ordering

$$m_Q \gg \Lambda_{\text{QCD}}, \quad |k| \ll m_Q, \quad \Gamma_Q \ll \Lambda_{\text{QCD}},$$

and the actual bound-state target [36, Theorem 19.2]. Its conclusion is physical occupancy, not only a large-mass label. The matter reception supplies the corresponding localized particle-persistence representatives [47, Criterion 10.3], [44, Theorem 6.1]. Let $\mathcal{K}_c(S), \mathcal{K}_b(S) \subset \mathcal{P}_S$ be those complete records in their licensed identity-retaining regimes. Both are nonempty, their union need not be disjoint, and no unique hadron is assigned to a quark flavor. Charm retains its less precise expansion control; top does not satisfy this same lifetime/hadronization conjunction. No infinite lifetime or vacuum one-particle vector follows.

This physical occupancy is preserved by the occupied symmetry assignment [52, Secs. 8–9]. It is not an identification of all HQET records with generic full-SM deformations. In particular the heavy flavor, spin and spin-flavor action records have the source's specialized incidences, while the matched-current record retains its distinct materialized matching locus. Equality-induced branch maps are used only on their actual fibers. The matched amplitude remains the qualified QCD/HQET observation described below.

Theorem 7.5 (Complete reception of the persistence branch). *Restrict the source particle equivalence $\sim_\Pi$ to $\mathcal{P}_S$, and let $\mathcal{B}_{\text{pers}}(S)$ be its admitted neutral composite-persistence classes. There are actual maps*

$$\begin{aligned} q_S : \mathcal{P}_S &\longrightarrow \mathcal{B}_{\text{pers}}(S), & q_S(\kappa) &= [\kappa]_\Pi, \\ k_S = q_S \pi_S : D_{\text{pers}}(S) &\longrightarrow \mathcal{B}_{\text{pers}}(S), \\ \overline{\chi}_S k_S(i_S \kappa) &= [\chi_\kappa]. \end{aligned} \tag{79}$$

The map q_S is onto the complete independently typed persistence branch. Every member has its actual identity-continuation witness for the operational channel. This family is nonempty on both occupied heavy subdomains.

Proof. The original particle-role proposition supplies the equivalence relation with the full comparison tuple, and its persistence theorem constructs the standing-bearing class from the stated antecedents [44, Proposition 5.2, Theorem 6.1]. The independently admitted composite status fixes the color-neutral physical boundary type. Thus restriction defines q_S , with exactly those admitted classes as its codomain. Every such class has an actual representative satisfying Definition 7.4, and every such representative is included independently of flavor, mass order or a later detector choice. This proves surjectivity at this physical grade, not a decision procedure for a trial tensor.

An equivalent representative transports χ through its original boundary comparison. Hence its witness class $[\chi]$ is unchanged and $\overline{\chi}_S$ is well defined. Equality of χ alone is not used to infer equality of the complete particle class. Composing with the already checked full-record reader proves the remaining maps.

For each κ , the source licenses the null admissible transport/restoration chain as identity continuation. The actual input $i_S \kappa$, this history, and the indicated class $[\kappa]_\Pi$ with its original witness form the productive persistence-record channel of BCRS–IIC [5, Sec. 8]. The physical charm and bottom instances above prove nonemptiness. This is an admitted operational comparison of an existing physical configuration, not a claim that its dynamics is trivial or that decay never occurs. $\square$

The full attached record is $(x, k_S(x))$, with its original incident maps. Keeping x matters: its preparation, momentum, mixed response or time-dependent Higgs observations need not be functions of a particle-persistence class. The theorem attaches a physical boundary without replacing the source by that quotient. The witness χ has classification and persistence authority; it is not assigned probability one, a scattering amplitude, a pole or a mass atom. Other supplied hadronic admission records enter by their own complete witnesses, not by copying a proton-specific history onto every hadron.

Corollary 7.6 (Identity-role exhaustion and actual failures). *Within the realized field-mediated regime, an admissible candidate ground of particle identity is received by the persistence-class route or has its original failed identity claim. An actual source-proved failure of stable localization excludes the particle-identifiable persistence assertion in that regime.*

Proof. The particle source exhausts primitive-object, label-only, hidden-core, record-only, transport-free and cross-domain imported identity [44, Theorem 18.1]. A route carrying the complete stable persistence witness reduces to the configuration above; otherwise its proposed identity fails for the source-proved reason. Its necessary-condition theorem gives $\text{ParticleIdent} \Rightarrow \text{StableLoc}$ [44, Proposition 5.1]; an actual contrary witness therefore refutes that persistence claim. A missing localization proof is not such a witness. Neither statement excludes a non-particle field configuration or supplies a scattering classification. $\square$

7.3 Full-source covariance of persistence boundaries

The persistence comparison has an explicit nontrivial family of lawful source presentations. Use the physical right-handed convention [35] and write the complete Yukawa interaction as

$$\mathcal{L}_Y = -\overline{Q}_L Y_u \tilde{H} u_R - \overline{Q}_L Y_d H d_R - \overline{L}_L Y_e H e_R + \text{h.c.}$$

Choose constant $g = (V_Q, V_u, V_d, V_L, V_e) \in \text{SU}(3)^5$ on the generation multiplicities, transform each fermion by its corresponding V , and set

$$Y'_u = V_Q Y_u V_u^*, \quad Y'_d = V_Q Y_d V_d^*, \quad Y'_e = V_L Y_e V_e^*. \quad (80)$$

Gauge, metric and Higgs fields are unchanged. This acts within equal gauge representations, not between right-handed charm and bottom fields of different hypercharge.

Let r_g be this invertible substitution on the complete source presentation, including dual source tensors, existing preparations, renormalized insertions, endpoints and boundary witnesses. Define gS by transporting those existing objects. For every supplied multilinear functional the construction is

$$W_{gS,n}(r_g A_1, \dots, r_g A_n) = W_{S,n}(A_1, \dots, A_n). \quad (81)$$

Equivalently compose the original functional with r_g^{-1} in each slot. Its preparation, normalization and product/contact prescription are transported together. A represented operation A with domain D_A has transported operation $r_g A r_g^{-1}$ on $r_g D_A$. Here r_g denotes the corresponding presentation-carrier isomorphism, not the five generation matrices acting on arbitrary Hilbert vectors. The underlying physical Hilbert space and state can be identical.

Theorem 7.7 (Full-source persistence square). *The substitution (80) preserves the literal full source and constructs bijections $F_g : \mathcal{P}_S \rightarrow \mathcal{P}_{gS}$ and $b_g : \mathcal{B}_{\text{pers}}(S) \rightarrow \mathcal{B}_{\text{pers}}(gS)$ such that*

$$q_{gS} F_g = b_g q_S, \quad \overline{\chi}_{gS} b_g = r_g^\partial \overline{\chi}_S. \quad (82)$$

Every actual partial history h is transported by

$$h_g = F_g h F_g^{-1}, \quad D_{h_g} = F_g D_h.$$

All composed domains, identity comparisons, changed classes and source-defined failures are retained. The occupied heavy subfamilies are carried to the same physical source roles in the new presentation.

Proof. The kinetic and gauge operations are generation blind. For example

$$\overline{Q}'_L Y'_u \tilde{H} u'_R = \overline{Q}_L V_Q^* V_Q Y_u V_u^* \tilde{H} V_u u_R = \overline{Q}_L Y_u \tilde{H} u_R.$$

The other Yukawa and all bosonic terms are preserved in the same way. This retains the quantum Higgs field, not only a vacuum representative. The constant determinant-one choice has vanishing generation trace in the gauge-squared and gravitational Jacobian terms. The original gauge and global anomaly data are still retained. If non-Abelian flavor backgrounds are present, they and their counterterms undergo the same coordinate substitution; their anomalies are not declared zero. General anomalous Abelian rephasings require the original theta and mass-phase transport and are not included in this restricted family.

The substitution preserves every field, adjoint and source relation. Its inverse preserves and reflects each actual domain and failure. Equation (81) therefore uses the same correlation values in the new coordinates; a positive quadratic sum remains the same sum. An existing operator or continued channel retains its authority, while a merely formal insertion is not promoted to one. No unitary implementation of a broken physical flavor symmetry is assumed. This is the

full passive source action, not invariance with the numerical Yukawas held fixed outside their stabilizer.

Apply this substitution to every entry of (77) and its retained comparison data. It preserves the antecedents of Definition 7.4. Every original full-tuple equivalence is transported to an equivalence; the inverse proves reflection. Hence $b_g[\kappa] = [F_g\kappa]$ is well defined and bijective, and evaluating the same transported witness proves both equations in (82). For histories h followed by k ,

$$D_{k_g h_g} = \{F_g x : x \in D_h, h(x) \in D_k\} = F_g D_{kh}, \quad k_g h_g = F_g(kh)F_g^{-1}.$$

This also preserves the first failed comparison and the actual endpoint, without adding any unlicensed history. Matrix multiplication proves $F_1 = I$ and $F_{g_2 g_1} = F_{g_2} F_{g_1}$.

To check the heavy roles rather than their labels, put $A_u^L = Y_u Y_u^*$ and $A_u^R = Y_u^* Y_u$, and similarly in the down sector. On a simple nonzero singular-value stratum their spectral projections are

$$p_i(A) = \prod_{j \neq i} \frac{A - \lambda_j I}{\lambda_i - \lambda_j}.$$

They transform by the corresponding left/right conjugation. At a degeneracy use the admitted cluster, not an unprovided individual eigenvector. In a temporary paired singular frame, $Y_u e_i^R = y_i e_i^L$ and $Y_u^* e_i^L = y_i e_i^R$, one has

$$(V_Q e_i^L)^* (V_Q Q_L) = e_i^{L*} Q_L, \quad (V_u e_i^R)^* (V_u u_R) = e_i^{R*} u_R.$$

Thus the heavy coefficient field in an actual matching chart, including its registered rest mass and velocity, is unchanged. The physical mass, width and hadronization predicates are invariant data of the transported source. These are flavor-role projections, not hadron spectral projections or physical bare-quark states. They prove the last assertion at the actual H06 point. $\square$

The full charged-current normalization is retained. In a simple paired mass frame,

$$V_{ij} = e_i^{u,L*} e_j^{d,L}, \quad |V_{ij}|^2 = \text{Tr}(p_i^{u,L} p_j^{d,L});$$

both endpoints transform by the same V_Q . Their overlap, the original $g_2/\sqrt{2}$ coefficient and the full W^+ current pairing are unchanged. Holding an endpoint fixed is a different source operation. For contrast, mixing right-handed charm and bottom by R at fixed hypercharge gives, with $D_\mu = \partial_\mu - ig_1 B_\mu Y_R$,

$$D_\mu(Rq_R) - R D_\mu q_R = -ig_1 B_\mu [Y_R, R] q_R, \quad Y_R = \text{diag}(2/3, -1/3).$$

Its arbitrary- B_μ coefficient does not vanish for a mixing R . The lawful source map above does not make that omitted-sibling square commute. Color singlet status alone also does not make $\bar{c}_L \gamma^\mu b_L$ a neutral gauge-invariant physical insertion: its electric charge is canceled by the actual charged weak endpoint.

7.3.1 Matching and differentiated source contacts

SYM-6's selected QCD/HQET matching keeps its actual current, external ports, common-velocity/recoil regime, scale, normalization, operator basis and registered order [36, H01–H06, R01]. Write that comparison at its supplied authority as

$$A_{\text{QCD}} = C_N M_N + \mathcal{R}_N. \quad (83)$$

When full amplitudes are supplied, $\mathcal{R}_N$ is their actual difference; for a registered expansion it is the formal remainder class at that order, not a newly supplied full function. An admitted effective basis change gives

$$M'_N = R M_N, \quad C'_N = C_N R^{-1}, \quad C'_N M'_N = C_N M_N.$$

The same source substitution transports the remainder and every external current index. This proves covariance of the qualified comparison, not an increase of its accuracy or an equality between an active-weak, fluctuating-Higgs full-SM amplitude and a QCD amplitude. The physical heavy boundary and its qualified matching observation can be retained together without identifying those two functions.

All source derivatives obey the complete product rule. Along a path of the constant matrices, $K_f = \dot{V}_f V_f^*$ gives $\dot{Y}'_u = K_Q Y'_u - Y'_u K_u$. At fixed original Y_u the mixed rule is

$$(Y'_u)_{ab} = (V_Q)_{ab} Y_u V_u^* + (V_Q)_a Y_u (V_u^*)_b + (V_Q)_b Y_u (V_u^*)_a + V_Q Y_u (V_u^*)_{ab}. \quad (84)$$

The down and lepton formulas have the corresponding factors. The original mixed Higgs/Yukawa action insertion is still

$$D_H D_Y S[k, Z] = - \int dV_g (\bar{Q}_L Z_u \tilde{k} u_R + \bar{Q}_L Z_d k d_R + \bar{L}_L Z_e k e_R + \text{h.c.}). \quad (85)$$

It transforms with the same cancellation as the original Yukawa word. Setting H to a constant before differentiation would remove this contact. The complete renormalized insertion additionally retains its mixing and counterterm derivatives from Section 3.

For any actual differentiable normalized response W along a fully transported presentation $z(a, b)$, including preparation, endpoints and prescription, the constant physical value gives

$$0 = DW[z_a], \quad 0 = D^2 W[z_a, z_b] + DW[z_{ab}].$$

At finite source-jet authority these are the corresponding algebraic identities, not a new transverse differentiability assertion. For an existing continuous multilinear response, finite matrix substitution does give a smooth family in its existing insertion slots. Differentiating $C'_N M'_N$ keeps both first terms and all four mixed second terms; their cancellation follows from differentiating $R^{-1} R = I$. The same derivative acts on the remainder at its original authority. Power counting alone supplies no uniform bound on its derivatives. An actually supplied analytic channel is identical on its detecting domain under this coordinate transport and consequently retains its whole continued germ. Neither χ nor a truncated matching formula creates such a germ.

7.4 Persistence endpoints of source processes

Use the fixed source's primitive operation graphs, with their joint operands, intrinsic parameters, exact domains, outputs and failure diagnostics [1, Sec. 2]. A finite use is an original application or a finite well-typed network of such applications and actual interface comparisons. A source-given infinite-dimensional or infinitary operation remains its original primitive graph; it is not replaced by finite samples. Shared operands remain shared. This records the same weak, Higgs or preparation input at several vertices, not copies of a quantum state or independently selected Hilbert factors.

Let $w : D_w \subset X \rightarrow Y$ be such an actual partial operation. When an original terminal interface supplies the complete persistence and neutral-composite witnesses of Definition 7.4, let

$$\gamma : Y_{\text{pers}} \longrightarrow \mathcal{P}_S$$

read those retained fields. If the output already is $\mathcal{P}_S$, this is the identity. No terminal reader is asserted for every process. Restricting an untyped output to a hoped-for successful subset does not construct γ or a hadronization dynamics.

Theorem 7.8 (Exact finite-process attachment). *Every original finite source use with this witnessed terminal interface has the actual boundary and readout operations*

$$\begin{aligned} D_{w,\text{pers}} &= \{x \in D_w : w(x) \in Y_{\text{pers}}\}, \\ K_w &= q_S \gamma w : D_{w,\text{pers}} \longrightarrow \mathcal{B}_{\text{pers}}(S), \quad R_w = \bar{\chi}_S K_w. \end{aligned} \quad (86)$$

These constructions receive all such uses, with their original joint incidences, continuation, terminal boundary and failures. A positive operational witness requires an actual $x \in D_{w,\text{pers}}$, its source process and its indicated boundary record; forming an empty pullback supplies no positive witness.

Proof. The terminal reader is the original witness-bearing interface map. Composing its partial graph with w and the everywhere-defined q_S gives exactly (86). Nothing is obtained by inverting the particle quotient.

For coverage, start with the actual graph at a native application, including all operand records. At a composition vertex graft the original common-interface maps and retain every shared edge. Induction over the finite source use preserves its evaluation, exact domains and all incident values. A supplied limit-defined operation is the same graph at its primitive vertex. For an operator application, use its actual domain and every successive-product domain; density of individual domains or the notation of a common core does not prove arbitrary product closure.

At the terminal witness-bearing interface, γ reads the given record, and Theorem 7.5 identifies exactly its particle class and witness. Conversely each constructed diagram reads back as those same native operations followed by γ and q_S . This proves all and only the claimed finite-use reception, not existence of an absent primitive process.

For a sequence $f_1, \dots, f_n$, the domain is explicitly

$$\left\{ x : \begin{array}{l} x \in D_{f_1}, \quad f_1(x) \in D_{f_2}, \quad \dots, \\ f_{n-1} \cdots f_1(x) \in D_{f_n}, \quad f_n \cdots f_1(x) \in Y_{\text{pers}} \end{array} \right\}. \quad (87)$$

Reassociation preserves these intermediate tests and the first failed vertex in the original evaluation order. If every source vertex succeeds but the final output has another type, retain that native outcome; it is neither a value of K_w nor an invented failed physical process. An actual source witness can prove persistence inapplicable; a merely unresolved question cannot. Thus the original diagnostic information survives. Nonemptiness is required exactly as stated; the identity processes of Theorem 7.5 give actual nonempty examples without proving production by every other use. $\square$

When input and output both have persistence records, retain

$$(q_S(\kappa_{\text{in}}), q_S(\kappa_{\text{out}}))$$

and the original process. A proved equality of these classes is identity preservation; a proved difference is a class-changing physical outcome. Decay is not renamed identity continuation. A joint final family requires its actual joint record and is not forced into a single-particle output. The absence of a retained particle role is not automatically nonexistence of the source output.

Proposition 7.9 (When a process descends to particle classes). *An actual $h : D_h \subset \mathcal{P}_S \rightarrow \mathcal{P}_S$ admits a partial operation on $q_S(D_h)$ with exact domain reflection and its retained failure readouts if and only if:*

1. D_h is a union of complete $\sim_{\Pi}$ classes;
2. equivalent successful inputs have equivalent outputs;

3. every retained failure diagnostic is constant on its corresponding excluded equivalence class.

Proof. A single quotient input has one definedness status, one successful output class, or its specified failure readout. Evaluating a descended operation therefore forces all three conditions. Conversely define $\bar{h}([\kappa]) = [h(\kappa)]$ on $q_S(D_h)$ and retain each common failure value outside that domain. The conditions prove representative independence and exact domain reflection. They do not hold for an arbitrary preparation-dependent dynamics merely because its endpoints have particle identities. Without them, the full-record operation (86) remains the correct object. $\square$

Let $F : S \rightarrow S'$ be an actual invertible full-source comparison, reflecting domains and all original witness relations. Transport of the complete equivalence tuple gives bijections F_P, F_B satisfying

$$q_{S'} F_P = F_B q_S, \quad \bar{\chi}_{S'} F_B = F_\chi \bar{\chi}_S.$$

For actual process and reader comparisons $F_Y w = w' F_X$ and $\gamma' F_Y = F_P \gamma$, direct substitution yields

$$\begin{aligned} F_X(D_{w, \text{pers}}) &= D_{w', \text{pers}}, \\ K_{w'} F_X &= F_B K_w, \quad R_{w'} F_X = F_\chi R_w. \end{aligned} \tag{88}$$

The inverse proves equality of domains, rather than only inclusion. The same vertexwise proof preserves class changes and first failures. The source presentations of Theorem 7.7 are constructed examples of this comparison. A general noninvertible postcomposition need not preserve productivity. No differentiability of a particle label, q_S or χ follows from these operational equations; actual differentiable responses retain their separately proved contacts and domains.

7.5 Realization and non-null history reconstruction

The physical significance of these histories cannot be reduced to a formal list of labels. The realization theorem for irreversible standing continuation forces a consequence-retaining persistence profile up to lawful same-domain equivalence [45, Theorems 21.1–22.1, Corollary 20.2]. The realized-persistence theorem supplies actual standing-sensitive reconstruction contrasts and a nonempty stable interior [46, Theorems 4.14, 9.10 and 10.2]. We now receive these structures on the same physical histories.

On the complete persistence/history interface just constructed, the minimal-realization antecedents have actual source witnesses. The admissibility gate is fixed; continuations are the licensed histories; standing, identity, transport and boundary readouts belong to the internal source; and the failed act retains its failure rather than being repaired by reindexing its lineage. The occupied heavy members supply actual realized persistence, not merely a capability to ask whether persistence is possible. Irreversibility here is the retained fail-closed consequence of the same evaluated act. It neither denies that a Hamiltonian time group has inverses nor gives an unstable hadron an infinite lifetime.

Let Hist_D be the original complete history interface in this realized domain, including its failure distinctions. Its licensed persistence-sensitive reconstructions are actual maps

$$r : \text{Hist}_D \longrightarrow O_r, \quad r \in \mathcal{F}_D.$$

The source constrains them by same-domain licensing, admissibility, continuation compatibility, standing-sensitive distinguishability, lawful composition, quotient/role invariance and fail-closedness [46, Definitions 4.1–4.10, Theorem 4.6]. They are not arbitrary tests chosen to create a contrast. Define

$$h \sim_{\text{Rec}} k \iff r(h) = r(k) \text{ for every } r \in \mathcal{F}_D.$$

Let I_H receive the original history graphs and their arguments, order, shared operands and first failures by the actual-source composition used in Theorem 7.8. An independently supplied complete history operation remains that primitive graph, even if it is not a finite sequence of time steps. Let $\text{Hist}_{\text{rec}} = I_H(\text{Hist}_D)$ be this exact interface image; reading the retained source records is the inverse on that image. For every original r , retain its graph and its injective output-role identification j_r , so that

$$r_{\text{rec}}(I_H h) = j_r(r(h)). \quad (89)$$

This identity is reception of the original map, not replacement by a coarse detector number. It needs no numerical table of that map's values.

Theorem 7.10 (Faithful non-null reconstruction of physical histories). *There is a bijection*

$$I_{\text{Rec}} : \text{Hist}_D / \sim_{\text{Rec}} \longrightarrow \text{Hist}_{\text{rec}} / \sim_{\text{Rec,rec}}, \quad [h] \longmapsto [I_H h], \quad (90)$$

preserving every original reconstruction readout, standing-sensitive contrast, continuation/composition domain, quotient comparison and fail-closed consequence. On this actually realized persistence domain, the reconstruction quotient is non-null and its induced persistence interior is nonempty and stable.

Proof. For $h, k \in \text{Hist}_D$, equality of all original r values implies equality of all received values by (89). Conversely equality of all received values implies $r(h) = r(k)$ for each r , since j_r is injective. Thus I_H preserves and reflects reconstruction equivalence. The induced map is well defined and injective; every received history lies in its stated image, proving surjectivity. No history representative or section of the particle quotient was chosen.

Composition and continuation commute because the received operations are their very original graphs on the exact domains, while each r satisfies its continuation and composition laws. The same reasoning transports admissibility, role/quotient comparisons and the original failure contrast. The realized-persistence theorem supplies genuinely distinct standing-relevant classes in the original quotient. Injectivity and readout preservation carry that distinction to the received one. More explicitly the pair map $([h], [k]) \mapsto (I_{\text{Rec}}[h], I_{\text{Rec}}[k])$ preserves the original differentiation structures and every lawful equivalence between them. It therefore transports the source's nonempty interior quotient and its reconstruction stability [46, Propositions 6.7 and 9.4]. Finally the irreversible-standing realization theorem fixes the same consequence-retaining profile up to its original equivalence [45, Theorem 21.1]. $\square$

The family-valued profile $(r(h))_{r \in \mathcal{F}_D}$ can be read componentwise; it introduces neither a Hilbert product nor a probability measure. Its stable interior is the source's quotient of standing-relevant distinctions, not an asserted manifold or a norm estimate. A zero value of one quantum response does not collapse this full history profile. Equally, its non-nullity is not a lower bound for a scattering amplitude or a nonzero pole. A restriction to a smaller family of histories must retain the actual realization/reconstruction antecedents before this non-nullity theorem is applied there; selecting only a successful endpoint does not preserve them automatically.

7.6 Global authority and the complete physical color comparison

The source's stronger Yang–Mills conclusions remain positive inputs, with their original objects and domains. The local construction supplies the sharp-local gauge-invariant theory, its vacuum representation, curvature differential-polynomial fields and gap [16, 18]. The nonlocal theorem fixes a predecessor local object $B(G)$ containing its algebra, distinguished vacuum and dynamics, and constructs

$$\Xi_G : \mathcal{E}(G) / \sim_{\text{def}} \longrightarrow \mathcal{S}_{\text{ext}}(G)$$

with actual sector representations π_ξ , vacuum-equivalent local restrictions away from the support, and unitary deformation comparisons [19, VI.10–VI.14]. This is not only a list of charge labels. Its inputs are the stated compact closed line/surface supports with the fixed theorem-input label sets, not all conceivable nonlocal charges, open endpoints or junction networks. Its stronger same-theory, same-domain completion and gap conclusions likewise retain the fixed local object and its state/dynamics [19, VIII.121–VIII.124].

Two different uses must be kept distinct. FCGS–II already receives the source-qualified theorem object and its hypotheses as nonlocal authority for the marked U/D targets [28, Sec. 6]. That use reindexes the authority coordinate; it is not a map from a constituent configuration into $\mathcal{E}(G)$ or an evaluation of π_ξ on a different physical source. There is no need to acquire this exact theorem authority again. If a conclusion actually invokes an extended-support physical operation, however, its full representation, localization, domains and transport must be retained. A charge label, screening quotient or compact local response cannot replace that operation. Conversely a citation to the theorem alone does not activate every imaginable extended-support request on the full SM.

For a proposed matched-base attachment i of the fixed local YM algebra, the dynamical GNS comparison uses actual identities

$$\omega_S(i(A)) = \omega_{\text{YM}}(A), \quad \alpha_t^S(i(A)) = i(\tau_t^{\text{YM}}(A)),$$

with the corresponding local inclusions and representation hypotheses [19, VIII.22, VIII.27]. A common color Lie algebra does not prove these equations. They are conditions of that particular attachment, not a requirement that every direct full-source physical construction pass through pure YM. The persistence maps proved above use the occupied full source directly. Similarly the exact full-matter electric screening quotient from SYM–9 retains its actual global form and matter charges, but is not an area/perimeter phase law or a universal confinement theorem [21, Theorem 9.1, Sec. 9.4].

The preceding results supply complementary parts of the physical interface. Theorem 7.2 supplies the source-side action and exact neutral-port quotient for all its admitted joint inputs. Theorem 7.5 covers the entire independently defined physical persistence branch, not only heavy examples. Theorems 7.8 and 7.10 retain every actual process with its original terminal witness and every furnished standing-sensitive reconstruction. Their histories are physically non-null and retain irreversible consequence; they are not merely formal record labels.

Complete color comparison combines source-operation reception with independent input coverage and physical-witness authority. The confinement-acquisition interface [5, Sec. 8] retains:

1. the independently characterized, nonempty, role-complete joint input domain with its common matter, Higgs, vacuum and boundary attachments;
2. comparison with actual source configuration operations for every terminally relevant class, not restriction to successful channels;
3. actual neutral physical boundaries and witnesses for the positive realizations, or exact source-proved exclusion or inapplicability;
4. the source’s phase-appropriate confinement and asymptotic-sector statement on the sectors that the conclusion invokes;
5. the actual persistence, stability, pole, spectral or scattering authority of each invoked physical boundary;
6. source-preserving continuation and gluing of all mixed data, domains, failure cases and boundary types; and

7. complete outcome comparison on that declared domain at the common matter/Higgs source, without hiding unresolved required authority as a zero channel.

All-state asymptotic completeness is required only when that stronger claim is made on a stated scattering domain. An occupied composite needs no invented birth history, and an unstable persistence record needs no invented stable mass atom. But neither may a persistence readout replace an independently required resonance or scattering map.

To state the distinction used in the complete application, let Use_S be the actual furnished source uses, and Req_T the original target's required input/output uses with their physical grades. Exact native reception proves, for every $u \in \text{Use}_S$,

$$\begin{aligned} \text{Dom}(\text{Rec}(u)) &= \text{Dom}(u), \\ \text{Eval}_{\text{rec}}(\text{Rec}(u), x) &= \text{transport}(\text{Eval}_S(u, x)). \end{aligned} \tag{91}$$

where intrinsic outputs, shared arguments, authority, continuation and failure fields are included, not only a measured scalar. The complete application matches every member of Req_T , at its original grade, with its actual received operation and physical witness or exact source-proved alternative. Theorem 6.11 supplies the independently formed input and licensed-operation comparison. Theorem 11.3 combines that comparison with the independently fixed physical declarations and their actual source-qualified outcomes. Neither proof defines the required family as the image of successful channels.

An entire operation or boundary family can be supplied intensionally by its source theorem, without a table of all its values; its actual graph, domain and authority must nevertheless be compared. This distinction introduces no extra colored sector, pole for every neutral tensor or universal hadron-production request. It also prevents absence of a counter-request from being mistaken for a coverage proof. The complete color-to-composite continuation and asymptotic classification are proved by the positive source/interface comparison, with the neutral-port and persistence results above at their exact grades. Theorem 11.5 places that full interface at the same occupied matter, Higgs, quantum and gravitational source; the inherited premetric unification and physical occupancy remain its starting data.

8 Physical stress, curvature response and Einstein reception

The occupied physical source also admits a direct metric-facing construction. It uses its original scalar detector, local stress, state and source prescription, without identifying them with the native Hamiltonian of Section 26. The shared premetric origin is unchanged. The comparison concerns the actual physical response and its continuation, rather than two independent foundations. Its action-level completion is Section 9; the full common-source statement is Theorem 11.5.

We first construct smooth all-test stress profiles from actual smeared-field preparations. An original curvature-dependent mass-source protocol then produces a separately represented conserved tensor with the full physical Higgs germ. The same preparation detects that tensor at first order and has positive excitation energy. Finally the vacuum stress gives an exact matched Einstein member, while differentiation of the complete source equation identifies the response required for physical backreaction.

8.1 The physical source and its complete metric insertion

Fix the occupied physical parameter point of Section 3, including its full gauge, Higgs, chiral and lepton inventory, global form, state and boundary convention, renormalization and improvement data, and physical continuation. Let $\mathcal{H}_{\text{ph}}$ be its positive Hilbert space, Ω its invariant vacuum, and $\mathcal{D}_0$ its joint invariant polynomial-field domain. The fields used below are the original mass

insertion O_ρ , the admitted total renormalized stress/current, and the finite smeared words used to prepare states. Their joint distributions, not merely their separate names, belong to this represented local-observable interface [34, Sections 8–9 and 12].

The actual metric action-principle identification is

$$F_g(h) = \frac{1}{2}\hat{T}(h), \quad W_{[\psi]}(h) = \frac{\langle \psi, F_g(h)\psi \rangle}{\|\psi\|^2}, \quad \hat{T}(h) = \int dV_\eta \hat{T}^{\mu\nu}(x) h_{\mu\nu}(x). \quad (92)$$

Here h is any real compact smooth covariant symmetric metric test. The selected coframe and its derived connection, composite mixing, source contacts, boundary and improvement convention are part of this equality. It is received at the physical source's admitted metric/current interface, not inferred from Poincare covariance or from a formal chiral source column [33, Sections 5–7, Theorem 6.1]. Every protocol in this fixed first-response class uses the same insertion and all compact tests. Its actual response graph is therefore the graph in (92), giving the direct-ledger source attachment of [25] before tensor representation.

The total Ward identity retains internal exchanges and any declared external current. In the source convention it reads

$$W_{[\psi]}(\mathcal{L}_X \eta) = -J_{[\psi]}(X)$$

for compact X . It is closed for the total physical stress at constant closed parameters. Trace anomalies need not vanish. The conversion of the variational stress to the physical translation charge is its separately admitted source comparison, including sign, normalization, temporal smearing and boundary limit.

For the physical scalar and momentum formulas use $\eta = \text{diag}(1, -1, -1, -1)$. The metric-response convention is $\delta S[k] = \frac{1}{2} \int dV_g T^{\mu\nu} k_{\mu\nu}$. The native $(-+++)$ presentation in Section 26 is not identified with this one by changing a metric sign alone: action, current and charge conventions must be transported together. The Einstein matching below is written using the actual coefficient of $T = t_* \eta$, so it does not guess that conversion.

8.2 Smooth stress from actual polynomial preparations

Theorem 8.1 (Smooth normalized physical source family). *Let $v_a = A_a \Omega$ be finitely many admitted field words with smooth compact tests, allowing $v_0 = \Omega$. Put*

$$v(c) = \sum_a c_a v_a, \quad q(c) = \|v(c)\|^2, \quad \psi(c) = v(c)/\sqrt{q(c)}, \quad q(c) > 0.$$

The stress matrix-element distributions $u_{ab}^{\mu\nu}(x) = \langle v_a, \hat{T}^{\mu\nu}(x) v_b \rangle$ have smooth polynomially bounded representatives. The actual normalized stress is

$$t_c^{\mu\nu}(x) = \frac{\sum_{a,b} \bar{c}_a c_b u_{ab}^{\mu\nu}(x)}{q(c)}. \quad (93)$$

It is smooth in spacetime and C^∞ in the real preparation coordinates with values in C_{loc}^∞ and distributions. Its source is exactly $W_{[\psi(c)]}(h) = \frac{1}{2} \int t_c : h$, with the original current, support and boundary data. The notation for the smooth representative does not define a point-field operator.

Proof. For a matrix element with m fields on the left and n on the right of the stress, use the actual joint tempered vacuum distribution. Change its coordinates to the stress position x and the relative positions $r_j = y_j - x$. This is an invertible linear change of a distribution, not restriction to a coincidence surface. Translation covariance and vacuum invariance make it translation invariant in x .

A distribution invariant under translations in that factor is $1_x \otimes w(r)$. For a direct proof, choose a product test $\chi(x)$ of integral one. Subtract successively the partial averages in the four x coordinates of an arbitrary test. Each difference has zero integral in one coordinate and is the derivative of its compact primitive. Translation invariance annihilates these derivatives, leaving precisely the pairing of a distribution w with the integral in x . The construction also applies to Schwartz tests with their primitive estimates. It does not assume that a general test is a finite sum of tensor products.

After smearing the other fields, the remaining distribution is represented by

$$u_{ab}(x) = w_{ab}(F_{ab,x}),$$

where $F_{ab,x}$ is the ordered product of their translated tests, with conjugation and reversed adjoint order on the bra side. The map $x \mapsto F_{ab,x}$ is smooth in the Schwartz topology. For a seminorm bounding the continuous distribution w_{ab} ,

$$\sup_{x \in K} |\partial_x^\alpha u_{ab}(x)| \leq C \sup_{x \in K} p_N(\partial_x^\alpha F_{ab,x}) < \infty.$$

Translations of fixed tests have at most polynomial growth in these seminorms, proving the corresponding global bound. Empty words give constant distributions. No expression $T(0)$, or a point-field vector, has entered.

Sesquilinearity gives (93). The positive Gram polynomial q has a smooth reciprocal on $q > 0$, locally uniformly on $q \geq \delta > 0$. This proves every preparation derivative in the displayed seminorms. Source and Ward identities hold on each actual matrix element, hence on their sum and normalized quotient. $\square$

The exact normalization contacts are useful. For $N_c(h) = \langle v(c), F_g(h)v(c) \rangle$,

$$\begin{aligned} \partial_s W_c(h) &= \frac{N_s(h)}{q} - \frac{N(h)q_s}{q^2}, \\ \partial_s \partial_t W_c(h) &= \frac{N_{st}(h)}{q} - \frac{N_s(h)q_t + N_t(h)q_s + N(h)q_{st}}{q^2} + \frac{2N(h)q_s q_t}{q^3}. \end{aligned} \quad (94)$$

Nonlinear preparation paths contribute their actual c_{st} terms. Moving tests add their derivatives; a moving source or prescription has its separate operator/form derivative. The homogeneous extension $2 \|v\|^2 W_{[v]}(h) = \langle v, \hat{T}(h)v \rangle$ satisfies the parallelogram identity and polarizes to the original stress form, as in the normalized-ray construction of [26, Definition 4.5 and Theorem 4.6]. These are actual smooth preparation families, not a smooth dependence on a still-unconstructed curved quantum state.

8.3 Curvature probes of the original mass source

Use the original scalar insertion of Definition 4.1, with its fixed real nonzero normalization and identity mixing:

$$O_\rho = nO_H + bI, \quad G_\rho = n^2 G_H + L.$$

Here L is the actual local extension/contact transport, not a fitted background. The physical detecting channel and its marked continuation are those of [20, Sections 9.1–9.8]. Write $O = O_\rho$, and use $\square = \eta^{\mu\nu} \partial_\mu \partial_\nu$, whose Fourier multiplier on e^{ipx} is $-p^2$. Fix the curvature convention by its covariant-metric variation

$$\delta R[k] = -R^{\mu\nu} k_{\mu\nu} + \nabla^\mu \nabla^\nu k_{\mu\nu} - \square_g(g^{\mu\nu} k_{\mu\nu}).$$

For compact k , let $g_t = \eta + tk$ in a small nondegenerate neighborhood and compose the original source as $\rho_{\sigma,t} = \rho_0 + \sigma R(g_t)$. This has the correct source dimension and compact relative support. At $t = 0$ it is the original source for every σ .

Theorem 8.2 (Original curvature contact and its full response). *For the complete relative source action F , with $F(\eta, \rho_0) = 0$, the mixed insertion is*

$$A_k = \partial_\sigma \partial_t F|_0 = O(L_\eta k), \quad L_\eta k = \partial^\mu \partial^\nu k_{\mu\nu} - \square(\eta^{\mu\nu} k_{\mu\nu}). \quad (95)$$

Use independent $\sigma_1, \sigma_2, t_1, t_2$ with $g = \eta + t_1 k_1 + t_2 k_2$ and $\rho = \rho_0 + (\sigma_1 + \sigma_2)R(g)$. For the same normalized $W = -i \log Z$,

$$\begin{aligned} W_{\sigma_1 \sigma_2 t_1 t_2}|_0 &= 2W_{\rho\rho}(L_\eta k_1, L_\eta k_2), \\ F_{\sigma_1 \sigma_2 t_1 t_2}|_0 &= 2F_{\rho\rho}(L_\eta k_1, L_\eta k_2), \\ \frac{W_{\sigma_1 \sigma_2 t_1 t_2} - \omega(F_{\sigma_1 \sigma_2 t_1 t_2})}{2i} &= \kappa_T(A_{k_1}, A_{k_2}). \end{aligned} \quad (96)$$

These identities hold at the actual finite source-jet authority as well. The right side additionally has its represented physical scalar meaning, independently of convergence of an infinite source series.

Proof. The σ derivative is $F_\rho[R(g_t)]$. Its t derivative consists of

$$F_{g,\rho}[k, R] + F_{\rho,\rho}[\rho_t, R] + F_\rho[R_t].$$

At the baseline $R = 0$ and $\rho_t = 0$, so only the last term survives. Volume and insertion-prescription derivatives multiply the same zero curvature in this contact. They have not been discarded from the source generally. The map L_η is a continuous compact-test map of differential order two without support enlargement, so its insertion and adjoint are on the original scalar domain.

For the fourth derivative, first differentiate twice in σ . The result is $W_{\rho\rho}[R, R]$, with coefficient one. At $R = 0$, the two metric derivatives must hit different curvature factors. There are exactly two such assignments. Any derivative of $W_{\rho\rho}$, or both derivatives on one curvature factor, leaves the other factor zero. The identical calculation applies to F , including its complete source contacts. Finally the normalized rule $W_{ab} = \omega(F_{ab}) + i\kappa_T(F_a, F_b)$ proves (96). In its partition expansion the two nontrivial connected blocks pair (σ_1, t_1) with (σ_2, t_2) , or their interchange. This also checks the factor two without suppressing the action contact. $\square$

The protocol is an external mass-source experiment at an unchanged baseline. It does not differentiate a newly chosen global vacuum. If a state or prescription is independently changed, the full source comparison and its contacts precede this pullback, as in Theorem 3.1. A fixed curvature coupling in a gravitational target remains fixed: σ here defines a probe, not a new value of that coupling or an equivalence between different nonminimal theories.

The formal adjoint in (95) gives the represented tensor

$$I^{\mu\nu} = 2(\partial^\mu \partial^\nu - \eta^{\mu\nu} \square)O, \quad I(k) = 2A_k, \quad \partial_\mu I^{\mu\nu} = 0, \quad \eta_{\mu\nu} I^{\mu\nu} = -6\square O. \quad (97)$$

The factor two follows from the stress convention. The conservation and trace are exact distributional identities. In particular $L_\eta(\mathcal{L}_X \eta) = 0$; no equation of motion or gauge-fixing condition is used. This tensor is the mixed curvature contact, not the entire baseline stress.

For the held scalar source word $S_u = \int dV_g R(g)u$, the corresponding curved coefficient is

$$T_u^{\mu\nu} = 2[-G^{\mu\nu}u + (\nabla^\mu \nabla^\nu - g^{\mu\nu} \square_g)u], \quad \nabla_\mu T_u^{\mu\nu} = R(g) \nabla^\nu u. \quad (98)$$

The first formula follows from the declared variation and compact integration by parts. For the second, the Bianchi identity kills ∇G , while the commutator on the scalar gradient gives $R^{\nu\mu} \nabla_\mu u$. Its combination with $-G^{\mu\nu} \nabla_\mu u$, including the factor two, is $R \nabla^\nu u$. Thus $u = \sigma O$ retains the external source exchange. Differentiating this identity retains connection, raised-index and volume contacts; a frozen divergence of δT_u is not its derivative. A genuinely renormalized O_g also has its own metric and prescription derivatives, including terms $R \delta O_g$ at higher order. The held-scalar calculation does not eliminate them or construct a curved state.

8.4 Physical continuation and a strict local preparation response

All vacuum vectors $I(k)\Omega = O(2L_\eta k)\Omega$ lie in the same physical Hilbert space as the original scalar port. Their finite mixed kernels, including admitted bounded physical observers, are positive by the squared-norm identity. Positivity is distinct from the time-ordered extraction in (96).

The tensor multiplier is $\Pi^{\mu\nu}(p) = 2(\eta^{\mu\nu}p^2 - p^\mu p^\nu)$. In the trace channel the complete operational germs are consequently

$$G_{O,I_{\text{tr}}}(z) = 6zG_\rho(z), \quad G_{I_{\text{tr}},I_{\text{tr}}}(z) = 36z^2G_\rho(z). \quad (99)$$

These are the products induced by the actual mass-source pullback. A separately normalized time ordering of differentiated fields can differ by its local equal-time or extension contacts; the actual comparison must be retained. In the detector convention above, both n and L remain in (99). Multiplication also retains the regular terms generated from the principal part, not just a pole location.

At the source-authorized nonzero simple Higgs pole p_H in the mass-squared variable, with original residue $R_\rho \neq 0$,

$$\text{Res}_{p_H} G_{I_{\text{tr}},I_{\text{tr}}} = 36p_H^2 R_\rho \neq 0, \quad G_\rho(z) = \frac{G_{I_{\text{tr}},I_{\text{tr}}}(z)}{36z^2}$$

on its marked neighborhood avoiding zero. The full existing germ is recovered; no new sheet, pole or spin-two state is inferred. The unordered scalar/trace spectral matrix is

$$\begin{pmatrix} 1 & 6s \\ 6s & 36s^2 \end{pmatrix} d\mu_\rho(s),$$

a positive outer product. Its spectral mass variable is not a complex resonance pole. On the nonzero bounded positive-mass band of Theorem 4.5, multiplication by $6s$ has bounded inverse. Hence the scalar-generated and trace-generated closed boundaries agree on that band. No inversion at the massless stratum is claimed.

Theorem 8.3 (A named nonzero physical preparation response). *There is a real compact scalar test f such that $\phi = D(f)\Omega \neq 0$, where $D = \square O_\rho$, and $\phi \perp \Omega$. The normalized same-source preparation*

$$\psi_s = \frac{\Omega + s\phi}{\sqrt{1 + s^2 \|\phi\|^2}} \quad (100)$$

lies in the common polynomial domain. With $k_0 = -(f/6)\eta$, it satisfies

$$I(k_0) = D(f), \quad \left. \frac{d}{ds} \langle \psi_s, I(k_0) \psi_s \rangle \right|_{s=0} = 2 \|\phi\|^2 > 0. \quad (101)$$

Its excitation energy is

$$\langle \psi_s, P^0 \psi_s \rangle = \frac{s^2 e_\phi}{1 + s^2 \|\phi\|^2} > 0 \quad (s \neq 0), \quad 0 < e_\phi = \langle \phi, P^0 \phi \rangle < \infty. \quad (102)$$

Proof. Theorems 4.4 and 4.5 construct the nonzero descendant as an actual scalar test pullback. A nonzero complex test can be split into real and imaginary parts. Translation invariance makes $\omega(D(f)) = 0$, giving orthogonality. The invariant polynomial domain contains both ϕ and $I(k_0)\phi$. For $k = f\eta$, $L_\eta k = -3\square f$; hence the specified k_0 gives $I(k_0) = D(f)$. Symmetry on that domain and the zero first derivative of the normalization give $2 \text{Re} \langle \phi, I(k_0)\Omega \rangle = 2 \|\phi\|^2$.

Smooth test translation proves every required generator domain and $\phi = -P_\mu P^\mu O_\rho(f)\Omega$. Thus ϕ has no null-mass spectral component. Forward spectrum implies $P^0 \geq 0$, with $p^0 = 0$ only at zero four-momentum; the nonzero vector therefore has strictly positive energy. Its first energy moment is finite by the proved generator domain. The vacuum is invariant, so the two cross energy terms vanish, proving (102). No spectral projection is applied to this preparation: such a projection need not preserve the joint local-field domain needed for its stress. $\square$

The vacuum mean of every compact $I(k)$ is zero because $\omega(O)$ is constant, yet its response kernel and (101) are nonzero. This is an explicit local metric/source response without a global-charge limit. It must not be conflated with a universally nonzero first derivative of the different baseline stress T .

On the source's actual stress-to-charge comparison domain, the same ψ_s , $s \neq 0$, has a compact metric test separating its total stress from the vacuum. To prove this, form the calibrated temporally smeared, spatially cut-off charge difference. Subtract the vacuum at each finite cutoff if the source uses a subtracted charge; do not subtract infinite total energies. Its limit is the strictly positive value in (102), so some finite cutoff already separates the source. Its actual stress-to-charge test map gives the desired compact metric test.

This conclusion for the named preparation retains the charge-domain condition. Generator membership alone would not prove every global current limit. Independently, the local joint-field domain supplies a weaker existential check: if every normalized finite-word ray had the same stress as the vacuum on all compact tests, polarization would make every stress smearing a scalar form. Its admitted charge-generation relation would then act trivially on all local fields, contradicting the nontrivial translation action of the scalar channel. Thus some finite-word preparation has nonconstant local stress even without assigning a new global-charge domain to the named curve.

8.5 The exact physical vacuum Einstein member

Theorem 8.4 (All-test vacuum source and Einstein matching). *The actual physical vacuum stress has the form*

$$\langle \Omega, \hat{T}_{\mu\nu}(x)\Omega \rangle = t_* \eta_{\mu\nu}, \quad W_*(h) = \frac{t_*}{2} \int dV_\eta \eta^{\mu\nu} h_{\mu\nu}, \quad (103)$$

where t_* is the fixed source's real coefficient, including its vacuum and identity convention. For the inherited Einstein law $G[g] + \Lambda_* g = \kappa_* T$, $\kappa_* \neq 0$, the pair (η, Ω) satisfies the complete source-attached equation on all compact tests if and only if

$$\Lambda_* = \kappa_* t_*. \quad (104)$$

Its lawful diffeomorphic and frame presentations, with the whole physical source transported, are presentations of the same solution.

Proof. Translation invariance makes the one-point stress distribution a constant tensor, by the distributional argument in Theorem 8.1. Proper connected Lorentz invariance makes a symmetric constant rank-two tensor proportional to the metric: rotations remove mixed components and fix the spatial diagonal, and one boost relates it to the temporal component. Hermiticity makes t_* real. The actual source insertion gives (103) on every test.

Since $G[\eta] = 0$, the Einstein residual is $(\Lambda_* - \kappa_* t_*)\eta$. Its vanishing on all compact tests is equivalent to (104). This is a full tensor equation, hence includes its normal, mixed and tangential constraint and evolution components. The physical quantum evolution is its original translation evolution with invariant vacuum ray; none of its fields, interactions or channels is reconstructed from the solved metric. Transport of metric, source, state, domains, spin lift, tests and boundary

preserves the equation by actual source naturality and Einstein covariance. A diffeomorphism does not turn flat geometry into physical curvature. $\square$

The vacuum is not a null source, even if its certified representative has $t_* = 0$: its correlations, physical poles and preparation responses remain. A trace anomaly can contribute to the actual coefficient; Poincare invariance does not set it to zero. No cutoff vacuum-energy sum is used in (104).

If the gravitational calibration has not yet been chosen over this fixed physical source, (104) selects a nonempty compatible member of the inherited Einstein parameter family, after which the joint point is frozen. It is not a prediction of the observed cosmological constant or equivalence of different joint points. For an already fixed Λ_*, κ_* , it is instead an exact test. A mismatch cannot be repaired by changing a parameter silently. Under the admitted homogeneous convention change

$$T' = T - Cg, \quad t'_* = t_* - C, \quad \Lambda'_* = \Lambda_* - \kappa_* C, \quad (105)$$

the residual $\Lambda_* - \kappa_* t_*$ is invariant. In the convention $T_{\text{vac}} = -\rho_R g$, the same matching says $\Lambda_R + \kappa_R \rho_R = 0$; the nonlinear Einstein theorem fixes this quotient-stable combination, not its numerical value [39, Theorems 23.1 and 27.3]. No infinite-volume constant density is treated as a finite Hamiltonian offset, and no physical response is normal-ordered away at fixed Λ_* .

8.6 The physical response required for backreaction

The matched member is an exact source-attached solution. Smooth preparation profiles alone, however, do not provide the metric-induced quantum response of a coupled physical source family. The distinction has a precise all-test equation.

For covariant k , put

$$P_\eta(k)^{\mu\nu} = \frac{1}{2}(\text{tr}_\eta k)\eta^{\mu\nu} - k^{\mu\nu}.$$

Then $D[\sqrt{|g|}g^{\mu\nu}]_\eta[k] = \sqrt{|\eta|}P_\eta(k)^{\mu\nu}$. Suppose an actual differentiable physical source/Einstein family through the matched member is given on a declared domain. Resolve its response into its fixed-frame metric-induced part $\delta_m W$, preparation part $\delta_p W$, and any explicitly varied original sibling source part $\delta_j W$, with all state, frame, test and prescription contacts assigned consistently. Define

$$\Xi(k, h) = \delta_m W(h) - \frac{t_*}{2} \int dV_\eta P_\eta(k)^{\mu\nu} h_{\mu\nu}. \quad (106)$$

This subtracts a known baseline density/index contact; it does not replace the unknown metric response by that contact.

Proposition 8.5 (Contact-complete linearized compatibility). *At the fixed matched κ_*, Λ_* , every actual differentiable coupled family satisfies*

$$\frac{1}{2\kappa_*} \int dV_\eta G'_\eta(k)^{\mu\nu} h_{\mu\nu} - \Xi(k, h) = \delta_p W(h) + \delta_j W(h) \quad (107)$$

for every compact metric test. The preparation term is already constructed by (94) on the actual finite-word families, including (100).

Proof. Differentiate the exact residual

$$\mathcal{R}(g, \text{source}; h) = \frac{1}{2\kappa_*} \int \sqrt{|g|} (G^{\mu\nu}[g] + \Lambda_* g^{\mu\nu}) h_{\mu\nu} d^4x - W_{g, \text{source}}(h).$$

At the flat matched point the baseline Einstein tensor vanishes. The remaining terms are its derivative, the cosmological contact $\Lambda_*/(2\kappa_*) \int P_\eta(k) : h$, and the three actual source derivatives. Substitution of (104) and (106) proves (107). Moving tests contribute their derivatives on both sides before cancellation. $\square$

One cannot set $\Xi = 0$ by differentiating (103) away from the family where it was proved. The physical in-out two-jet in a fixed prescription is

$$W_{\text{in-out};hk} = \omega_*(F_{hk}) + i\kappa_T(F_h, F_k),$$

with the actual moving-state and prescription terms where present. It retains the full stress insertions and metric, spin, density, curvature, mixing and source contacts. A real causal backreaction derivative uses its actual causal or closed-time prescription; a possibly complex in-out Hessian is not substituted for it. Proposition 26.7 proves this distinction by construction on the native source. It does not identify that source's kernel with the physical one here. Finite physical insertion jets retain precisely their finite-jet authority until an actual metric-differentiable physical family supplies the corresponding response in (106).

The Ward derivative is equally constrained. For fixed compact X ,

$$(\delta W)_{\text{fixed}}(\mathcal{L}_X \eta) + W_*(\mathcal{L}_X k) = 0 \quad (108)$$

on a closed actual family; moving X adds $W_*(\mathcal{L}_{\delta X} \eta)$, and declared exchange adds $-\delta J(X)$ on the right. Along a lawful transported vacuum $g_s = \chi_s^* \eta$, the complete source derivative is exactly $(t_*/2) \int P_\eta(\mathcal{L}_X \eta) : h$, while $G'_\eta(\mathcal{L}_X \eta) = 0$. Thus the presentation tangents satisfy the full equation with density, index and moving-test contacts. They do not settle transverse metric response.

The nonvacuum preparation gives a direct countercheck on extending the flat assignment without backreaction. At a compact separating stress test in its admitted charge domain,

$$\mathcal{R}(\eta, \psi_s; h) = -[W_{[\psi_s]}(h) - W_*(h)] \neq 0.$$

Thus changing the quantum preparation while keeping the matched flat metric does not define a coupled solution map on that curve. This is not a no-go theorem for a physical semiclassical or other Einstein realization; it identifies the actual response that must enter its construction.

The nonlinear source theory distinguishes a field-eligible Einstein member from a smooth hyperbolic Cauchy family and its compatible seed and flow [38, Definition 16.2 and Theorem 17.3]. The certified-zero vacuum identity branch of that theory is not a theorem about an arbitrary nonzero vacuum coefficient. Stationary solutions remain valid members of a dynamical theory, but one all-test stationary solution and its coordinate presentations do not by themselves construct that theory's coupled preparation and metric solution map.

The resulting represented stress family, nonzero curvature/contact response and complete Higgs germ share the same physical source as the matched Einstein member. Their contact-complete continuation equation fixes the response comparison for a coupled family when such a family is invoked. Theorems 9.2, 9.4 and 9.7 construct the original-source action, its canonical holding comparison and an occupied dynamical member at the original quantum-action order. Theorem 11.5 retains this member together with the complete physical interface. A universal transverse solution family, a particular lattice limit or a Wheeler–DeWitt quantization is not an additional premise of that result. The native coupled realization remains a distinct construction; it is not identified with the physical source by common field names.

9 The original physical source's causal Einstein action

The metric response in Section 8 is an operation of the occupied interacting Standard Model source. We now construct its source-varying gravitational action before restricting to a solution. This supplies the action and holding comparison beyond equality of vacuum tensors. The construction uses the original causal field prescription of Section 5; it does not substitute the native Hamiltonian, a pure-gauge vacuum or a newly normalized curved theory.

The common premetric architecture, gravitational role and full nonlinear minimal-response Einstein law are inherited [9, 39, 10, 38]. The action below is a variational representative of these existing roles with their actual physical source. It is not a new foundational coupling of separately founded quantum and gravitational theories. Its quantum construction has the original perturbative or retained-order authority, while the stronger represented and continued baseline observations remain the same objects. One compatible stationary member is proved. A solution through every changed preparation is not asserted.

9.1 One source, one coefficient prescription

Fix the physical source S_* of Section 3, including all color, weak, hypercharge, Higgs, chiral-fermion and Yukawa interactions, the global form, spin and boundary branch, counterterms, improvement convention and original preparation family. Write $\mathcal{A}_{S_*}$, $\star$, T , R and $\mathcal{E}_*$ for its original graded field carrier, ordinary product, chronological products, relative insertions and normalized interacting reconstruction. The gauge-fixed computational carrier and its physical quotient retain their distinct meanings.

Theorems 5.2 and 5.3 identify these maps with the original source prescription, rather than with a free-field coefficient expectation. In particular, with its complete interaction V ,

$$Q(F) = S_V(F) = S_0(V)^{-1} \star S_0(V + F), \quad Q(0) = 1. \quad (109)$$

All maps in this section use that same V and normalization. The selected physical regime and retained error convention are the ones of [20, Definitions 2.1–2.2] and [33, Definition 2.1].

Let $\mathfrak{J}$ be the ideal generated by the external compact source amplitudes. Inverses and logarithms with constant term one are taken in the $\mathfrak{J}$ -adic coefficient completion, or in its supplied retained-order quotient with the original remainder record. At each specified coefficient they require finitely many products. No numerical sum is assigned to an unproved formal series. In particular, metric-source amplitude is not an expansion about $\kappa = 0$, and no Standard Model parameter is changed by introducing this notation.

Use the original $(+---)$ metric and covariant-source convention of (92). Denote the complete local coefficient-action prescription by $S_{S_*}[g, \Phi; j]$, where Φ includes its original fields, ghosts and computational partners. For $g = \eta + k$, define the compact relative insertion

$$F[g, j] = S_{S_*}[g, \Phi; j] - S_{S_*}[\eta, \Phi; j_*], \quad F[\eta, j_*] = 0. \quad (110)$$

This is the difference of local densities on the original compact relative-support or finite-region envelope. It is not subtraction of infinite-volume actions. Inverse metrics, volume density, the normalized coframe, induced spin connection, composite mixing, curvature terms and source counterterms are expanded as the original coefficient functions. Their finite-order local contacts are retained.

Put $f_g(h) = D_g F[h]$ and $f_g(k, h) = D_g^2 F[k, h]$. These are source labels before the original interacting insertion map is applied. At the baseline,

$$R_V(f_\eta(h)) = F_g(h) = \frac{1}{2} \hat{T}(h), \quad \mathcal{E}_* R_V(f_\eta(h)) = W_*(h). \quad (111)$$

Its first represented insertion has exactly the physical domain already supplied in Section 8. This does not invert a possibly noninjective relative insertion map.

A moving original chronological prescription, product, reconstruction or BRST differential is differentiated as part of the same source comparison. One may work in a fixed coefficient frame and transport those original maps. This does not identify an independently chosen curved-background renormalization with (110); such a different construction would require its own comparison theorem.

9.2 Metric coordinates and the actual holding map

The original normalized kinetic/coframe packet fixes the metric argument of (110). A convenient formal coframe representative near the baseline is $e = (\eta^{-1}g)^{1/2}$. Its unit-constant-term series is η -self-adjoint and satisfies $e^T \eta e = g$. The comparison to any other allowed oriented coframe uses its original spin lift, not a change of metric value or a deletion of spin data.

For a canonical presentation write

$$g_{00} = N^2 - \gamma_{ij}\beta^i\beta^j, \quad g_{0i} = -\gamma_{ij}\beta^j, \quad g_{ij} = -\gamma_{ij}. \quad (112)$$

The time-gauge coframe is $e^0 = N dt$, $e^a = e_i^a(dx^i + \beta^i dt)$, with $\gamma = e^T e$. The original spin/frame comparison and

$$\Psi = (\det \gamma)^{1/4} \psi_{\text{time-gauge}} \quad (113)$$

give the Weyl half-density coordinates. The inverse, adjoint, antifield and dual-source maps are transported together; the Higgs, gauge and test-density components use their original tensor types. No independent torsion or additional fermion is introduced.

Denote this invertible local field substitution by $\Phi = M_g \Phi_{\text{can}}$, and its map on the coefficient carrier by ι_g . Define the canonical-coordinate prescription by transporting the original one:

$$\mu_{\text{can},g}(\iota_g X, \iota_g Y) = \iota_g(X \star Y), \quad \mathcal{E}_{\text{can},g}(\iota_g X) = \mathcal{E}_*(X). \quad (114)$$

Chronological, adjoint and BRST maps are pulled back by the same rule. Original Jacobian, anomalous normalization and local mixing terms belong to the transported prescription; coordinate notation does not remove them. In particular a physical trace anomaly does not vanish by this construction.

Proposition 9.1 (Complete multiplier-coordinate source response). *At $(N, \beta, \gamma) = (1, 0, \delta)$, the tangent $v = (n, b, k)$ has*

$$(h_v)_{00} = 2n, \quad (h_v)_{0i} = -b_i, \quad (h_v)_{ij} = -k_{ij}, \quad (115)$$

and its complete original source insertion is

$$F_g(h_v) = \hat{T}^{00}(n) - \sum_i \hat{T}^{0i}(b_i) - \frac{1}{2} \sum_{i,j} \hat{T}^{ij}(k_{ij}). \quad (116)$$

This continuous linear test map is invertible onto all compact symmetric metric tests. For $w = (m, c, \ell)$, the second coefficient contact is

$$\begin{aligned} C(v, w)_{00} &= 2nm - 2b \cdot c, \\ C(v, w)_{0i} &= -k_{ij}c^j - \ell_{ij}b^j, \\ C(v, w)_{ij} &= 0. \end{aligned} \quad (117)$$

Every furnished second source derivative consequently contains the term $D_g S[C(v, w)]$ in addition to $D_g^2 S[h_v, h_w]$.

Proof. Differentiate (112). The inverse first test map is $n = h_{00}/2$, $b_i = -h_{0i}$, $k_{ij} = -h_{ij}$; hence no component is lost. Stress symmetry combines its two mixed entries, giving (116) from (111). A second differentiation gives (117); the chain rule gives the stated additional insertion. All state, prescription and field holding terms remain inside the original source derivatives. Moreover

$$D_v \sqrt{|g|} = n + \frac{1}{2} \text{tr } k, \quad D_v g^{-1} = -\eta^{-1} h_v \eta^{-1}.$$

Thus the inverse-metric convention $-\frac{1}{2} \sqrt{|g|} T_{\mu\nu} \delta g^{\mu\nu}$ equals the covariant-metric convention $+\frac{1}{2} \sqrt{|g|} T^{\mu\nu} h_{\mu\nu}$. This conversion does not change the separately supplied signed stress-to-time-charge map. All tests are spacetime smearings; no sharp equal-time stress operator is used. $\square$

The field-holding identity is equally explicit. With the full graded derivative and the original boundary terms,

$$D_g(S_{S_*} \circ M_g)[h] = (D_g S_{S_*}[h])_{\Phi_{\text{held}}} + D_{\Phi} S_{S_*}[(D_g M_g[h])\Phi_{\text{can}}]. \quad (118)$$

Moving original state or prescription data contributes its own derivative. Pure coordinate contributions cancel in complete evaluations by (114); a genuine source or preparation change does not.

9.3 Closed-time variation and causal source composition

Write $Q_g = Q(F[g, j])$. For real acts its original unitary normalization gives $Q_g^* = Q_g^{-1}$ at the supplied grade. Define

$$K_*(g_+, j_+; g_-, j_-) = \mathcal{E}_*(Q_-^* \star Q_+), \quad W_{\text{CTP}} = -i \log K_*. \quad (119)$$

At equal branches $K_* = 1$; the logarithm is fixed by zero constant term. The two-branch field products exist before $\mathcal{E}_*$. They are not reconstructed from a scalar in-out functional.

Theorem 9.2 (Original retarded stress from the two-branch action). *The held insertion*

$$A_g(h) = -iQ_g^{-1} \star D_g Q_g[h] = R_{V+F[g, j]}(f_g(h)) \quad (120)$$

is Hermitian. For $g_{\pm} = g \pm \epsilon h/2$, with the other source data equal on the two branches,

$$D_- W_{\text{CTP}}[g; h] := \left. \frac{d}{d\epsilon} W_{\text{CTP}}(g + \epsilon h/2; g - \epsilon h/2) \right|_0 = \mathcal{E}_*(A_g(h)). \quad (121)$$

At the baseline it is the original all-test metric source $W_*(h)$. In a fixed original coefficient frame its metric derivative is

$$D_k A_g(h) = R_{V+F}(f_g(k, h)) + i\{T_{V+F}(f_g(k), f_g(h)) - R_{V+F}(f_g(k)) \star R_{V+F}(f_g(h))\}. \quad (122)$$

The noncontact term has the original retarded support; the first term and every local extension retain the complete source contacts.

Proof. Differentiate $Q_g^* \star Q_g = 1$. It gives $(Q_g^{-1} \star DQ_g)^* = -Q_g^{-1} \star DQ_g$, proving Hermiticity. Substitution of (109) cancels the two baseline inverse factors in $Q_g^{-1} DQ_g$, giving the original $R_{V+F}(f_g(h))$, not a newly defined stress operator.

At the diagonal, the plus derivative of K_* is $i\mathcal{E}_*(A_g(h))/2$. The derivative of the inverse on the minus branch, whose tangent is $-h/2$, gives the same contribution. Hence $D_- K_* = i\mathcal{E}_*(A_g(h))$. The logarithmic denominator is one, proving (121). Branch exchange conjugates K_* , so it sends W_{CTP}^* to $-W_{\text{CTP}}$; the diagonal difference derivative is real. Equation (111) identifies its baseline source on every compact test.

Differentiating the same inverse-relative-source expression gives (122), equivalently (34) with the full contact $f_g(k, h)$. Its chronological and ordered products coincide when the input variation is outside the causal past of the output, by the original causal factorization. At coincidence their original extension remains. The result is therefore the actual retarded susceptibility, not an in-out Hessian multiplied by a discontinuous step function. Metric density, inverse-index, spin, coframe, improvement and counterterm terms belong to $f_g(k, h)$. Moving ADM coordinates add the contact of (117) before applying the formula. $\square$

For a moving original product or reconstruction one differentiates the complete expression, for example

$$D_k \mathcal{E}(A) = \dot{\mathcal{E}}(A) + \mathcal{E}(D_k A), \quad D_k(X \star Y) = \dot{X} \star Y + X \star \dot{Y} + \dot{\mu}(X, Y).$$

Chronological and calibration maps have their corresponding terms. The moving BRST identity is $d_g A_i(g) = 0 \Rightarrow d_0 \dot{A}_i = -\dot{d} A_i$, not a declaration that each differentiated summand is a frozen cocycle. Proposition 5.5 preserves the full physical readout and its original representative independence. No gauge-fixed mixed expression is thereby given an unproved positive-Hilbert operator domain.

Proposition 9.3 (Causal increments and their fixed holding sign). *For original relative acts F, G, H , put $Q_F(G) = Q(F)^{-1} \star Q(F + G)$. Then*

$$Q_F(G) \star Q_{F+G}(H) = Q_F(G + H), \quad Q_F(0) = 1. \quad (123)$$

When F is later than G , causal factorization gives $Q(F + G) = Q(F) \star Q(G)$. Along an admitted formal source path the left generator is

$$i\dot{Q}(F) \star Q(F)^{-1} = -Q(F) \star A_F(\dot{F}) \star Q(F)^{-1}. \quad (124)$$

Proof. The middle inverse factors cancel in (123). The causal equality is the original factorization on its declared supports. Finally (120) gives $\dot{Q} = iQ \star A_F(\dot{F})$; multiplication by i proves (124), including its minus sign and conjugation. Differentiation of these operations supplies their coefficientwise causal continuation and all higher contacts. The source-path parameter is not physical time by renaming. $\square$

9.4 Canonical source action with metric and boundary contacts

The preceding source action has an actual canonical-coordinate pullback, not merely a current with the same name. Apply (112)–(114) to its complete local coefficient density on a finite slab. At each supplied coefficient it has finite jet order. Let Z_A denote all coordinates varied in this presentation: matter and metric/coframe coordinates, together with any source coordinates actually varied. An external control held fixed is instead a held jet coefficient; its later variation uses the full functional derivative. Thus Z_A is not just the matter multiplet.

Theorem 9.4 (The original canonical holding identity). *Put $Q_A^r = \partial_t^r Z_A$, with maximal temporal order m for the coefficient under consideration. Define the spatial Euler derivative and temporal boundary coefficients by*

$$\begin{aligned} \frac{\delta_{\text{sp}} L}{\delta Q_A^j} &= \sum_{\alpha} (-\partial_x)^\alpha \frac{\partial L}{\partial (\partial_x^\alpha Q_A^j)}, \\ P_A^r &= \sum_{s=0}^{m-r-1} (-\partial_t)^s \frac{\delta_{\text{sp}} L_{\text{can}}}{\delta Q_A^{r+s+1}}, \quad 0 \leq r < m. \end{aligned} \quad (125)$$

With the original graded left/right derivative conventions and spatial boundary current, the canonical density is

$$H_{\text{can}} = \sum_{A,r} P_A^r \partial_t Q_A^r - L_{\text{can}}. \quad (126)$$

Write its integrated identity as $S_{\text{can}} = \Theta_I - C_{\text{src},I} + B_I$. With the same canonical fields and jet coordinates held,

$$D_N C_{\text{src},I}[n] = D_N \Theta_I[n] + D_N B_I[n] - D_N (S_{S_*} \circ M_g)[n]. \quad (127)$$

The last term is the original normal source insertion and the explicit field-holding derivative in (118), including its original equation-of-motion and boundary contacts.

Proof. First integrate spatial derivatives of each variation by parts, retaining their boundary current. Repeated temporal integration by parts gives

$$\delta L_{\text{can}} = \sum_A E_A(L_{\text{can}}) \delta Z_A + \partial_t \left(\sum_{A,r} P_A^r \delta Q_A^r \right) + \partial_i \theta^i.$$

The coefficient of each δQ_A^r is the finite telescoping sum in (125); its signs come from the temporal integrations by parts. Performing the calculation with the original Grassmann ordering gives the graded identity without assuming a nondegenerate fermion velocity Hessian. Introducing multipliers for $Q_A^{r+1} = \partial_t Q_A^r$ and eliminating them recovers precisely the original Euler equations. These constrained jet coordinates are bookkeeping, not new physical fields or a new Hilbert representation.

Equation (126) rewrites that same action as kinetic one-form minus Hamiltonian term, with its boundary contribution. Differentiation at the declared holding convention gives (127); (118) supplies the last term's actual source meaning. Applying the transported original chronological, relative-source and reconstruction maps preserves this coefficient identity and its contacts. It is the local microscopic coefficient density which is converted, not a local Legendre transform of the generally nonlocal W_{CTP} . $\square$

All source curvature and higher-derivative counterterms remain. They contribute to momenta of the corresponding varied metric/source coordinates and to boundary terms; held external jets retain their full later functional derivative. In the coupled action the metric momentum is the sum of geometric and source contributions. A naked Einstein–Hilbert ADM momentum cannot replace it. First-order Weyl sectors retain their original primary constraints. No zero-fermion reduction or independent torsion is used, and no coefficient is resummed into an extra physical pole.

If the actual kinetic, boundary and holding derivatives vanish in a particular presentation, the normal canonical row reduces to $-W(h_n)$, not $+T^{00}(n)$. Otherwise all terms of (127) remain. Angular coframe and spin contributions are decided by the full density, even for a symmetric metric lift. This is an action/variational comparison, not a global Hilbert–ADM symplectomorphism or a represented quantum local-lapse constraint algebra.

9.5 The original physical time and a strict source response

At $(g, j) = (\eta, j_*)$, the field, prescription and relative maps restrict to the original identity. The original total-stress charge map, including its signed normalization, temporal smearing, spatial boundary/improvement and clock comparison, is therefore unchanged. The source's Noether-generation theorem identifies that map with its actual self-adjoint H_0 , not with a guessed uncentered integral of T^{00} [34, Chapter 12]. We use $\hbar = 1$, $H_0 \Omega = 0$, $H_0 \geq 0$, with the original phase convention and no infinite-volume vacuum-energy subtraction.

Lemma 9.5 (Finite time continuation and natural word domains). *Let $U(t) = e^{-itH_0}$ and $\alpha_t(A) = U(t)^* A U(t)$. For every actual finite word $P = A_n \cdots A_1$ with its natural successive-action domain,*

$$\alpha_t(P) = \alpha_t(A_n) \cdots \alpha_t(A_1) = U(t)^* P U(t), \quad D(\alpha_t(P)) = U(t)^* D(P). \quad (128)$$

For multi-time words all offsets are translated together; their relative times and original failures are preserved.

Proof. Insert $UU^* = I$ between successive factors. The first domain membership is equivalent under U ; induction gives each subsequent membership and then both identities. Closures,

extensions and graph differentiability are asserted only where the original operation supplies them. The group law shifts every time offset by t , without changing relative separations. Vacuum word Gram products are unchanged by the same unitary. $\square$

For a positive smooth time-only lapse, $U_N(t, s) = \exp[-iH_0 \int_s^t N(r)dr]$ is the original proper-time continuation. On its graph-domain histories the normalized-ray action is

$$\mathcal{A}_N[\psi] = \int_I \left\{ \frac{i}{2} (\langle \psi, \dot{\psi} \rangle - \langle \dot{\psi}, \psi \rangle) - N \langle \psi, H_0 \psi \rangle \right\} dt. \quad (129)$$

Compact variations give the Schrödinger ray equation; the original phase section gives its vector equation. The sphere form is presymplectic and its phase is not an extra physical degree of freedom. At a held history, $D_N \mathcal{A}_N[\nu] = - \int_I \nu \langle \psi, H_0 \psi \rangle dt$. This fixes the time-action sign but does not turn a spacetime source test into a sharp local Hamiltonian.

Lemma 9.6 (A nonzero normal source/preparation derivative). *There is a real compact smooth f such that, with $n_* = \partial_0 f$, $h_* = 2n_* dt \otimes dt$,*

$$L(n) = \hat{T}^{00}(n), \quad \phi = L(n_*)\Omega \neq 0, \quad a = \|\phi\|^2 > 0, \quad e = \langle \phi, H_0 \phi \rangle > 0. \quad (130)$$

The vector lies on the original joint local-field and translation generator domains, with $\phi \perp \Omega$. For $\psi_s = (\Omega + s\phi)/\sqrt{1 + s^2 a}$ and $c = \langle \phi, L(n_)\phi \rangle$,*

$$W_{[\psi_s]}(h_*) - W_*(h_*) = \frac{2sa + s^2 c}{1 + s^2 a}, \quad \partial_s W_{[\psi_s]}(h_*) \Big|_0 = 2a > 0. \quad (131)$$

Proof. Work in the original vacuum-generated joint-field sector, with its actual relative locality and local cyclicity, not in unrelated charged sectors. Suppose $L(\partial_0 f)\Omega = 0$ for every real compact f . Test translation and covariance give $H_0 L(f)\Omega = 0$. In the forward cone $p^0 = 0$ implies $p = 0$, so these vectors are translation invariant; vacuum uniqueness is not required.

For any translation b , the symmetric local smearing $L(f_b - f)$ annihilates Ω . Its closure annihilates $B\Omega$ for local words in the spacelike complement of a bounded region containing both supports, by relative locality on the original joint domain. Those vectors are dense by local cyclicity. The kernel of a closed operator is closed, so that closure is zero. Thus $L(f_b) = L(f)$ on the joint domain. Translate f_b spacelike to any fixed finite local word; locality then makes $L(f)$ commute with that word.

All corresponding finite normal stress-charge smearings would therefore have zero commutator. The original signed charge and boundary comparison would make H_0 's local-field action trivial, contrary to the nonzero positive-energy Higgs descendant in Theorem 4.5. This uses the admitted charge-generation comparison, not conservation alone or a new charge limit for ϕ . Hence the asserted f exists.

Test differentiation makes ϕ a time-generator image of $L(f)\Omega$, up to its fixed imaginary unit/sign. Smooth compact tests give the required generator domains; the vector has no zero-energy component, and thus finite strictly positive first energy moment. Its vacuum mean is zero because the constant vacuum stress pairs with a compact time derivative. Expanding its normalized matrix element gives (131). $\square$

On $\psi_s(t) = U(t)\psi_s$, the distinct held time-action test is

$$\partial_s^2 D_N \mathcal{A}_N[\nu] \Big|_0 = -2e \int_I \nu(t) dt. \quad (132)$$

Indeed its energy is $s^2 e / (1 + s^2 a)$, since the vacuum and cross energy terms vanish. A compact endpoint-fixing time redescription has zero elapsed-time change and is not the nonzero test in (132). Equations (131) and (132) retain different holding maps; their comparison is (127).

9.6 The coupled action and a once-frozen Einstein member

For the inherited nonzero curvature normalization, the finite-region Einstein variational representative is

$$\begin{aligned} S_{\text{EH}}[g] &= \frac{1}{2\kappa_*} \int \sqrt{|g|} (R - 2\Lambda_*) d^4x, \\ D_g S_{\text{EH}}[h] &= -\frac{1}{2\kappa_*} \int \sqrt{|g|} (G^{\mu\nu} + \Lambda_* g^{\mu\nu}) h_{\mu\nu} d^4x. \end{aligned} \quad (133)$$

Compact interior variations remove the displayed boundary variation, not its original boundary ledger. The sign agrees with the inverse to covariant metric conversion in Proposition 9.1. The minimal first-curvature result is the full nonlinear Einstein form, not a linearized weak-field equation [39, Sections 13–19, Theorem 30.1].

Construct, before imposing a solution,

$$\Gamma_* = S_{\text{EH}}[g_+] - S_{\text{EH}}[g_-] + W_{\text{CTP}}[g_+, j_+; g_-, j_-]. \quad (134)$$

Its all-test diagonal equation is

$$D_- \Gamma_*[g; h] = -\frac{1}{2\kappa_*} \langle G[g] + \Lambda_* g, h \rangle_g + \mathcal{E}_*(A_g(h)) = 0. \quad (135)$$

The source is precisely the original causal insertion, with its normalization and all contacts. Its source equations are the original quantum field equations and Ward identities at their supplied grade, not classical matter equations substituted for them. The holding comparison is already constructed in Theorem 9.4.

Apply an original compact diffeomorphism/redescription act to both branches, fields, tests and source/preparation endpoints. The metric term is the divergence of the extracted stress; the field terms give its original equation-of-motion contacts, and other terms give the declared external exchange and boundary contribution. Thus its reconstructed source obeys $W_g(\mathcal{L}_X g) = -J_g(X)$ in that original ledger. Differentiation retains, for fixed X ,

$$(\delta W)_{\text{fixed}}(\mathcal{L}_X g) + W_g(\mathcal{L}_X \delta g) = -\delta J(X),$$

with the additional moving- X term when needed. This follows from the same action-principle integration by parts and the moving Ward comparison, so it agrees with the Bianchi identity. Trace and flavor anomalies are not set to zero. On a closed invariant vacuum and compact interior tests the appropriate exchange terms vanish. The invertible ADM test map supplies all normal, shift and tangential equations together, not constraints selected after discarding other components.

Theorem 9.7 (A compatible physical member at one joint point). *Let t_* be the actual original vacuum-stress coefficient of (103). Above the fixed physical source its gravitational calibration has not been numerically selected. Choose once a permitted $\kappa_* \neq 0$ in the inherited normalization family and freeze*

$$J_{\text{joint}} = (S_*, \kappa_*, \Lambda_*), \quad \Lambda_* := \kappa_* t_*. \quad (136)$$

Then (η, Ω) , with its original physical time continuation and observation algebra, is a stationary member of the source-coupled dynamical action (134). The quantum action and causal continuation have the original retained grade; the stronger represented baseline source and physical channels remain intact.

Proof. The nonzero curvature normalization, source conversion and homogeneous coefficient are the inherited family [39, Corollary 23.3 and Remark 30.2]. The fixed physical source does not

prescribe the separate classical target's compact topology, scalar clock or zero-offset parameter tuple. Choosing (136) over this unselected calibration therefore changes no Standard Model parameter, state, normalization or source. The joint point is fixed before any comparison of outputs.

At the baseline $F = 0$, $Q_g = 1$ and $G[\eta] = 0$. Theorem 8.4 and (111) give, for every compact h ,

$$D_- \Gamma_*[\eta; h] = -\frac{\Lambda_*}{2\kappa_*} \langle \eta, h \rangle_\eta + \frac{t_*}{2} \langle \eta, h \rangle_\eta = 0.$$

All tensor components vanish, including the normal/shift constraint and tangential evolution components. The source remains the original interacting one, obeying its original equations and Ward identities. Its continuation is Lemma 9.5; the vacuum ray is stationary but the field action is not trivial. Lawful coordinate and frame presentations transport the whole source, tests, domains and boundary data together.

The action and its source domain were constructed before this restriction. The metric is varied in (134), the inherited gravitational kinetic term remains nonzero, and Theorems 9.2 and 9.4 identify its source-varying causal and canonical response with the original physical source. Thus the solution belongs to that constructed dynamical theory, rather than being an independently matched flat tensor placed next to a fixed-background QFT. $\square$

The normalization selection preserves authority. If the source record has $t_* = t_N + r_N$, the joint record is exactly $\Lambda_* = \kappa_* t_N + \kappa_* r_N$, with the transported error domain. A formal coefficient record stays formal; an actual represented value and its supplied coefficient certificate are both retained. No sum, positivity or numerical remainder bound is added. For previously fixed (κ_0, Λ_0) , the same member instead passes if and only if $\Lambda_0 = \kappa_0 t_*$. Under a lawful homogeneous convention change,

$$T' = T - Cg, \quad t'_* = t_* - C, \quad \Lambda'_* = \Lambda_* - \kappa_* C,$$

the residual $\Lambda_* - \kappa_* t_*$ is invariant [39, Theorem 27.3]. This is neither retuning an already fixed point nor predicting the observed cosmological constant.

Corollary 9.8 (The coupled action is not blind to preparation). *The original preparation of Lemma 9.6 has the nonzero represented mixed action response*

$$\partial_s D_- \Gamma_{[\psi_s]}[\eta; h_*] \Big|_0 = 2a > 0 \tag{137}$$

on its original represented first-response restriction. The graded action retains the coefficient certificate of this same equality, not a positivity assertion for every truncated coefficient.

Proof. The source preparation is explicitly the original normalized operation. In coefficient notation, with $B_s = 1 + sR_V(f_\eta(h_*))$, it is

$$\mathcal{E}_s(X) = \frac{\mathcal{E}_*(B_s^* \star X \star B_s)}{\mathcal{E}_*(B_s^* \star B_s)}.$$

Its denominator has unit constant term and, on the represented common domain, is the actual norm $1 + s^2 a$. No new product domain or curved state is inferred. The geometric action is preparation independent, so (131) and (121) give (137). Equation (127) retains the corresponding $-2a$ normal source contribution together with its actual kinetic, boundary and holding derivatives; the complete pullback recovers the displayed positive action-normal response. $\square$

Nontrivial coupling is also visible before reconstruction. The original nonzero Higgs kinetic coefficient c_H multiplies $\sqrt{|g|}g^{\mu\nu}(D_\mu H)^\dagger D_\nu H$. At $h = 2n dt \otimes dt$, its principal variation is

$$-c_H n \left\{ (D_0 H)^\dagger D_0 H + \sum_i (D_i H)^\dagger D_i H \right\}.$$

The original normalized kinetic coefficient is nonzero; its field monomials have not been replaced by a vacuum scalar. Gauge, potential, fermion, spin and counterterm terms remain in the same action. At fixed J_{joint} , keeping η while changing Ω to ψ_s therefore generally fails (135). The preparation is still physical; it is not declared inadmissible to protect the stationary member.

No transverse solution map through all preparations follows from this member theorem. At fixed κ_* , linearization retains the actual retarded susceptibility (122) and local metric counterterms in the leading response operator. They cannot be reassigned to lower order to obtain an inverse without proof. The all-test stationary member of an actual source-coupled theory does not require that additional universal inverse, a finite curved positive-state family or a Wheeler–DeWitt quantization.

9.7 Identity of the baseline record and common color operations

With the minus branch at the baseline, an original plus-branch probe satisfies

$$K_*(\eta, j; \eta, j_*) = \mathcal{E}_*(S_V(F[\eta, j])) = Z_{\text{original}}(j). \quad (138)$$

This follows from $S_V(0) = 1$ and the original coefficient identity. More strongly, include the baseline source record by identity: its Hilbert space, states, represented operators, natural word domains and failures, positive kernels, physical boundaries, marked poles and entire authorized germs are unchanged. Formal source-jet equality is not used to reconstruct those stronger records. Both-branch variation retains the original mixed ordinary and chronological field products before reconstruction.

The source and response incidence is also preserved at the common premetric level. For an admitted common record a , take its original quantum/source restriction $\Pi_Q(a)$, its normalized kinetic metric, and its original compact act labels. Evaluate the displayed Einstein term and original CTP operation on these data. This adjoins their evaluation graph without a new source variable or structural carrier. Complete source/act comparisons transport each coefficient and its domain through the existing gravitational presentation [9, 25, 38]. On the member their continuations agree by the proved canonical holding, source composition and time identities. Neither the equation $\Gamma = 0$, a solved metric nor a clock constructs the upstream continuation category. Actual preparations, off-shell source acts and lawful presentation changes remain distinct.

Theorem 9.9 (Same-source conjunction for the retained color operations). *The accepted compact color/Higgs source operations, the actually attached neutral color ports and the original witnessed composite-persistence/process operations admit a source-preserving lift to the action object at J_{joint} . Their intrinsic values, exact domains, boundary grade, failures and actual continuation are unchanged. Every jointly admitted finite composition has the same property. This is a conjunction of the constructed operations, not a proof of a larger required-domain coverage statement.*

Proof. Let x_* denote the constructed source/action object and $r_Q(x_*) = S_*$ its complete quantum restriction. For an actual original operation O define

$$\tilde{O}(x, a) = O(r_Q(x), a), \quad D_{\tilde{O}} = \{(x, a) : a \in D_O(r_Q(x))\}. \quad (139)$$

This receives an already supplied graph; it does not manufacture its operation. The concrete source identity is checked in three cases.

For an admitted completed color insertion with raw source label f_C and an original physical scalar port with raw label f_f , the lifted source derivative is literally

$$\begin{aligned} A_f &= R_V(f_f), & B_C &= R_V(f_C), \\ D_{f;C} &= R_V(f_{fC}) + i\{T_V(f_C, f_f) - B_C \star A_f\}. \end{aligned} \quad (140)$$

It uses exactly the $V, T_V, R_V, \star, \mathcal{E}_*$ of (119). The completed-composite normalization and product/domain authority precede the lift. These are raw source labels; their original insertion normalization is applied once by the displayed maps, not applied again to an already normalized insertion. The original field-valued Hermitian response, its connected real readout and its distinct complex sesquilinear polarization are therefore unchanged. Products such as $\mathcal{E}_*(A_f^* \star B_C \star A_g)$ are the same original products, not newly supplied scalar moments. The mixed contacts f_{fC} , f_{fg} , f_{fgC} , moving prescription/BRST/preparation and source-dual calibration terms remain exactly once. The physical Higgs port and its complete baseline germ are included by identity; the compact mixed response is not renamed a new continued resonance.

For an actual neutral color port, the original space $\mathcal{H}_{\text{obs}}$, its state and translation action, input representation R_C , intertwiner J and Haar projector are unchanged. Hence its established equation $J = JP_C$ and its actual continuation remain the same at x_* . Operational neutral boundaries likewise retain their whole joint input tensor and boundary comparison. No separate projection is applied to each constituent, and a nonlinear history is not made linear by averaging. A different boundary color action is not silently identified with the proper action on $\mathcal{H}_{\text{obs}}$.

For a witnessed composite-persistence record the full original local profile, standing, identity-memory and boundary witness are part of $r_Q(x_*)$. Its original inclusion, quotient q_S and witness reader are unchanged. For an actual process w with terminal witness reader γ ,

$$K_w = q_S \gamma w, \quad D_{(w, \text{pers})} = \{a \in D_w : w(a) \in D_\gamma\}$$

is exactly the same partial operation. Intermediate domains, class changes, first failure and the nonempty occupied records remain. No persistence record is replaced by a Hilbert vector, and a process lacking a terminal witness acquires none here.

Finally the domain of two successive original operations is $\{a \in D_{O_1} : O_1(a) \in D_{O_2}\}$. Substitution of (139) gives its exact pullback, output and failure. Induction proves the finite-network statement with shared operands retained as one record. The source comparison and boundary squares transport vertex by vertex, as in the original projection/process reception [3, 1]. Joint admission is required; separate argument domains are not assumed to intersect. $\square$

Original preparations remain part of this same source/action object. Where $P\Omega$ has its actual nonzero norm and product-domain witnesses, its normalized state is $\mathcal{E}_P(X) = \mathcal{E}_*(P^*XP)/\mathcal{E}_*(P^*P)$. At a coefficient-only authority the original graded counterpart is used instead, without inventing positivity or domains. At the fixed joint point its flat residual is

$$D_- \Gamma_P[\eta; h] = -\frac{\Lambda_*}{2\kappa_*} \langle \eta, h \rangle_\eta + \frac{1}{2} \mathcal{E}_P(\hat{T}(h)).$$

It vanishes for the proved vacuum member, not automatically for every retained preparation or process. Thus common source identity does not multiply stationary gravitational solutions.

The result is one original-source causal Einstein action, explicit canonical/source holding, a once-frozen compatible physical member, and identity-preserving reception of the retained electroweak, quantum and color operations. It supplies the sufficient dynamical member at the physical source's declared quantum-action grade. Theorems 10.4–10.9 extend the comparison to the entire original physical-interface diagram and its compatible witness families. Theorem 11.5 then places the complete original color-boundary acquisition at this same action object and frozen joint point. It uses the actual source-qualified witnesses, not a promotion of a phase label, an unresolved authority or a zero response to a productive boundary.

10 Complete physical interfaces and exact source comparison

A physical boundary retains the authority by which it is observed. A compact source response, a represented spectral channel, a persistent composite history, an incoming or outgoing asymptotic attachment, and a marked resonance germ are therefore different objects. The source may supply several of these on related carriers; the comparison must retain their actual maps and domains, rather than replace them by one scalar readout. We first give a quantitative spectral construction on the represented physical source. We then compare the entire original physical-interface diagram with its inclusion in the common action object of Section 9.

Throughout, S_* is the occupied full interacting source of Section 3. Its parameters, global form, spin and boundary data, preparation, normalization and original continuation authority are fixed. In particular, x_* and J_{joint} are the object and joint point already constructed in Theorem 9.7, with $r_Q(x_*) = S_*$. The inclusion of the original baseline record in (138) retains its represented domains and continued channels by identity. This is stronger than equality of formal source derivatives and does not construct the same spectral or analytic objects at every additional off-shell metric.

10.1 The represented source and a fixed spectral boundary family

Let $\mathcal{H}_{\text{obs}}$, Ω and $U(t) = e^{-itH}$ be the original vacuum-observable representation, normalized vacuum and time translations, with $H \geq 0$. The original proper-color action is the identity on this space and it reduces translations by Lemma 7.1. The gauge-fixed computational field algebra and separately typed charged sectors are not all declared operators on this vacuum-observable space [34, Sections 8–11].

For its Wightman presentation let $\mathfrak{W}_*$ be the original test-word $*$ -algebra: finite sums of finite ordered words in the supplied fields and adjoints, with their original test completion, independent spacetime arguments and contraction incidences. A coincident composite requires its own original normalization; it is not obtained by restricting arbitrary separated arguments to a diagonal.

Lemma 10.1 (The original vacuum word domain). *With the original positive word functional ω_* , put $\mathcal{N} = \{b : \omega_*(b^*b) = 0\}$. Then $\mathcal{N}$ is a left ideal and the map*

$$\mathfrak{W}_*/\mathcal{N} \longrightarrow \mathcal{H}_{\text{obs}}, \quad [b] \longmapsto b\Omega$$

is an isometry with dense range. It identifies the supplied word actions and their adjoints on the original common invariant domain.

Proof. The squared norm of the displayed vector is $\omega_*(b^*b)$, and polarization gives its existing mixed word kernels. If $b \in \mathcal{N}$, positivity and Cauchy–Schwarz, applied to b and a^*ab , give $\omega_*(b^*a^*ab) = 0$; all these are admitted finite words. Thus $ab \in \mathcal{N}$. Left multiplication descends to the quotient and intertwines its actual field action with $b\Omega$; the $*$ -identity gives the adjoint relation. Cyclicity of the original observable representation gives density. No new state is selected by this construction. $\square$

For the local-observable alternative use the original represented operation algebra and its actual product domains; bounded local products have the domain property immediately. A source that has supplied only a form distribution does not gain an unbounded invariant word domain from the lemma. In particular, a color singlet carrying an electric or boundary charge still requires its actual endpoint; color averaging alone does not put it in $\mathcal{H}_{\text{obs}}$.

Before choosing an input, fix the spectral measure E_H and the family

$$P_R = E_H([0, R]), \quad \mathcal{H}_R = \text{Ran } P_R, \quad a_R : \mathcal{H}_R \hookrightarrow \mathcal{H}_{\text{obs}}, \quad R \in \mathbb{N}_{>0}. \quad (141)$$

Here a_R is the inclusion, and we also allow $a_\infty = I$ on $\mathcal{H}_{\text{obs}}$. The projections increase strongly to the identity and commute with $U(t)$. The boundary authority is positive-energy spectral response of this original source. Neither locality of P_R , finite dimension of $\mathcal{H}_R$, nor a particle interpretation of $\mathcal{H}_R$ is asserted.

Let V be a finite-dimensional original joint input space and $j : V \rightarrow \mathcal{H}_{\text{obs}}$ its actual realized linear insertion map. It is supplied before applying a spectral projection, on its original word or graph domain. If necessary its actual invariant input projection is included in j ; weak, Higgs and all other indices remain. Finitely many such input maps can be concatenated on their finite direct sum, preserving their common source and mixed matrix entries. Define

$$J_R(t) = a_R^* U(t) j, \quad G = j^* j, \quad G_R = j^* P_R j, \quad (142)$$

$$\begin{aligned} \mu(\Delta) &= j^* E_H(\Delta) j, \\ K_R(t, s) &= j^* U(t)^* P_R U(s) j = \int_{[0, R]} e^{-i(s-t)E} d\mu(E). \end{aligned} \quad (143)$$

The inner product is antilinear in its first argument. These matrix entries are Hilbert pairings, not an application of an unbounded composite adjoint to a projected vector. The kernel is positive: its finite quadratic form is $\|\sum_k P_R U(t_k) j x_k\|^2$. All unbounded insertions occur on their original domains before the terminal projection. No assertion that P_R preserves those domains is needed; a subsequent unbounded operation retains its own successive-action domain.

Theorem 10.2 (Simultaneous finite-window detection). *For every such finite family j , some finite R satisfies*

$$\ker(P_R j) = \ker j. \quad (144)$$

If $j \neq 0$, let λ_ be the least positive eigenvalue of G , and put*

$$\delta_R = \text{Tr}(j^*(I - P_R)j).$$

Then $\delta_R \downarrow 0$. For $0 < \epsilon < 1$, every R with $\delta_R \leq \epsilon \lambda_$ satisfies*

$$(1 - \epsilon)G \leq G_R \leq G. \quad (145)$$

The same R preserves all linear null relations and all nonzero combinations of this input family at every common time translation. For $j = 0$, every channel is exactly zero and no positive normalization is asserted.

Proof. Choose an orthonormal basis $(e_l)_{l=1}^d$ of V . Strong convergence gives

$$\delta_R = \sum_{l=1}^d \|(I - P_R)j e_l\|^2 \longrightarrow 0.$$

The positive operator $G - G_R$ has norm at most its trace. Both G and G_R vanish on $N = \ker j$. On $N^\perp$, $G \geq \lambda_* I$, so

$$0 \leq G - G_R \leq \delta_R I \leq \epsilon G.$$

Extending by zero on N proves (145). Its lower bound gives $\ker G_R = \ker G$, which is (144). The choice $\epsilon = 1/2$ supplies a finite R . Commutation with $U(t)$ leaves both Gram matrices unchanged under common translation. The zero-family case is immediate. $\square$

The family of boundaries was fixed before the input; only the finite witness R depends on the whole fixed input family. It need not be numerically known for the existence theorem. There is no single asserted R for all configurations or all test families. The matrix amplitudes obey the separate quantitative estimate

$$\|K_\infty(t, s) - K_R(t, s)\| \leq \|j^*(I - P_R)j\| \leq \delta_R, \quad t, s \in \mathbb{R}. \quad (146)$$

Indeed $I - P_R$ can be inserted on both sides of $U(s - t)$, whose norm is one. The estimate does not assert exact equality of amplitudes, residues or states; the exact conclusion is equality of the null relation.

For maps $j_a : V_a \rightarrow \mathcal{H}_{\text{obs}}$ with independent anchored times t_a , the block Gram difference satisfies

$$\|K_\infty((t_a)) - K_R((t_a))\| \leq \sum_a \|(I - P_R)j_a\|_{\text{HS}}^2. \quad (147)$$

To prove it, set $L((x_a)) = \sum_a U(t_a)j_ax_a$. The difference is $L^*(I - P_R)L$, bounded by the squared Hilbert–Schmidt norm of its tail. This gives the displayed sum, uniformly in the times. The null-relation theorem applies to each fixed joint family; a uniform null-relation assertion for all independently varying time tuples would additionally need a uniform positive singular-value bound. For an actual $jx \neq 0$,

$$\nu_x(\Delta) = \frac{\langle jx, E_H(\Delta)jx \rangle}{\|jx\|^2}$$

is its original spectral probability and $\nu_x([0, R]) \geq 1 - \epsilon$ under (145). A spectral response probability does not by itself classify that preparation as a hadron.

Corollary 10.3 (One current boundary for a whole descendant family). *For any finite original compact-test family $f_1, \dots, f_m$, put $j_D(c) = \sum_l c_l D(f_l)\Omega$. One boundary from the independently current-generated family $\mathcal{H}_K(B_n)$ of Theorem 4.5 preserves exactly $\ker j_D$ under every common time translation, with the relative Gram and tail bounds above. The statement includes dependent and all-zero test families.*

Proof. Theorem 4.4 gives $D(f)\Omega = -P_\mu P^\mu O_\rho(f)\Omega$, so every such vector has no massless component. The bands

$$B_n = \{p \in \bar{V}_+ : 1/n \leq p^2 \leq n, p^0 \leq n\}$$

increase to its entire spectral support. On B_n , the multipliers $-i/P^0$ and $-P^2$ recover $E(B_n)D(f)\Omega$ from $E(B_n)K^0(f)\Omega$. They are bounded there. Hence the orthogonal projection onto the current-generated boundary obeys

$$P_{\mathcal{H}_K(B_n)}D(f)\Omega = E(B_n)D(f)\Omega.$$

These boundaries reduce translations. The finite Gram proof applies with their projections, which converge to the identity on $\text{Ran } j_D$. The original nonzero Higgs germ supplies a nonzero test family when needed, by Theorem 4.5; a complex pole is not used as a spectral atom. $\square$

Equation (17) gives this an interaction-sensitive meaning: the represented descendant is the total Higgs kinetic, potential, quark-and-lepton Yukawa, Euler and contact combination. The corollary does not separately represent or give a nonzero spectrum to each Yukawa summand.

10.2 Spectral comparisons and their actual source contacts

Suppose a lawful full-source unitary comparison T and an input comparison S satisfy $T\Omega = \Omega'$, $TU(t) = U'(t)T$, and $Tj = j'S$. Spectral calculus gives $TP_R = P'_R T$. With the unitary restriction $T_R : \mathcal{H}_R \rightarrow \mathcal{H}'_R$, the channel square and matrix congruences are

$$T_R J_R(t) = J'_R(t) S, \quad (148)$$

$$G = S^* G' S, \quad G_R = S^* G'_R S, \quad \mu(\Delta) = S^* \mu'(\Delta) S. \quad (149)$$

These are congruences, not operator-intertwining identities for an arbitrary nonunitary S . If S is unitary, the Gram spectra, λ_* and δ_R agree. Complete source, boundary, weak/Higgs and

anomaly data travel in this comparison; equal field labels at different coupling points do not supply it.

When a genuinely differentiable represented family j_s and spectral projection $P_R(s)$ is supplied in a unitary Hilbert frame, the full differential is

$$dG_R = (dj)^* P_R j + j^* (dP_R) j + j^* P_R (dj). \quad (150)$$

At any furnished order with labelled derivative set I ,

$$\partial_I G_R = \sum_{I_1 \sqcup I_0 \sqcup I_2 = I} (\partial_{I_1} j)^* (\partial_{I_0} P_R) (\partial_{I_2} j). \quad (151)$$

The blocks may be empty. Both formulas follow by distributing each derivative to one of the three factors; dj includes the actual insertion and preparation derivatives. A nonisometric frame adds the original inner-product and connection contacts. Sharp spectral cutoffs can fail to be differentiable when spectral weight crosses the cutoff. Thus these identities use supplied represented differentiability, not a consequence of the fixed-source spectral theorem. The coefficientwise compact-source calculus remains valid at its own grade without becoming a derivative of an unprovided positive spectral measure.

10.3 The whole original physical-interface diagram

Use the fixed source universe, or a small skeleton, of the original realization construction [1, 5]. Let $\mathcal{G}_*$ retain all its independently admitted original input sorts and joint configurations, including those without a positive channel; every originally authorized physical boundary with its type, source and continuation authority; every actual primitive partial graph with its entire intrinsic parameter; and the original interface maps, source relations, equality witnesses, fixed siblings and domain/failure tests. Whole spectral, limiting and continued maps are legitimate primitives with their original infinite-dimensional arguments [1, Definition of the received operational target].

This definition is fixed by original source admission, not by a favorable result of the receiving construction. A boundary label without its required authority does not become an authorized object. If its unresolved status is part of the original target, that status remains distinct from a defined zero or a source-proved exclusion. Likewise a boundary object's identity arrow is not a preparation map from every configuration. The independently typed persistence input of Theorem 7.5 is a genuine positive case, not permission to reverse every boundary quotient.

For an original sort X , define

$$i_X : X \longrightarrow X^{\text{ret}}, \quad i_X(a) = (x_*; a), \quad p_X(x_*; a) = a. \quad (152)$$

The semicolon records incidence with the fixed action object, not a new independent source variable. The complete original a , topology, norm, adjoint and structural maps remain. Thus $p_X i_X = I$ and $i_X p_X = I$. For joint inputs restrict these identifications to the actual joint incidence subset, not a product of independently compatible marginals. For an original $e : D_e \subset X \rightarrow Y$, put

$$D_e^{\text{ret}} = i_X(D_e), \quad e^{\text{ret}} i_X = i_Y e, \quad \text{Graph}(e^{\text{ret}}) = (i_X \times i_Y) \text{Graph}(e). \quad (153)$$

Original diagnostics on failed arguments are copied by their own injective output maps. This is the original graph evaluated under $r_Q(x_*) = S_*$, not a new operation stipulated to have its values.

Theorem 10.4 (Exact original diagram comparison). *Let $\mathcal{R}_*$ be the original source-witnessed realization category, with its actual partial composition and original source relations. Equations (152)–(153) give inverse functors*

$$\begin{aligned} \mathbf{I} : \mathcal{R}_* &\longrightarrow \mathcal{R}_*^{\text{ret}}, & \mathbf{P} : \mathcal{R}_*^{\text{ret}} &\longrightarrow \mathcal{R}_*, \\ \mathbf{PI} &= I, & \mathbf{IP} &= I. \end{aligned} \quad (154)$$

They preserve and reflect original equations, complete operation graphs, domains, failures, physical types and authority, fixed siblings and original continuation. The same holds for the complete original finite operational networks.

Proof. The sort maps are inverse, and (153) proves each primitive statement, including the actual warrant rather than only its scalar value. For $e : D_e \subset X \rightarrow Y$ and $f : D_f \subset Y \rightarrow Z$,

$$\begin{aligned} D_{fe} &= \{a \in D_e : e(a) \in D_f\}, \\ D_{f^{\text{ret}}e^{\text{ret}}} &= i_X(D_{fe}), \quad f^{\text{ret}}e^{\text{ret}}i_X = i_Zfe. \end{aligned} \tag{155}$$

Each intermediate domain is tested in its original order, so the first failed vertex and its diagnostic agree. This does not assert composability of arbitrary densely defined operators.

For a finite admitted source network, use an original evaluation order and the corresponding domain pullbacks. A shared edge carries one $i_X(a)$ to every consumer. The inverse therefore reflects all equality and sibling conditions as well as preserving them. A source-supplied loop, whole evolution or infinitary mechanism is retained as its original primitive, not constructed from finite samples by this induction.

Before quotienting, the original graph and its retained graph have inverse path maps: vertices, edges, full operands and warrants are all copied. Each defining source relation is copied with its proof on its actual domain. Reflexivity, symmetry, transitivity and composition closure of those relations correspond in both directions. An independently established analytic or operator identity is a whole retained relation with its original domain, not an inference from finitely many tests. Hence the generated congruences correspond, and the inverse path maps descend to (154). $\square$

The retained category is not asserted to be full in the enlarged category containing new metric-response arrows. It has the same original source relations, not all equations obtained by evaluating only on the stationary (η, Ω) member. Reading back an original graph recovers every original preparation and multi-time argument. It neither reconstructs an arbitrary curved extension from one point nor gives a global inverse on the entire gravitational category.

10.4 All boundary channels, balanced gluing and histories

Let $\mathcal{C}_*$ and $\mathcal{B}_*$ denote the original configuration and physical-boundary categories, and use $A : \mathcal{C}_* \rightarrow \mathcal{R}_*$ and $B_{\text{in}} : \mathcal{B}_* \rightarrow \mathcal{R}_*$ for their actual interface functors. On the color interface, $\mathcal{C}_*$ is the same independently formed category $\mathcal{C}_*$ of (43); a configuration domain means its original object domain with the stated incidence and operation data, not a different set of successful inputs. The original source-witness channel profunctor is

$$\mathcal{K}_*(C, B) = \text{Hom}_{\mathcal{R}_*}(A(C), B_{\text{in}}(B)). \tag{156}$$

Its Hom set can be empty. An existing zero linear map is instead an actual arrow with a defined zero value [5, Composite-continuation and source-witness profunctor theorems].

Theorem 10.5 (Every original channel and its realized gluing). *Let I_C, I_B be the restrictions of $\mathbb{I}$ to the two interfaces. For every original pair (C, B) ,*

$$\mathcal{K}_*(C, B) \longrightarrow \mathcal{K}_*^{\text{ret}}(I_C C, I_B B), \quad k \longmapsto \mathbb{I}k \tag{157}$$

is a bijection, natural contravariantly in C and covariantly in B . It preserves and reflects evaluated domains and original productive, nonproductive and failure records. For another originally supplied boundary interface it induces an isomorphism of the balanced composition modules and commutes with realized gluing.

Proof. Copy the original interface arrows. Then $!A = A^{\text{ret}}I_C$ and $!B_{\text{in}} = B_{\text{in}}^{\text{ret}}I_B$. Full faithfulness onto the retained category gives the bijection, including empty Hom sets. For $u : C' \rightarrow C$, $v : B \rightarrow B'$,

$$!(B_{\text{in}}(v)kA(u)) = B_{\text{in}}^{\text{ret}}(I_B v)(!k)A^{\text{ret}}(I_C u).$$

This is exactly the two profunctor actions and preserves every originally admitted pre- and postcomposition.

In a balanced product send $[k, \ell]$ to $[!k, !\ell]$. The relation $(vk, \ell) \sim (k, \ell v)$ goes to its corresponding relation by the preceding identity. The inverse reads back the pair and the same relation. This proves an isomorphism of the balanced sets or, with the original linear structure, their modules and coends. Moreover,

$$\text{glue}^{\text{ret}}([!k, !\ell]) = !(k\ell).$$

Equation (155) gives its complete domain and first failure. Distinct factorizations can still obey additional source equalities, and some larger Hom-set arrows may not factor through the stated intermediate boundary. Thus the theorem identifies the actual gluing image; it does not claim injectivity or surjectivity of gluing beyond its original theorem.

A productive represented port retains its actual input, nonzero output and norm; a productive persistence port retains its actual history and indicated boundary. Their inverse images give the same witnesses, proving reflection as well as preservation. A later noninvertible readout may annihilate a productive channel; no stronger monotonicity is asserted. $\square$

Corollary 10.6 (Complete original context and history comparison). *For the complete original family of jointly admitted source contexts, two retained inputs are observationally equivalent if and only if their original inputs are. The exact domain-reflecting observation quotients are isomorphic. The same holds for the full licensed history-reconstruction family and its non-null stability conclusion in Theorem 7.10.*

Proof. Every original context has its retained context, and every retained source context reads back to an original one by Theorem 10.4. Their totalized output and failure records are related by injective output maps. Equality for all original contexts is therefore equivalent to equality for all retained contexts. This proves both quotient directions without choosing representatives. Applying the same argument to the actual history readers preserves their reconstruction classes, continuation and original non-null stable interior. Additional gravitational contexts may refine a comparison; they are not identified with the original context family. $\square$

10.5 Concrete comparison at the original physical boundary types

These theorems act on actual boundary objects, not just names of allowed types. The following applications specify the whole graphs that are retained.

For persistence, Theorems 7.5, 7.7 and 7.8 give the independent domain $\mathcal{P}_{S_*}$, quotient q_{S_*} , and process with its actual terminal reader γ . Retain the complete input alongside its quotient. Then

$$\begin{aligned} q^{\text{ret}}i_{\mathcal{P}} &= i_B q_{S_*}, \\ K_w^{\text{ret}}i_X &= i_B q_{S_*} \gamma w, \quad D_{K_w^{\text{ret}}} = i_X \{a \in D_w : w(a) \in D_\gamma\}. \end{aligned} \tag{158}$$

The equalities follow by composing the exact graph squares. All history, restoration, identity/class change and first-failure data remain. Descending an arbitrary process to classes alone still uses the original saturation, output-class and failure hypotheses of Proposition 7.9; no inverse of q_{S_*} is chosen. Persistence is not relabelled as an eigenstate or a scattering amplitude.

For a supplied self-adjoint H , retain the same spectral measure and every originally invoked measurable functional calculus. In particular,

$$D(f(H)) = \left\{ u : \int |f(\lambda)|^2 d\langle u, E_H(\lambda)u \rangle < \infty \right\} \quad (159)$$

is unchanged. The spectral integral gives the same $f(H)$, resolvent and projections, not values reconstructed from finitely many moments. In a lawful unitary frame T , $E_{THT^{-1}}(\Delta) = TE_H(\Delta)T^{-1}$ and the domain is $TD(f(H))$, by that same integral. Any actual bound-state subspace and its original physical attachments persist. The comparison does not infer that a chosen spectral projection is nonzero, discrete or stable without its source certificate.

For an original stable asymptotic interface, retain its actual incoming/outgoing spaces, preparations j_C , continuation U and attachment isometries $a_{\text{in}}, a_{\text{out}}$. All belong to the same source and state. Then the whole maps

$$J_{C,\text{out}} = a_{\text{out}}^* U j_C, \quad S_{\text{out},\text{in}} = a_{\text{out}}^* U a_{\text{in}} \quad (160)$$

are unchanged, with every successive domain. Identity on the named boundary space is onto. More generally a unitary comparison on that named space gives the adjoint square; a non-onto boundary isometry does not suffice. If an attachment is an original strong limit $F_\alpha u \rightarrow Fu$, then

$$\|T(F_\alpha u - Fu)\| = \|F_\alpha u - Fu\|$$

preserves and reflects its whole convergence domain. This receives an existing wave operator or limit, not an unprovided one. Any original completeness theorem is retained on its entire stated scattering domain; none is inferred for a claim that did not invoke that authority.

For a physical resonance, retain the original gauge-invariant detecting channel, source ports, normalization, connected continued domain or sheet, marked pole and all regular/contact terms. Its actual function satisfies

$$G^{\text{ret}}(z; i_u u, i_v v) = G_*(z; u, v) \quad (161)$$

on that whole domain. The functions and admitted contour integrands are literally equal, so their Laurent coefficients, regular germs and continuation along original overlap paths agree. A zero residue at chosen ports remains zero. This includes the Higgs and W/Z germs of Section 4; it does not construct a new mixed resonance from a compact source derivative, turn an unstable pole into a stable spectral atom, or extend a flat-source germ to an arbitrary curved metric.

For other originally authorized neutral channels the same graph argument applies with their own physical authority. A nonlinear source operation remains its actual equivariant partial graph; lawful orbit descent uses its original domain invariance and equivariance, not a nonlinear Haar-averaging formula. The finite list of boundary types describes authority, not a finite list of all possible values of a physical functional.

10.6 Physical signatures before branching and actual readers

The exact source realization of Theorems 6.9, 6.10 and 6.11 allows the original structural interfaces to be compared with the physical maps just retained. Keep each full presentation and its lawful marking, not a selected word or preferred root order. Write r_m for its complete source re-presentation and rb_m for inverse readback.

The source-first operator primitive contains the actual data $(O, Q_{\text{BRST}}, \text{class/action})$, together with target, physical quotient, anomaly, vacuum, operator locus and scale/order data. The state, local-operator and deformation complexes are different source categories. The accidental-operator primitive contains $(O_i, d_i, \text{charges, effect set})$ on its registered operator-order domain. Extended line, surface, defect and other operator objects keep their own categories [37, Local-operator

and accidental-operator primitive definitions]. Thus an operator is an operand, not the word “operator” attached to an otherwise structural coordinate.

The independently extracted physical signature is

$$\pi_{\text{cl}}([z]) = (D, \kappa, F, I, S, V). \quad (162)$$

Here D is the original target/domain fingerprint, κ is proposition kind, F its physical family, I its exact interface, S the retained physical features and V the typed value/status data. Extraction uses the original class coordinates before branch assignment, not branch identifiers, role, residue, liveness or endpoint status. Independently, each source-clause branch has its signature $\pi_{\text{br}}(c)$. The original branch map is characterized by

$$\beta([z]) = c \iff \begin{cases} c \text{ is source-available at the exact locus,} \\ \pi_{\text{cl}}([z]) = \pi_{\text{br}}(c). \end{cases} \quad (163)$$

The original range and unique-match theorems supply this map on their occupancy-relevant domain and preserve the complete assignment predicate. They also retain the non-occupancy branch where present [52, Sections 8–9]. Source availability alone is not memberwise matching.

An operation-specific support license keeps its own meaning: carrying boundary data is not constructing a preparation-to-boundary channel, and typing an operator is not evaluating its physical expectation. The physical reader must therefore be compared as an actual map in addition to (162) [52, Section 10].

Theorem 10.7 (Physical-signature and actual-reader matching). *For every primitive $M : D_M \subset X \rightarrow Y$ of the original received source-use grammar, the complete source re-presentation has*

$$M_m r_X = r_Y M \quad \text{on } D_M, \quad D_{M_m} = r_X(D_M), \quad (164)$$

with inverse readback of the full record. At every invoked class-normal-form locus,

$$\pi_{\text{cl}}([\text{rb}_m z_m]) = \pi_{\text{cl}}([z]). \quad (165)$$

Consequently the equality-induced branch and its complete assignment predicate agree on their original domains. Physical vertices retain their actual reader, normalization and authority, not merely that signature. Every license retains its original operation kind.

Proof. For a local field/source primitive, Theorem 6.9 gives, before reconstruction,

$$\begin{aligned} T_0^m(r_m A_1, \dots, r_m A_n) &= r_m T_0(A_1, \dots, A_n), \\ (r_m A) \star_m (r_m B) &= r_m (A \star B), \\ V_m &= r_m V, & \mathcal{E}_m r_m &= \mathcal{E}_*, \\ S_{V_m}^m(r_m f) &= r_m S_V(f). \end{aligned} \quad (166)$$

Every field leg, family index, Yukawa tensor, counterterm, diagonal extension, full interaction and preparation is transported. These are identities of the original algebra-valued prescription, not inferences from scalar moments. The current-leg refinement of Theorem 6.10 restricts this same source at a marked U/D incident slot. Both weak components retain the same family marking; Higgs and all other Yukawa/source incidences remain. Its reduct is the original ρ_f , with the specified θ and complete ancestral record, not an identification of two different full realization spaces.

Readback therefore recovers the actual O, Q_{BRST} , class/action and every original target/status coordinate. If the source differential moves, differentiate that original identity, including its $\dot{Q}_{\text{BRST}}$, product and prescription terms; do not impose frozen closure on each derivative separately.

For an accidental operator the same readback retains its actual operator, dimension, full charge action and registered effect/order set, not only an eigenvalue list [51].

Apply the original extraction after this recovery. Its arguments are identical, which proves (165) without consulting β first. Original source guards and normalization keep the same locus. The range theorem and exact signature equality in (163) then give the same branch; every inspected compatibility coordinate gives the same assignment predicate. No graph coordinate is silently added to the signature definition.

For an original path, tensor, support, phase, history or whole physical primitive, Theorem 6.11 and Theorem 10.4 transport its entire graph, which proves (164). If such a vertex has an original class classification, the same coordinate proof applies; otherwise its actual graph comparison suffices. A composite network is not forced into one artificial class-normal-form object. The retained license still concerns the same operation. The concrete physical cases below establish that the readout is the original physical map, rather than a source constraint substituted for it. $\square$

For compact source readers use raw original source labels f_C, f_f , with $A_f = R_V(f_f)$, where A_f is the actual scalar mass detector. The derivative is

$$D_{f;C} = R_V(f_f C) + i\{T_V(f_C, f_f) - R_V(f_C) \star A_f\}. \quad (167)$$

Its normalization is applied once by these original maps. For real Hermitian detector directions let

$$Q_C(f, g) = \frac{1}{2}\{D_{f;C} \star A_g + A_g \star D_{f;C} + A_f \star D_{g;C} + D_{g;C} \star A_f\}.$$

In a fixed product and reconstruction frame the connected response is

$$\mathcal{E}_*(Q_C(f, g)) - \mathcal{E}_*(D_{f;C})\mathcal{E}_*(A_g) - \mathcal{E}_*(A_f)\mathcal{E}_*(D_{g;C}). \quad (168)$$

Substituting (166) in each term gives $D_{rf;rC}^m = r_m D_{f;C}$ and equality of the reconstructed responses. The same substitution applies to each moving product, reconstruction, BRST, rule and preparation contact because it is a derivative of that same original source identity. The original detector conversion, including n^{-2} and its derivatives, is transported, not selected anew. Joint detector contacts f_{fg} and f_{fgC} remain their additional original terms as in Theorem 5.6.

For complex tests the unsymmetrized kernel is instead $D_{f;C}^* \star A_g + A_f^* \star D_{g;C}$, with its original antilinear/linear source duals. Its Hermitian symmetrization is real-bilinear on the realified test spaces, not complex sesquilinear. The field-valued operation attaches the physical detector before $\mathcal{E}_*$; it is neither an arbitrary scalar decorated with a physical name nor a newly continued mixed resonance. The completed neutral-word branch retains its independently authorized composite and product normalization; admission of a bare P does not supply a new coincident P^*P prescription.

For persistence the exact identities $\gamma_m r_Y = r_P \gamma$ and $q_m r_P = r_B q_{S^*}$, combined with (164), prove (158) and reflect its full domain. For whole spectral, continued and asymptotic readers use (159)–(161). They compare the actual entire functions, limits, spaces and domains, not a finite collection of samples. Negative and source-status operations have their same inputs, proof and sector scope. Neither a support license nor a status comparison changes an unresolved antecedent into a productive physical reader.

10.7 All original native uses and their common operands

The independently fixed original grammar has six constructor families: source, restriction, representation and mixed incidence; declared local law, variations and jets; paths, networks and invariant tensors; global descent, screening and actually received support/phase maps; supplied

observable, state, graph, continuation and physical-port maps; and incidence, domain, authority, obstruction and failure tests [5, Reception of finite native uses]. Each family contains all its admitted intrinsic arguments. An entire operator or limiting mechanism can be one primitive; an arbitrary new operation on a carrier with the same field names is not thereby admitted.

Theorem 10.8 (Complete original native-use comparison). *Every original finite use at every admitted intrinsic argument has a receiving network with the same full result, exact domain, joint incidence, operation kind, physical authority and first failure. These data are preserved and reflected. Whole intrinsic mechanisms remain whole primitives. An actual original positive physical witness, or a source-proved alternative, is retained with its input, process and authority-bearing boundary.*

Proof. Apply the original six-family exhaustion theorem. The first three families use the actual full-source and rooted-current constructions of Theorems 6.9–6.11, with all original constituent and support incidences. The fourth uses its original global/support/phase graphs with their branch and grade. The fifth uses the physical reader comparison of Theorem 10.7, including the complete compact, persistence and whole spectral/analytic cases. The sixth keeps the exact original tests and diagnostics. No physical vertex is omitted because it is not a monomial.

For each vertex v , use its actual $M_v : D_v \subset X_v \rightarrow Y_v$. The network domain is the original iterated pullback of these domains along its common-interface equations. The product of the complete r maps preserves and reflects every equation. Restriction to that pullback is therefore an inverse comparison on the full network domain, not just an injection on successful evaluations. Concretely,

$$D_{g_m f_m} = r_X \{a \in D_f : f(a) \in D_g\}, \quad g_m f_m r_X = r_Z g f.$$

Adjoin vertices in their original admitted evaluation order. A shared operand has one transported value for every weak/Higgs, source and preparation consumer. A whole evolution, loop or limit retains its intrinsic graph and domain rather than being solved by this finite induction.

The same domain test fails at the same first vertex with its original diagnostic. Exact zero, proved obstruction, inactive proposal, undefined act and unresolved authority remain distinct. Original path relations are preserved with their source warrants and reflected by inverse readback. At a classified vertex the physical signature agrees before branching by Theorem 10.7; network compatibility is still the actual conjunction of its incidence equations, not one invented class signature. Finally every component of an original witness is transported and read back. This proves the last assertion without creating a missing witness. $\square$

10.8 Phase, source qualifications and physical continuation

The original synchronized physical sector assignment distinguishes stable massive, stable massless, resonance, confined-field and composite-asymptotic records on the supplied physical state and spectrum [34, Chapters 11 and 16]. It is an actual source object, not a classifier inferred from a finite tensor's covariance. The theorems above preserve its complete domain and interpretation.

The chiral and infrared construction supplies its source-qualified physical content as well. Its confining QCD vacuum uses registered physical input 8.1; the stabilizer follows from its nonzero chiral witness. The heavy-hadron construction carries its mass/residual momentum, scale/lifetime and bound-state assumptions, populating c/b and excluding top from that particular long-lived hadron predicate [36, Theorems 8.1 and 19.2]. Its source-to-observable paths retain exact source and destination, map, warrant, preserved-data and assumption fields [36, Theorems 20.1–21.1]. A path may start at the Standard Model source or a registered conventional source; a decoupling or EFT matching is not an invertible representative change. The physical records remain valid with those qualifications, not exact generic full-SM amplitudes obtained by reversing a restriction.

For the selected neutral physical interfaces, the actual comparison of Theorem 7.2 has $JR_C(g) = J$, hence $J = JP_C$. Retaining the full source action, port and domain preserves this exclusion for every admitted joint input and its actual continuations. It does not remove the singlet part of $3 \otimes \bar{3}$, identify screening with confinement, or derive the source's phase from the projector alone. Whole phase-qualified records and their actual maps supply that meaning. A lawful change of one port retains the full pointed substitutions and fixed-sibling comparison; simultaneous covariance of the whole tensor is not substituted for an experiment with the other port held fixed.

All admitted original preparations and source acts coexist in x_* . They are not all stationary Einstein histories. An excitation can change the stress without changing its legitimacy as a physical preparation, and the fixed joint gravitational parameters are not adjusted to compensate. The source/action identity preserves that distinction rather than reducing the observation algebra to the vacuum's one-point values.

10.9 Complete compatible witness families

Keep an original request family $\mathcal{Q}$ independently fixed by the target. A request retains its input/configuration, source branch, fixed matter/Higgs and preparation siblings, required physical output type and grade, domains, continuation and permitted outcomes. Let $\mathcal{W}_*(q)$ be the actual original-source solution packets for that request. Each packet contains all demanded incidence maps, physical boundary authority, channel or history, normalization, domain and gluing equations, and the original evidence for its asserted outcome. A productive request includes its nonzero witness. A response-only request need not be made productive, spectral or resonant; its exact zero can be a legitimate response. A missing operation or unresolved authority is not made a solution packet.

Let $\mathcal{W}_{\text{ret}}(q)$ use the same request on the originally indexed retained diagram. It is not the set of all possible packets using new metric arrows or stronger curved-source authorities.

Theorem 10.9 (Complete compatible witness-family transport). *For every original q , the diagram maps induce inverse maps*

$$\begin{aligned} L_q : \mathcal{W}_*(q) &\longrightarrow \mathcal{W}_{\text{ret}}(q), \\ R_q : \mathcal{W}_{\text{ret}}(q) &\longrightarrow \mathcal{W}_*(q). \end{aligned} \tag{169}$$

They preserve and reflect boundary grade, source equations, domains, failures, original continuation and productive/nonproductive distinctions, and commute with original presentation changes and request restrictions. They restrict to inverse maps on complete jointly compatible families, with

$$(Lw)_q = L_q(w_q), \quad (Rw^{\text{ret}})_q = R_q(w_q^{\text{ret}}), \quad RL = I, \quad LR = I. \tag{170}$$

This comparison alone does not assert that an empty original solution set is inhabited.

Proof. Apply Theorem 10.4 to every packet component and Theorem 10.5 to its channels and gluing. Each exact domain and first failure is preserved in both directions. An original graph equality is copied with its actual domain and warrant; a retained equality is an original one or a consequence of those relations, so readback preserves its derivation. Equality of values solely at the stationary member inserts no new relation.

The boundary components are the actual objects described above. Represented graphs keep their physical inner products and domains; resonances keep their entire functions and sheets; asymptotic maps keep their limits; persistence keeps the full history and boundary. An asserted exclusion retains its original input and proof. Hence the request predicate is evaluated on exactly the same full packet. A common incidence or gluing equation sees the same shared operand, not

independent marginal replacements. Fixed siblings remain fixed, and genuine variations retain their actual mixed contacts and source authority. Therefore both maps preserve being a solution of the same request.

Their composites recover every originally indexed primitive, operand, relation, authority and witness. This proves (169). Composition of the original source comparisons and request restrictions proves naturality. For a complete family, the coordinate formulas in (170) are specified maps, not fresh choices of representatives or witnesses. Every compatibility equation between packets is one of the preserved common-interface equations. The maps consequently restrict to the compatible families and remain inverse there. $\square$

There is thus no separate assembly obstruction merely because an original channel is spectral, asymptotic or resonant rather than a compact response. The independently formed input and exact native-use comparison provide the other leg of the application. Theorem 11.5 applies these maps to the original physical declarations and their actual witnesses at the same x_* , retaining one source and the once-frozen joint point.

11 A common physical realization

The preceding constructions now meet at one source. The comparison includes the complete original finite configuration domain, its actual physical uses, the full electroweak channels, and the source-dependent Einstein action. It is stronger than agreement of marginal states or lists of masses: the source restrictions identify the operations, their domains, their common operands, and their continuation.

We first identify the physical declarations to which the color and composite result applies. This fixes its quantifiers independently of the realization. We then prove the complete physical interface and place its whole compatible family at the same action/source point as the quantum and gravitational constructions.

11.1 The original physical declarations

Fix the occupied interacting source S_* and the physical target T_{ph} of Definition 1.1. Its field inventory, chiral and weak representations, Higgs and Yukawa incidences, global form, spin and boundary data, anomalies, vacuum, renormalized prescription, preparation classes, and continuation are retained. In particular, the independently formed configuration domain C_* is the full original source fibre of Theorem 6.1, not the image of successful physical channels.

Definition 11.1 (Original physical-use declarations). Let Λ_{ph} denote the original physical and source operations invoked at T_{ph} , with their source-qualified domains. A declaration λ includes its source and target node, operation kind, operand sorts, intrinsic parameter, joint interfaces, domain, output type, physical authority, and diagnostic. Write its actual graph as

$$g_\lambda : D_\lambda \subseteq X_\lambda \longrightarrow Y_\lambda. \quad (171)$$

An intrinsic parameter may be a whole operator, response family, spectral calculus, marked analytic germ, asymptotic attachment or history mechanism. An original use is an admitted graph, restriction, comparison, or finite witnessed composition of such declarations, with every shared operand and intermediate domain retained.

This is the received operational target of [1, Definition *Received operational target*], specialized to the fixed source, not a new selection of required uses. The complete native operation classification in [5, Proposition on color-native reception] has six families: source restrictions

and mixed incidences; declared local source laws and variations; paths, networks, tensors and evaluations; global descent, phase and closed-sector operations; actual observable, state, operator, continuation and physical-port operations; and exact domain, authority, obstruction and failure tests.

These are constructors for all the admitted intrinsic arguments, not a finite list of numerical outputs. An originally supplied non-finite mechanism remains its whole primitive graph; finite composition does not reduce it to a polynomial or a finite jet. Equally, a declaration that carries boundary data or types an operator is evaluated as that operation. Its name does not supply a physical boundary reader.

The relevant physical declarations have the following meanings.

1. A compact original color or matter insertion at the physical Higgs ports uses the full-interaction source prescription and the contact-complete calibrated reader of Theorem 5.6. It is a response on its entire admitted insertion and test domain, including its exact zeros.
2. A localized composite-persistence input uses the whole independently typed domain and witness of Theorem 7.5. A process uses its actual terminal witness reader and the exact domain in Theorem 7.8. The occupied heavy-hadron records give nonempty physical instances.
3. A supplied spectral, stable asymptotic, or continued resonance primitive retains its original source theorem, full domain, preparation and port, spectral or limiting prescription, and marked sheet. Its complete comparison is furnished by Theorems 10.4 and 10.5.
4. The phase and sector declarations retain the source's synchronized physical classification. At the selected neutral interface, Theorem 7.2 supplies the actual exclusion of a productive freely colored port. Exact source-proved impossibility, inapplicability and failure remain separate outcomes with their original premises.
5. A structural or source-constraint use retains its actual constraint, typing, transport or support graph. Its typing or constraint operation alone is not a physical reader; a subsequent physical output uses the actual admitted physical vertex and its domain.

The independently extracted physical signatures and occupied assignment are supplied by [52, Sections 8–9]. Their operation-specific source licenses distinguish construction, witnessing, transport, constraint and fixing. The local BRST primitive begins with the original actual operator and BRST class/action data; the accidental-symmetry primitive begins with the registered operators, dimensions, charges and effect set [37, 51]. Theorem 10.7 matches these actual operands and signatures before branch assignment, and matches the physical graph in addition to its signature.

Source nodes are not amalgamated by this notation. The physical core and its BRST/electroweak formulation retain their admitted relation. A QCD/HQET-matched record retains its matching order and directed source path. In particular, the chiral vacuum result uses registered physical input 8.1, the confining QCD branch with nonzero chiral witness; its stabilizer is then derived. The heavy-hadron record uses its registered scale/lifetime and bound-state target, and the source path retains those assumptions [36, Input 8.1, V01–V02, H06 and R01]. These are positive source results at those grades, not an unconditional reverse-decoupling theorem for generic full-SM amplitudes.

Input coverage and positive realization. The original confinement interface makes two different demands. Its configuration and operation comparison covers the independent role-complete domain, including every terminally relevant class. Its positive-output clause requires an actual boundary and witness for each positively claimed realization, and its authority clause

requires the independent authority of each boundary type invoked [5, Definition *Complete target-relative confinement acquisition*].

It does not assign a productive hadron, pole, scattering state or promoted quark parent to every pre-boundary configuration. FCGS–II distinguishes pre-boundary admission from physical descent and states the additional conditions for a positive parent claim [28, Definition of κ_{col} and positive-parent conditions]. Those conditions remain valid when such a claim is made; they do not make every source configuration terminal. Conversely, an original physical declaration cannot be removed because its evaluation is difficult. All declarations in Definition 11.1 retain their complete original domains and propositions.

11.2 Exact comparison of physical uses

Keep the full original source-marked presentation category $\mathcal{F}_*$. The configuration interpretation of Theorem 6.11 has the readback equation

$$J : \mathcal{F}_* \longrightarrow \mathcal{P}_*, \quad E : \mathcal{P}_* \longrightarrow \mathcal{C}_*, \quad EJ = e, \quad (172)$$

where e reads the original configuration and licensed operation record. No word or realization is selected from a coarser configuration class. In particular, Theorem 6.10 retains the full ancestral record, the actual current field realization and their specified common-reduct comparison.

For each primitive of (171), Theorem 10.7 identifies the receiving graph by maps r_X, r_Y with inverse readback on the retained records:

$$D_\lambda^{\text{rec}} = r_X(D_\lambda), \quad g_\lambda^{\text{rec}} r_X = r_Y g_\lambda. \quad (173)$$

The source, node, domain, normalization, authority and diagnostic belong to the record on both sides. Thus, writing rb for the full readback, one has

$$\begin{aligned} \text{rb}\{(\lambda, r_X a, g_\lambda^{\text{rec}}(r_X a)) : \lambda \in \Lambda_{\text{ph}}, a \in D_\lambda\} \\ = \{(\lambda, a, g_\lambda(a)) : \lambda \in \Lambda_{\text{ph}}, a \in D_\lambda\}. \end{aligned} \quad (174)$$

At a classified primitive, the complete physical signature agrees before the source-available branch is chosen. At a physical vertex, (173) identifies the actual reader as well; signature equality alone would not prove (174).

Proposition 11.2 (Physical declarations and complete native uses). *The interpretation (172) and graph comparison (173) identify every original physical/source declaration at T_{ph} , and all its admitted finite native uses, on the exact original domains. Every invoked positive realization retains its actual source witness. Every source-proved alternative retains its branch and warrant. The full pre-boundary configuration domain remains included, whether or not a configuration is positively declared terminal.*

Proof. Theorem 6.9 identifies the original coefficient products, full interaction V , relative-source operation and interacting reconstruction $\mathcal{E}_*$ before scalar evaluation. Thus the compact physical use is the same source operation as in Theorem 5.6, not merely a formula with a matching label. In a fixed original rule frame its field-valued insertion derivative is

$$D_{f;C} = R_V(F_{fC}) + i\{T_V(F_C, F_f) - R_V(F_C) \star R_V(F_f)\}. \quad (175)$$

The scalar-port reader retains its original nonzero detector normalization and contact-complete n^{-2} calibration. If the original variation moves the product, reconstruction, renormalized rule, BRST representative, preparation or detector, their derivatives occur exactly once, as in Proposition 5.5 and Theorem 10.7. Consequently the full reader square holds on every original

admitted insertion/test tuple. An exact zero is a defined response, not a proof of a different positive channel or of impossibility.

For a persistence use the actual quotient q_{S_*} , process w , and terminal witness reader γ give

$$K_w = q_{S_*} \gamma w, \quad D_{K_w} = \{a \in D_w : w(a) \in D_\gamma\}. \quad (176)$$

Theorems 7.5 and 7.8 supply these maps on their full independently typed domains. The complete input and history remain alongside their quotient. An empty process pullback is not a productive realization; the source's nonempty persistence population is independently supplied.

For each other whole physical primitive, Theorems 10.4 and 10.5 identify its complete original graph and authority. An unbounded spectral function keeps its integrability domain; a continued channel keeps its whole marked germ and ports; an asymptotic map keeps its original limiting domain. The original source theorem supplies the object and the comparison theorem identifies its use. It does not create a new channel from the existence of a formal graph. The phase, negative and constraint cases likewise use their actual source maps and warrants.

These cases exhaust the original primitive operation families. For composable vertices $f : D_f \subseteq X \rightarrow Y$ and $g : D_g \subseteq Y \rightarrow Z$, the exact composite is

$$\begin{aligned} D_{g^{\text{rec}} f^{\text{rec}}} &= r_X \{a \in D_f : f(a) \in D_g\}, \\ g^{\text{rec}} f^{\text{rec}} r_X &= r_Z g f. \end{aligned} \quad (177)$$

A finite network has the iterated pullback of its vertex domains along the original interface equations. Each repeated operand has one transported value. Theorem 10.8 preserves and reflects those pullbacks, fixed siblings, source relations and first failures. Induction proves (174) for every admitted finite use. An originally supplied whole evolution, limit or history mechanism remains a primitive with its own domain; the induction asserts no new fixed-point or convergence theorem.

Finally, a positively invoked witness consists of its original physical input, actual response/process/port and authority-bearing boundary. The preceding source cases carry all these components together, and their readback is the original witness. This is not a positivity inference from graph existence: the positive proposition and witness are the actual source conclusion, while the graph comparison identifies their previously specified configuration/use. The same argument retains an exact source-proved alternative. $\square$

The exhaustive operation grammar is essential here. It prevents an original physical primitive from being omitted merely because its intrinsic parameter was not displayed in a short table. It also prevents a stronger output from being introduced without a source comparison through a familiar operator name. Actual represented boundaries retain their stronger authority; valid coefficientwise source operations do not thereby become represented fields.

11.3 The complete color and composite interface

Theorem 11.3 (Complete physical color and composite interface). *At the occupied original source S_* and fixed T_{ph} , the complete original finite-configuration interpretation, physical-use comparison, and source constructions above supply the target-relative confinement and asymptotic-sector interface of [5]. Its domain is the independently formed role-complete $\mathcal{C}_*$; its physical conclusions have the original invoked authority. It includes exact color/composite continuation and the exclusion of isolated-quark/gluon physical output maps at the stated physical interfaces. It does not assert a positive parent for every pre-boundary configuration or uninvoked all-state asymptotic completeness.*

Proof. We verify the seven source requirements in the definition of complete target-relative confinement acquisition.

Independent input. The source fibre and finite constructors are fixed before any channel-success test. The occupied source supplies actual current up/down, conjugate, gauge/support and weak/Higgs data. Theorems 6.10 and 6.11 retain their common color, matter, Higgs, global, boundary, vacuum and ancestral/current incidences. Thus the domain is nonempty and role-complete, not an image of successful probes.

Complete realized operation comparison. Equation (172) covers every original configuration and licensed arrow. Proposition 11.2 adds the exact physical-signature and operation comparison (173)–(177). Every originally terminally relevant subclass is included without selecting it by receiving success. Structural/support primitives remain at their native grade; physical uses reach their actual physical reader.

Positively claimed boundaries and outcomes. The original positive declarations are those of Definition 11.1, not a new terminality assignment to all $\mathcal{C}_*$. The compact physical responses have (175) and their actual Higgs endpoint. The complete persistence and process witnesses have (176). Other invoked physical types retain their actual whole objects and witnesses under Theorem 10.5. Proposition 11.2 carries these at every original argument of the asserted proposition. Exact impossibility or inapplicability retains its source proof; neither a missing attachment nor an unproved positive existence statement is replaced by zero.

Phase and freely colored ports. The occupied source's synchronized phase/sector classification is retained with its actual physical meaning [34, 36, 52]. At the selected neutral interface the actual color action and admitted intertwiner satisfy

$$AR_C(g) = A, \quad A = AP_C, \quad A(1 - P_C) = 0. \quad (178)$$

Theorem 7.2 derives this comparison from the original output action and port, not from a bare input projector or a mass gap. It excludes a productive isolated fundamental, antifundamental or adjoint port into that neutral physical interface. The projection acts on the full joint input and does not remove neutral composite responses constituentwise. Together with the actual confined-field, composite and other sector declarations, this is the asserted phase-qualified exclusion of isolated-color physical output maps. Registered QCD/HQET records retain their source assumptions, matching direction and physical type; no additional charged or global boundary sector is introduced.

Independent boundary authority. Compact response, persistence, represented spectral measure, stable asymptotic attachment and unstable continued pole remain distinct typed objects. Their complete domains and authority are retained by (173). A supplied continued channel keeps its full germ, not only a pole list or finite jet. Persistence is not relabelled as scattering. Every invoked asymptotic map retains its original limiting authority. All-state asymptotic completeness is not an assertion of this theorem, so its separate all-state premise is not introduced.

Continuation, gluing and mixed inputs. Theorem 10.5, Theorem 10.8, and (176)–(177) identify the actual continuation and gluing operations, including intermediate domains, failure and boundary type. The balanced-module comparison preserves the original realized gluing image; it does not assert that every ambient channel factors through a chosen boundary. All weak/Higgs, flavor, preparation and source consumers use the same operands. Genuine variations retain their joint contacts. No marginal sources are spliced.

Source-complete outcomes and common matter/Higgs incidence. Equation (174) identifies the original declarations and outcomes, and (177) identifies their complete admitted compositions. A constraint is evaluated by its constraint operation, a physical response by its actual reader, and an asserted positive boundary by its source witness. Missing required authority is never hidden as a zero. The original full input domain contains pre-boundary uses, but does not declare an additional positive boundary for every such input. The complete matter/Higgs and source incidences are already those of Theorem 6.11; the common action/source comparison is made explicitly below. This verifies the complete interface at the original domains and authority

layers. □

This result uses the positive source assignment and actual physical operations, not merely source licenses alone. If a different target asserts a new particle boundary, positive parent, stronger product domain or additional scattering coverage, it has its own source obligation. Such an extension does not follow from reception of the present graphs, and is not silently made here.

11.4 Restriction and lift at the common action

Keep the once-selected point of Theorem 9.7,

$$J_{\text{joint}} = (S_*, \kappa_*, \Lambda_*), \quad \Lambda_* = \kappa_* t_*, \quad \kappa_* \neq 0, \quad (179)$$

where t_* is the original vacuum stress coefficient. The gravitational calibration was selected in its admitted fibre before being frozen. No new parameter choice is made for a color preparation.

The source-dependent action was constructed before selecting its stationary member. In the notation of Section 9,

$$\begin{aligned} S_{g,j} &= S_V(F[g, j]), \\ K_{\text{CTP}} &= \mathcal{E}_*(S_{g_-, j_-}^* \star S_{g_+, j_+}), \\ \Gamma &= S_{\text{EH}}[g_+] - S_{\text{EH}}[g_-] - i \log K_{\text{CTP}}. \end{aligned} \quad (180)$$

The relative action F is the original metric/coframe, gauge, chiral-fermion, Higgs/Yukawa and source prescription, with its normalization and contacts. These are the actual algebra-valued operations of Theorem 5.3. The action has its original quantum order; its independently supplied stronger baseline physical records are not reduced to that order.

Let x_* be this common action/source object with the constructed member (η, Ω) . Its restrictions are:

1. $r_Q(x_*) = S_*$, retaining the entire original quantum source and forgetting only the additional gravitational action/calibration coordinates;
2. the baseline restriction b to $g = \eta, j = j_*$, where $F = 0$ and $S_V(0) = 1$;
3. $r_G(x_*)$, retaining Γ , its full all-test source extraction, holding/time-action comparison and the member at (179).

In particular,

$$K_{\text{CTP}}(\eta, j; \eta, j_*) = Z_{\text{original}}(j), \quad b(A_{x_*}) = A_{S_*} \quad (181)$$

for every original represented or continued baseline record A , with its complete domain and authority. This is identity of the retained object, not equality of a few coefficient jets or a partial trace from an independently formed tensor-product state.

For every admitted compact metric test h , the member equation is

$$D_{\text{diff}}\Gamma(\eta)[h] = -\frac{\Lambda_*}{2\kappa_*}\langle \eta, h \rangle + \frac{t_*}{2}\langle \eta, h \rangle = 0. \quad (182)$$

The Einstein kinetic coefficient and complete source response remain in the action. The strict original preparation response of Corollary 9.8 prevents (182) from being merely a source-independent fixed background with a matching tensor.

Proposition 11.4 (Exact original-operation lift). *The complete original configuration, source-operation and physical-interface family has a lift to x_* with inverse readback. It retains every source equation, exact partial domain, common operand, failure, physical type and authority. At a rooted constituent it retains both full source records and their existing common-reduct comparison.*

Proof. For an original operation $M : D_M \subseteq X \rightarrow Y$, set

$$i_X(a) = (x_*, a), \quad D_M^\uparrow = i_X(D_M), \quad M^\uparrow(i_X a) = i_Y M(a). \quad (183)$$

Equivalently, on the actual source-family pullback domain, $M^\uparrow(x, a) = M(r_Q(x), a)$. The inverse reads the entire original argument and its warrant, not just the value of an observation on (η, Ω) . Whole operators, channels, limits and history mechanisms keep their original graphs.

The actual current coefficient/source operations are precisely $V, \star, S_V, \mathcal{E}_*$ in (180), by Theorem 6.9. The current field leg is the realization of Theorem 6.10. For a full ancestral record A_f , actual current restriction $\hat{\rho}_f$, and specified comparison θ_f , the rooted lift is

$$(A_f, \hat{\rho}_f, \theta_f) \mapsto (x_*, A_f, \hat{\rho}_f, \theta_f). \quad (184)$$

The two full target fingerprints are not identified. Each consumer uses its original full record, and the common root, kind, weak/Higgs and source data compare by the same θ_f . Reading back returns the whole triple.

For $f : D_f \subseteq X \rightarrow Y$ and $g : D_g \subseteq Y \rightarrow Z$,

$$D_{g^\uparrow f^\uparrow} = i_X\{a \in D_f : f(a) \in D_g\}, \quad g^\uparrow f^\uparrow i_X = i_Z g f. \quad (185)$$

Thus each intermediate domain and its first failure is the original one. A shared edge carries one $i_X(a)$. The full physical-use matching in Proposition 11.2 and induction on the original finite network prove the complete statement, with no new physical vertex inferred from metadata. An originally supplied non-finite mechanism stays a primitive on its furnished domain.

All original source relations carry their same warrants and read back to the same equations. On physical boundaries these maps are exactly the inverse retained source functors of Theorem 10.4. They are not an inverse on the entire enlarged gravitational category: new metric-response arrows need not be original quantum arrows. Consequently (172) remains valid after the lift. This establishes the complete input-to-interface comparison at the common point. $\square$

11.5 Simultaneous physical realization

Theorem 11.5 (One common physical source and realization). *The occupied original Standard Model source, its complete color/composite interface, its physical W , Z and Higgs channels, and its source-compatible dynamical Einstein realization are simultaneously realized at x_* and the single fixed point J_{joint} .*

Their comparison preserves the original field inventory, interaction, branch, global form, spin and boundary, source and parameter data, preparation and observation domains, and every invoked authority layer. The color/asymptotic records used by the interacting quantum realization are the same records as those of Theorem 11.3. No independent sector state, source fit, root realization or further parameter choice is used in their conjunction.

Proof. The original source construction and Theorem 9.7 give x_* in the intersection of the physical electroweak, received interacting-source, and dynamical-gravity domains. These domains are the inverse images of the actual source-projection domains, not abstract sets known separately to be nonempty.

For the Higgs channel, retain the original $O_\rho = nO_H + b\mathbf{1}$, $n \neq 0$, with its local contact subtraction, physical ports, sheet and full continued germ from Theorem 4.3. The original W/Z records keep their current, charged-sector or physical-amplitude attachment and endpoint factors. Equation (181) leaves these objects unchanged. It does not reconstruct a resonance from a compact source jet or replace it by a stable mass eigenstate.

The original full field prescription, interacting reconstruction, gauge/anomaly, vacuum and renormalization data are exactly $r_Q(x_*)$. All originally supplied operator, continued and asymptotic records are retained there with their full authority. The gravitational object is the source-dependent action (180) with its actual holding/time-action comparison and the occupied member (182).

Let w_{col} be the complete compatible interface family supplied by Theorem 11.3 on this very S_* . Proposition 11.4 identifies its entire independent input and operation family at x_* . Theorem 10.5 gives the actual channel bijections

$$K_*(C, B) \simeq K_*^\uparrow(I_C C, I_B B), \quad k \longmapsto I_R k. \quad (186)$$

They preserve both interface actions, exact domains and physical authority. Their balanced composition map sends $[k, l]$ to $[I_R k, I_R l]$, commutes with realized gluing and retains its exact image. No additional factorization through a chosen boundary is asserted.

Theorem 10.9 applies these specified maps componentwise to the whole compatible family:

$$L(w_{\text{col}})_q = L_q((w_{\text{col}})_q), \quad RL(w_{\text{col}}) = w_{\text{col}}. \quad (187)$$

No new independent choice is made for each q . All joint equations and fixed siblings hold because they are equations of the same shared source operands. A persistence witness remains a history with its boundary reader; a residue remains part of its full continued channel; an exclusion retains its original sector and premises. Required-domain coverage is provided by Theorem 11.3 and the exact input comparison, not by declaring the receiving diagram to be the required domain.

It follows that x_* also belongs to the color acquisition domain, the inverse image under r_Q of its actual original source domain. The color/asymptotic columns of the interacting quantum source are precisely these records, not another quantum realization. The projection-intersection theorem of [3, Section 8, theorem on projection descent] therefore evaluates the whole dependent physical output family at this single point.

Every named target coordinate is unchanged: the complete field inventory and interaction are those of S_* ; the branch, global form, spin and boundary are retained; all original physical parameters stay fixed; κ_*, Λ_* are the once-selected values of (179); and each construction keeps its actual authority. The mixed constraints use the same source maps and arguments. This proves simultaneous realization, rather than an intersection inferred from separately nonempty marginal domains. $\square$

11.6 Mixed response and the meaning of the common member

The lift preserves the source's actual variations, not only its baseline values. For an originally admitted physical source act c and metric act h , the same relative action F , in a fixed original coefficient frame, gives

$$\begin{aligned} A_g(h) &= R_{V+F}(f_h), \\ D_c A_g(h) &= R_{V+F}(f_{ch}) \\ &\quad + i\{T_{V+F}(f_c, f_h) - R_{V+F}(f_c) \star R_{V+F}(f_h)\}. \end{aligned} \quad (188)$$

The mixed seagull f_{ch} and all metric-density, coframe, normalization and BRST contacts are those of the original source. If its prescription, product, reconstruction, port or holding map moves, the corresponding original derivatives are retained exactly once, as in Theorems 5.6 and 9.4. Reading (188) through (183) gives the identical field-valued expression before $\mathcal{E}_*$ and the same physical readout at its original grade. A color/Higgs source act is therefore not an independent current appended to gravity. A structural-only act is not promoted to a differentiable field insertion.

The baseline time continuation $U(t) = e^{-itH_0}$, its represented domains and full multi-time arguments remain those of Lemma 9.5. Histories, limiting attachments and continued germs retain their original domains. The coefficientwise metric formula does not extend them to arbitrary off-shell metrics. The independently formed upstream continuation remains prior to these projections [9, 38, 39]; neither the stationary member nor its quantum time flow is renamed as that construction.

Finally, the common source contains its entire original preparation and operation classes. This does not mean that each excited preparation is another flat Einstein solution. With κ_*, Λ_* fixed, a change of state generally changes the all-test stress. No retuning or compensating metric is asserted for every such preparation. The theorem supplies the genuine source-coupled dynamical action, its occupied invariant member, and the complete original physical source on the same target. The separate regulated constructions that follow provide additional existence, domain, response and comparison theorems; their method-specific refinement questions are not antecedents of Theorem 11.5.

Part II

Constructive interacting source realizations

The finite spatial realization gives explicit interacting Hamiltonians, physical domains and controlled observations. Its refinements provide complementary existence and response theorems. Throughout this part, statements about continuum identification concern this particular realization; the common physical source of Part I is already constructed at its original source authority.

12 An occupied quantum gauge–Higgs–matter regulator

The source-first physical realization of Theorem 11.5 is complemented by an explicit finite-spatial construction. We realize the complete gauge, Higgs and matter inventory on one state space, with a common operator domain and mixed observations, including nonzero physical Yukawa response and feedback on the Higgs coordinate. This is an occupied interacting quantum theory at its stated resolution. Identifying a refining family of these theories with the original physical source is a distinct comparison, formulated at the end of the section.

The local Yang–Mills construction supplies a useful precedent with a precise proof mechanism. Lane A first constructs positive finite mixed families, controls replacement of one insertion against every fixed word in the remaining slots, and proves compatibility on overlaps. Only then does it form the inductive-union state [17, Appendix B, Theorems B.11–B.13 and Corollary B.17]. The already constructed local quantum theory and its nonzero spectral witness are supplied by the local theorem and reconstruction companion [16, Theorem I.1][18, Theorems M.1, M.3 and M.6]. The following finite regulator uses the complete matter and Higgs alphabet, with its joint positive source constructed directly.

12.1 The product group, fields and physical subspace

Fix a finite connected cubical graph with vertex set V , oriented edge set E , and a finite plaquette family $\mathcal{P}$. Require at least one genuine non-backtracking plaquette with independently variable holonomy. Fix

$$G_0 = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1), \quad G = G_0 / \langle c \rangle, \quad c = (e^{2\pi i/3} I_3, -I_2, e^{i\pi/3}).$$

The compact Abelian factor acts by integer characters z^q , where $q = 6Y$. The full internal representation list is

$$\begin{aligned} Q &= (3, 2)_1, & u^c &= (\bar{3}, 1)_{-4}, & d^c &= (\bar{3}, 1)_2, \\ L &= (1, 2)_{-3}, & e^c &= (1, 1)_6, & H &= (1, 2)_3. \end{aligned} \quad (189)$$

There are three copies of each matter representation and one Higgs doublet. Fractional hypercharge remains available as the notation $Y = q/6$; it is not used as a fractional power of z .

Lemma 12.1 (Exact descent and a faithful Wilson representation). *Every representation in (189) descends through G . Their direct sum σ is faithful as a representation of G .*

Proof. The action of c is respectively

$$e^{2\pi i/3}(-1)e^{i\pi/3}, \quad e^{-2\pi i/3}e^{-4\pi i/3}, \quad e^{-2\pi i/3}e^{2\pi i/3}, \quad (-1)e^{-i\pi}, \quad e^{2\pi i}, \quad (-1)e^{i\pi},$$

all equal to one. Conversely, write an element of the common kernel as (C, S, z) . The Higgs equation gives $S = z^{-3}I_2$, hence $z^6 = 1$. The d^c equation gives $C = z^2I_3$. These elements are precisely the powers of c . Thus the common kernel in G_0 is exactly $\langle c \rangle$, and the descended direct sum is faithful. This is the common-product descent used by the global-form source [21, Theorem 6.1]. $\square$

The bosonic configuration manifold is

$$X = G^E \times (\mathbb{C}^2)^V.$$

Let W be the finite one-particle space consisting, at each vertex, of the three matter copies in (189), each with a two-component spin index. Set $\mathcal{F} = \bigwedge W$ and

$$\mathcal{H}_{\text{kin}} = L^2(G^E, dU) \otimes L^2(\mathbb{R}^{4|V|}, dH) \otimes \mathcal{F}. \quad (190)$$

The finite CAR generators are denoted by ψ_R for each matter species R . Higgs multiplication and $-i$ times Higgs differentiation give the usual CCR on the smooth domain specified below. Link multiplication and invariant differentiation give the group-coordinate and electric operators. This is the Hamiltonian finite-link setting [23], with all representations taken on the joint quotient rather than on independently selected color links.

The local gauge group $K = G^V$ acts on configurations by

$$U_{xy} \mapsto k_x U_{xy} k_y^{-1}, \quad H_x \mapsto \rho_H(k_x) H_x.$$

Its bosonic Hilbert action is pullback, $(\mathcal{U}_b(k)f)(x) = f(k^{-1}x)$, and its Fock action is $\mathcal{U}_f(k) = \Gamma(R(k))$. We fix the forward field convention

$$\mathcal{U}(k)^{-1} \psi_R \mathcal{U}(k) = R(k) \psi_R. \quad (191)$$

The annihilation column has this transformation by second quantization. Higgs and link multiplication obey the same forward convention; adjoint CAR fields transform contragrediently. Thus no matter multiplet is replaced by its conjugate to establish invariance.

Definition 12.2 (Physical regulator space). The physical Hilbert space and its orthogonal projection are

$$\mathcal{H}_{\text{phys}} = \mathcal{H}_{\text{kin}}^K, \quad P_K = \int_K \mathcal{U}(k) dk.$$

Physical bounded observables preserve this Gauss-invariant space. The reflected-observable algebra below is its parity-even part. Elementary charged fields act on $\mathcal{H}_{\text{kin}}$ and need not preserve $\mathcal{H}_{\text{phys}}$. Their compressed operators are not asserted to obey the original CAR or CCR.

12.2 One common Hamiltonian and its domain

Choose positive electric coefficients and let T_G be the resulting positive invariant Laplacian on G^E . Put $T_H = -\Delta_H$, choose $\lambda_H > 0$ and $a_H \in \mathbb{R}$, and define

$$\mathbf{H} = T_G + T_H + \lambda_H \sum_x |H_x|^4 + a_H \sum_x |H_x|^2 + W_G + W_H + F_{\text{hop}} + \mathbf{Y}. \quad (192)$$

Here

$$W_G = \sum_{p \in \mathcal{P}} \beta_p (\dim \sigma - \text{Re Tr } \sigma(U_p)), \quad W_H = \sum_{xy} \kappa_{xy} |H_x - \rho_H(U_{xy}) H_y|^2,$$

with $\beta_p, \kappa_{xy} > 0$. The finite hopping operator is a sum of

$$\psi_{R,x}^\dagger (\rho_R(U_{xy}) \otimes S_{xy}) \psi_{R,y}$$

and its adjoint. Each S_{xy} is a fixed spin hopping matrix and commutes with the internal action. The coefficient record is part of this regulator target.

For clarity, the Yukawa contraction is fully displayed. With i, j generation indices, A a color index, a, b weak indices and α, β spin indices, set

$$\begin{aligned} Z_Y = \sum_{x,i,j} \epsilon_{\alpha\beta} [(y_u)_{ij} \epsilon_{ab} Q_i^{Aa\alpha} u_{j,A}^c{}^\beta H^b \\ + (y_d)_{ij} Q_i^{Aa\alpha} d_{j,A}^c{}^\beta \bar{H}_a + (y_e)_{ij} L_i^{a\alpha} e_j^{c\beta} \bar{H}_a], \quad \mathbf{Y} = Z_Y + Z_Y^\dagger. \end{aligned} \quad (193)$$

All fields in a term are at x . CAR products use the displayed order. The integer Abelian sums are $1 - 4 + 3 = 0$, $1 + 2 - 3 = 0$ and $-3 + 6 - 3 = 0$. Color uses the $3-\bar{3}$ pairing; weak indices use ϵ or the conjugate-Higgs pairing. Equation (191) therefore proves gauge invariance. The matrix-valued multiplication operator $\mathbf{Y}(H)$ is Hermitian, parity even, linear in real Higgs coordinates and satisfies

$$\|\mathbf{Y}(H)\| \leq C|H|. \quad (194)$$

In particular, all mixed terms belong to one operator on one Hilbert space.

Let $\mathcal{D}$ consist of functions smooth in the compact link coordinates and Schwartz in the real Higgs coordinates, with uniform Schwartz bounds for every mixed derivative, tensored with $\mathcal{F}$. Smooth link coefficients, the polynomial Higgs coefficients and all displayed differential expressions preserve $\mathcal{D}$. Haar averaging preserves it. Its physical part is $P_K \mathcal{D}$.

Theorem 12.3 (Self-adjoint physical occupation). *The Hamiltonian (192) has a unique self-adjoint semibounded realization associated with its specified closed form. It commutes with the local gauge action and restricts to a self-adjoint operator on a nonzero $\mathcal{H}_{\text{phys}}$. This restriction has compact resolvent, a lowest eigenvalue with a nonzero finite-dimensional eigenspace, and a finite heat trace for every positive inverse temperature. Thus it has exact unitary dynamics and a faithful stationary Gibbs state on its represented bounded physical algebra.*

Proof. Begin with the positive closed form of

$$T_G - \Delta_H + \lambda_H \sum_x |H_x|^4 + 1.$$

Its domain is the intersection of the kinetic first-order Sobolev domains with the space on which $\int \sum_x |H_x|^4 \|u\|^2$ is finite. For every $\varepsilon > 0$,

$$r^2 \leq \varepsilon r^4 + C_\varepsilon, \quad r \leq \varepsilon r^4 + C_\varepsilon.$$

Consequently the quadratic Higgs terms and (194) have arbitrarily small relative form bound. W_G is bounded by compactness and the finite plaquette sum. F_{hop} is bounded because the link

representations are unitary and $\mathcal{F}$ is finite dimensional. The small-relative- form perturbation theorem gives a closed semibounded form and its unique associated self-adjoint operator. In particular,

$$\mathbf{H} \geq T_G - \Delta_H + \frac{1}{2}\lambda_H \sum_x |H_x|^4 - C. \quad (195)$$

Uniqueness here concerns this specified form, not different counterterm or regulator records.

All form terms and their common domain are gauge invariant. Uniqueness of the associated resolvent makes it commute with each $\mathcal{U}(k)$ and hence with P_K . The physical space therefore reduces $\mathbf{H}$. Independently, the normalized vector

$$\psi_0(U, H) = c_0 e^{-\sum_x |H_x|^2} \otimes |0\rangle \quad (196)$$

is gauge invariant: its bosonic function is radial and link-independent, and the Fock vacuum is invariant. It proves that the physical space is nonzero.

Smooth cutoff and mollification show that $\mathcal{D}$ is a form core. Integration by parts identifies its differential operator with the form realization. Since the expression preserves $\mathcal{D}$, this domain is contained in every operator-power domain. The same statements hold after Haar averaging.

For compactness, a form-bounded sequence has uniformly small Higgs tails, since for large R its squared norm outside $|H| \leq R$ is bounded by a constant times R^{-4} . On each bounded Higgs set, Rellich compactness, compactness of G^E and finite-dimensionality of $\mathcal{F}$ give a convergent subsequence. A diagonal choice and the tail estimate give convergence in the full Hilbert norm. The form embedding is compact, so the resolvent is compact, also on the reducing physical space. Nonemptiness then gives a lowest eigenvalue and a nonzero finite-dimensional ground eigenspace.

The lower form bound also dominates, up to positive constants and a scalar shift, a compact-manifold Laplacian plus a finite-dimensional harmonic oscillator. Their heat traces are finite. Eigenvalue min-max comparison therefore gives $\text{Tr } e^{-\beta \mathbf{H}_{\text{phys}}} < \infty$ for every $\beta > 0$. The normalized Gibbs operator has strictly positive eigenvalues on every physical spectral vector. Its state is stationary and faithful on every represented bounded physical subalgebra. $\square$

For subsequent constructions fix a scalar shift c_1 such that

$$H_0 = \mathbf{H}_{\text{phys}} + c_1 I \geq I. \quad (197)$$

The shift does not change the mixed dynamics or any commutator below. The physical space is infinite dimensional: independent radial Higgs functions times the Fock vacuum are physical.

Lemma 12.4 (Nonzero gauge response). *The vector (196) has strictly positive expectation of W_G .*

Proof. For a unitary matrix, $\dim \sigma - \text{Re Tr } \sigma(U)$ is nonnegative and vanishes only when $\sigma(U) = I$. Faithfulness and independent variation of a genuine plaquette therefore make its contribution positive on an open set. Haar measure has full support, and the Gaussian is nonzero almost everywhere. The integral of this nonnegative function is strictly positive. $\square$

12.3 A mixed physical state and its backreaction

Theorem 12.5 (Nonzero Yukawa occupation and Higgs response). *Suppose at least one entry of the Yukawa matrices is nonzero. With ψ_0 normalized as above, put*

$$\chi = \mathbf{Y}\psi_0, \quad \phi = \chi / \|\chi\|, \quad \eta = (\psi_0 + \phi)/\sqrt{2}.$$

Then χ is nonzero and physical, η is normalized and physical, and

$$\langle \eta, \mathbf{Y}\eta \rangle = \|\mathbf{Y}\psi_0\| > 0. \quad (198)$$

For the total finite fermion number N_F ,

$$\langle e^{-itH}\psi_0, N_F e^{-itH}\psi_0 \rangle = 2 \|\mathbf{Y}\psi_0\|^2 t^2 + o(t^2). \quad (199)$$

Moreover, for the physical radial Higgs coordinate $R_H = \sum_x |H_x|^2$, the Yukawa contribution to its Heisenberg acceleration at zero is $-4Y$. Its expectation in η is nonzero.

Proof. The annihilation terms of Z_Y kill the Fock vacuum. Its adjoint creates two distinct species modes with a nonzero linear Higgs coefficient. For a nonzero matrix entry fix a vertex, a generation pair, a color/weak component and opposite spin components. The two spin contractions occupy different wedge basis vectors; up, down and lepton terms use distinct species sectors. Thus CAR antisymmetry does not cancel every coefficient. Multiplication by the nonvanishing Gaussian gives $\|\chi\| > 0$.

Gauge invariance of $\mathbf{Y}$ and ψ_0 makes χ physical. It has fermion number two, whereas ψ_0 has number zero. The two vectors are orthogonal. Since $\mathbf{Y}$ changes number by two, its diagonal expectations in ψ_0 and ϕ vanish, while $\langle \psi_0, \mathbf{Y}\phi \rangle = \|\chi\|$. Self-adjointness supplies the conjugate cross term and proves (198).

The vector ψ_0 belongs to the Hamiltonian domain, so its unitary orbit has the first-order norm expansion $e^{-itH}\psi_0 = \psi_0 - itH\psi_0 + o(t)$. All other terms preserve fermion number, and the hopping kills the Fock vacuum. Since N_F is bounded and annihilates ψ_0 , only the number-two component $\mathbf{Y}\psi_0$ contributes at order t^2 . This gives (199).

In real Higgs coordinates of dimension $d = 4|V|$,

$$[T_H, R_H] = -4H \cdot \partial_H - 2d, \quad [\mathbf{Y}, [T_H, R_H]] = 4(H \cdot \partial_H \mathbf{Y}) = 4Y.$$

The final equality uses the degree-one homogeneity of $\mathbf{Y}$. Every other matter potential commutes with R_H ; the Yukawa part of the negative double commutator is therefore $-4Y$.

This is an actual derivative statement on the displayed states. The coercive form estimate gives $\|R_H u\|^2 \leq C \langle u, H_0 u \rangle$ after a sufficient shift, so $R_H H_0^{-1/2}$ is bounded. Both ψ_0 and ϕ are in $P_K \mathcal{D}$, hence in every Hamiltonian-power domain. Graph-norm differentiability of their unitary orbits validates the second derivative of the R_H expectation at zero and its double-commutator formula. Global invariance of the Schwartz domain under time evolution is not needed. $\square$

All three matrices may be simultaneously nonzero, for example invertible diagonal matrices. The corresponding pair sectors then give occupation witnesses in all charged Yukawa channels. The construction has nonzero gauge response, Higgs support and mixed fermionic response in one physical Hilbert realization. It does not identify each component with an asymptotic particle.

12.4 Positive temporal data on the same source

Proposition 12.6 (Mixed Gram kernels and even-string reflection). *For any physical vector ξ , bounded physical operators A_i and real t_i , the kernel*

$$K_{ij} = \langle A_i e^{-it_i H_0} \xi, A_j e^{-it_j H_0} \xi \rangle$$

is positive semidefinite. Let Ω_0 be a physical ground vector and $L = H_0 - E_0 I \geq 0$. For positive times the kernel

$$K_{ij}^E = \langle A_i \Omega_0, e^{-(t_i + t_j)L} A_j \Omega_0 \rangle$$

is positive semidefinite. The same holds for reflected ordered strings from the parity-even bounded physical algebra, with reflection defined by reversing order, taking adjoints and negating the time labels.

Proof. The first two matrices are Gram matrices, using respectively $A_i e^{-it_i H_0} \xi$ and $e^{-t_i L} A_i \Omega_0$. For $0 \leq t_1 \leq \dots \leq t_n$, define the transfer vector

$$v_{\mathbf{A}, \mathbf{t}} = e^{-t_1 L} A_1 e^{-(t_2 - t_1)L} A_2 \dots e^{-(t_n - t_{n-1})L} A_n \Omega_0.$$

The reflected negative-time string is its adjoint transfer expression. Pairing two strings is therefore the inner product of their transfer vectors. Every finite reflected quadratic form is the squared norm of their corresponding linear combination. The restriction to even observables avoids identifying this reflection with an unproved graded Euclidean fermion prescription. $\square$

These are actual positive mixed data from a constructed source; they can enter the represented-source comparison of BCRS-3 [3]. A finite spatial graph does not supply four-dimensional Euclidean covariance or a continuum OS theorem.

12.5 A mixed extension of the existing Yang–Mills source

The following comparison shows that the supplied quantum source can also support an explicit new mixed interaction without assuming its resulting correlations.

Proposition 12.7 (Bounded entangling quantum extension). *Let $(\mathcal{H}_Y, \Omega_Y, H_Y)$ be the constructed Yang–Mills vacuum source. There is a bounded self-adjoint element A of its represented gauge-invariant quasilocal observable algebra with $\langle \Omega_Y, A \Omega_Y \rangle = 0$ and $A \Omega_Y \neq 0$. On a two-mode auxiliary fermion Fock space put $B = a_1^\dagger a_2^\dagger + a_2 a_1$ and $N = a_1^\dagger a_1 + a_2^\dagger a_2$. For $g \neq 0$ and $\omega > 0$,*

$$H_g = H_Y \otimes I + I \otimes \omega N + gA \otimes B$$

is self-adjoint on the uncoupled domain and bounded below. For $\Psi = \Omega_Y \otimes |0\rangle$,

$$\langle e^{-itH_g} \Psi, (I \otimes N) e^{-itH_g} \Psi \rangle = 2g^2 \|A \Omega_Y\|^2 t^2 + o(t^2).$$

For all sufficiently small nonzero t the evolved pure state is entangled between the two factors.

Proof. The bounded quasilocal observable algebra acts cyclically in its vacuum representation. If every self-adjoint element sent Ω_Y into its own ray, so would every element of that algebra. Cyclicity would make $\mathcal{H}_Y$ one dimensional, contradicting the nonzero centered witness [16, Theorem I.1][18, Theorems M.3 and M.6]. Choose such an observable and subtract its expectation. The perturbation $gA \otimes B$ is bounded self-adjoint, giving the asserted domain and lower bound. Norm differentiability at zero gives

$$e^{-itH_g} \Psi = \Psi - itgA \Omega_Y \otimes |12\rangle + o(t).$$

Boundedness of N and $N|0\rangle = 0$ prove the number formula. Write the evolved vector as $\chi_0(t) \otimes |0\rangle + \chi_2(t) \otimes |12\rangle$. Then $\|\chi_2(t)\|^2 = g^2 \|A \Omega_Y\|^2 t^2 + o(t^2)$ and $\langle \chi_0(t), \chi_2(t) \rangle = o(t)$ because A is centered. The reduced two-by-two density matrix has determinant $g^2 \|A \Omega_Y\|^2 t^2 + o(t^2) > 0$. Its mixedness proves entanglement of the pure joint state. $\square$

The auxiliary pair is a gauge singlet. This example preserves the Yang–Mills marginal at time zero; it does not preserve the uncoupled later automorphism or claim a local Lorentz-covariant Standard Model extension. Its changed Hamiltonian is not automatically a gap-preserving same-domain completion.

12.6 Compatible mixed limits and their source comparison

Theorem 12.8 (Compatible mixed limits and algebraic reconstruction). *Let $\mathcal{A}_1 \subset \mathcal{A}_2 \subset \dots$ be unital star algebras with common specified relations and a positive-time reflection structure, and set $\mathcal{A} = \bigcup_n \mathcal{A}_n$. Suppose actual regulated joint representations and positive unital states supply representatives of every fixed capwise word, with:*

1. *a finite limiting expectation for every word, independent of the containing cap;*
2. *vanishing linear, star, product and quotient-relation defects inside every required mixed context;*
3. *an actual reflection-safe representative for every fixed positive-time polynomial, with vanishing reflected quadratic-form error.*

Then the limits define a unique compatible positive unital state on $\mathcal{A}$, reflection positive on its stated positive-time algebra. Its algebraic GNS representation has a common invariant polynomial domain. Uniqueness is relative to the specified limiting mixed source data.

Proof. Every finite linear combination, product or star expression belongs to a containing cap. The first two hypotheses make its value independent of cap and presentation and make all specified quotient relations valid in the limiting functional ω . Normalization passes from the regulated states. For every $P \in \mathcal{A}$,

$$\omega(P^\dagger P) = \lim_{\varepsilon \rightarrow 0} \omega_\varepsilon(P_\varepsilon^\dagger P_\varepsilon) \geq 0.$$

Thus every finite mixed Gram matrix is positive. The third hypothesis gives the same conclusion for reflected positive-time matrices. The positive-functional Cauchy–Schwarz inequality makes $\{P : \omega(P^\dagger P) = 0\}$ a left ideal. Quotienting by it and completing gives GNS, with left multiplication on the dense polynomial classes. Since each word lies in one cap, its limiting value is fixed; hence the extension is unique. $\square$

This theorem derives positivity at the union from actual regulated data; it does not assume a complete mixed kernel. No estimate uniform in the number of generators is needed merely to form this algebraic union. Limits and compatibility for every fixed mixed word are needed. Continuum field closure, regularity, covariance, locality and the appropriate physical quantum identities require their further domain and limit estimates, as in the separate OS dossier [18, Theorems M.1–M.3].

The finite regulator above establishes joint occupation and mixed positivity at a fixed graph. Ground or Gibbs states on different graphs need not restrict compatibly. Finite-site chiral representation inventory does not prove continuum chirality, absence of doubling, continuum anomaly cancellation, renormalization convergence or asymptotic completeness. Identification of this refining gauge–Higgs–chiral family with the original physical source requires those properties and their exact source comparisons. This is a condition on the regulated route, not a prerequisite for Theorem 11.5. The occupied regulator and the compatible-limit theorem give distinct constructive results.

13 Common asymptotic sources from regulated quantum states

The regulated constructions admit an actual limiting positive source without first assuming convergence of every mixed correlation. Bounded observations can be placed in a common sequence algebra and their states can be taken to a cluster point. Explicit sampling maps then identify continuum equal-time canonical relations inside that algebra. These results construct a

common native cluster source, while leaving its field regularity, chiral dynamics and physical source comparisons to distinct theorems.

This distinction also appears in the local Yang–Mills construction. Its bounded mixed families precede its local insertion, graph-domain and reconstruction estimates [17, Appendix D, Corollary D.8 and Theorem D.9] [18, Theorems M.1–M.3]. The compactness argument below concerns the first of these tasks. It does not import the pure-gauge estimates for the additional matter and Higgs fields.

13.1 A positive state on a common bounded algebra

Fix a sequence of the regulators of Section 12, with meshes $a_n \rightarrow 0$ and finite spatial boxes exhausting $\mathbb{R}^3$. The coefficient record is specified at every regulator; no assertion about its renormalized physical limit is implicit in this choice. Write $\mathcal{H}_{\text{kin},n}$ and $\mathcal{H}_{\text{phys},n}$ for the kinematical and physical Hilbert spaces. A normalized physical vector ξ_n defines a state on either the physical bounded algebra or the kinematical bounded algebra, in the latter case by

$$\omega_n(A) = \langle \xi_n, A\xi_n \rangle.$$

The kinematical extension is needed when charged CAR fields are retained: their physical compressions need not satisfy the original CAR.

For a sequence of unital represented C^* -algebras $\mathfrak{B}_n$, define

$$\begin{aligned} \mathfrak{B} &= \left\{ (A_n) : A_n \in \mathfrak{B}_n, \sup_n \|A_n\| < \infty \right\}, \\ \mathfrak{J}_0 &= \{(A_n) \in \mathfrak{B} : \|A_n\| \rightarrow 0\}, \quad \mathfrak{B}_{\text{as}} = \mathfrak{B}/\mathfrak{J}_0. \end{aligned} \tag{200}$$

The quotient class of a sequence is denoted by $[A_n]$.

Theorem 13.1 (Common bounded cluster source). *The regulated states have a nonempty set of asymptotic cluster states on $\mathfrak{B}_{\text{as}}$. For any free ultrafilter $\mathcal{U}$ on $\mathbb{N}$, one such state is*

$$\omega_{\mathcal{U}}([A_n]) = \lim_{\mathcal{U}} \omega_n(A_n). \tag{201}$$

It assigns all bounded mixed words simultaneously. If a mixed expectation has an ordinary limit, every cluster state takes that value. Exact relations and relation defects tending to zero in norm hold in the quotient.

Proof. The bounded product is a unital C^* -algebra in the supremum norm, and $\mathfrak{J}_0$ is a closed two-sided star ideal. The scalar sequence in (201) is bounded by $\sup_n \|A_n\|$. Ultralimits are linear on bounded scalar sequences and preserve real nonnegativity. They therefore give a positive normalized functional on the product. The estimate $|\omega_n(A_n)| \leq \|A_n\|$ makes it vanish on $\mathfrak{J}_0$, so it descends to the quotient.

Equivalently, let e_n be the state $e_n((A_j)) = \omega_n(A_n)$ on $\mathfrak{B}$. The weak-star compact state space contains the nonempty intersection

$$\bigcap_{N \geq 1} \overline{\{e_n : n \geq N\}}^{w*}.$$

Every element of this intersection annihilates $\mathfrak{J}_0$. This description retains the entire cluster-state set without selecting a distinguished ultrafilter. Ordinary scalar limits have the same value at all tail cluster points. A norm-vanishing relation belongs to the quotient ideal, which proves the last assertion. $\square$

No compatible embeddings of the ground-state Hilbert spaces are required. Initially, however, an observation is a bounded sequence, not an independently identified continuum field. Such identification will be supplied below for a specified canonical algebra. Cluster existence does not imply independence of the coefficient record or cluster point.

On a countably generated observation algebra, an ordinary subsequence can be used: diagonalize the bounded expectations on a countable norm-dense star-polynomial set and extend by the common state norm bound. The corresponding sequential assertion for an arbitrary nonseparable algebra is not needed here.

13.2 Continuum equal-time canonical observations

Take cubic cells of volume a_n^3 , with centers x_v . For a smooth compactly supported test function with the relevant internal indices, set

$$(J_n f)_v = a_n^{3/2} f(x_v). \quad (202)$$

Riemann sums give

$$\langle J_n f, J_n g \rangle \longrightarrow \int_{\mathbb{R}^3} \langle f(x), g(x) \rangle dx, \quad \sup_n \|J_n f\| < \infty. \quad (203)$$

For a smooth local gauge transformation k , sampled at the same centers, there is also the exact identity

$$J_n(R(k)f) = R(k_n)J_n f. \quad (204)$$

We use the observable action $\gamma_k(A) = \mathcal{U}_n(k)A\mathcal{U}_n(k)^{-1}$. Its action on an antilinearly smeared annihilation operator is the action $R(k)$ on the test function. This agrees with the inverse-conjugation forward-field convention of (191).

Theorem 13.2 (A continuum CAR source). *Let W_{int} be the internal space consisting of the three matter copies in (189), with their two spin components; $\dim_{\mathbb{C}} W_{\text{int}} = 90$. In the kinematical quotient define*

$$c(f) = [a_n(J_n f)], \quad f \in C_c^\infty(\mathbb{R}^3; W_{\text{int}}).$$

These classes satisfy

$$\{c(f), c(g)\} = 0, \quad \{c(f), c(g)^*\} = \langle f, g \rangle_{L^2} I, \quad \|c(f)\| = \|f\|_{L^2}. \quad (205)$$

They extend to the continuum CAR algebra over $L^2(\mathbb{R}^3; W_{\text{int}})$. The cluster state restricts to a positive state of this algebra, with the stated gauge covariance and gauge-invariant state. Its CAR representation is noncommutative.

Proof. Finite CAR gives $\|a_n(J_n f)\| = \|J_n f\|$ and an exact zero for the first anticommutator. The second anticommutator differs from the asserted limit by $(\langle J_n f, J_n g \rangle - \langle f, g \rangle_{L^2})I$, whose norm tends to zero. The quotient norm is the limsup of the component norms, giving the last equality in (205) and the extension from smooth tests to L^2 . Equation (204) proves covariance. Invariance of every physical vector state gives invariance of every mixed polynomial expectation, hence of the restricted state.

A normalized test generates the one-mode CAR matrix algebra. Its unital representation is nonzero and therefore faithful, so its image is not commutative. The same reasoning applies to each finite orthonormal test family. No charged operator was compressed to the physical Hilbert space in this argument. $\square$

The conclusion concerns equal-time canonical relations and the internal representation inventory. It does not identify the chiral dispersion or the physical charged sectors of a continuum dynamics.

For Higgs observations use the real phase-test space

$$\mathcal{S} = C_c^\infty(\mathbb{R}^3; \mathbb{R}^4) \oplus C_c^\infty(\mathbb{R}^3; \mathbb{R}^4), \quad \sigma((f, g), (f', g')) = \langle f, g' \rangle_{L^2} - \langle g, f' \rangle_{L^2}.$$

Let Q_n and Π_n denote the actual finite Higgs coordinate and momentum operators.

Theorem 13.3 (Joint gauge–Higgs–matter bounded source). *The unitaries*

$$W_n(v) = \exp(i\{Q_n(J_n f) + \Pi_n(J_n g)\}), \quad v = (f, g),$$

define classes $W(v) = [W_n(v)]$ satisfying

$$W(v)W(w) = e^{-i\sigma(v,w)/2}W(v+w), \quad W(v)^* = W(-v). \quad (206)$$

The same cluster state jointly evaluates these classes, the CAR classes and any fixed family of bounded gauge-loop sequences. The sampled local gauge action preserves the joint algebra and leaves its state invariant.

Proof. Every Weyl generator has norm one. Its finite product law uses the symplectic form σ_n formed with J_n . Equation (203) gives $\sigma_n(v, w) \rightarrow \sigma(v, w)$, so the scalar phase defect tends to zero in norm. This proves (206) without invoking norm continuity of the Weyl map. Bosonic and CAR operators commute at every regulator, and their joint words belong to the same quotient algebra. Wilson-loop traces divided by the representation dimension are also bounded sequences. They are evaluated by this very state, not by a separately selected gauge functional.

The Higgs representation is orthogonal on its four real components and acts canonically on position and momentum. Equation (204) gives the required exact intertwining. Gauge-loop invariance and physical-vector invariance finish the proof. $\square$

Graph paths approximating a loop are part of its source comparison; independence of different path prescriptions is not asserted. The abstract Weyl relations also do not imply regularity of the limiting state. In particular, self-adjoint smeared Higgs fields require continuity of their Weyl groups in the resulting representation.

For a bounded spatial region O , take sequences of gauge-invariant parity-even operators supported on cells and links contained in O . Their quotient algebras are isotonic. Algebras for disjoint regions commute, since their tensor factors and even CAR subalgebras commute before the limit. This proves equal-time locality. It is distinct from causal locality under a relativistic time evolution.

13.3 Continuous physical time and positive energy

Choose physical ground vectors Ω_n and put

$$L_n = H_{\text{phys},n} - E_{\text{phys},n}I \geq 0, \quad \alpha_t^n(A) = e^{itL_n}Ae^{-itL_n}. \quad (207)$$

The inequality is on the physical Hilbert space. The bottom of the Hamiltonian in every charged kinematical sector has not been identified with $E_{\text{phys},n}$.

Inside the physical bounded product define

$$\mathfrak{B}_{\text{uc}} = \left\{ (A_n) : \limsup_{t \rightarrow 0} \sup_n \|\alpha_t^n(A_n) - A_n\| = 0 \right\}. \quad (208)$$

It is a unital norm-closed star algebra, invariant under each α_t . We use its quotient by $\mathfrak{B}_{\text{uc}} \cap \mathfrak{J}_0$.

Proposition 13.4 (Uniform time smearing). *For any bounded physical sequence (A_n) and $f \in L^1(\mathbb{R})$, the weak-operator integrals*

$$B_n = \int_{\mathbb{R}} f(t) \alpha_t^n(A_n) dt$$

exist and belong to $\mathfrak{B}_{uc}$. In particular,

$$\begin{aligned} \sup_n \|B_n\| &\leq \|f\|_{L^1} \sup_n \|A_n\|, \\ \sup_n \|\alpha_s^n(B_n) - B_n\| &\leq \|f(\cdot - s) - f\|_{L^1} \sup_n \|A_n\|. \end{aligned}$$

Proof. Strong continuity of the finite unitary groups gives continuous bounded matrix elements. Integration defines a bounded sesquilinear form and thus a bounded operator with the first estimate. Changing the integration variable gives the second estimate. Translation continuity in L^1 proves the assertion. No norm continuity of the unsmeared bounded orbit is required. $\square$

Theorem 13.5 (Positive-energy physical cluster dynamics). *The ground-state cluster functional on the uniformly continuous physical quotient is stationary. Its GNS representation admits a strongly continuous implementing group $U(t)$ fixing the cyclic vector, with*

$$U(t) = e^{itL_{as}}, \quad L_{as} \geq 0.$$

Proof. Ground-state invariance passes to the cluster functional. Consequently

$$U(t)\pi(A)\Omega = \pi(\alpha_t(A))\Omega$$

defines an isometry on the cyclic domain with inverse $U(-t)$. Equation (208) proves strong continuity there and hence on its completion.

For $A \in \mathfrak{B}_{uc}$, the finite correlation

$$C_n(t) = \langle A_n \Omega_n, e^{itL_n} A_n \Omega_n \rangle$$

is the Fourier transform, with the displayed plus sign, of a finite positive measure supported in $[0, \infty)$. These functions are uniformly bounded and equicontinuous, because

$$|C_n(t) - C_n(s)| \leq \|A_n\| \|\alpha_t^n(A_n) - \alpha_s^n(A_n)\|.$$

For a Schwartz function h , integration against h commutes with the cluster limit: truncate its integrable tails uniformly, then approximate on a compact interval by Riemann sums uniformly in n . If

$$\hat{h}(E) = \int_{\mathbb{R}} h(t) e^{itE} dt, \quad \text{supp } \hat{h} \subset (-\infty, 0),$$

each integral $\int h(t) C_n(t) dt$ is zero. The same is true for the limiting correlation of $\pi(A)\Omega$ under $U(t)$. The spectral theorem therefore gives no negative spectral support on this vector. The cyclic vectors are dense, so the negative spectral projection of the generator is zero. $\square$

Ordered parity-even positive-time strings retain their reflected positivity as well. Every finite reflected matrix is the Gram matrix of transfer vectors built from the actual e^{-tL_n} and bounded physical insertions. Its quadratic forms are nonnegative and have nonnegative cluster limits. These are temporal source statements; neither four-dimensional Euclidean covariance nor a graded charged-fermion reflection prescription follows from them.

Time smearing can also enlarge spatial support. Without a uniform propagation estimate, Proposition 13.4 does not prove that every small spatial region has the required nontrivial time-continuous observable algebra. The positive energy theorem concerns physical invariant observations, not all charged kinematical CAR sectors.

13.4 Regularity and nontrivial mixed limits

Bounded-state compactness does not preserve regularity by itself. For example, regular Gaussian states of a canonical pair with position variance n satisfy

$$\omega_n(e^{itQ}) = e^{-nt^2/2}.$$

Every cluster state on their common Weyl algebra takes the value one at $t = 0$ and zero at every $t \neq 0$. This is a nonregular limit of regular states.

A uniform estimate for a specified smeared field changes the conclusion. If

$$\sup_n \|\varphi_n(v)\xi_n\|^2 \leq C_v < \infty, \quad (209)$$

functional calculus gives

$$\|(e^{is\varphi_n(v)} - I)\xi_n\|^2 \leq s^2 C_v.$$

Thus the limiting characteristic function is continuous on that Weyl subgroup. The Weyl relations extend this continuity to the cyclic Weyl-polynomial domain. Distributional fields and coincident local products require further test-seminorm and common-domain estimates. The finite graph form bound alone does not make its constants uniform on a refining interacting ground-state family.

For the explicit mixed vector of Theorem 12.5, some of these estimates can already be proved. They also permit an exact test of the local persistence of that vector's mixed occupation.

Theorem 13.6 (Regular reference-time fields and local dilution). *At each regulator take site-independent Yukawa matrices, not all zero, and the product seed $\psi_{0,n}$ of (196). Put*

$$\zeta_n = \frac{Y_n \psi_{0,n}}{\|Y_n \psi_{0,n}\|}, \quad \eta_n = \frac{\psi_{0,n} + \zeta_n}{\sqrt{2}}.$$

The states defined by η_n obey, for every fixed real phase test $v = (f, g)$,

$$\begin{aligned} \varphi_n(v) &= Q_n(J_n f) + \Pi_n(J_n g), \\ \|\varphi_n(v)\eta_n\| &\leq (1 + \sqrt{2}) \left(\frac{1}{4} \|J_n f\|^2 + \|J_n g\|^2 \right)^{1/2}. \end{aligned} \quad (210)$$

Their limiting Weyl states are therefore regular. Fixed-order linear-field moments also have bounds independent of the number of vertices.

Let S_n be a vertex region, N_n the total vertex count, and $p_n = |S_n|/N_n$. The reduced states of η_n and $\psi_{0,n}$ on its kinematical factors satisfy

$$\|\rho_{\eta,n}^{S_n} - \rho_{\psi_{0,n}}^{S_n}\|_1 = \sqrt{p_n + p_n^2}, \quad (211)$$

where $\|T\|_1 = \text{Tr}|T|$. In exhausting boxes, a fixed bounded physical region has $p_n \rightarrow 0$. Every uniformly bounded local observation there consequently has the product-seed limiting value. In particular, the equal-time Weyl and CAR restrictions are respectively

$$\omega(W(f, g)) = \exp\left[-\frac{1}{2} \left(\frac{1}{4} \|f\|_{L^2}^2 + \|g\|_{L^2}^2 \right)\right] \quad (212)$$

and the empty CAR Fock state. Initial bounded connected correlations between the seed's distinct gauge, Higgs and matter factors disappear locally.

Proof. In oscillator coordinates adapted to the Gaussian $e^{-\sum q_j^2}$, let $b_j = q_j + \frac{1}{2}\partial_{q_j}$. Then $b_j \psi_{0,n} = 0$, $q_j = (b_j + b_j^*)/2$ and $-i\partial_{q_j} = -i(b_j - b_j^*)$. Each term of $Y_n \psi_{0,n}$ contains one linear

Higgs coordinate and two created fermions. Its bosonic support is therefore in the number-one subspace, while that of η_n is in the number-zero and number-one subspaces. Write

$$\varphi_n(v) = b(u_n) + b(u_n)^*, \quad \|u_n\|^2 = \frac{1}{4} \|J_n f\|^2 + \|J_n g\|^2.$$

The annihilation and creation bounds on these subspaces give (210). Equations (203) and (209) give regularity. Applying the same bounds successively on the number-at-most- $j+1$ subspaces gives, for a fixed power m ,

$$\|\varphi_n(v)^m \eta_n\| \leq \prod_{j=1}^m (\sqrt{j} + \sqrt{j+1}) \|u_n\|^m.$$

Fixed products of different linear fields have the corresponding product bound.

For the local statement, write $Y_n = \sum_v Y_{n,v}$. The vectors $Y_{n,v} \psi_{0,n}$ occupy mutually orthogonal vertex fermion-pair sectors. Site independence makes their norms the same positive number c_n . If $\zeta_{n,v} = Y_{n,v} \psi_{0,n} / c_n$, then

$$\zeta_n = N_n^{-1/2} \sum_v \zeta_{n,v}.$$

The pairs have even fermion parity, so the region and complement can be separated without a grading sign. For $0 < p_n < 1$, denote their normalized excitation sums by ζ_S and ζ_{S^c} . Then

$$\zeta_n = \sqrt{p_n} \zeta_S \otimes 0_{S^c} + \sqrt{1-p_n} 0_S \otimes \zeta_{S^c}.$$

The link seed is unchanged. Tracing the complement gives, on the span of $0_S, \zeta_S$,

$$\rho_{\eta,n}^{S_n} - \rho_{0,n}^{S_n} = \begin{pmatrix} -p_n/2 & \sqrt{p_n}/2 \\ \sqrt{p_n}/2 & p_n/2 \end{pmatrix}.$$

Its eigenvalues are $\pm \sqrt{p_n + p_n^2}/2$, proving (211); the endpoints follow directly. The bound applies to every bounded local invariant observation with such a kinematical lift. No tensor factorization of the Gauss-invariant Hilbert space is used.

For a fixed physical region in growing boxes, its vertex fraction tends to zero. Trace-norm convergence gives the product-seed limiting values on bounded local words. The seed has Higgs variances $1/4$ and 1 in position and momentum and the empty fermionic vacuum. Riemann sums yield (212) and the stated CAR restriction. Factorization of the seed across the retained factors gives the connected-correlation conclusion. $\square$

This theorem concerns the displayed reference-time states. It does not imply that their interacting Hamiltonians have free continuum dynamics. Evolution can generate further mixed correlations, and identifying their limit requires uniform dynamical estimates. The theorem establishes both the regular equal-time data and the precise failure of this initial occupation witness to provide a nonzero local continuum interaction by itself.

A nonzero finite response likewise need not survive. For fixed bounded comparison sequences A_n, B_n , the stronger inequality

$$|\omega_n(A_n B_n) - \omega_n(A_n) \omega_n(B_n)| \geq \eta > 0 \tag{213}$$

does pass to every retained cluster state. Ultralimits preserve products of bounded scalar sequences and the inequality is closed. A strictly positive value at every regulator, with no uniform lower bound, has no such implication.

The acquired canonical algebra also does not settle the species content of its dynamics. The finite construction permits specified spin hopping matrices. The symmetric derivative choice in a free translation-invariant background,

$$D_a(k) = \sum_{j=1}^3 \sigma_j \frac{\sin(ak_j)}{a},$$

has eight zeros on the spatial Brillouin torus. Their linearization orientations are $\prod_j \cos(ak_j)$, with four signs of each kind. This is a test of an allowed kinetic choice, not a statement that every interacting regulator has that limiting spectrum. It shows why the internal CAR inventory is not itself a doubler-free chiral dynamics theorem.

Finally, equal-time commutativity and continuous time do not give the required causal propagation estimate. Nor does the common bounded state identify its gauge restriction with the inherited Yang–Mills state, or supply renormalized coincident gauge–Higgs–Yukawa operators. The bounded cluster-source set is positively occupied. Identifying this native cluster realization with the original physical target requires uniform field and propagation estimates, the appropriate chiral current identities, a surviving mixed response on the same selected source, and exact comparisons to the inherited source observations. These conditions concern the native comparison, not the completed source-first realization of Theorem 11.5.

14 Invariant response in the physical ground state

The interaction vector of Section 12 proves that the regulated physical algebra has a nonzero Yukawa response. A ground-state statement requires a different argument. We obtain local scalar moment bounds, a strictly positive invariant response, and an explicit positive-energy spectral interval in the ground state of the complete regulated gauge–Higgs–matter Hamiltonian. The estimates are uniform in spatial volume, with their coefficient dependence displayed.

14.1 The ground density and engineering normalization

Use periodic cubic boxes, with at least three vertices in each direction. Write $N = |V|$; there are $3N$ positively oriented nearest-neighbor edges and $3N$ plaquettes. A site has an integer lattice label x and physical location ℓx . The coefficient record is translation invariant. All representations, Higgs incidences and CAR contractions are those of (189) and (193).

We denote lattice spacing by ℓ_n , so that $\ell_n = a_n$ in the mesh notation of Section 13. It is distinct from the homogeneous scale factor a used in the gravitational evolution. At one mesh suppress n and write X_x, P_x for the four real Higgs coordinates and their canonical momenta. One candidate engineering normalization in three spatial dimensions is

$$X_x = \ell^{3/2} \varphi_x^{\text{lat}}, \quad P_x = \ell^{3/2} \pi_x^{\text{lat}}, \quad c_x = \ell^{3/2} \psi_x^{\text{lat}},$$

where c_x is a normalized finite CAR mode. These equalities define lattice field normalizations, not an already constructed continuum field. The unitary scalar dilation

$$X_x = \sqrt{\ell} q_x, \quad P_x = \ell^{-1/2} p_x$$

and the coefficient choices

$$\begin{aligned} \lambda_H &= \lambda \ell^{-3}, & \kappa_H &= \kappa \ell^{-2}, & a_H &= \mu \ell^{-2}, \\ y_R &= \bar{y}_R \ell^{-3/2}, & S_R &= \bar{S}_R \ell^{-1}, & \beta_G &= \beta \ell^{-1} \end{aligned} \tag{214}$$

with each electric coefficient scaled by ℓ^{-1} give exactly

$$\begin{aligned} H_\ell &\simeq \ell^{-1} \bar{H}_N(\vartheta_\ell), \\ \bar{H}_N &= T_G + T_H + \lambda \sum_x |q_x|^4 + \mu \sum_x |q_x|^2 \\ &\quad + W_G + W_H + F_{\text{hop}} + Y(q). \end{aligned} \tag{215}$$

Here $T_H = \sum_x |p_x|^2$, with the unit real-Laplacian coefficient of (192); W_G, W_H have coefficients β, κ . Every term of the original Hamiltonian is retained. The bare record ϑ_ℓ may depend on ℓ .

Let E_N be the lowest physical eigenvalue of $\bar{H}_N$, let $\mathcal{G}_N$ be its ground eigenspace, and put

$$P_{0,N} = \mathbf{1}_{\{E_N\}}(\bar{H}_N), \quad d_N = \dim \mathcal{G}_N, \quad \rho_N = d_N^{-1} P_{0,N}, \quad \bar{L}_N = \bar{H}_N - E_N I. \quad (216)$$

Theorem 12.3 gives $0 < d_N < \infty$. Translations commute with the physical Hamiltonian and therefore with $P_{0,N}$, so ρ_N is stationary and translation invariant. No choice of a translation-invariant pure ground vector is required. Write $\langle A \rangle_N = \text{Tr}(\rho_N A)$, using the associated form expectation for positive unbounded expressions.

For spectral arguments a convenient purification is

$$\mathcal{K}_N = \mathcal{H}_{\text{phys},N} \otimes \overline{\mathcal{G}_N}, \quad \Omega_N = d_N^{-1/2} \sum_{j=1}^{d_N} e_j \otimes \bar{e}_j, \quad A \mapsto A \otimes I. \quad (217)$$

It represents exactly $\text{Tr}(\rho_N A)$ and is independent of the orthonormal ground basis. In this representation $\bar{L}_N$ means $\bar{L}_N \otimes I$ and annihilates Ω_N . The auxiliary conjugate space represents a mixed state; it is not an additional physical sector. Bounded-state compactness and positive-energy cluster dynamics from Section 13 apply to these densities: their proofs use positivity and the ground spectral support, both of which hold in (217).

14.2 Local moments with explicit volume-independent constants

At a site write

$$Y_x(q_x) = \sum_{r=1}^4 q_{x,r} M_r, \quad y_* = \left(\sum_{r=1}^4 \|M_r\|^2 \right)^{1/2},$$

where the M_r are the Hermitian local Fock matrices determined by the full Yukawa contractions. Then $\|Y_x(q_x)\| \leq y_* |q_x|$. Let h_* bound the norm of the complete hopping operator on one edge, uniformly over its link and orientation. For the displayed hopping convention a computable choice is

$$h_* = 2 \max_j \sum_R 3 \dim(R) \|\bar{S}_{R,j}\|_1, \quad \|S\|_1 = \text{Tr} \sqrt{S^* S},$$

or its exact smaller edge norm. Generation and internal multiplicities are included, while the spin indices belong to $\bar{S}_{R,j}$. Consequently $F_{\text{hop}} \geq -3N h_* I$.

Put $\mu_{\pm} = \max(\pm\mu, 0)$, $d_{\sigma} = \dim \sigma$, and define

$$\begin{aligned} D &= \frac{\mu_-^2}{\lambda} + \frac{3}{4} y_*^{4/3} \lambda^{-1/3} + 3h_*, \\ U(s) &= 4s + \frac{3\lambda}{2s^2} + \frac{\mu_+ + 6\kappa}{s} + 3\beta d_{\sigma}, \quad s > 0, \\ C &= U(1) + D, \quad M_4 = \frac{2C}{\lambda}. \end{aligned} \quad (218)$$

Theorem 14.1 (Uniform physical-ground moments). *For $\lambda > 0$, $\kappa, \beta \geq 0$ and positive electric coefficients, the ground density (216) satisfies at every site*

$$\langle |q_x|^4 \rangle_N \leq M_4, \quad \langle |p_x|^2 \rangle_N \leq C, \quad \langle |q_x|^2 \rangle_N \leq \sqrt{M_4}. \quad (219)$$

The constants are independent of N . The nonnegative boundary values of κ, β are allowed for these estimates; positive values give the coupled source already specified above.

Proof. For $r \geq 0$, maximizing the respective differences gives

$$\mu_- r^2 \leq \frac{\lambda}{4} r^4 + \frac{\mu_-^2}{\lambda}, \quad y_* r \leq \frac{\lambda}{4} r^4 + \frac{3}{4} y_*^{4/3} \lambda^{-1/3}.$$

The Wilson and Higgs-gradient forms are nonnegative. Thus the full physical form obeys

$$\overline{H}_N \geq T_G + T_H + \frac{\lambda}{2} \sum_x |q_x|^4 - ND. \quad (220)$$

For the upper energy bound use the normalized physical trial vector with constant link wavefunction, empty Fock vacuum and scalar factor $\exp(-s \sum_x |q_x|^2)$. In four real dimensions direct Gaussian integration gives, per site,

$$\langle T_H \rangle_{\text{tr}}/N = 4s, \quad \langle |q_x|^2 \rangle_{\text{tr}} = s^{-1}, \quad \langle |q_x|^4 \rangle_{\text{tr}} = \frac{3}{2s^2}.$$

Each scalar edge has expectation $2\kappa/s$: the independent centered scalar factors make its cross term zero. Hopping and Yukawa expectations vanish in this trial vector, not as operators. Electric energy is zero. Every summand of the faithful σ is a nontrivial irreducible representation of the joint quotient, hence its Haar character integral vanishes. The distinct edges of a genuine plaquette make its holonomy Haar distributed in the trial measure. Its Wilson expectation is exactly βd_σ . The variational principle therefore gives $E_N/N \leq U(s)$.

Taking the expectation of (220) in ρ_N and using $s = 1$ yields

$$\left\langle T_H + \frac{\lambda}{2} \sum_x |q_x|^4 \right\rangle_N \leq NC.$$

Translation invariance gives the first two local bounds. The last follows from Cauchy–Schwarz. All expectations are finite on the physical ground form domain. $\square$

For example, on a coefficient family satisfying

$$\lambda_{\min} \leq \lambda \leq \lambda_{\max}, \quad |\mu| \leq M, \quad \kappa \leq K, \quad \beta \leq B_*, \quad y_* \leq Y_*, \quad h_* \leq H_*,$$

one may replace C by

$$\begin{aligned} C_* &= 4 + \frac{3}{2}\lambda_{\max} + M + 6K + 3B_*d_\sigma + \frac{M^2}{\lambda_{\min}} \\ &\quad + \frac{3}{4}Y_*^{4/3}\lambda_{\min}^{-1/3} + 3H_*. \end{aligned} \quad (221)$$

No upper bound on the positive electric coefficients is needed. Outside such a controlled family the dependence $C(\vartheta_\ell)$ must be retained; in particular an increasing β is not absorbed into a mesh-independent constant.

14.3 A positive invariant scalar spectral response

Fix a site and let

$$r = |q_x|^2, \quad b = \frac{r}{1+r}, \quad m_b = \frac{1}{2C(1+2\sqrt{M_4})^4}, \quad M_2 = (2\sqrt{C} + 8)^2. \quad (222)$$

Multiplication by b is bounded, parity even and gauge invariant. Its bounded first derivatives imply that b and b^2 preserve the kinetic form domain; both also preserve the quartic-weight domain. This is an admitted invariant scalar observation of the physical response sort, without using an elementary charged field [4, Sec. 12].

Theorem 14.2 (Ground response, second moment and variance). *On the purified physical ground source,*

$$\langle b\Omega_N, \bar{L}_N b\Omega_N \rangle = \left\langle \frac{4r}{(1+r)^4} \right\rangle_N \geq m_b > 0, \quad (223)$$

$$\|\bar{L}_N b\Omega_N\|^2 \leq M_2, \quad (224)$$

$$\text{Var}_{\rho_N}(b) \geq \frac{m_b^2}{M_2}. \quad (225)$$

The first equality is initially an energy-form identity. In fact $b\Omega_N$ belongs to the operator domain, so the displayed operator expressions are well defined.

Proof. The four-dimensional vector-valued Hardy inequality gives

$$\langle r^{-1} \rangle_N \leq \langle |p_x|^2 \rangle_N \leq C.$$

To see its constant directly, expand $\|\nabla u + (q/|q|^2)u\|^2 \geq 0$ and integrate the cross term using $\text{div}(q/|q|^2) = 2/|q|^2$. Cutoffs near zero and infinity, followed by form-core approximation, give the inequality on the ground form domain. All remaining coordinates, Fock indices and the finite purification space are Hilbert-valued spectators.

Markov's inequalities now give

$$\Pr(r \leq \delta) \leq \delta C, \quad \Pr(r \geq R) \leq M_4/R^2.$$

Take $\delta = (4C)^{-1}$ and $R = 2\sqrt{M_4}$. The shell $\delta < r < R$ has probability at least one half. Its endpoints are ordered because $1 \leq \langle r^{-1} \rangle_N \langle r \rangle_N \leq C\sqrt{M_4}$. On this shell, $4r/(1+r)^4 \geq 4\delta/(1+R)^4$, giving the lower bound m_b .

For an individual ground vector e_j , test its weak eigen-equation with $b^2 e_j$. Every multiplication and CAR term commutes with the scalar b , and link differentiation does not differentiate it. The scalar kinetic product rule therefore yields

$$q_{\bar{H}_N}(be_j, be_j) - E_N \|be_j\|^2 = \|(\nabla b)e_j\|^2.$$

Averaging over the full ground projection proves (223), since

$$\nabla b = \frac{2q_x}{(1+r)^2}, \quad \Delta b = \frac{8}{(1+r)^3}.$$

Both multipliers are bounded, with $\|\nabla b\|_\infty \leq 1$ and $\|\Delta b\|_\infty \leq 8$. The same weak equation gives

$$\bar{L}_N be_j = -2\nabla b \cdot \nabla e_j - (\Delta b)e_j. \quad (226)$$

Its right side lies in the Hilbert space. Form-domain preservation and the defining representation theorem for the closed form then place be_j in the operator domain, proving (226) as an operator equality.

Apply this identity in the purification. The averaged local kinetic norm, rather than that of an arbitrary vector in a degenerate ground space, is bounded by C . Hence

$$\|\bar{L}_N b\Omega_N\| \leq 2\sqrt{C} + 8,$$

which proves (224). Finally set $\eta_b = (b - \langle b \rangle_N)\Omega_N$. Then $\bar{L}_N \eta_b = \bar{L}_N b\Omega_N$ and $\|\eta_b\|^2 = \text{Var}_{\rho_N}(b)$. Thus

$$m_b^2 \leq |\langle \eta_b, \bar{L}_N \eta_b \rangle|^2 \leq \text{Var}_{\rho_N}(b) M_2.$$

This proof uses neither a positive scalar wavefunction nor uniqueness of the ground vector. $\square$

Corollary 14.3 (An explicit occupied spectral interval). *Let $\mu_{b,N}(\Delta) = \|\mathbf{1}_\Delta(\bar{L}_N)b\Omega_N\|^2$. Then*

$$\mu_{b,N} \left(\left[\frac{m_b}{4}, \frac{4M_2}{m_b} \right] \right) \geq \frac{m_b^2}{8M_2}. \quad (227)$$

Proof. The measure has total mass at most one. Below $m_b/4$ its first moment is at most $m_b/4$. Above $4M_2/m_b$ its first moment is at most $m_b/4$, by the second-moment bound. The first moment inside the displayed interval is therefore at least $m_b/2$. Dividing by the upper endpoint proves (227). Possible zero-energy weight does not affect this argument. $\square$

Proposition 14.4 (A physical finite-volume scalar pole). *At each regulator the same physical source has a scalar resolvent channel with a nonzero pole at an isolated positive excitation energy.*

Proof. Let $\hat{P}_0 = P_{0,N} \otimes I$ in the purification and put $\zeta_b = (1 - \hat{P}_0)b\Omega_N$. It is nonzero by (223). The finite multiplicity of the purification and compact resolvent give discrete positive energies λ_j with no finite accumulation. Their weights $w_j = \|\mathbf{1}_{\{\lambda_j\}}(\bar{L}_N)\zeta_b\|^2$ have finite positive sum. Consequently

$$G_{b,N}(z) = \langle \zeta_b, (\bar{L}_N - z)^{-1} \zeta_b \rangle = \sum_j \frac{w_j}{\lambda_j - z} \quad (228)$$

is locally normally convergent off the discrete spectrum. At any λ_j with $w_j > 0$ it has residue $-w_j \neq 0$. All vectors and operators are representations of this same ground source. The full zero-energy projection, not merely the chosen cyclic vector, has been removed. $\square$

The pole is a finite-volume invariant scalar excitation. Its identification with the physical Higgs would require the corresponding current, normalization and continuation comparison; a real discrete excitation is not an unstable second-sheet resonance [20, Secs. 2.2 and 9][4, Secs. 6 and 12].

14.4 Physical energy and scalar comparison

Ground-energy subtraction in (215) gives $L_\ell = \ell^{-1}\bar{L}_N$. The spectral measures of the same observation therefore obey the exact pushforward law

$$\mu_{b,\ell}^{\text{phys}}([\varepsilon, E]) = \mu_{b,N}([\ell\varepsilon, \ell E]). \quad (229)$$

The interval guaranteed by (227) becomes $[m_b/(4\ell), 4M_2/(m_b\ell)]$ in physical energy. For uniformly controlled coefficients it moves to the ultraviolet, not to a fixed physical mass.

For a Schwartz time test h , define

$$v_\ell(h) = \int_{\mathbb{R}} h(t) e^{itL_\ell} (b - \langle b \rangle_N) \Omega_N dt, \quad \hat{h}(E) = \int_{\mathbb{R}} h(t) e^{itE} dt.$$

The spectral theorem gives

$$\|v_\ell(h)\|^2 = \int_{[0,\infty)} |\hat{h}(\bar{E}/\ell)|^2 d\mu_{b,N}^{\text{cen}}(\bar{E}), \quad (230)$$

where the centered measure may still have zero-energy weight from ground degeneracy. A filter supported in a fixed positive compact frequency interval probes dimensionless energies of order ℓ . This is a different estimate from (227). For example, rescaling time on one fixed finite graph leaves every ground bound above unchanged, but its first positive excitation energy becomes arbitrarily large. Such a fixed-frequency filter eventually sees zero excitation. This tests the strength of the estimates; it is not a refining-box construction.

A physical scalar comparison can use an admitted renormalized invariant composite or current, with its spatial and temporal test prescriptions. The bare normalization alone gives $|\varphi_x^{\text{lat}}|^2 = \ell^{-2}|q_x|^2$; centering, mixing with the identity and the jointly transported source normalization must be specified before its physical interpretation. A sufficient nonzero finite-energy comparison for an actual preparation $B_\ell(f)$ is

$$\liminf_{\ell \rightarrow 0} \left\| \mathbf{1}_{[\varepsilon, E]}(L_\ell) B_\ell(f) \Omega_N \right\|^2 > 0, \quad 0 < \varepsilon < E < \infty, \quad (231)$$

or an equivalent nonzero time-smeared Gram entry. Its sibling words, current ports and physical clock must share the same comparison. Theorems 14.1–14.3 do not imply (231). This is a survival condition for the refining native family, not an identification of its finite scalar pole with the original physical Higgs germ.

14.5 The ground-state Yukawa sign and accessibility

Scale all three Yukawa matrices by one real g , holding the gauge, Higgs and fermion hopping inputs fixed, and write

$$\overline{H}_N(g) = \overline{H}_N(0) + gY_0.$$

This is a fixed-sibling comparison of the actual full regulator. Its common form domain is independent of g .

Proposition 14.5 (Exact ground interaction sign). *The physical ground energy $E_N(g)$ is even and concave. For every physical ground density at g ,*

$$g\langle Y_0 \rangle_g \leq E_N(g) - E_N(0) \leq 0. \quad (232)$$

Proof. The physical unitary $e^{i\pi N_F/2}$ fixes every bosonic and number-preserving hopping term. Each Yukawa term contains two annihilators or two creators, so the same unitary reverses Y_0 . It follows that $E_N(g) = E_N(-g)$. The variational infimum of affine functions of g is concave and finite by quartic coercivity. Evenness and concavity imply $E_N(g) \leq E_N(0)$. Evaluating $\overline{H}_N(0)$ in a ground density at g gives $E_N(0) \leq E_N(g) - g\langle Y_0 \rangle_g$. $\square$

Proposition 14.6 (A sufficient ground-accessibility criterion). *Let e_0 be a normalized physical ground vector of $\overline{H}_N(0)$ with definite total fermion number. If $Y_0 e_0 \neq 0$, then $E_N(g) < E_N(0)$ for every $g \neq 0$.*

Proof. Number conservation at $g = 0$ permits the stated ground choice and gives $\langle e_0, Y_0 e_0 \rangle = 0$. Quartic form control implies $Y_0 e_0 \in \mathcal{H}_{\text{phys}}$. Put

$$v = (\overline{H}_N(0) - E_N(0) + 1)^{-1} Y_0 e_0.$$

The resolvent is strictly positive, so $v \neq 0$ and $\langle Y_0 e_0, v \rangle > 0$. It commutes with fermion number; the number changes in $Y_0 e_0$ therefore make $v \perp e_0$. Moreover v belongs to the operator domain. The Rayleigh numerator of $e_0 - t \operatorname{sgn}(g)v$, relative to $E_N(0)$, is

$$-2t|g| \langle Y_0 e_0, v \rangle + t^2 \left(q_{\overline{H}_N(0)}(v, v) - E_N(0) \|v\|^2 + g \langle v, Y_0 v \rangle \right).$$

The second coefficient is finite by the form bound. For sufficiently small $t > 0$ the quotient is strictly below $E_N(0)$. $\square$

Ground accessibility cannot be replaced by the statement that Y_0 is a nonzero physical operator. At every site fill all Q and L modes and leave u^c, d^c, e^c empty. Each annihilating Yukawa term is killed by an empty singlet, and each adjoint by a filled doublet. The three generations and

two spins give 36 filled Q modes of integer charge 1 and 12 filled L modes of charge -3 . Their Abelian weights cancel, while their non-Abelian determinants are one. This is a gauge-singlet Fock vector. Tensoring it with an invariant scalar/link core function gives a physical vector annihilated by Y_0 , without asserting that it is the ground state.

An extensive estimate $E_N(g) - E_N(0) \leq -cNg^2$ would additionally give $-g\langle Y_0 \rangle_g / N \geq cg^2$ by (232). The finite accessibility criterion does not supply its uniform susceptibility and remainder bounds. The invariant scalar spectral response and the strictly Yukawa-dependent mixed response are thus distinct questions.

14.6 Undressed fields and invariant comparisons

The independent weak-center transformation at a site reverses q_x and fixes q_y for $y \neq x$. Gauge invariance of the physical ground density consequently gives $\langle q_{x,r} q_{y,s} \rangle_N = 0$ for $x \neq y$. For a real smooth compactly supported test f , its undressed engineering position smearing is

$$\Phi_\ell(f) = \sum_x \ell^{3/2} f(\ell x) \cdot X_x = \sum_x \ell^2 f(\ell x) \cdot q_x.$$

The diagonal moment bound therefore yields

$$\langle \Phi_\ell(f)^2 \rangle_N \leq \ell \sqrt{M_4(\vartheta_\ell)} \left(\ell^3 \sum_x |f(\ell x)|^2 \right). \quad (233)$$

When $\ell \sqrt{M_4(\vartheta_\ell)} \rightarrow 0$, these raw position smearings vanish in ground-state norm and their Weyl expectations tend to one. If the matched nondegenerate momentum Weyl extension is retained on the kinematical algebra, its limiting state is then nonregular in that direction: every position Weyl operator fixes the cyclic vector, while conjugation by it multiplies a nontrivial momentum Weyl expectation by a freely variable nontrivial phase. That expectation must be zero away from the origin.

These undressed coordinates do not preserve the Gauss-invariant Hilbert space. Equation (233) consequently does not exclude the invariant scalar response proved above, a neutral current channel, or a properly attached charged sector. The physical comparisons use their actual invariant or dressed observations [20, Sec. 9][3, Sec. 8].

14.7 Fixed physical frequencies of a bounded scalar average

An invariant observable permits a stronger, source-native frequency test than the undressed calculation. Let $f \in C_c^\infty(\mathbb{R}^3; \mathbb{R})$ and define on the same ground family

$$F_\ell(f) = \ell^3 \sum_x f(\ell x) b_x, \quad S_{p,\ell}(f) = \ell^3 \sum_x |f(\ell x)|^p \quad (p = 1, 2). \quad (234)$$

Fixed supports eventually lie inside the exhausting boxes without crossing the periodic seams. Riemann sums give $S_{p,\ell}(f) \rightarrow \|f\|_{L^p}^p$. The average is a self-adjoint invariant bounded multiplier with $\|F_\ell(f)\| \leq S_{1,\ell}(f)$. In this subsection Ω_ℓ denotes the purified full ground density at that mesh and ω_ℓ its state. No uniform coefficient bound is imposed.

Theorem 14.7 (Physical scalar sum rule). *For the full interacting physical generator $L_\ell = \ell^{-1} \bar{L}_N$,*

$$\langle F_\ell \Omega_\ell, L_\ell F_\ell \Omega_\ell \rangle = \ell^5 \sum_x |f(\ell x)|^2 \omega_\ell \left(\frac{4r_x}{(1+r_x)^4} \right) \quad (235)$$

$$\leq \frac{27}{64} \ell^2 S_{2,\ell}(f). \quad (236)$$

Consequently, for every $\varepsilon > 0$,

$$\left\| \mathbf{1}_{[\varepsilon, \infty)}(L_\ell) F_\ell \Omega_\ell \right\|^2 \leq \frac{27}{64\varepsilon} \ell^2 S_{2,\ell}(f). \quad (237)$$

Proof. The site scalar derivatives are independent coordinates, so

$$|\nabla F_\ell|^2 = \ell^6 \sum_x |f(\ell x)|^2 |\nabla b_x|^2.$$

At every finite graph F_ℓ has bounded first and second derivatives and preserves the form domain. All potential and CAR terms commute with it, and the electric derivatives act only on links. Testing the ground weak equation with F_ℓ^2 times each ground vector gives the same scalar kinetic form identity as in Theorem 14.2. Averaging those identities in the full ground density and multiplying by ℓ^{-1} proves (235). The maximum of $4r/(1+r)^4$ is $27/64$, attained at $r = 1/3$, proving the bound. Finally use $\mathbf{1}_{[\varepsilon, \infty)}(L_\ell) \leq L_\ell/\varepsilon$ in quadratic-form order. $\square$

On the controlled coefficient family, the Hardy-shell lower bound also gives $m_b \ell^2 S_{2,\ell}(f)$ for the first moment. Thus this response can be strictly positive at finite resolution while its fixed positive-frequency weight tends to zero. The upper bound needs neither that lower estimate nor a ground-gap assumption.

For a Schwartz time test with $\text{supp } \hat{h} \subset [\varepsilon, E]$, set

$$A_\ell(h, f) = \int_{\mathbb{R}} h(t) \alpha_t^\ell(F_\ell(f)) dt.$$

This actual bounded sequence belongs to the uniformly time-continuous physical algebra, since

$$\|\alpha_s^\ell(A_\ell) - A_\ell\| \leq S_{1,\ell}(f) \|h(\cdot - s) - h\|_{L^1}.$$

Stationarity of the ground source gives $A_\ell \Omega_\ell = \hat{h}(L_\ell) F_\ell \Omega_\ell$. If D_ℓ is any admitted mixed physical word or bounded current preparation with $\sup_\ell \|D_\ell\| \leq M_D$, then

$$|\langle D_\ell \Omega_\ell, \mathbf{1}_{[\varepsilon, E]}(L_\ell) F_\ell \Omega_\ell \rangle| \leq M_D \ell \sqrt{\frac{27 S_{2,\ell}(f)}{64 \varepsilon}}, \quad (238)$$

$$|\langle D_\ell \Omega_\ell, A_\ell(h, f) \Omega_\ell \rangle| \leq M_D \|\hat{h}\|_\infty \ell \sqrt{\frac{27 S_{2,\ell}(f)}{64 \varepsilon}}. \quad (239)$$

These follow directly from the spectral theorem and Cauchy–Schwarz. They compare a scalar-created ground response with arbitrary uniformly bounded sibling preparations on the same source. They do not apply to $F_\ell D_\ell \Omega_\ell$ as though $D_\ell \Omega_\ell$ were a ground vector.

Proposition 14.8 (Bounded scalar unitaries and normalization). *For real τ , let $U_\ell(\tau, f) = e^{i\tau F_\ell(f)}$. Then*

$$\langle U_\ell \Omega_\ell, L_\ell U_\ell \Omega_\ell \rangle = \tau^2 \langle F_\ell \Omega_\ell, L_\ell F_\ell \Omega_\ell \rangle. \quad (240)$$

In particular the positive-frequency norm bound (237) holds for $U_\ell \Omega_\ell$ with its right side multiplied by T^2 , uniformly for $|\tau| \leq T$. For the centered average $F_\ell^c = F_\ell - \omega_\ell(F_\ell)I$, every normalization $|Z_\ell| = o(\ell^{-1})$ satisfies

$$\|\mathbf{1}_{[\varepsilon, E]}(L_\ell) Z_\ell F_\ell^c \Omega_\ell\| \longrightarrow 0.$$

Proof. For a normalized ground vector u , conjugating the scalar kinetic form by $e^{i\tau F_\ell}$ gives

$$q_{\overline{H}_N}(e^{i\tau F_\ell} u, e^{i\tau F_\ell} u) - E_N = \tau J(u) + \tau^2 \|(\nabla F_\ell)u\|^2.$$

All other full Hamiltonian terms commute with the scalar unitary. The real coefficient $J(u)$ is initially retained; a fermionic ground wavefunction need not be real. The ground variational principle makes this polynomial nonnegative for both signs of τ , so $J(u) = 0$. Average over the ground density and multiply by ℓ^{-1} to obtain (240). The spectral projection estimate then proves the unitary bound. Finally centering changes no positive-energy projection, and (237) bounds the squared normalized response by $|Z_\ell|^2 \ell^2 27 S_{2,\ell}(f)/(64 \varepsilon)$, which tends to zero at the stated rate. $\square$

Thus this particular unrenormalized bounded scalar average is not a nonzero positive-frequency interpolating preparation, even when used through bounded unitary functions. The conclusion leaves zero-energy fluctuations and other physical source words intact. A normalization of order ℓ^{-1} or larger only escapes this upper-bound obstruction; it does not prove survival or define a uniformly bounded source element. An invariant renormalized spacetime composite or current may use a different, independently justified preparation. Its existence and common-source comparison are not excluded by this result.

15 Mass-coefficient insertions and static scalar response

A physical composite can be specified by its response to an actual source, rather than by assigning a particle name to a bounded observation. On the regulated source we apply this principle to the quadratic Higgs coefficient of the full Hamiltonian (215). The resulting invariant scalar insertion has a nonvanishing static susceptibility. Its equal-time excitation norm nevertheless diverges under spatial refinement on the controlled coefficient family. These two statements concern different moments of the same positive spectral measure.

15.1 The mass-source derivative

Retain the periodic cubic source, its complete interaction record, and the physical clock of Section 14. Put $r_x = |q_x|^2$. In the engineering normalization already fixed there, $\varphi_x^{\text{lat}} = \ell^{-1}q_x$. For a real $f \in C_c^\infty(\mathbb{R}^3)$, vary the Higgs-potential mass-squared coefficient by $tf(\ell x)$, with every other input held fixed. The dimensionless quadratic coefficient becomes $\mu_x(t) = \mu + t\ell^2 f(\ell x)$, and hence

$$\begin{aligned} H_\ell(t) &= H_\ell + tC_\ell(f), \\ C_\ell(f) &= \ell \sum_x f(\ell x) r_x = \ell^3 \sum_x f(\ell x) (\varphi_x^{\text{lat}})^* \varphi_x^{\text{lat}}. \end{aligned} \quad (241)$$

This is a literal coefficient derivative. It does not identify a bare quadratic coordinate with a pole mass or change the received Higgs carrier [28, 20]. Multiplication by $C_\ell(f)$ is gauge invariant and self-adjoint on its maximal multiplication domain. At each finite graph the positive quartic term absorbs every bounded quadratic coefficient variation, so (241) has one common semibounded form domain.

The normalized full ground density and its purification remain (216)–(217). On the unpurified physical space set

$$E_\ell = \ell^{-1}E_N, \quad L_\ell = H_\ell - E_\ell I = \ell^{-1}\bar{L}_N, \quad P_\ell = P_{0,N}, \quad Q_\ell = I - P_\ell.$$

In the purified space use $\hat{P}_\ell = P_\ell \otimes I$ and $\hat{Q}_\ell = Q_\ell \otimes I$; operators in scalar expectations act on the first factor. Thus the full zero-energy space, including ground degeneracy, is retained explicitly. Write Ω_N for the purified ground vector.

For boxes exhausting space, fixed supports eventually avoid their periodic seams. Define

$$S_{2,\ell}(f) = \ell^3 \sum_x |f(\ell x)|^2 \longrightarrow \|f\|_{L^2(\mathbb{R}^3)}^2. \quad (242)$$

The constants C, M_4, y_* are those of (218). In addition to (219), vector-valued Hardy in four real dimensions and Cauchy–Schwarz give

$$\langle r_x^{-1} \rangle_N \leq C, \quad \langle r_x \rangle_N \geq C^{-1}, \quad \langle \sqrt{r_x} \rangle_N \leq M_4^{1/4}. \quad (243)$$

Indeed $1 \leq \langle r_x \rangle_N \langle r_x^{-1} \rangle_N$. All local bounds refer to the translation-invariant ground density, not to an arbitrarily chosen vector in a degenerate ground space.

15.2 Polynomial insertions and scalar dilation

For real site weights w_x , put

$$F_w = \sum_x w_x r_x, \quad D_w = \frac{1}{2} \sum_x w_x (q_x \cdot p_x + p_x \cdot q_x). \quad (244)$$

The second operator generates independent unitary dilations of the four real scalar coordinates at each site. Such dilations commute with the orthogonal real form of the local Higgs gauge action. They act within the physical Hilbert space and do not rescale physical time.

The required unbounded-operator statements follow from the finite source, as we now verify. After a sufficiently large fixed-graph scalar shift, write $\bar{H}_N = T + W$, where T is the positive sum of the complete site and link Laplacians and W is Hermitian matrix multiplication. The quartic lower bound permits $W \geq 1 + c \sum_x r_x^2$. Its second coordinate derivatives have growth at most $C_0(1 + \sum_x r_x)$; compact-link derivatives of the scalar-edge terms have the same bound. Integration by parts on the invariant smooth/Schwartz core gives

$$2 \operatorname{Re} \langle Tu, Wu \rangle = 2 \sum_j a_j \langle \partial_j u, W \partial_j u \rangle - \left\langle u, \left(\sum_j a_j \partial_j^2 W \right) u \right\rangle,$$

with invariant vector fields for the compact-link factors. The first sum is nonnegative. The last term is bounded by $\varepsilon \|Wu\|^2 + C_\varepsilon \|u\|^2$. Expansion of $\|(T + W)u\|^2$ therefore yields

$$\|Tu\| + \|Wu\| \leq C_{\text{gr}} (\|\bar{H}_N u\| + \|u\|). \quad (245)$$

The constant is finite at each fixed graph; no volume uniformity is claimed for it. Complete kinetic summands on distinct factors commute and are nonnegative, so each is controlled by T .

Lemma 15.1 (The physical operator domain). *The self-adjoint form realization of the shifted finite Hamiltonian has domain $\operatorname{Dom} T \cap \operatorname{Dom} W$. The invariant smooth/Schwartz domain is an operator core, including after restriction to the Gauss-invariant physical space.*

Proof. The product Laplacian has its second-order Sobolev domain. The bounds $1 + c \sum_x r_x^2 \leq W \leq C_1(1 + \sum_x r_x^2)I$ identify the multiplication domain with the corresponding quartic weighted L^2 space. Smooth scalar cutoffs ζ_R , equal to one on $|q| \leq R$ and zero outside $|q| \leq 2R$, approximate a vector in the intersection in both graph norms. The kinetic commutator contains only first and second cutoff derivatives, bounded by $O(R^{-1})$ and $O(R^{-2})$, against its first derivatives and itself. Coordinate smoothing with a partition of unity on the compact-link manifold then approximates in H^2 ; on the fixed enlarged scalar support, W is bounded. Thus the smooth/Schwartz domain is dense in the intersection in the sum graph norm. Equation (245) extends there and makes $T + W$ closed.

It remains to identify this closed operator with the form realization H_F . If $u \in \operatorname{Dom} H_F$ and $g = H_F u$, the weak equation is $Tu = g - Wu$. Since W is locally bounded, local elliptic regularity gives $u \in H_{\text{loc}}^2$. Consequently $\zeta_R u$ belongs to the intersection and

$$(T + W)\zeta_R u = \zeta_R g + [T, \zeta_R]u, \quad \|[T, \zeta_R]u\| \leq C_2(R^{-1} \|\nabla_q u\| + R^{-2} \|u\|).$$

The derivative on the right is globally square integrable because u is in the form domain. The graph estimate is therefore uniform in R . Fatou gives $Wu \in L^2$, and the weak equation then gives $Tu \in L^2$. The maximal-domain characterization of the self-adjoint product Laplacian implies $u \in \operatorname{Dom} T$. Conversely, for a vector in the intersection, integration by parts and form-core approximation give the defining form identity for H_F with value $(T + W)u$. This proves domain equality and the operator-core assertion. Haar averaging commutes with T , W and the form realization and preserves the test core, so the same approximations prove the physical-space statement. $\square$

In four real scalar dimensions another integration by parts gives

$$\int r_x |\nabla_x u|^2 = \operatorname{Re} \langle r_x u, -\Delta_x u \rangle + 4 \|u\|^2. \quad (246)$$

Equations (245)–(246) control $q_x \cdot p_x$, hence D_w , by the Hamiltonian graph norm. Completion of the operator core puts every ground vector in $\operatorname{Dom} D_w$. The fourth scalar moments also put $F_w \Omega_N$ in the Hilbert space.

On the common core the exact identity is

$$[\bar{H}_N, F_w] = -4iD_w. \quad (247)$$

Every nonscalar-kinetic term commutes with F_w : the Yukawa and scalar-edge terms are coordinate multipliers, and neither the electric terms nor CAR hopping differentiates q . Testing against the operator core gives the distributional equation $\bar{H}_N F_w \Omega_N = E_N F_w \Omega_N - 4iD_w \Omega_N$. Its right side is a Hilbert vector. Self-adjointness of the core closure therefore proves

$$F_w \Omega_N \in \operatorname{Dom} \bar{L}_N, \quad \bar{L}_N F_w \Omega_N = -4iD_w \Omega_N. \quad (248)$$

Theorem 15.2 (First and third scalar spectral moments). *Define*

$$K = 4C + 16\lambda M_4 + (4|\mu| + 48\kappa)\sqrt{M_4} + y_* M_4^{1/4}. \quad (249)$$

For every real finite site-weight family,

$$\langle F_w \Omega_N, \bar{L}_N F_w \Omega_N \rangle = 4 \sum_x w_x^2 \langle r_x \rangle_N, \quad (250)$$

$$\begin{aligned} F_w \Omega_N &\in \operatorname{Dom} \bar{L}_N^{3/2}, \\ \|\bar{L}_N^{3/2} F_w \Omega_N\|^2 &\leq 8K \sum_x w_x^2. \end{aligned} \quad (251)$$

Proof. Insert (248) in the first scalar product and integrate by parts in the first-order expression. The real part of the weighted derivative gives $4 \sum_x w_x^2 \langle r_x \rangle_N$; the imaginary part vanishes by self-adjointness. The preceding graph bounds and polynomial moments justify compact cutoffs and their removal. In particular, the proof does not require the unverified test vector $F_w^2 \Omega_N$ in the form domain.

Let $U_t = \exp(itD_w)$. Its coordinate action gives

$$U_t^* q_x U_t = e^{-tw_x} q_x, \quad U_t^* p_x U_t = e^{tw_x} p_x.$$

It preserves the physical form domain. The full energy expectation $\mathcal{V}_w(t) = q_{\bar{H}_N}(U_t \Omega_N, U_t \Omega_N)$ is a finite sum of exponentials in t , with finite coefficients given by the kinetic, quartic, quadratic and Yukawa moments. The ground variational principle gives $\mathcal{V}_w(t) \geq E_N = \mathcal{V}_w(0)$, hence $\mathcal{V}_w'(0) = 0$. If $\mathcal{D}_w = \sum_x w_x q_x \cdot \partial_{q_x}$ acts on coordinate multipliers and V is the full multiplication part,

$$\mathcal{V}_w''(0) = 4 \sum_x w_x^2 \langle |p_x|^2 \rangle_N + \langle \mathcal{D}_w^2 V \rangle_N.$$

The scalar quartic, quadratic and linear Yukawa contributions are $16\lambda w_x^2 r_x^2$, $4\mu w_x^2 r_x$, and $w_x^2 Y_x$. For an edge, write $c_{xy} = q_x \cdot \rho_H(U_{xy}) q_y$ in real coordinates. Its exact contribution is

$$\kappa(4w_x^2 r_x + 4w_y^2 r_y - 2(w_x + w_y)^2 c_{xy}).$$

Using $|c_{xy}| \leq (r_x + r_y)/2$ and $(w_x + w_y)^2 \leq 2(w_x^2 + w_y^2)$ bounds its absolute expectation by $8\kappa\sqrt{M_4}(w_x^2 + w_y^2)$. Six edge incidences per site give the coefficient $48\kappa\sqrt{M_4}$ in (249). The other local moment bounds now imply

$$0 \leq \mathcal{V}_w''(0) \leq K \sum_x w_x^2. \quad (252)$$

Gauge electric, Wilson and fermion-hopping terms are unchanged by the scalar test unitary. Their explicit derivatives vanish; their contributions to the full ground state have not been removed.

Finally, because $\bar{L}_N \Omega_N = 0$,

$$\left\| \bar{L}_N^{1/2} \frac{(U_t - I)\Omega_N}{t} \right\|^2 = \frac{\mathcal{V}_w(t) - E_N}{t^2} \rightarrow \frac{1}{2} \mathcal{V}_w''(0).$$

The Hilbert difference quotient converges to $iD_w \Omega_N$. Lower semicontinuity of the closed positive form proves that this vector belongs to its form domain and has squared form norm at most $\mathcal{V}_w''(0)/2$. Equation (248) then gives (251). No twice-differentiable orbit in form norm or formal triple commutator has been assumed. $\square$

15.3 Equal-time excitation and physical scaling

Take $w_x = \ell f(\ell x)$, so $F_w = C_\ell(f)$ and $\sum_x w_x^2 = \ell^{-1} S_{2,\ell}(f)$. The physical moments are

$$\begin{aligned} M_{1,\ell} &:= \langle C_\ell(f) \Omega_N, L_\ell C_\ell(f) \Omega_N \rangle \geq \frac{4}{C} \ell^{-2} S_{2,\ell}(f), \\ M_{3,\ell} &:= \left\| L_\ell^{3/2} C_\ell(f) \Omega_N \right\|^2 \leq 8K \ell^{-4} S_{2,\ell}(f). \end{aligned} \tag{253}$$

The corresponding first-moment upper bound is $M_{1,\ell} \leq 4\sqrt{M_4} \ell^{-2} S_{2,\ell}(f)$.

Theorem 15.3 (Equal-time positive-excitation lower bound). *With $\Psi_\ell(f) = \hat{Q}_\ell C_\ell(f) \Omega_N$,*

$$\|\Psi_\ell(f)\|^2 \geq \frac{2\sqrt{2}}{C^{3/2}\sqrt{K}} \ell^{-1} S_{2,\ell}(f). \tag{254}$$

The same lower bound holds for $\|(C_\ell(f) - c_\ell I)\Omega_N\|^2$ for every scalar c_ℓ . For nonzero f , the positive-excitation norm diverges on every refining coefficient family with C, K uniformly bounded.

Proof. If all sampled weights vanish, the assertion is immediate. Otherwise removing the entire zero-energy space leaves both positive moments unchanged. For the positive spectral measure $\nu_\ell(\Delta) = \|\mathbf{1}_\Delta(L_\ell)\Psi_\ell(f)\|^2$, Hölder gives

$$M_{1,\ell} \leq \nu_\ell((0, \infty))^{2/3} M_{3,\ell}^{1/3}.$$

Thus $\|\Psi_\ell(f)\|^2 \geq M_{1,\ell}^{3/2} / \sqrt{M_{3,\ell}}$. Substitution of (253) proves the constant in (254). The orthogonal projection $\hat{Q}_\ell$ annihilates every scalar multiple of Ω_N and cannot increase norm. Equation (242) proves the refinement assertion. $\square$

Consequently an equal-time family obtained by multiplication by a scalar Z_ℓ can have bounded source-vector norms only if $|Z_\ell| = O(\sqrt{\ell})$ on this controlled branch. Fixed nonzero normalizations, and logarithmic factors with $|Z_\ell|^2/\ell \rightarrow \infty$, fail this necessary test. A factor of order $\sqrt{\ell}$ is not thereby sufficient or a physically justified renormalization. The bound concerns the total positive spectral norm; it does not locate that norm in any fixed energy interval.

15.4 A nonvanishing static susceptibility

At a fixed finite source the excited spectrum is separated from zero. On the unpurified space let

$$R_\ell = Q_\ell(L_\ell|_{\text{Ran } Q_\ell})^{-1}Q_\ell, \quad \hat{R}_\ell = R_\ell \otimes I.$$

The inverse is restricted to the indicated positive subspace. It is bounded at this regulator; a uniform spectral gap is not needed. Define the actual reduced-resolvent response

$$M_{-1,\ell}(f) = \left\langle C_\ell(f)\Omega_N, \widehat{R}_\ell C_\ell(f)\Omega_N \right\rangle = \int_{(0,\infty)} \frac{d\nu_\ell(\omega)}{\omega}. \quad (255)$$

Theorem 15.4 (Static response of the complete ground cluster). *The source (241) satisfies*

$$M_{-1,\ell}(f) \geq \frac{2}{C^2 K} S_{2,\ell}(f). \quad (256)$$

Let $P_\ell(t)$ be the analytic continuation of the isolated full ground cluster on the unpurified physical Hilbert space, and put

$$\mathfrak{e}_\ell(t) = d_N^{-1} \text{Tr}(H_\ell(t)P_\ell(t)).$$

Then

$$\begin{aligned} \mathfrak{e}'_\ell(0) &= \langle C_\ell(f) \rangle_N, \\ \chi_\ell(f) &:= -\mathfrak{e}''_\ell(0) = 2M_{-1,\ell}(f) \geq \frac{4}{C^2 K} S_{2,\ell}(f). \end{aligned} \quad (257)$$

In particular the *liminf* of this static response is positive for nonzero f on a uniformly controlled coefficient branch.

Proof. The lower bound is trivial when every sampled weight is zero. For a nonzero weight family, Cauchy–Schwarz in the positive spectral measure, with factors $\omega^{-1/2}$ and $\omega^{3/2}$, gives $M_{1,\ell}^2 \leq M_{-1,\ell} M_{3,\ell}$. Equation (253) proves (256); the mesh powers cancel.

Quadratic multiplication is relatively H_ℓ -bounded with arbitrarily small relative bound, by (245) and the quadratic/quartic Young inequality. The affine family is therefore analytic on its common operator domain. A resolvent contour around the isolated ground eigenvalue supplies $P_\ell(t)$ of constant finite rank for small t . The neighborhood may depend on the regulator. Its cluster mean is analytic even if the individual ground levels split. Differentiating the contour at zero gives

$$P'_\ell(0) = -R_\ell C_\ell(f)P_\ell - P_\ell C_\ell(f)R_\ell.$$

The products are well defined: $C_\ell(f)P_\ell$ is bounded and finite rank, and the other term is its appropriate adjoint composition. Since $P'_\ell(t)$ is off diagonal relative to $P_\ell(t)$ and $H_\ell(t)$ commutes with that projection, $\text{Tr}(H_\ell(t)P'_\ell(t)) = 0$. Consequently

$$\mathfrak{e}''_\ell(0) = -\frac{2}{d_N} \text{Tr}(P_\ell C_\ell(f)R_\ell C_\ell(f)P_\ell).$$

By (217) this is exactly $-2M_{-1,\ell}(f)$. All traces in this calculation are on the unpurified space; the tensor identities defining $\widehat{P}_\ell, \widehat{R}_\ell$ identify them with the ground-density expectations. $\square$

The theorem concerns the mean of the continued rank- d_N cluster, not a twice-differentiable choice of the lowest split level. At each finite source the same response has the positive-time form

$$M_{-1,\ell}(f) = \int_0^\infty \left\langle \Psi_\ell(f), e^{-tL_\ell} \Psi_\ell(f) \right\rangle dt.$$

This follows directly from the spectral theorem on the fixed physical clock. The lower bound neither supplies a continuum upper bound nor permits interchange of refinement with this infinite-time integral. For example, spectral mass of order ℓ^{-1} at energy of order ℓ^{-1} has an order-one inverse first moment while leaving every fixed finite energy window. The present estimates do not decide its location.

15.5 Source normalization and the spacetime comparison

The susceptibility above belongs to the affine coefficient protocol (241). If a source prescription instead has the local expansion

$$H_\ell(t) = H_\ell + tC_\ell(f) + \frac{1}{2}t^2J_\ell + o(t^2)$$

on the same controlled domain, its cluster second derivative is $\langle J_\ell \rangle_N - 2M_{-1,\ell}(f)$. Thus a quadratic identity counterterm changes that response by its scalar second derivative. Linear centering leaves the reduced-resolvent fluctuation unchanged, but nonlinear source changes require their own derivative comparison. These contact contributions cannot be discarded when the full source is compared.

The equal-time obstruction does not exclude a renormalized four-dimensionally smeared invariant composite. The local Yang–Mills construction provides a concrete precedent: its curvature insertions use spacetime tests, source-owned graded mixing, and summable insertion estimates on common word domains [31, Props. F.235 and F.245A, Thm. F.246, Lem. F.247]. Those estimates belong to its actual gauge source. Applying that method to the mixed Higgs–matter source requires the corresponding mixed insertions and bounds, rather than an identification of the two states by their common gauge labels.

Once such actual source-word kernels are available, the invariant-composite reconstruction of Section 22 and [4] provides the appropriate common operator and domain comparison. The present theorems give a source-derived mass insertion, a quantitative equal-time test, and a nonvanishing static response on the complete regulated family. Section 16 constructs its spacetime insertion, and Theorem 21.11 gives the fixed-source recovery of its bounded calibrations. Identifying a refining family with the original physical source additionally requires the corresponding mixed-word and source-normalization comparison; it is not a relabeling of a finite-regulator scalar pole.

16 Spacetime mass response and physical spectral weight

The regulated mass-coefficient derivative admits a physical spacetime insertion without assuming a continuum limit of its correlations. Its natural energy estimate also separates two operations that behave differently under refinement: conjugation by physical time evolution and damping by a positive, but extensively shifted, Hamiltonian.

Write $a = \ell$ for the mesh and n_a for the number of vertices. The full dimensionless Hamiltonian, its lowest physical eigenvalue and its ground-subtracted physical generator are denoted by $\overline{H}_a$, $\overline{E}_a$ and $L_a = a^{-1}(\overline{H}_a - \overline{E}_a)$. All the gauge, Higgs, CAR and Yukawa terms of Section 12 are retained. Let $D_a, C_a^{\text{mom}}, M_{4,a}$ denote the coercivity and moment constants of the ground-state construction, so that

$$\begin{aligned} \overline{H}_a + n_a D_a &\geq T_G + \sum_x p_x^2 + \frac{\lambda_a}{2} \sum_x r_x^2, \\ \overline{E}_a + n_a D_a &\leq n_a C_a^{\text{mom}}, \quad \omega_a(r_x^2) \leq M_{4,a}, \quad r_x = |q_x|^2. \end{aligned} \quad (258)$$

Here and below the full ground density is represented by Ω_a in its purification when necessary; each physical operator acts on the first factor. Put

$$C_a(f) = a \sum_x f(ax) r_x, \quad S_a(f) = a^3 \sum_x |f(ax)|^2, \quad I_a(f) = a^3 \sum_x |f(ax)|. \quad (259)$$

The insertion $C_a(f)$ is not the moment constant C_a^{mom} .

16.1 The common energy domain and physical time

Lemma 16.1 (Extensive coercive energy). *For*

$$A_a = a^{-1}(\overline{H}_a + n_a D_a) = L_a + e_a, \quad e_a = a^{-1}(\overline{E}_a + n_a D_a),$$

one has

$$A_a \geq m_a I, \quad m_a = 3(2\lambda_a)^{1/3} \frac{n_a}{a} \leq e_a \leq C_a^{\text{mom}} \frac{n_a}{a}. \quad (260)$$

Moreover, on $V_a = \text{Dom } A_a^{1/2}$,

$$\|C_a(f)A_a^{-1/2}\| \leq k_a(f) := \sqrt{\frac{2S_a(f)}{\lambda_a}}. \quad (261)$$

Proof. For one four-component scalar and $w > 0$, completing the square gives $(\lambda_a/2)r^2 \geq w^2 r - w^4/(2\lambda_a)$, whereas the sum of four oscillator quadratic forms gives $p^2 + w^2 r \geq 4w$. The choice $w = (2\lambda_a)^{1/3}$ gives $p^2 + (\lambda_a/2)r^2 \geq 3(2\lambda_a)^{1/3}$. This form inequality holds also for vector-valued wavefunctions. Sum over vertices, use (258), and restrict to the reducing physical subspace. The upper bound is the existing physical variational estimate.

Pointwise Cauchy–Schwarz gives $|C_a(f)|^2 \leq a^2(\sum |f(ax)|^2) \sum r_x^2$. Since $(\lambda_a/2) \sum r_x^2 \leq aA_a$ in form sense, $\|C_a(f)u\|^2 \leq k_a(f)^2 \|A_a^{1/2}u\|^2$ on V_a . This proves the multiplication-domain inclusion and the bounded extension in (261). $\square$

Theorem 16.2 (Actual spacetime insertion). *For $F \in C_c^\infty(\mathbb{R} \times \mathbb{R}^3)$ the expression*

$$J_a(F)u = \int_{\mathbb{R}} e^{itL_a} C_a(F_t) e^{-itL_a} u \, dt, \quad F_t(x) = F(t, x), \quad u \in V_a, \quad (262)$$

is a Hilbert-space Bochner integral and satisfies

$$\|J_a(F)A_a^{-1/2}\| \leq \sqrt{2/\lambda_a} \int_{\mathbb{R}} \sqrt{S_a(F_t)} \, dt. \quad (263)$$

It is symmetric and closable for real F . With $\hat{h}(E) = \int h(t) e^{itE} \, dt$ and $Q_a = I - P_a$, where P_a is the complete physical ground projection,

$$J_a(hf)\Omega_a = \hat{h}(L_a)C_a(f)\Omega_a, \quad Q_a J_a(hf)\Omega_a = \hat{h}(L_a)Q_a C_a(f)\Omega_a. \quad (264)$$

Proof. The group generated by L_a preserves V_a and its $A_a^{1/2}$ norm, because $A_a = L_a + e_a$. Strong continuity in that norm and the finite smooth coefficient sampling imply Hilbert-norm continuity of the integrand. Its compact time support and (261) prove the Bochner assertion and bound. For real tests each conjugated multiplier has symmetric matrix elements on V_a ; the integrable bound allows integration of that identity. A symmetric densely defined operator is closable. Finally $e^{-itL_a}\Omega_a = \Omega_a$ and the spectral theorem give (264). The same argument extends separated tests to $h \in L^1(\mathbb{R})$. $\square$

For $F^s(t, x) = F(t - s, x)$, physical time translates $J_a(F)$ to $J_a(F^s)$. The fundamental theorem of calculus yields

$$\|(J_a(F^s) - J_a(F))A_a^{-1/2}\| \leq |s| \sqrt{2/\lambda_a} \int \sqrt{S_a(\partial_t F_t)} \, dt.$$

Thus the derivative is $-J_a(\partial_t F)$ in the indicated relative operator norm. These bounds are uniform on a controlled refining branch for fixed tests. They are not bounds on the vacuum-relative domain of $L_a + 1$: $\|A_a^{1/2}\Omega_a\|^2 = e_a$ is extensive. Neither self-adjointness of every spacetime composite nor an invariant domain for arbitrary unbounded words has been used.

16.2 A sharp test of global energy damping

Proposition 16.3 (Norm collapse above the half-power). *For $s \geq 0$, $\alpha > 1/2$ and $t > 0$,*

$$\|C_a(f)(A_a + s)^{-\alpha}\| \leq k_a(f)m_a^{1/2-\alpha}, \quad (265)$$

$$\|C_a(f)e^{-tA_a}\| \leq k_a(f)(et)^{-1/2}e^{-m_at/2}. \quad (266)$$

On branches with λ_a bounded below and $k_a(f)$ bounded, both expressions tend to zero. The same holds for adjoints and for every fixed word containing one such factor and otherwise uniformly bounded physical factors.

Proof. Factor out $C_a(f)A_a^{-1/2}$. For $x \geq m_a$, $\sqrt{x}/(x+s)^\alpha \leq m_a^{1/2-\alpha}$, while $\sqrt{x}e^{-tx} \leq (et)^{-1/2}e^{-m_at/2}$ follows by maximizing $\sqrt{x}e^{-tx}$. Functional calculus and the product norm bound prove all statements. $\square$

For bounded measurable g on positive time the operator-norm integral $T_a(g, f) = \int_0^\infty g(t)C_a(f)e^{-tA_a} dt$ exists. Writing $G(x) = \int_0^\infty g(t)e^{-tx} dt$ gives $|G(x)| \leq \|g\|_\infty/x$ and therefore

$$\|T_a(g, f)\| \leq k_a(f)\|g\|_\infty m_a^{-1/2}. \quad (267)$$

For time-dependent spatial tests the corresponding bound is $\sqrt{2/\lambda_a} \sup_t \sqrt{S_a(F_t)} \sqrt{2\pi/(em_a)}$. It follows by integrating (266); its $t^{-1/2}$ singularity is integrable. Norm continuity away from zero follows from the same bounded factorization. For k insertions separated by independently integrated positive heat times, bounded time weight G , and bounded physical siblings $D_0, \dots, D_k$, the identical argument gives

$$\left\| \int G(\mathbf{t}) D_0 C_a(f_1) e^{-t_1 A_a} \dots D_{k-1} C_a(f_k) e^{-t_k A_a} D_k d\mathbf{t} \right\| \leq \|G\|_\infty \prod_{j=0}^k \|D_j\| \prod_{i=1}^k k_a(f_i) \left(\frac{2\pi}{em_a} \right)^{k/2}. \quad (268)$$

All such insertion slots are zero in the norm-asymptotic algebra.

Proposition 16.4 (The half-power threshold and the ground state). *For nonnegative f , the bounded operator $B_a = C_a(f)A_a^{-1/2}$ satisfies*

$$\sqrt{S_a(f)/\lambda_a} \leq \|B_a\| \leq \sqrt{2S_a(f)/\lambda_a}. \quad (269)$$

Nevertheless, writing $v_a = a^3 n_a$ and $g_a = 3(2\lambda_a)^{1/3}$,

$$\|B_a \Omega_a\|^2, \|B_a^* \Omega_a\|^2 \leq \frac{M_{4,a}}{g_a} \frac{I_a(f)^2}{v_a}. \quad (270)$$

Proof. For the lower norm bound, use physical link-constant, Fock-vacuum test vectors that are separately radial in the scalars. At sampled sites with $f(ax) > 0$ concentrate a fixed-width radial annulus at radius $R\sqrt{f(ax)}$. Keep fixed radial vectors elsewhere. As $R \rightarrow \infty$, the leading squared insertion norm is $a^2 R^4 (\sum f(ax)^2)^2$ and the leading A_a form is $a^{-1} \lambda_a R^4 \sum f(ax)^2$. Kinetic annular costs, quadratic and edge terms are lower order. The hopping and Yukawa expectations vanish on these test vectors, without removing them as operators. The variational ratio proves the lower bound.

The actual ground moments instead give $\|C_a(f)\Omega_a\| \leq \sqrt{M_{4,a}} a^{-2} I_a(f)$. Since $A_a \Omega_a = e_a \Omega_a$ and $A_a \geq e_a I$, both norms in (270) are at most $e_a^{-1/2} \|C_a(f)\Omega_a\|$. Use $e_a \geq g_a n_a/a$ and square. $\square$

The preparations vanish as physical volume exhausts space on the controlled branch. This is not a statement that $B_a D_a \Omega_a$ vanishes for every bounded sibling D_a . Nor does the estimate prove vanishing when physical volume is fixed. The distinction between source-vector and operator-norm limits is essential here.

Finally $e^{-tL_a} = e^{te_a} e^{-tA_a}$. Vacuum-normalized Euclidean transfer therefore removes precisely the extensive damping used above. A real-time scalar energy shift cancels in conjugation, but does not cancel in an unnormalized heat insertion. The source-owned local flow and mixing in the Yang–Mills construction [31, 17] cannot be replaced by this global heat multiplier.

16.3 Dynamic susceptibility and the contact contribution

Let $\Psi_a = Q_a C_a(f) \Omega_a$ and $\nu_a(B) = \|\mathbf{1}_B(L_a) \Psi_a\|^2$. The mass-insertion domain theorem gives a finite measure with third moment. The full ground eigenspace is removed; the remaining finite source gap need not be uniform. With $M_j = \int E^j d\nu_a(E)$, the established constants obey

$$M_1 \geq \frac{4}{C} a^{-2} S_a(f), \quad M_3 \leq 8K a^{-4} S_a(f), \quad M_1 \leq 4\sqrt{M_4} a^{-2} S_a(f). \quad (271)$$

We use positive susceptibility, the negative of the linear response to adding $+j(t)C_a(f)$ to the Hamiltonian. Define

$$g_a(z) = 2 \int \frac{E}{E^2 + z} d\nu_a(E), \quad z \in \mathbb{C} \setminus (-\infty, 0]. \quad (272)$$

It is holomorphic by compact-uniform domination. For $s > 0$ it is the Laplace transform, at s , of $2\mathbf{1}_{t>0} \int \sin(Et) d\nu_a(E)$, evaluated as $g_a(s^2)$. Also $g_a(0) = 2M_{-1}$ at a finite regulator, and, for $\text{Im } z \neq 0$,

$$-\frac{\text{Im } g_a(z)}{\text{Im } z} = 2 \int \frac{E}{|E^2 + z|^2} d\nu_a(E) \geq 0.$$

Theorem 16.5 (Positive imaginary-frequency response). *For $z \geq 0$ and $S_a(f) > 0$,*

$$g_a(z) \geq \frac{2M_1^2}{M_3 + zM_1} \geq \frac{4S_a(f)}{C^2 K + (C/2)za^2}. \quad (273)$$

If all sampled weights vanish, $g_a(z) = 0$ instead and the first fraction is not used. Consequently, on a branch with $C \leq \bar{C}$ and $K \leq \bar{K}$, every fixed $z \geq 0$ satisfies $\liminf_a g_a(z) \geq 4\|f\|_2^2 / (\bar{C}^2 \bar{K})$.

Proof. Cauchy–Schwarz applied to $E = \sqrt{E/(E^2 + z)} \sqrt{E(E^2 + z)}$ gives the first inequality. For fixed $d > 0$, $z \geq 0$, the function $x^2/(d + zx)$ increases for $x > 0$. Substitute the lower M_1 and upper M_3 bounds in (271). Zero sampled weights give the trivial case. The Riemann sums give the final assertion. $\square$

The Euclidean kernel $k_a(t) = \int e^{-E|t|} d\nu_a(E)$ has, for a Schwartz test h , the ordinary positive quadratic pairing

$$\int \overline{h(t)} k_a(t - u) h(u) dt du = \frac{1}{2\pi} \int |\hat{h}(s)|^2 g_a(s^2) ds. \quad (274)$$

The finite-source gap and finite measure justify this Fourier identity. Fatou’s lemma gives the positive lower limit $4\|h\|_2^2 \|f\|_2^2 / (\bar{C}^2 \bar{K})$. For positive-time tests the reflected pairing is different:

$$\int_0^\infty \int_0^\infty \overline{h(t)} k_a(t + u) h(u) dt du = \left\| \int_0^\infty h(t) e^{-tL_a} \Psi_a dt \right\|^2. \quad (275)$$

It is the latter expression that is a reflected physical source norm. For example, $\nu_a = a^{-1} \delta_{a-1}$ gives $g_a(z) = 2/(1 + a^2 z)$ and $k_a \rightarrow 2\delta_0$ in tempered distributions. The ordinary pairing tends to $2\|h\|_2^2$, but the reflected norm vanishes when the time support stays a fixed distance from zero. Every fixed Schwartz real-time preparation and every fixed finite positive-energy band also vanishes. This example refutes that spectral inference; it is not an identification of the actual interacting source limit.

16.4 A response test that controls both spectral tails

Fix $0 < z_1 < z_2 < z_3$ independently of the regulator and put

$$\mathcal{B}_a = \frac{1}{z_3 - z_1} \left\{ \frac{z_2 g_a(z_2) - z_1 g_a(z_1)}{z_2 - z_1} - \frac{z_3 g_a(z_3) - z_2 g_a(z_2)}{z_3 - z_2} \right\}. \quad (276)$$

Two divided differences of $z/(E^2 + z)$ give the exact positive representation

$$\mathcal{B}_a = 2 \int \frac{E^3 d\nu_a(E)}{(E^2 + z_1)(E^2 + z_2)(E^2 + z_3)} \geq 0. \quad (277)$$

It is also the squared norm of the actual bounded spectral filter $\sqrt{2}L_a^{3/2} \prod_j (L_a^2 + z_j)^{-1/2} \Psi_a$. Adding $c + b/z$ to g_a leaves it unchanged, even for regulator-dependent b, c . Arbitrary higher derivative-contact polynomials are not canceled by this test.

Theorem 16.6 (Quantitative finite physical band). *If the actual source values on a refining tail satisfy $g_a(z_2) \leq M < \infty$ and $\mathcal{B}_a \geq \eta > 0$, choose fixed $0 < \delta < R$ with*

$$\delta^2 \leq \frac{\eta z_1 z_3}{4M}, \quad R^2 \geq \frac{4M}{\eta}, \quad r_{\max} = (\sqrt{z_1} + \sqrt{z_3})^{-2}.$$

Then

$$\nu_a([\delta, R]) \geq \frac{\sqrt{z_2} \eta}{2r_{\max}} > 0. \quad (278)$$

Proof. Put $d\mu_a = 2E(E^2 + z_2)^{-1} d\nu_a$ and $r(E) = E^2/((E^2 + z_1)(E^2 + z_3))$. Its maximum is $r_{\max}$, as differentiation with respect to E^2 shows. The mass of μ_a is at most M , and $\mathcal{B}_a = \int r d\mu_a$. Below δ , $r \leq \delta^2/(z_1 z_3)$; above R , $r \leq R^{-2}$. Each tail contributes at most $\eta/4$. The middle therefore has μ_a mass at least $\eta/(2r_{\max})$. Since $2E/(E^2 + z_2) \leq 1/\sqrt{z_2}$, the conclusion follows. $\square$

A fixed smooth spectral cutoff equal to one on $[\delta, R]$ and supported in $(\delta/2, 2R)$ then gives uniformly bounded, nonvanishing source vectors. The upper bound follows from the positive minimum of $2E/(E^2 + z_2)$ on that larger compact band; its inverse Fourier test is Schwartz. Mixed Gram entries with bounded sibling preparations are controlled by Cauchy–Schwarz and can enter the common cluster kernel simultaneously. This does not bound an unbounded insertion after every sibling word. The hypotheses $g_a(z_2) \leq M$ and $\mathcal{B}_a \geq \eta$ are not supplied by (273): the contact example has $\mathcal{B}_a = 2a^2/\prod_j (1 + a^2 z_j) \rightarrow 0$. Thus the criterion identifies a stronger spectral estimate for continuation and particle identification along this native refinement. It does not condition the independently constructed physical channels of Theorem 11.5.

17 Bounded Higgs observations with exact mass-source recovery

The failure of global energy damping motivates a different construction: regularize the observation locally while retaining the full quantum Hamiltonian, physical time and state. Fix two auxiliary coefficients $\kappa_F, \lambda_F > 0$, independent of mesh, volume and the desired response. At each configuration of the original source let

$$\frac{d\Phi_x}{d\tau} = \frac{\kappa_F}{a^2} \sum_{y \sim x} (\rho_H(U_{xy}) \Phi_y - \Phi_x) - \lambda_F |\Phi_x|^2 \Phi_x, \quad \Phi_x(0) = a^{-1} q_x. \quad (279)$$

Here U_{xy} transports the Higgs fibre at y to that at x ; the reverse link is its inverse. Links are held fixed during the auxiliary flow, but remain quantum configuration variables of the

resulting multiplication operators. The parameter τ is not physical time. In real four-component coordinates this is the negative gradient of

$$V_F(\Phi; U) = \frac{\kappa_F}{2a^2} \sum_{\{x,y\}} |\Phi_x - \rho_H(U_{xy})\Phi_y|^2 + \frac{\lambda_F}{4} \sum_x |\Phi_x|^4, \quad (280)$$

where each unoriented edge is counted once. It is not the gradient of the full quantum Hamiltonian. Its omission of the bare quadratic and Yukawa terms from this declared observation flow does not remove them from the state or from physical evolution.

17.1 Uniform amplitude and the physical observation algebra

Theorem 17.1 (Covariant forward flow). *Every finite initial configuration has a unique solution for all $\tau \geq 0$. On finite-time local coordinate charts the solution depends smoothly on the initial scalars and links. It is gauge covariant and obeys, for $\tau > 0$,*

$$\max_x |\Phi_x(\tau)|^2 \leq \frac{R_0}{1 + 2\lambda_F \tau R_0} \leq \frac{1}{2\lambda_F \tau}, \quad R_0 = \max_x |a^{-1}q_x|^2. \quad (281)$$

For two solutions with the same links, both the Euclidean site norm and the maximum site norm of their difference are nonincreasing.

Proof. The finite polynomial vector field gives local existence and uniqueness. Unitarity of the links implies, for $r_x = |\Phi_x|^2$,

$$r'_x \leq \frac{\kappa_F}{a^2} \sum_{y \sim x} (r_y - r_x) - 2\lambda_F r_x^2.$$

The upper right derivative of the finite maximum therefore satisfies $D^+ R \leq -2\lambda_F R^2$. Scalar comparison gives (281), including the identically zero solution. There is no finite-time blowup, so continuation is global. The right side of (279) transforms covariantly under every local gauge transformation; uniqueness gives the asserted covariance. Variational ODEs on a bounded neighborhood of any fixed trajectory give the stated smooth dependence.

For the difference of two solutions use $\text{Re}(u - v)^\dagger(|u|^2 u - |v|^2 v) \geq 0$. Summing the covariant edge Dirichlet identity proves Euclidean contraction. Applying the maximum argument to the squared pointwise difference proves maximum-norm contraction. These comparisons keep the links fixed. $\square$

Define the actual invariant multiplication operators

$$B_{a,\tau,x} = |\Phi_x(\tau)|^2, \quad Z_{a,\tau,xy} = \Phi_x(\tau)^\dagger \rho_H(U_{xy}) \Phi_y(\tau). \quad (282)$$

Their norms are at most $(2\lambda_F \tau)^{-1}$. The scalar is nonnegative and self-adjoint, and the real and imaginary parts of the edge observation are invariant. They preserve the original physical space and parity. The elementary Φ_x is gauge covariant, not a physical charged operator on the Gauss-invariant space. For real compact smooth f put

$$B_{a,\tau}(f) = a^3 \sum_x f(ax) B_{a,\tau,x}, \quad \|B_{a,\tau}(f)\| \leq \frac{I_a(f)}{2\lambda_F \tau}. \quad (283)$$

Fixed-orientation nondifferentiated edge sums have the analogous bound. The source evaluating their words still includes every gauge, Higgs, CAR and Yukawa interaction.

Theorem 17.2 (Exact physical norm). *For $f \geq 0$,*

$$\|B_{a,\tau}(f)\|_{\text{phys}} = \frac{I_a(f)}{2\lambda_F\tau}. \quad (284)$$

For nonzero f the sequence therefore has a nonzero class in the norm-asymptotic algebra at every fixed positive flow time.

Proof. At trivial links and constant initial data $\Phi_x(0) = Rv$, $|v| = 1$, the edge term vanishes and $\Phi_x(\tau) = Rv/\sqrt{1 + 2\lambda_F\tau R^2}$. Its observation tends to the upper bound in (283) as $R \rightarrow \infty$. For any finite large R , smooth dependence gives an open set of configurations with values arbitrarily close to this one. Thus a single measure-zero link configuration is not being used as an essential-supremum witness.

The near-maximum level set is gauge invariant. A nonzero smooth function of the observation supported in it, multiplied by a sufficiently large invariant compact radial scalar cutoff and the invariant Fock vacuum, is a physical square-integrable test vector. Its normalized expectation approaches the bound. This proves the physical norm equality. The Riemann sum for $I_a(f)$ then gives the nonzero asymptotic norm. These norm-testing vectors do not replace the actual ground state. $\square$

Theorem 17.3 (Excitation of the full finite ground density). *For real sampled $f \not\equiv 0$ and $\tau > 0$,*

$$Q_a B_{a,\tau}(f) P_a \neq 0. \quad (285)$$

This holds without a simple ground eigenvalue or a positive ground wavefunction.

Proof. At fixed links the map from initial scalar coordinates to finite-time flow coordinates is a local smooth diffeomorphism. Its variational Jacobian solves a linear ODE with invertible initial value, so its determinant is nonzero at every finite initial point and time. In the output coordinates the observation is the nonconstant real quadratic polynomial $a^3 \sum f(ax)|y_x|^2$. Every level set has Lebesgue measure zero: fix all but one real component with nonzero squared coefficient; the level equation has at most two solutions in that component. Fubini applies. Inverse local diffeomorphisms preserve null sets on smaller coordinate neighborhoods, and a countable cover gives the result on the entire initial-coordinate space. Fubini over links now shows that the multiplication operator has no ambient nonzero L^2 eigenvector.

If $Q_a B_{a,\tau}(f) P_a = 0$, its bounded self-adjoint restriction to the nonzero finite-dimensional ground space has an eigenvector. That contradicts the multiplication result. The complete ground density consequently has a nonzero excited source vector. $\square$

The compact resolvent gives positive spectral weight at a positive energy at each finite regulator. This neither fixes that energy nor gives a uniform lower bound as the regulator changes. No kinetic sum rule was used: the flowed observation depends on quantum links, whose electric derivatives cannot be discarded in such a calculation.

17.2 Common physical time and bounded mixed words

For $h \in L^1(\mathbb{R})$ define

$$\mathcal{A}_{a,\tau}(h, f) = \int_{\mathbb{R}} h(t) e^{itL_a} B_{a,\tau}(f) e^{-itL_a} dt. \quad (286)$$

This is a strong operator integral, defined by Bochner integration after application to a Hilbert vector. It need not be an operator-norm Bochner integral. The bounded strongly continuous

orbits give

$$\begin{aligned}\|\mathcal{A}_{a,\tau}(h, f)\| &\leq \|h\|_1 \frac{I_a(f)}{2\lambda_F\tau}, \\ \|\alpha_s^a(\mathcal{A}_{a,\tau}) - \mathcal{A}_{a,\tau}\| &\leq \|h(\cdot - s) - h\|_1 \frac{I_a(f)}{2\lambda_F\tau}.\end{aligned}\tag{287}$$

The second inequality follows by a change of variable in the strong integral. Thus these sequences belong to the existing uniformly physical-time-continuous algebra. On the stationary purification,

$$\mathcal{A}_{a,\tau}(h, f)\Omega_a = \widehat{h}(L_a)B_{a,\tau}(f)\Omega_a.\tag{288}$$

Every fixed word of such labels, their adjoints and uniformly bounded physical siblings is bounded by the product of these constants and the sibling norms. The common cluster-state construction therefore retains their joint positivity, gauge and parity identities and positive-energy physical dynamics. Charged CAR generators are not silently compressed into this physical algebra; only its actual admitted invariant CAR words are used here.

A fixed test independent of the mesh preserves the finite excitation whenever the sampled spatial test is not identically zero. For $\sigma > 0$ take

$$h_\sigma(t) = (4\pi\sigma)^{-1/2}e^{-t^2/(4\sigma)}, \quad \widehat{h}_\sigma(E) = e^{-\sigma E^2}.$$

The spectral multiplier is injective, so

$$Q_a\mathcal{A}_{a,\tau}(h_\sigma, f)\Omega_a = e^{-\sigma L_a^2}Q_aB_{a,\tau}(f)\Omega_a \neq 0.\tag{289}$$

Its inverse is unbounded on the full spectrum. A common fixed test with nonzero finite responses is therefore not, by itself, a positive lower bound in a fixed physical band or a nonzero limiting ground representation.

17.3 Recovery of the original mass insertion

Theorem 17.4 (Zero-flow source comparison at fixed regulator). *For each fixed finite source, real f , and $u \in V_a$,*

$$B_{a,\tau}(f)u \longrightarrow C_a(f)u \quad (\tau \downarrow 0).\tag{290}$$

For $h \in L^1(\mathbb{R})$ the same vectors satisfy

$$\mathcal{A}_{a,\tau}(h, f)u \longrightarrow J_a(hf)u.\tag{291}$$

Proof. Pointwise continuity at zero gives $\Phi_x(\tau) \rightarrow a^{-1}q_x$. The powers in the sampled observation give exactly $a^3a^{-2} \sum f(ax)r_x = C_a(f)$. For all $\tau \geq 0$, the initial maximum bound implies

$$|B_{a,\tau}(f)| \leq a \left(\sum_x |f(ax)| \right) \sum_y r_y.$$

The square of this dominating multiplier is integrable against $|u|^2$: $(\sum r_y)^2 \leq n_a \sum r_y^2$ and $u \in V_a$. Dominated convergence proves (290). The same inequality and the coercive form give a bound from V_a to the Hilbert space independent of τ at this fixed regulator. Apply the convergence to $e^{-itL_a}u$, use invariance of the A_a form norm, and dominate by that bound times $|h(t)|$. Bochner dominated convergence proves (291). $\square$

These are strong limits on each form-domain vector, not relative operator-norm limits. They do not interchange zero flow and refinement, or control every product of unbounded insertions. The nearest-neighbor generator does not imply compact positive-flow support or relativistic locality.

The quantitative spatial approximation is proved in Theorems 18.2 and 18.4 and Corollary 18.5. The engineering current $\Phi^\dagger D_i \Phi$ has an additional inverse mesh factor, so its uniform norm does not follow from the amplitude cap. The flow supplies bounded scalar and nondifferentiated link observations; source-owned mixing and normalization must still compare them with the mass and current operators in a common refining quantum source.

18 Spatial localization of the bounded Higgs source

Positive auxiliary flow produces bounded observations on the full interacting regulated source. We prove that these observations admit regulator-uniform approximation by physical cylinder operators in finite spatial boxes. The comparison holds in operator norm, and therefore inside every fixed bounded mixed word. It concerns the spatial dependence of an observation, not the propagation speed of the quantum dynamics.

Throughout this section write $\kappa = \kappa_F$ and $\lambda = \lambda_F$ for the fixed positive auxiliary coefficients in (279). They are distinct from the quantum scalar-potential coefficients. Every quantum interaction, the physical generator L_a , and the state evaluating these observations remain unchanged. For brevity put $U_{xy} = \rho_H(U_{xy})$, with transport from y to x .

18.1 A local observation and nonlinear boundary control

Choose a rectangular vertex box Λ within a non-wrapping coordinate patch of the periodic cubic graph. Write its vertices as $x = (j_1, j_2, j_3)$, $1 \leq j_i \leq n_i$, with $n_i \geq 3$. Retain the actual initial scalars and internal links, and define the Dirichlet observation flow

$$\frac{d\Psi_x}{d\tau} = \frac{\kappa}{a^2} \left(\sum_{\substack{y \sim x \\ y \in \Lambda}} U_{xy} \Psi_y - 6\Psi_x \right) - \lambda |\Psi_x|^2 \Psi_x, \quad \Psi_x(0) = a^{-1} q_x. \quad (292)$$

Every missing neighbor has value zero. This is the gradient flow of the positive internal edge and quartic functional, with the additional term $\kappa |\Psi_x|^2 / (2a^2)$ for each edge leaving the box. It uses only the scalar coordinates in Λ and its internal quantum links. It does not specify another Hamiltonian or another state.

The zero boundary values are gauge covariant. Uniqueness and the maximum argument of Theorem 17.1 give the same covariance and amplitude cap for this finite flow as for the full flow Φ :

$$|\Phi_x(\tau)|, |\Psi_x(\tau)| \leq (2\lambda\tau)^{-1/2}, \quad \tau > 0. \quad (293)$$

Consequently $|\Psi_x|^2$ and $\Psi_x^\dagger U_{xy} \Psi_y$ on internal edges are bounded gauge-invariant cylinder multiplication operators on the original physical Hilbert space.

Lemma 18.1 (Cubic difference absorption). *For real vectors, or complex vectors with their real inner product,*

$$\operatorname{Re}[(u - v)^\dagger (|u|^2 u - |v|^2 v)] \geq \frac{1}{4} |u - v|^4. \quad (294)$$

Proof. Set $m = (u + v)/2$ and $d = u - v$. Expansion of the left side gives $|m|^2 |d|^2 + 2(\operatorname{Re}(m^\dagger d))^2 + |d|^4/4$. The first two terms are nonnegative. $\square$

Write $\Delta v_x = \sum_{y \sim x} (v_y - v_x)$ for the scalar graph Laplacian. For $x \in \Lambda$ put $w_x(\tau) = |\Phi_x(\tau) - \Psi_x(\tau)|$. At a missing neighbor use $w_y = |\Phi_y|$ only as boundary data. Norm differentiation, unitarity and (294) give

$$D^+ w_x \leq \frac{\kappa}{a^2} \sum_{y \sim x} (w_y - w_x) - \frac{\lambda}{4} w_x^3, \quad w_x(0) = 0. \quad (295)$$

At zero difference this is the upper directional derivative of the norm; it also follows by regularizing the norm and passing to the limit. No evolution equation for the exterior boundary notation is needed.

Write

$$c_0 = \sqrt{2/\lambda}, \quad C_0 = \sqrt{24\kappa/\lambda}.$$

The two amplitude caps give $w \leq c_0/\sqrt{\tau}$, including the boundary data. Dropping the nonpositive diagonal and cubic terms in (295), and integrating the six positive neighbor terms from zero, gives

$$w_x(\tau) \leq \min \left\{ \frac{c_0}{\sqrt{\tau}}, \frac{12\kappa c_0 \sqrt{\tau}}{a^2} \right\} \leq \frac{C_0}{a}. \quad (296)$$

The two expressions intersect at $\tau = a^2/(12\kappa)$. Their minimum is nowhere larger than that intersection value. The singular boundary majorant is integrable at zero; in particular, no upper bound on the initial exterior scalar amplitude has been assumed.

Define the six integer face depths $d_{i,-}(x) = j_i$ and $d_{i,+}(x) = n_i + 1 - j_i$. The notation $\sum_F$ below sums over these six faces.

Theorem 18.2 (Uniform nonlinear spatial comparison). *For every actual configuration, every enclosing finite regulator and every $\tau > 0$,*

$$|\Phi_x(\tau) - \Psi_x(\tau)| \leq b_x := \frac{C_0}{a} \sum_F \frac{1}{d_F(x)}, \quad x \in \Lambda. \quad (297)$$

If all six physical face depths satisfy $ad_F(x) \geq R > 0$, then

$$|\Phi_x(\tau) - \Psi_x(\tau)| \leq \frac{6C_0}{R}. \quad (298)$$

Proof. On the first vertex layer at least one depth is one, so (296) gives $w_x \leq C_0/a \leq b_x$. At every remaining vertex all depths are at least two. The exact one-dimensional second difference obeys

$$\frac{1}{d-1} + \frac{1}{d+1} - \frac{2}{d} = \frac{2}{d(d^2-1)} \leq \frac{8}{3d^3}.$$

Each face term varies along only its own coordinate. Therefore

$$\frac{\kappa}{a^2} \Delta b_x \leq \frac{8\kappa C_0}{3a^3} \sum_F d_F(x)^{-3}, \quad \frac{\lambda}{4} b_x^3 \geq \frac{\lambda C_0^3}{4a^3} \sum_F d_F(x)^{-3}.$$

Since $\lambda C_0^2/4 = 6\kappa \geq 8\kappa/3$, b is a stationary supersolution of (295). It dominates the zero initial difference and the first-layer boundary values for all times. The finite cooperative comparison principle proves (297); equivalently, apply the first-positive-maximum argument to $w - b$. If no remaining interior vertices exist, the first-layer estimate already proves the result. Summing six inverse physical depths gives (298). $\square$

The estimate is uniform in every scalar amplitude and every quantum link configuration. The nonlinear first-layer estimate is essential: directly applying a heat kernel to arbitrary unbounded exterior initial data would not give this conclusion.

18.2 Physical operator norms and simultaneous words

Let f be a real spatial test whose supported vertices have physical buffer R in Λ . With $I_a(f)$ from (259), define

$$B_{a,\tau}^\Lambda(f) = a^3 \sum_x f(ax) |\Psi_x(\tau)|^2.$$

Both the full and local observations have norm at most $I_a(f)/(2\lambda\tau)$. Their pointwise difference is bounded by the field difference times the sum of the two amplitude caps. Thus

$$\|B_{a,\tau}(f) - B_{a,\tau}^\Lambda(f)\| \leq \varepsilon_{\tau,R}(f) := \frac{24\sqrt{3\kappa}}{\lambda\sqrt{\tau}R} I_a(f). \quad (299)$$

This bound holds already on the kinematical Hilbert space and hence after restriction to the original Gauss space. For the scalar observation it may be replaced by its minimum with $I_a(f)/(2\lambda\tau)$, because both squared amplitudes lie pointwise in the interval $[0, (2\lambda\tau)^{-1}]$.

For an internal edge,

$$|\Phi_x^\dagger \mathbf{U}_{xy} \Phi_y - \Psi_x^\dagger \mathbf{U}_{xy} \Psi_y| \leq (2\lambda\tau)^{-1/2} (w_x + w_y).$$

Consequently a nondifferentiated edge smearing satisfies (299) with its edge-test absolute-weight sum, provided both endpoints of every tested edge have buffer R . These are uniform configuration bounds on actual quantum operators. No ground-state factorization or classical probability model has been inserted.

Proposition 18.3 (Replacement in every fixed bounded context). *Let $F_1, \dots, F_m$ be fixed-flow scalar or edge observations, and F_i^Λ their local versions. Suppose both versions have norm at most M_i and $\|F_i - F_i^\Lambda\| \leq \varepsilon_i$. For retained bounded physical siblings $D_0, \dots, D_m$, with $\|D_j\| \leq d_j$,*

$$\begin{aligned} & \|D_0 F_1 D_1 \cdots F_m D_m - D_0 F_1^\Lambda D_1 \cdots F_m^\Lambda D_m\| \\ & \leq \left(\prod_{j=0}^m d_j \right) \sum_{i=1}^m \varepsilon_i \prod_{k \neq i} M_k. \end{aligned} \quad (300)$$

Proof. Telescope one changed slot at a time. Each term contains one difference and otherwise original or local factors with the same individual norm bounds. Submultiplicativity gives (300). $\square$

The bound applies to matrix elements, preparations after any fixed bounded source word, and expectations in the full actual ground or Gibbs density. Taking physical buffers $R_j = 2^j R_0$ at fixed positive flow time makes the spatial replacement errors summable. Successive local approximants differ by at most the sum of their errors relative to the same full observation. On exhausting physical boxes, these families therefore lie in the spatial quasilocal C^* -completion and its norm-asymptotic image. No unidentified comparison of unbounded operators across meshes is needed for this assertion.

If a bounded physical operator D has kinematical coordinate support disjoint from the local box, including its internal links, then it commutes with the local multiplier. Hence

$$\|[B_{a,\tau}(f), D]\| \leq 2 \|D\| \varepsilon_{\tau,R}(f). \quad (301)$$

The commuting coordinate actions are restricted to the physical space; a tensor factorization of that space is not assumed.

For $h \in L^1(\mathbb{R})$, use the actual physical dynamics in (286) to define the corresponding full and local time-smeared observations. Unitarity and their strong operator integrals give

$$\|\mathcal{A}_{a,\tau}(h, f) - \mathcal{A}_{a,\tau}^\Lambda(h, f)\| \leq \|h\|_1 \varepsilon_{\tau,R}(f). \quad (302)$$

The difference has a time-translation bound with $\|h(\cdot - s) - h\|_1$ in place of $\|h\|_1$. Proposition 18.3 applies with these additional L^1 factors. A physically time-evolved cylinder need not remain a cylinder: this transports a spatial observation replacement by the common dynamics, not by a presumed finite-speed or spacelike-commutativity law.

18.3 A sharper short-flow bound

The nonlinear barrier supplies uniformly bounded data on an inner spatial boundary. A scalar heat comparison then gives an exponential improvement without reference to unbounded exterior amplitudes.

Theorem 18.4 (Exit-probability localization). *Suppose $ad_F(x) \geq R$ for all six faces and $a \leq R/4$. Set $m = \lfloor R/(2a) \rfloor$ and $r = ma$, so $R/4 \leq r \leq R/2$. Let T be the first exit time from the inner cube $|y_i - x_i| < m$ of the scalar nearest-neighbor walk whose six oriented jump rates are each κ/a^2 . Then*

$$\begin{aligned} w_x(\tau) &\leq \frac{12C_0}{R} \Pr_x(T \leq \tau) \\ &\leq \frac{12C_0}{R} \min \left\{ 1, 6 \exp \left(-\frac{R^2}{64\kappa\tau + 8aR} \right) \right\}. \end{aligned} \quad (303)$$

Proof. Every boundary vertex of the inner cube has original physical face depths at least $R - r \geq R/2$. Theorem 18.2 therefore bounds w there by $12C_0/R$ for all times. Inside, its initial value is zero. Drop only the nonpositive cubic term in (295). The function $(12C_0/R) \Pr_y(T \leq \tau)$ solves the resulting finite heat equation with zero interior initial data and constant boundary value $12C_0/R$. Parabolic comparison proves the first inequality. The random walk represents this scalar comparison equation; it is not a change of quantum measure.

Until exit, the walk lifts to the ordinary unrestricted cubic walk. Each physical coordinate X_i has exponential martingale

$$\exp \left\{ \theta X_i(t) - \frac{2\kappa}{a^2} (\cosh(a\theta) - 1)t \right\}.$$

The nonnegative-martingale maximal inequality, or stopping at the positive level r or time τ , gives

$$\Pr \left(\sup_{t \leq \tau} X_i(t) \geq r \right) \leq \exp \left\{ -\theta r + \frac{2\kappa\tau}{a^2} (\cosh(a\theta) - 1) \right\}.$$

For $0 \leq u < 1$, the power series gives $\cosh u - 1 \leq u^2/[2(1-u)]$. Choose $\theta = r/(2\kappa\tau + ar)$, for which $a\theta < 1$. The exponent is at most $-r^2/(4\kappa\tau + 2ar)$. The negative excursion has the same bound; union over three coordinates and two signs yields

$$\Pr_x(T \leq \tau) \leq \min \{ 1, 6e^{-r^2/(4\kappa\tau + 2ar)} \}.$$

The stopped exponential is integrable at this bounded stopping time; its level-crossing overshoot is bounded by one lattice step. Finally $r^2/(4\kappa\tau + 2ar)$ increases with r . Substitution of $r \geq R/4$ gives $R^2/(64\kappa\tau + 8aR)$ and proves the result. $\square$

Multiplying (303) by the two amplitude caps proves the sharper physical operator estimate

$$\begin{aligned} \|B_{a,\tau}(f) - B_{a,\tau}^\Lambda(f)\| &\leq \varepsilon_{a,\tau,R}^{\exp}(f), \\ \varepsilon_{a,\tau,R}^{\exp}(f) &:= \frac{48\sqrt{3\kappa}}{\lambda\sqrt{\tau}R} I_a(f) \min \left\{ 1, 6e^{-R^2/(64\kappa\tau + 8aR)} \right\}. \end{aligned} \quad (304)$$

The same bound holds for the edge composites when both endpoints have buffer R . It may be combined by taking the minimum with the earlier algebraic and appropriate amplitude bounds. The bounded-word and physical-time estimates apply with this sharper error.

Corollary 18.5 (Summable localization along a joint refinement). *Fix $R > 0$, $\tau_0 > 0$ and $c_* > 0$. Let $\tau_j = \tau_0 4^{-j}$ and choose refining meshes satisfying $a_j \leq c_* \sqrt{\tau_j}$. For fixed compact tests and eventually enclosing physical boxes, the spatial replacement errors in (304) are summable in j . The same is true for every fixed finite word of such shrinking-flow observations with uniformly bounded siblings, and for its fixed L^1 physical-time smearings.*

Proof. Eventually $a_j \leq R/4$, and

$$64\kappa\tau_j + 8a_jR \leq (64\kappa\sqrt{\tau_0} + 8c_*R)\sqrt{\tau_j}.$$

The compact-test Riemann sums are bounded. Thus the error is at most a constant times $\tau_j^{-1/2} \exp(-b/\sqrt{\tau_j})$, where

$$b = \frac{R^2}{64\kappa\sqrt{\tau_0} + 8c_*R} > 0.$$

In terms of j this is a constant times $2^j \exp(-b'2^j)$ and is summable. In a fixed word of length m , one changed slot contributes this error; each of the remaining at most $m - 1$ shrinking-flow slots contributes only a factor bounded by a constant times τ_j^{-1} . Every resulting fixed polynomial in τ_j^{-1} times the same exponential is summable. Apply Proposition 18.3 and (302). $\square$

These are sums of norm differences between two observation prescriptions on each same regulated Hilbert space. They do not assert a common operator space across different meshes, or that either raw word has a bounded or Cauchy limit. The stated mesh-flow relation is retained; no arbitrary-rate joint limit is claimed.

18.4 Sensitivity to initial scalars and quantum links

Let $p_\tau^a(x, y)$ be the scalar heat kernel generated by $(\kappa/a^2)\Delta$ on the full periodic graph. Same-link norm comparison, using monotonicity of the reaction, gives

$$\begin{aligned} |\Phi_x(\tau; q) - \Phi_x(\tau; q')| &\leq a^{-1} \sum_y p_\tau^a(x, y) |q_y - q'_y|, \\ |D_{q_y} \Phi_x(\tau)[v]| &\leq a^{-1} p_\tau^a(x, y) |v|. \end{aligned} \tag{305}$$

Indeed the norm difference is a subsolution of the scalar heat equation with initial data $a^{-1}|q - q'|$; the finite comparison principle proves the first inequality, and differentiation proves the second. The engineering factor a^{-1} is essential. The nonlinear boundary argument above does not discard it; it avoids requiring a bound on those initial data.

For an actual smooth link variation at fixed initial scalars, write $\sigma_e = \|\delta U_e\|$. The inverse-oriented derivative has the same norm. In real four-component coordinates, differentiation of the flow gives a linear equation with nonnegative reaction matrix $\lambda(|\Phi_x|^2 I + 2\Phi_x \Phi_x^T)$ and endpoint forcing $(\kappa/a^2)\delta U_{xy} \Phi_y$. Scalar norm comparison and Duhamel therefore give

$$|\delta \Phi_x(\tau)| \leq \frac{\kappa}{a^2} \sum_{e=(u,v)} \sigma_e \int_0^\tau \frac{p_{\tau-s}^a(x, u) + p_{\tau-s}^a(x, v)}{\sqrt{2\lambda s}} ds. \tag{306}$$

The sum is over the varied unoriented links. Each endpoint forcing is bounded by its link derivative times the amplitude cap; the positive reaction can be dropped in the scalar comparison. The endpoint singularity $s^{-1/2}$ is integrable. This proves (306) without omitting the quantum-link dependence of the observation.

If every link variation has norm at most σ , the degree-six bound and the same integral imply

$$\begin{aligned} \|\delta\Phi(\tau)\|_{\max} &\leq \frac{12\kappa\sigma}{a^2} \sqrt{\frac{\tau}{2\lambda}}, \\ \max_x |\delta(|\Phi_x(\tau)|^2)| &\leq \frac{12\kappa\sigma}{\lambda a^2}. \end{aligned} \quad (307)$$

These derivative bounds retain inverse-mesh powers. Regulator-uniform localization in the bounded-observation norm is not a uniform electric-energy, derivative-current or insertion-activity bound. The estimates do not exclude a stronger reaction or connection argument.

18.5 Spatial replacement and scale renormalization

Spatial localization at a given flow time must be distinguished from changing that flow time. The latter difference already has a sharp obstruction in the raw operator norm.

Proposition 18.6 (Raw dyadic scale separation). *For nonnegative f with $I_a(f) > 0$ at a fixed finite source,*

$$\begin{aligned} \|B_{a,\tau/4}(f) - B_{a,\tau}(f)\|_{\text{phys}} &\geq \frac{3I_a(f)}{2\lambda\tau}, \\ \|B_{a,\tau/4}(f) - B_{a,\tau}(f) - zI\|_{\text{phys}} &\geq \frac{3I_a(f)}{4\lambda\tau}, \quad z \in \mathbb{C}. \end{aligned} \quad (308)$$

Proof. At trivial links and constant initial field $\Phi_x(0) = Rv$, $|v| = 1$, the exact solution is $Rv/\sqrt{1 + 2\lambda\tau R^2}$. The smeared dyadic difference tends as $R \rightarrow \infty$ to

$$I_a(f) \left(\frac{2}{\lambda\tau} - \frac{1}{2\lambda\tau} \right) = \frac{3I_a(f)}{2\lambda\tau}.$$

At zero initial scalars the difference is zero. Smooth dependence at every finite R gives open near-level neighborhoods approaching both endpoints. As in Theorem 17.2, invariant radial cutoffs and the invariant Fock vacuum produce nonzero physical square-integrable vectors supported there. Thus these are physical essential-spectrum norm witnesses, not isolated null-measure configurations. The first inequality follows. For any complex z , at least one of its distances from the two endpoints is half their separation or larger, giving the second inequality. $\square$

This is a statement about the actual operator norm, not the ground-source spectral distribution. In particular, Corollary 18.5 makes fixed-buffer spatial truncation errors summable while (308) still separates successive raw flow scales. A source-owned renormalization can seek Hilbert and graph norms instead; the norm lower bound does not exclude that construction.

The distinction matches the load-bearing Yang–Mills insertion method. Its finite engineering-dimension blocks, reference Gram normalization, canonical one-shell transport and summable renormalized remainder precede the common field-domain conclusion [31, Appendix F, Definition F.242, Theorem F.246 and Lemma F.247]. Its bounded-factor estimate further uses the matching-scale insertion activity and the same finite-cap source, rather than boundedness alone [31, Appendix F, Proposition F.245A]. The present result supplies actual local bounded-flow observations and their bounded-context spatial replacement on the full mixed quantum source. It does not supply that source’s graded scale-mixing matrices, electric-derivative activity estimates or zero-flow graph bounds along refinement. The fixed-source graph and insertion calibrations of Section 21 have a different domain and do not replace these cross-mesh estimates.

Nor is positive auxiliary flow itself a proof of reflection positivity or causal spacetime locality. The primary Yang–Mills construction explicitly distinguishes its flowed probes from the sharp OS-positive base [31, Appendix F, pp. 225–226]. Here the comparison walk governs auxiliary

spatial flow, whereas physical time remains generated by the unchanged L_a . Renormalized common mass/current insertions, nonzero continuum ground response and physical particle identity remain distinct from the spatial comparison just proved.

19 Physical-frequency bounds and scale calibration

A positive imaginary-frequency response need not retain spectral weight in a fixed physical energy interval. For the native family we obtain upper bounds for two actual response protocols and identify the spectral scaling that a physical channel would require. The first bound is pointwise in the retained source and applies to the positive-flow Higgs composite. The second applies to the literal mass insertion after integration over a specified coefficient interval. Neither changes the physical clock or selects a critical bare trajectory. We use the engineering spacing ℓ , equal to the mesh a_n in Section 13, and distinct from the homogeneous gravitational scale factor.

19.1 A uniform response bound at positive flow time

Fix the auxiliary coefficients $\kappa_F, \lambda_F > 0$ and $\tau > 0$ of Section 17. Write $B = B_{\ell,\tau}(f)$ for its actual scalar observation, for a real spatial test f , and set

$$\begin{aligned} b_+ &= \frac{\ell^3}{2\lambda_F\tau} \sum_x \max(f(\ell x), 0), & b_- &= \frac{\ell^3}{2\lambda_F\tau} \sum_x \max(-f(\ell x), 0), \\ W_{\ell,\tau} &= b_+ + b_- = \frac{I_{1,\ell}(f)}{2\lambda_F\tau}, & I_{1,\ell}(f) &= \ell^3 \sum_x |f(\ell x)|. \end{aligned} \quad (309)$$

The flow cap gives $-b_-I \leq B \leq b_+I$. Retain the full ground projection P_ℓ , its complement Q_ℓ , the physical generator L_ℓ , and the purification Ω_N of P_ℓ/d_N . All operators in purified expressions act on the first factor. Define

$$\begin{aligned} \psi_{\ell,\tau} &= (Q_\ell \otimes I)B\Omega_N, \\ \nu_{\ell,\tau}(\Delta) &= \|\mathbf{1}_\Delta(L_\ell)\psi_{\ell,\tau}\|^2, \\ g_{\ell,\tau}(z) &= 2 \int_{(0,\infty)} \frac{E}{E^2 + z} d\nu_{\ell,\tau}(E), \quad z > 0. \end{aligned} \quad (310)$$

Theorem 19.1 (Upper response of the flowed mass observation). *The actual full ground source satisfies*

$$\begin{aligned} \nu_{\ell,\tau}((0, \infty)) &\leq \frac{1}{4}W_{\ell,\tau}^2, \\ 0 \leq g_{\ell,\tau}(z) &\leq \frac{W_{\ell,\tau}^2}{4\sqrt{z}} = \frac{I_{1,\ell}(f)^2}{16\lambda_F^2\tau^2\sqrt{z}}. \end{aligned} \quad (311)$$

These upper bounds are uniform along refinement for a fixed compact test, positive τ , and positive z . If the sampled test is nonzero, both $g_{\ell,\tau}(z)$ and its three-frequency detector (277) are strictly positive at this finite source.

Proof. Put $m = \text{Tr}((P_\ell/d_N)B)$. Since $(Q_\ell \otimes I)m\Omega_N = 0$,

$$\|\psi_{\ell,\tau}\|^2 \leq \|(B - mI)\Omega_N\|^2 = \text{Var}_{P_\ell/d_N}(B).$$

For a self-adjoint operator with spectrum in $[l, u]$, positivity of $(B - lI)(uI - B)$ gives

$$\text{Var}(B) \leq (u - m)(m - l) \leq \frac{1}{4}(u - l)^2.$$

This holds in any state. Apply it with $l = -b_-$ and $u = b_+$. The spectral function $2E/(E^2 + z)$ is at most $1/\sqrt{z}$, which proves the response bound. Theorem 17.3 gives $Q_\ell B P_\ell \neq 0$. Thus the positive spectral measure is nonzero, and both response filters are strictly positive on $(0, \infty)$. Finally the sampled L^1 norms are bounded Riemann sums. No simple ground eigenvalue is required. $\square$

For this flowed observation, the upper-response hypothesis in Theorem 16.6 is therefore proved. Its lower-response hypothesis for this refining flowed family is a positive detector bound uniform in refinement. Finite-source strict positivity does not give that bound. The zero-flow comparison of Theorem 17.4 also does not transfer (311) to the literal mass insertion: its right side grows as τ^{-2} , and the established recovery is on each fixed-regulator form-domain vector.

19.2 An integrated bound for the literal mass response

On a periodic graph with N sites, vary the actual quadratic coefficient profile while retaining every other interaction:

$$\overline{H}_N(s) = \overline{H}_N(0) + sF_w, \quad F_w = \sum_x w_x |q_x|^2, \quad \mu_x(s) = \mu_0 + sw_x. \quad (312)$$

The real weights are fixed for this variation. Let $s \in I = (s_-, s_+)$, a bounded interval, and choose μ_* bounding every $|\mu_x(s)|$ on a slightly larger compact interval. Retain the complete local constants y_*, h_* and d_σ from Section 14, and define

$$\begin{aligned} D_* &= \frac{\mu_*^2}{\lambda} + \frac{3}{4} y_*^{4/3} \lambda^{-1/3} + 3h_*, \\ U_* &= 4 + \frac{3\lambda}{2} + \mu_* + 6\kappa + 3\beta d_\sigma, \\ C_* &= U_* + D_*, \quad R_* = \sqrt{\frac{2C_*}{\lambda}}. \end{aligned} \quad (313)$$

The summed lower-form and Gaussian trial estimates from Section 14 require no translation invariance. They give, in every normalized ground vector or density of (312),

$$\begin{aligned} \sum_x \langle |q_x|^4 \rangle &\leq R_*^2 N, \\ \sum_x \langle |q_x|^2 \rangle &\leq N R_*, \\ |\langle F_w \rangle| &\leq N \|w\|_\infty R_*. \end{aligned} \quad (314)$$

Indeed the lower operator inequality and the variational upper energy hold in each ground vector; Cauchy–Schwarz over sites and the state gives the second line. The profile may therefore be inhomogeneous without replacing its state by a translated one.

Let $e_N(s) = \inf \text{spec } \overline{H}_N(s)$, and denote by $\chi_N(s; w)$ the positive reduced-resolvent susceptibility of F_w in its full ground density, as in Theorem 15.4. It is finite at each fixed source, including a parameter at which ground levels meet.

Theorem 19.2 (Source-parameter integral bound). *For the complete interacting family (312),*

$$\int_I \chi_N(s; w) ds \leq 2N \|w\|_\infty R_*. \quad (315)$$

If every $w_x \geq 0$, the factor two can be omitted. For $H_\ell(s) = \ell^{-1} \overline{H}_N(s)$, take the physical mass-coefficient insertion $C_\ell = \ell F_w$ at each base parameter, by varying s to $s + \ell^2 t$. Its actual response obeys, for $z \geq 0$,

$$\int_I g_\ell(z; s, w) ds \leq 2V_\ell \|w\|_\infty R_*, \quad V_\ell = \ell^3 N. \quad (316)$$

Again the factor two is absent for nonnegative weights. The bounds divided by N and V_ℓ , respectively, are uniform in the number of sites and mesh when the displayed coefficient bounds are uniform.

Proof. The variational infimum of affine functions is concave. The slope bound (314) makes e_N locally Lipschitz, with both one-sided slopes in $[-N \|w\|_\infty R_*, N \|w\|_\infty R_*]$. For nonnegative weights the lower endpoint is zero. Its positive distributional curvature measure $-D^2 e_N$ therefore has mass on I at most the corresponding slope range.

We compare this measure with the full-cluster response, without selecting a differentiable lowest level through a crossing. At each fixed s , common-domain analytic perturbation gives a continuation of the isolated full ground cluster. Its mean has the expansion proved in Theorem 15.4:

$$m_s(t) = e_N(s) + t \langle F_w \rangle_s - \frac{1}{2} t^2 \chi_N(s; w) + O_s(t^3). \quad (317)$$

The smallest eigenvalue is no greater than that mean, so $e_N(s+t) \leq m_s(t)$ for sufficiently small real t . At a differentiability point, the variational inequalities from both signs give $e'_N(s) = \langle F_w \rangle_s$. A concave function has a finite second-order expansion almost everywhere. At such a point comparison with (317) yields

$$0 \leq \chi_N(s; w) \leq -e''_N(s).$$

The density $-e''_N$ is the absolutely continuous part of the positive curvature measure. Integrating it is bounded by the total measure, which also retains any crossing atoms or singular part. This proves (315).

The susceptibility in this integral is measurable. The family has an analytic common resolvent and continuous lowest energy. Its ground projection is the strong limit of $\exp[-t(\overline{H}_N(s) - e_N(s))]$ as $t \rightarrow \infty$. Bounded spectral approximations to the reduced inverse, followed by bounded scalar cutoffs of F_w , express the finite-rank response trace as limits of measurable expressions. The proved ground-vector domains justify removal of those cutoffs. Thus no choice of measurable individual ground eigenvectors is needed, and isolated parameter values do not invalidate the integral estimate.

Finally spectral scaling gives

$$g_\ell(z; s, w) = \ell^3 \overline{g}_N(\ell^2 z; s, w) \leq \ell^3 \chi_N(s; w).$$

The inequality uses $\epsilon/(\epsilon^2 + \ell^2 z) \leq 1/\epsilon$ on the positive dimensionless spectrum. Multiplication of (315) by ℓ^3 proves (316). $\square$

For $w_x = f(\ell x)$, Markov's inequality gives the concrete exceptional-set estimate

$$\left| \left\{ s \in I : \frac{g_\ell(z; s, w)}{V_\ell} > M \right\} \right| \leq \frac{2 \|f\|_\infty R_*}{M}. \quad (318)$$

Fatou also bounds the integral of $\liminf_\ell g_\ell(z; s, w)/V_\ell$ by $2 \|f\|_\infty R_*$. Hence almost every parameter has a parameter-dependent refining subsequence with bounded normalized response. This does not give a bound along the entire sequence at a previously fixed parameter, and it does not prescribe a new parameter or average different theories into a physical state. On a fixed physical torus the bound is mesh-uniform for the actual profile. For expanding boxes it is extensive; a volume-independent bound for a fixed compact test requires an additional local susceptibility estimate.

19.3 The exact dimensionless spectral window

Return to the translation-invariant coefficient record, which may depend on ℓ . For a nonzero sampled test put

$$F_\ell = \sum_x f(\ell x) |q_x|^2, \quad n_f(\ell) = \sum_x |f(\ell x)|^2, \quad S_\ell = \ell^3 n_f(\ell).$$

$$w_\ell(\Delta) = \frac{1}{n_f(\ell)} \left\| \mathbf{1}_\Delta(\bar{L}_N)(Q_\ell \otimes I) F_\ell \Omega_N \right\|^2. \quad (319)$$

Here w_ℓ is a positive spectral measure normalized by $n_f(\ell)$, not necessarily a probability measure or a site weight. The physical identities $C_\ell = \ell F_\ell$ and $L_\ell = \ell^{-1} \bar{L}_N$ imply, for every positive Borel set J ,

$$\nu_\ell(J) = \ell^2 n_f(\ell) w_\ell(\ell J) = \ell^{-1} S_\ell w_\ell(\ell J). \quad (320)$$

The actual susceptibility and detector consequently satisfy

$$\frac{g_\ell(z)}{S_\ell} = 2 \int \frac{\epsilon}{\epsilon^2 + \ell^2 z} dw_\ell(\epsilon), \quad (321)$$

$$\frac{\mathcal{B}_\ell}{S_\ell} = 2\ell^2 \int \frac{\epsilon^3}{\prod_{j=1}^3 (\epsilon^2 + \ell^2 z_j)} dw_\ell(\epsilon). \quad (322)$$

These are finite-source spectral changes of variable, with every interaction retained. For nonzero fixed f , S_ℓ tends to a positive limit. A preparation in a fixed physical band $[\delta, R]$ therefore has norm bounded above and away from zero precisely when its dimensionless weight $w_\ell([\ell\delta, \ell R])$ is of order ℓ . The condition concerns its actual matrix elements, not the mere presence of some unrelated low eigenvalue.

The established moments become

$$\frac{4}{C} \leq \int \epsilon dw_\ell \leq 4\sqrt{M_4},$$

$$\int \epsilon^3 dw_\ell \leq 8K, \quad \int \epsilon^{-1} dw_\ell \geq \frac{2}{C^2 K}. \quad (323)$$

They provide neither an upper inverse-moment estimate nor a lower weight in the shrinking interval. A uniform dimensionless response gap would exclude that interval, but no such gap is assumed or proved for the complete interacting family.

For example, an actual bound on the inverse third moment would give

$$\mathcal{B}_\ell \leq 2\ell^2 S_\ell \int \epsilon^{-3} dw_\ell. \quad (324)$$

Thus a uniform bound on that moment would force the detector to vanish, whereas a positive detector limit requires it to grow at least as fast as ℓ^{-2} . This follows directly from (322); the required inverse-moment bound is not supplied by (323).

For the positive-flow observation, whose definition itself depends on ℓ , the corresponding identity is simply $\nu_{\ell,\tau}(J) = \bar{\nu}_{\ell,\tau}(\ell J)$. Its bounded total weight must retain positive weight at dimensionless energies of order ℓ to survive in a fixed positive physical band. Theorem 19.1 supplies the upper response, not that infrared occupation estimate.

19.4 Bare coordinates and physical calibration

The exact source dictionary

$$\mu_x(t) = \mu_x + t\ell^2 f(\ell x) \quad (325)$$

shows that keeping a dimensionless μ fixed under refinement does not keep its bare physical quadratic coordinate fixed; the latter is μ/ℓ^2 . A fixed dimensionless record can nevertheless support a finite physical channel if its growing-volume interacting source already has the low-energy response scaling in (320). A gapless phase or an appropriately critical record is not excluded. If that scaling is absent, an actual calibrated trajectory and its source comparison are needed; engineering dimensions alone do not construct them.

The source (215) already permits ϑ_ℓ to vary. Fixing every bare coefficient is an additional specialization, not an implication of carrier identity. For a trajectory along which C or K grows, the coefficient dependence in the earlier estimates must be retained rather than applying their uniformly bounded special case.

A brief classical check fixes the normalization. On the homogeneous scalar body with trivial links,

$$\bar{V}(q) = \lambda|q|^4 + \mu|q|^2, \quad T_H = p^2.$$

For $\mu > 0$, the Hessian at the origin is $2\mu I$, so Hamilton's equations give the linearized physical frequency $2\sqrt{\mu}/\ell$. For $\mu < 0$, a homogeneous minimum has $|q_0|^2 = -\mu/(2\lambda)$; its real radial Hessian is -4μ , giving frequency $\sqrt{-8\mu}/\ell$. The factor follows from the unit coefficient of p^2 , not a kinetic coefficient of one half. A prescribed finite classical amplitude $v = |q_0|/\ell$ would therefore require $\mu(\ell) = -2\lambda\ell^2 v^2$ in this polynomial comparison. This is not an interacting critical-mass calculation or a quantum Higgs pole. Quantum matching can shift the critical bare coordinate and couple it to the full interaction record.

The source-preserving scale-transport theorem gives the appropriate identity criterion: the complete response, operator and matching data must persist, not the raw numerical values of bare coordinates [32, Theorem 11.2]. The renormalized trace and universality constructions retain relevant deformations, operator mixing, phase and source comparisons [33, Theorems 6.1 and 15.3]. Their critical examples do not determine the trajectory of this mixed family. The bounds proved here therefore sharpen the actual source tests without assigning a critical surface, selecting a pole, or changing the accepted Higgs and matter identities.

20 Invariant Higgs and matter-hypercharge currents

The scalar mass insertion and the Higgs current are different responses of the same regulated source. We construct the latter by changing covariant-edge phases while keeping all other inputs fixed. This realizes the fixed-sibling source comparison of BCRS, rather than introducing a separate current kernel [30, 4].

Use periodic cubic graphs with at least three vertices in each direction, one positive orientation per nearest-neighbor edge, and Higgs-edge coefficient $\kappa > 0$. For $e = (x, y)$, U_e transports y to x . In the engineering normalization of Section 14, put

$$z_e = q_x^\dagger \rho_H(U_e) q_y, \quad b_{R,e} = \sum_{\text{generations}} \psi_{R,x}^\dagger (\rho_R(U_e) \otimes \bar{S}_e) \psi_{R,y}. \quad (326)$$

Spin and internal indices are contracted in the second expression. It contains the full inherited representation and generation inventory. The Hamiltonian has the terms $\kappa a^{-1} |q_x - \rho_H(U_e) q_y|^2$ and $a^{-1} (b_{R,e} + b_{R,e}^\dagger)$, together with every original gauge, scalar-potential and Yukawa term.

20.1 Source derivatives and contact terms

For a real sampled edge profile A_e , define the Higgs probe

$$H_a^H(tA) = H_a + \frac{\kappa}{a} \sum_e \left\{ |q_x - e^{itaA_e} \rho_H(U_e) q_y|^2 - |q_x - \rho_H(U_e) q_y|^2 \right\}. \quad (327)$$

The external phase commutes with the gauge representation. No fermionic, gauge or Yukawa term has been deleted, and no independent conserved Higgs phase symmetry is asserted. Its derivatives are

$$\begin{aligned} J_a^H(\mathbf{A}) &= 2\kappa \sum_e \mathbf{A}_e \operatorname{Im} z_e, \\ K_a^H(\mathbf{A}) &= 2\kappa a \sum_e \mathbf{A}_e^2 \operatorname{Re} z_e. \end{aligned} \quad (328)$$

For the complete matter-hypercharge probe use the inherited integer characters $q_R = 6Y_R$, $q_H = 3$. Replace the Higgs phase by $e^{i3ta\mathbf{A}_e}$ and each hopping pair by

$$a^{-1} \left(e^{iq_R t a \mathbf{A}_e} b_{R,e} + e^{-iq_R t a \mathbf{A}_e} b_{R,e}^\dagger \right).$$

The gauge Hamiltonian and local potential and Yukawa coefficients remain fixed. This is a background matter-current probe, not a replacement of the entire dynamical gauge connection. Its derivatives are

$$\begin{aligned} J_a^Y(\mathbf{A}) &= \sum_e \mathbf{A}_e J_e^Y, & J_e^Y &= 3J_e^H - 2 \sum_R q_R \operatorname{Im} b_{R,e}, \\ J_e^H &= 2\kappa \operatorname{Im} z_e, & K_a^Y(\mathbf{A}) &= a \sum_e \mathbf{A}_e^2 \left(18\kappa \operatorname{Re} z_e - 2 \sum_R q_R^2 \operatorname{Re} b_{R,e} \right). \end{aligned} \quad (329)$$

Proposition 20.1 (Common current-source domain and response). *Each probe has the original physical form and operator domains locally in its real source parameter, compact resolvent and an analytic isolated continuation of the full ground cluster near zero. If $\mathcal{R}_a = Q_a(L_a|_{Q_a})^{-1}Q_a$, its cluster mean obeys*

$$e'_J(0) = \omega_a(J), \quad e''_J(0) = \omega_a(K) - 2 \langle J\Omega_a, (\mathcal{R}_a \otimes I)J\Omega_a \rangle. \quad (330)$$

Proof. For real parameter every scalar edge square is nonnegative. Its changes and all derivatives are quadratic multipliers with bounded smooth link coefficients; CAR changes are bounded at the fixed graph. The quartic coercivity and operator-domain estimate of Section 15 absorb those multipliers with arbitrarily small relative form and operator bound. Thus the physical common domains and compact resolvent persist. The local complex Taylor expansion is analytic on that common operator domain. The separated finite ground cluster has its usual contour projection; differentiating its trace mean twice gives (330), with the actual second derivative K . The existing fourth moments ensure $J\Omega_a$ is a Hilbert vector. No uniform-in-mesh cluster neighborhood is asserted. $\square$

In particular, total curvature is not minus the paramagnetic susceptibility when the contact expectation $\omega_a(K)$ is nonzero. This distinction matters even before a continuum limit is taken.

Theorem 20.2 (Occupied finite physical current channels). *For every nonzero sampled $\mathbf{A}$ on the specified graph,*

$$Q_a J_a^H(\mathbf{A}) P_a \neq 0, \quad Q_a J_a^Y(\mathbf{A}) P_a \neq 0. \quad (331)$$

Each current therefore has a finite-volume positive-energy spectral pole with nonzero residue in the actual full ground density.

Proof. At fixed links $J_a^H(\mathbf{A})$ is a nonconstant real scalar polynomial. Select an edge with nonzero coefficient. No other positive edge has the same unordered endpoints; its q_x, q_y bilinear cannot cancel against terms with different coordinate pairs. For every unitary transport $\operatorname{Im}(q_x^\dagger \rho_H(U_e) q_y)$ is a nonzero real bilinear form, as a choice $q_x = i\rho_H(U_e) q_y$ shows.

At those links J_a^Y is three times that scalar polynomial times the Fock identity, plus a finite Hermitian matrix independent of the scalar coordinates. At a fixed eigenvalue its scalar-coordinate support lies in a finite union of null polynomial level sets. Fubini excludes a nonzero ambient L^2 eigenvector, and hence a physical one. The same argument covers J_a^H . Ground vectors belong to the current domain. If $Q_a J P_a = 0$, the Hermitian restriction of J to the finite nonzero ground space would have an eigenvector, a contradiction.

For $\psi_J = (Q_a \otimes I)J\Omega_a \neq 0$, compact resolvent gives

$$G_J(z) = \langle \psi_J, (L_a - z)^{-1} \psi_J \rangle = \sum_j \frac{\|\mathbf{1}_{\{E_j\}}(L_a)\psi_J\|^2}{E_j - z}. \quad (332)$$

The series is locally normally convergent off the discrete spectrum. Some positive E_j has nonzero weight, and the residue there is minus that weight. This proves the finite-source assertion. $\square$

A real finite-volume current pole is not an identified unstable Z or Higgs resonance. The theorem neither supplies a uniform physical frequency band nor attaches the charged W boundary channel.

20.2 Full-source charge exchange

At a vertex define the scalar phase generator

$$\mathbf{Q}_{H,x} = q_1 p_2 - q_2 p_1 + q_3 p_4 - q_4 p_3. \quad (333)$$

It satisfies $[\mathbf{Q}_H, H_{\text{complex}}] = H_{\text{complex}}$ in the forward convention (191). It is gauge invariant. Write the three local annihilating Yukawa terms as Z_u, Z_d, Z_e with the incidences of (193). Direct differentiation on the common physical smooth/Schwartz core gives

$$\dot{\mathbf{Q}}_{H,x} = a^{-1} \left(\sum_{e \text{ out } x} J_e^H - \sum_{e \text{ in } x} J_e^H \right) + \frac{2}{a} \text{Im}(Z_u - Z_d - Z_e)_x. \quad (334)$$

Indeed $[\mathbf{Q}_{H,x}, z_e] = -z_e$ at the initial vertex and $+z_e$ at the final vertex. The Yukawa commutators have signs $+, -, -$ for Z_u, Z_d, Z_e , respectively; their adjoints have the opposite signs. The kinetic, scalar potential, gauge and CAR hopping terms commute with this scalar charge. In this orientation the outward Higgs flux is $-J_e^H/a$.

The complete forward matter-hypercharge generator is

$$\mathbf{Q}_{Y,x} = 3\mathbf{Q}_{H,x} - \sum_R q_R N_{R,x}. \quad (335)$$

The sign follows from $[N_R, \psi_R] = -\psi_R$ and the forward gauge convention. The actual Yukawa integer sums are $1 - 4 + 3 = 0$, $1 + 2 - 3 = 0$ and $-3 + 6 - 3 = 0$; all torques cancel. The outgoing number derivative is $\dot{N}_{R,x} = 2a^{-1} \text{Im } b_{R,e}$, with the opposite incoming sign. Hence

$$\dot{\mathbf{Q}}_{Y,x} = a^{-1} \left(\sum_{e \text{ out } x} J_e^Y - \sum_{e \text{ in } x} J_e^Y \right). \quad (336)$$

These identities have an actual domain interpretation. The weighted gradient estimate (246) bounds each angular generator by the Hamiltonian graph norm:

$$\|\mathbf{Q}_{H,x}u\| \leq c_x(\|H_a u\| + \|u\|).$$

The proved operator-core density extends that estimate to $\text{Dom } H_a$. Core vectors lie in $\text{Dom } H_a^2$, because the polynomial/smooth differential expression preserves the core. Their unitary orbits

are therefore differentiable in the H_a graph norm. Composing with the graph-bounded charge and differentiating its expectation at zero gives exactly the displayed commutator. The number operators are bounded; the right sides are quadratic scalar multipliers, bounded CAR terms and the linear Yukawa torque, all controlled in that graph norm. Thus (334)–(336) are justified initial expectation derivatives as well as core identities. They do not assume a distributional all-time Ward identity or a continuum anomaly theorem. Electric flux and the local Gauss constraint remain part of the same physical source; no independent global charge sector has been added.

20.3 Invariant-current normalization and joint reconstruction

Let $\phi_x = a^{-1}q_x$ and use the actual forward difference $D_i^a \phi_x = (\rho_H(U_e)q_y - q_x)/a^2$. Its invariant current is

$$\mathcal{O}_i^a = i(\phi_x^\dagger D_i^a \phi_x - (D_i^a \phi_x)^\dagger \phi_x), \quad a^3 \mathcal{O}_i^a = -2 \operatorname{Im} z_e. \quad (337)$$

Consequently $J_a^H(\mathbf{A}) = -\kappa \sum_e a^3 \mathbf{A}_e \mathcal{O}_i^a$. This is an exact finite-mesh normalization and signed comparison with the invariant channel $iH^\dagger \overleftrightarrow{D}_i H$ in SYM-8 [20]. On matched smooth classical profiles its forward difference converges to that current. No convergence on rough quantum configurations or of current kernels is inferred.

At a fixed regulator the current is self-adjoint matrix multiplication on its maximal physical domain. Its nonreal resolvents are bounded. For $H_a + c \geq 1$ the graph estimate also makes $B_J = J(H_a + c)^{-1}$ bounded, and $J\Omega_a = (E_a + c)B_J\Omega_a$. Enlarge the actual unital adjoint/product-closed algebra by these labels, the current and Hamiltonian resolvents, both signs of the dynamics, and the pre-existing bounded invariant observations. Its full joint ground-state kernel is positive on this native ground-state source. It includes the sibling cross entries, rather than only the current's diagonal correlation.

The joint-kernel and closed-operator reconstruction theorems of BCRS-II EW/H [4] therefore identify the actual cyclic realization. For the possibly degenerate density its exact space is

$$\mathcal{H}_{\text{cyc}} = \overline{\{(D \otimes I)\Omega_a : D \in \mathcal{A}_{\text{joint}}\}} \subset \mathcal{H}_{\text{phys}} \otimes \overline{\operatorname{Ran} P_a}. \quad (338)$$

Adjoint resolvent labels make this closure reducing for the amplifications of both self-adjoint operators. Their full restricted closed domains are reconstructed there. The finite second factor realizes the density, not a new physical sector. There is no claim of cyclicity for all of the unamplified ambient physical space, or that a ground vector lies in every unbounded current-word domain. Uniform bounds for these current labels, their physical normalization and mixed spacetime kernels along refinement remain separate comparisons.

20.4 Static curvature and excitation energy

The distinction persists even for simultaneous source parameters. Let K be a strictly positive real $m \times m$ matrix and, on $\mathbb{C}^{m+1}$, take

$$H_a(0) = a^{-1} \operatorname{diag}(0, I_m), \quad C_{i,a} = a^{-1/2} \sum_j (K^{1/2})_{ji} (|j\rangle\langle 0| + |0\rangle\langle j|).$$

The affine family $H_a(s) = H_a(0) + \sum_i s_i C_{i,a}$ has the simple lowest eigenvalue

$$e_a(s) = \frac{1 - \sqrt{1 + 4a s^T K s}}{2a}, \quad D^2 e_a(0) = -2K. \quad (339)$$

The remaining coupled eigenvalue has the plus sign; $m - 1$ levels stay at a^{-1} . Expanding the square root gives $e_a(s) = -s^T K s + a(s^T K s)^2 - 2a^2(s^T K s)^3 + \dots$. There is a common complex neighborhood of every fixed bounded real parameter set for small a , with uniformly

bounded derivatives of every fixed order and uniformly strictly negative Hessian. Nevertheless the source measure in direction v is $(v^T K v / a) \delta_{a-1}$. Every fixed physical band becomes empty, simultaneously in all directions. The susceptibility and the three-frequency test are

$$g_a(z) = \frac{2K}{1 + a^2 z}, \quad \mathcal{B}_a = \frac{2a^2 K}{\prod_j (1 + a^2 z_j)}.$$

This finite matrix family is a counterexample to a response implication, not a replacement Standard Model source. It shows why even uniform analytic multichannel curvature cannot substitute for the physical-frequency estimate of Theorem 16.6. The actual interacting source is also matrix-valued in its fermions; scalar ground-wavefunction log-concavity is not justified merely by a convex scalar potential. Source-preserving scale transport must retain the physical current and its mixed matrix elements, not only a normalization convention.

21 Dressed tion

The regulated realization admits more than a list of separately occupied sectors. Its physical fields, relative preparations, eliminated degrees of freedom and original source derivatives can be compared on one Hilbert space. This section constructs that comparison without removing the Yukawa interaction. All statements are at a fixed finite physical mesh, or a stated finite chain of meshes. They supplement the directly occupied physical-source construction; they do not identify that source with an unspecified regulator limit.

The construction has three parts. First, neutral physical letters and the ordered fermionic history are defined on the common source. Second, dressed spatial inclusions retain the complementary dynamics and its exact feedback. Finally, graph and source calibrations recover the original insertions and their broken-current spectral response. The resulting comparisons retain both the interaction and the physical clock.

We use the complete quotient-group, four-real-component Higgs and three-generation CAR source of Section 12. Inner products are antilinear in the first variable. The physical generator is always $H = a^{-1} \bar{H}$. A scalar c in $A = H + cI \geq I$ is an analytic norm reference, not a changed physical clock or cosmological term. Unless a comparison explicitly moves its preparation, c and all input ports are fixed. Scalar radii are denoted $s_x = |q_x|$, with $R_2 = \sum_x s_x^2$ and $R_4 = \sum_x s_x^4$.

21.1 A common domain and the original source alphabet

Theorem 21.1 (Complete native source domain). *Let a finite geometric and coupling neighborhood have the strict positive electric, lapse, spatial-metric and quartic margins used in Sections 12 and 25.4. Include the ordered gauge and scalar shifts and both spatial and temporal spin terms. On its physical space put*

$$K = I + T_0 + R_4, \quad V = D(K^{1/2}), \quad D = D(T_0) \cap D(R_4),$$

where T_0 is the sum of the complete unweighted link Casimirs and scalar Laplacians. Every member has operator domain D , with locally equivalent graph norms. For every separately finite collection I of source or metric directions, its actual coefficient derivative H_I is a smooth bounded map $D \rightarrow \mathcal{H}_{\text{phys}}$. Consequently, with one common positive shift,

$$R_s = (H_s + c)^{-1} : \mathcal{H}_{\text{phys}} \rightarrow D, \quad \chi_s = \frac{R_s \Omega}{\|R_s \Omega\|}, \quad \Omega \neq 0, \quad f_i(s) = H_i(s) \chi_s \quad (340)$$

are smooth in $\mathcal{L}(\mathcal{H}_{\text{phys}}, D)$, D and $\mathcal{H}_{\text{phys}}$, respectively. This is not the assertion $H_I : V \rightarrow \mathcal{H}_{\text{phys}}$ for every metric insertion.

Proof. The closed-domain statement is the operator-core theorem 15.1, with the literal geometric ordering retained. Here is the additional alphabet check. A scalar second derivative is controlled by its complete Laplacian. For link generators P_α , the complete Casimir commutes with them and dominates P_α^2 ; applying this bound twice gives $\|P_\alpha P_\beta u\| \leq \|T_0 u\|$, including a noncommuting pair on one link. Multiplication by a quartic coordinate is controlled by R_4 ; every quadratic or linear matrix multiplier is controlled by $I + R_4$. Wilson, hopping, Clifford and temporal-spin coefficients are bounded on the compact link and finite Fock factors.

For a gauge momentum the actual ordered expression is $(FP + PF)/2 = FP - (i/2)XF$, where $P = -iX$. Both F and XF have bounded compact-link coefficients. Scalar momenta also retain their ordering term: on the core, integration by parts gives

$$\| |q_y| \nabla_{q_x} u \|^2 \leq \| |q_y|^2 u \| \| \Delta_{q_x} u \| + C \mathbf{1}_{x=y} \| u \|^2.$$

Interpolation against R_4 and Young's inequality give the requisite graph bound. Differentiating the finite external inverse metrics, determinants, endpoint averages, Clifford factors, path transports, lowered shifts and temporal coframe connection leaves this same alphabet, with its exact coefficient contacts. Taylor remainders obey these graph estimates too. Completion from the invariant core proves smoothness in $\mathcal{L}(D, \mathcal{H})$. The Banach-space inverse theorem applied to $H_s + c : D \rightarrow \mathcal{H}$ gives the first claim in (340). Injectivity of R_s makes the normalization nonzero. Product differentiation, for example

$$(f_i)_{ab} = H_{iab}\chi + H_{ia}\chi_b + H_{ib}\chi_a + H_i\chi_{ab},$$

then proves all the claimed finite jets. $\square$

At fixed mesh the engineering mass direction is $a \sum_x N_x v_x f(ax) |q_x|^2$, whereas a Yukawa direction is $a^{-1} \sum_x N_x Y_x(z)$. These two insertions do map V to $\mathcal{H}$. The static metric derivative of the latter coefficient vanishes at fixed lapse in half-density coordinates; a lapse variation n_x contributes $a^{-1} \sum_x n_x Y_x(z)$. Neither statement sets its propagated metric response to zero. The full time-dependent domain and weak mixed amplitude calculus is developed in the metric-continuation section.

21.2 Canonical neutral fermions on the physical space

Theorem 21.2 (Physical neutral CAR and the Higgs frame). *For each lepton generation and spin, define, almost everywhere in the Higgs coordinate,*

$$n_x = \frac{\epsilon L_x q_x}{|q_x|}, \quad n_{x,\delta} = \frac{\epsilon L_x q_x}{\sqrt{|q_x|^2 + \delta^2}}.$$

The n_x are bounded odd Gauss-invariant operators with the exact canonical anticommutators. If $N_a(f) = a^{3/2} \sum_x \overline{f(ax)} n_x$, then

$$\|N_a(f)\|^2 = I_{2,a}(f) := a^3 \sum_x |f(ax)|^2, \quad \|N_a(f)u\|^2 + \|N_a(f)^* u\|^2 = I_{2,a}(f) \|u\|^2. \quad (341)$$

Both n_x and n_x^ preserve the energy form domain V ; their smoothed versions converge on every fixed vector in that form norm. Preservation of the operator domain D is not asserted and generally fails.*

There is also an exact a.e. Higgs-frame unitary. With $\Omega(q) = (\tilde{q}, q)/|q| \in \text{SU}(2)$, use its vacuum-fixing Fock lift $\mathcal{W}$ on each weak doublet. The transformed generator is precisely $\hat{H} = \mathcal{W}^{-1} H \mathcal{W}$, with domain $\mathcal{W}^{-1} D$. Its scalar momentum is $p + \mathcal{A}$, where $\mathcal{A} = -i \mathcal{W}^{-1} \partial_q \mathcal{W}$; this angular connection is retained, not set to zero. The Yukawa multiplier becomes its radial expression with the original complex matrices unchanged.

Proof. The coefficient of each lepton annihilator is a unit vector at $q \neq 0$. The CAR and the smeared norm therefore follow pointwise, including cross-site and cross-generation anticommutators. The zero set has Lebesgue measure zero; no neighborhood has been excised. Under $q \mapsto z^3 S q$ and $L \mapsto z^{-3} S L$ the numerator is invariant, so these are physical operators rather than charged kinematic letters.

In four real scalar dimensions Hardy's inequality is $\|s_x^{-1} u\|^2 \leq \|\nabla_{q_x} u\|^2$. The coefficient derivative obeys $\sum_\mu |\partial_\mu(q/s)|^2 = 3/s^2$; product differentiation gives the safe estimate $\|\nabla_{q_x}(n_x u)\|^2 \leq 8\|\nabla_{q_x} u\|^2$. The other scalar derivatives, all link derivatives and the quartic multiplier commute with this local coefficient. The same Hardy bound dominates the smoothed derivatives; dominated convergence proves convergence in V on a fixed input. It does not control a second scalar derivative: $\Delta(q/s)$ has size s^{-2} , whose square has the divergent radial integral $\int_0 ds/s$. A Gaussian vacuum input already detects this operator-domain failure.

The identity

$$\Omega(z^3 S q) = S \Omega(q) \text{diag}(z^{-3}, z^3)$$

proves the frame transformation. Its second quantization is pointwise unitary and hence gives the displayed exact domain and momentum conjugation. In the residual group write $A = z^{-2} C \in \text{U}(3)$. The neutral lepton, charged lepton and dressed left quarks transform respectively as $1, \det A, (\det A)^{-1} A, A$ for n, ℓ, u_L, d_L ; the conjugate species retain their corresponding all-left representations. Thus the frame is a residual-group description of the same source, not removal of its charged matter or gauge incidence. Substitution of $q = \Omega(q)(0, s)^T$ in the original Yukawa contractions proves the final claim. $\square$

The uniform ground-state kinetic estimates of Theorem 14.1 also give, for the same preparations,

$$\|(N_{a,\delta}(f) - N_a(f))u_a\|^2 + \|(N_{a,\delta}(f)^* - N_a(f)^*)u_a\|^2 \leq C\delta^2 I_{2,a}(f).$$

Indeed the square of the coefficient difference is at most δ^2/s^2 , and the CAR remove cross-site terms in the sum of the two norms. This retains physical neutral equal-time letters in the common bounded-kernel construction. A right-hand word still needs the form or graph domain required by its next insertion; boundedness alone is not that permission.

21.3 Fermionic histories retain their sign and temporal memory

Proposition 21.3 (Ordered conditional trace). *Write the actual finite source as $H = T + V_b I + F(X)$, where $F(X)$ is its centered quadratic Majorana matrix and X is the full gauge-Higgs configuration. In a form-Trotter history, keep the ordered Fock factor*

$$\mathcal{T} = e^{-\delta F(X_n)/2} e^{-\delta F(X_{n-1})} \dots e^{-\delta F(X_1)} e^{-\delta F(X_0)/2}.$$

The full Gauss factor is included inside its final trace: in the following formula replace $\mathcal{T}$ by that complete ordered lift when a gauge factor is present. If $\mathcal{T}^{-1} c \mathcal{T} = R c$ for the Majorana column c , then

$$(\text{Tr}_F \mathcal{T})^2 = \det(I + R).$$

The sign is fixed by the actual ordered Spin lift, not by a positive square root. On $\det(I + R) \neq 0$, with $C_R = (I - R)(I + R)^{-1}$, distinct Majorana insertions satisfy

$$\tau_F(\mathcal{T} c_{i_1} \dots c_{i_{2r}}) = \tau_F(\mathcal{T}) \text{Pf}[(C_R)_{i_\alpha i_\beta}],$$

and odd traces vanish. At the singular chart boundary the unnormalized finite trace, not its normalized Pfaffian factors separately, is continued.

Proof. For $F = (i/4) \sum A_{rs} c_r c_s$, with real antisymmetric A and $\{c_r, c_s\} = 2\delta_{rs}$, one has $[F, c] = -iAc$ and the Euclidean factor induces e^{-itA} . Multiplication preserves the original order. Conjugation on the Clifford algebra has trace $\sum_k \text{Tr}(\wedge^k R) = \det(I + R)$. In its matrix realization this trace is $\text{Tr } \mathcal{T} \text{Tr } \mathcal{T}^{-1}$; reversion on the even Spin lift preserves the scalar coefficient and identifies the two traces. The identity holds along the actual analytic product from the identity, thereby fixing its sign through zeros. The actual all-left gauge representation has zero total abelian charge and traceless semisimple generators, so its vacuum-fixing Fock action is the canonical lift; an extra arbitrary gauge phase is not allowed.

Trace cyclicity and $c_j \mathcal{T} = \mathcal{T}(Rc)_j$, followed by the Clifford relation, give the contraction $2(I + R)^{-1} = I + C_R$. Induction is the displayed Pfaffian recurrence. Parity gives the odd result. Analyticity of the original finite trace supplies the singular-chart interpretation. Finally the form-Trotter product is that of the actual lower-semibounded source; the scalar heat kernel and the entire ordered matrix factor are integrated together. $\square$

Independent tracing of time slices changes this source. For a constant nonzero quadratic matrix with positive durations $\sum_j t_j = \beta$,

$$\tau_F \prod_j e^{-t_j F} = \tau_F e^{-\beta F} = \prod_\ell \cosh(\beta \epsilon_\ell / 2) > 1, \quad \log \prod_j \tau_F e^{-t_j F} \leq \frac{1}{2} \tau_F(F^2) \beta \max_j t_j \rightarrow 0.$$

This is a finite-duration memory calculation, not an assertion that a constant path has positive path measure.

Proposition 21.4 (Actual Euclidean insertion expansion). *At the fixed source choose $A \geq I$ with $A \geq (\lambda/2a)R_4$ as forms. If $Y = a^{-1} \sum_x Y_x(q_x)$ and $\|Y_x(q_x)\| \leq y_* |q_x|$, then*

$$\|YA^{-1/4}\| \leq B_Y := y_*(2/\lambda)^{1/4} N^{3/4} a^{-3/4}.$$

The ordered Dyson coefficients $\mathcal{D}_n(\beta)$ for $e^{-\beta(A+zY)}$ satisfy

$$\|\mathcal{D}_n(\beta)\| \leq \frac{[B_Y(4e)^{-1/4} \Gamma(3/4) \beta^{3/4}]^n}{\Gamma(1 + 3n/4)}.$$

The series $\sum_n (-z)^n \mathcal{D}_n$ converges in operator norm uniformly on compact complex z sets. For the mass insertion $C_a(f) = a \sum_x f(ax) |q_x|^2$ one similarly has $\|C_a(f)A^{-1/2}\| \leq \sqrt{2I_{2,a}(f)/\lambda}$ and the analogous Dirichlet-integral expansion with exponent $1/2$.

Proof. Pointwise Cauchy–Schwarz gives $Y^2 \leq y_*^2 a^{-2} N^{3/2} R_4^{1/2}$. The Loewner–Heinz inequality applied to the coercive form bound proves the first estimate. Functional calculus then gives $\|Ye^{-tA}\| \leq B_Y(4et)^{-1/4}$. Integrate the product of these bounds over the ordered n -simplex:

$$\int_{t_j \geq 0, \sum t_j \leq \beta} \prod_{j=1}^n t_j^{-1/4} dt = \frac{\Gamma(3/4)^n \beta^{3n/4}}{\Gamma(1 + 3n/4)}.$$

It is integrable at every face and gives an entire majorant. Duhamel iteration first on the core and then by the form semigroup construction identifies this series with the actual perturbed semigroup. The mass estimate follows from $C_a(f)^2 \leq I_{2,a}(f)R_4/a$ and the identical argument. The physical answer includes $e^{\beta c}$; the shift has not been discarded. The estimate is fixed-source and does not claim a mesh-uniform Yukawa coefficient. $\square$

21.4 An actual dressed quantum inclusion

Take a periodic coarse lattice with even side $n \geq 4$, $N = n^3$ sites and spacing a , and its refinement with spacing $a/2$. In each eight-site block rooted at $2x$, choose the tree whose parent removes the first nonzero offset coordinate. Let h_v transport a fine vertex to its root and set $z_v = \rho_H(h_v)q_v$. Complete Haar coordinates consist of $7N$ tree paths, $3N$ selected cross edges y (from $2x + e_i$ to $2x + 2e_i$), and $14N$ other edges η . Their measures are independent normalized Haar measures by successive left and right translations; the coordinate change is invertible. An orthogonal eight-site scalar transform gives

$$Z_x = 8^{-1/2} \sum_{v \in x} z_v = \sqrt{2} q_x^c, \quad z_v = q_x^c/2 + r_v,$$

and seven relative four-real-component coordinates ζ per block. The coarse CAR mode is the transported sum with coefficient $8^{-1/2}$; all relative CAR modes are prepared empty. In the forward convention its Fock embedding is $F(h) = \Gamma(h)^{-1}F(1)$.

Let $\sigma > 0$, $\tau \geq 0$, and all $\beta_p > 0$. Set $\chi_\sigma(\zeta) = (\pi\sigma^2)^{-7N} e^{-|\zeta|^2/(2\sigma^2)}$. The genuine fine Wilson multiplier is

$$W(y, \eta) = \sum_p \beta_p [17 - \Re \chi_\Sigma(U_p)], \quad 0 \leq W \leq M = 34 \sum_p \beta_p,$$

where Σ is the faithful original quotient representation. Define one joint denominator, with every overlapping plaquette retained,

$$Z_\tau(y) = \int e^{-\tau W(y, \eta)} d\eta, \quad r_\tau(y, \eta) = Z_\tau(y)^{-1/2} e^{-\tau W(y, \eta)/2}, \quad J_\tau = 2^{-N} r_\tau \chi_\sigma F(h).$$

The final expression means multiplication followed by the pullback of the coarse wavefunction in these coordinates; it is a map of full Hilbert spaces, not a tensor product of postulated physical blocks.

Theorem 21.5 (Normalized physical inclusion and retained interaction). *The map J_τ is a number-preserving, Gauss-equivariant isometry, and preserves the smooth-link/Schwartz-scalar physical core. Its exact physical Yukawa compression, including all real parameter directions and block averages of fine profiles, is*

$$J_\tau^* B_f J_\tau = B_c, \quad B_f = 2Y_f/a, \quad B_c = Y_c/a. \quad (342)$$

For a normalized link-independent radial scalar and empty-CAR coarse core vector ϕ , put $S = \langle \phi, R_{2,c} \phi \rangle$, $P = J_\tau J_\tau^*$, $Q = I - P$, and

$$\kappa_Y = 6\|Y_u\|_F^2 + 6\|Y_d\|_F^2 + 2\|Y_e\|_F^2.$$

Then

$$\|QB_f J_\tau \phi\|^2 = \frac{\kappa_Y}{a^2} (7S + 56N\sigma^2). \quad (343)$$

It is strictly positive when $\kappa_Y > 0$ and is a lower bound for the squared complementary full-generator response on this seed.

Proof. The scalar dilation has Jacobian 4^N , canceled by $|2^{-N}|^2$. The relative Gaussian and Fock embedding are normalized, while $e^{-\tau M} \leq Z_\tau \leq 1$ and $\int r_\tau^2 d\eta = 1$ at every y . This proves isometry. Telescoping tree paths makes W independent of the tree coordinates; simultaneous root gauge changes in y, η preserve its integral. This and the forward Fock convention prove full Gauss covariance. All fixed-graph link derivatives of r_τ are bounded, and it is scalar coordinate independent, proving core preservation.

Each retained Yukawa pair has two factors $8^{-1/2}$; its mean scalar is $q_c/2$ and there are eight sites. The factor is therefore $1/2$, exactly canceled by the physical fine clock $2/a$. The calculation

is independent of η , so the dressing changes neither it nor its parameter directions. The relative scalar second moment is $14N\sigma^2$. The original empty-CAR pair identity is $\|Y_x(q)|0\rangle\|^2 = \kappa_Y|q|^2$; different sites and wedges are orthogonal. It follows that

$$\|B_f J_\tau \phi\|^2 = \frac{\kappa_Y}{a^2}(8S + 56N\sigma^2), \quad \|PB_f J_\tau \phi\|^2 = \frac{\kappa_Y}{a^2}S.$$

Subtracting proves (343). Every other Hamiltonian term preserves number and hence cannot cancel this number-two component. $\square$

Proposition 21.6 (Magnetic response and its normalization contacts). *The compressed Wilson multiplier is $m_\tau(y) = \langle W \rangle_{\tau,y}$ and $\partial_\tau m_\tau = -\text{Var}_{\tau,y}(W) < 0$. At a sufficiently large finite τ it is a nonconstant physical coarse multiplier. For a general preparation action $S(y, \eta)$, its exact derivatives at fixed fiber coordinates are*

$$X \log Z = -\langle XS \rangle, \quad Xr = -\frac{1}{2}(XS - \langle XS \rangle)r, \quad X\langle F \rangle = \langle XF \rangle - \text{Cov}(F, XS).$$

An original fine link derivative D instead obeys the chain rule

$$D \log r = -\frac{1}{2}DS + \frac{1}{2}\langle d_y S \rangle(Dy). \quad (344)$$

In particular its denominator term and all induced fiber derivatives belong to the original electric energy.

Proof. Differentiate the single compact conditional integral. To prove strict variance, choose a fine plaquette whose lower vertex is odd in its two directions. Its two outgoing perpendicular edges are extra coordinates e, f and occur together in no other elementary plaquette. Haar integration projections P_e, P_f give $(I - P_e)(I - P_f)W = -\beta_p \Re \chi_\Sigma(U_p)$. Character orthogonality for the six original summands gives $\int (\Re \chi_\Sigma)^2 = 4$. Since this is an orthogonal projection annihilating constants, $\text{Var}_{0,y}(W) \geq 4\beta_p^2$. The conditional density is at least $e^{-\tau M}$, so $\text{Var}_{\tau,y}(W) \geq 4\beta_p^2 e^{-\tau M} > 0$.

For coarse dependence, use an actual two-by-two sheet spanning a coarse plaquette. Give one boundary selected edge a nonidentity quotient element g_0 , the others identity. Peeling the four faces and applying the Hilbert–Schmidt triangle inequality gives, for every η ,

$$W(y_{\text{bad}}, \eta) \geq b := \frac{\beta_{\text{min, sheet}}}{8} \|\Sigma(g_0) - I\|_{\text{HS}}^2 > 0.$$

At $y_{\text{good}} = 1$, the all-identity fiber has zero action. Thus $\delta = \text{Haar}\{\eta : W(y_{\text{good}}, \eta) \leq b/4\} > 0$. The normalized sublevel estimate is

$$m_\tau(y_{\text{good}}) \leq b/2 + (M/\delta)e^{-\tau b/4}, \quad m_\tau(y_{\text{bad}}) \geq b.$$

Choosing $\tau > (4/b) \log[2M/(b\delta)]$ separates the two values. Smooth functions of the invariant m_τ , multiplied by the scalar/vacuum seed, supply physical test vectors supported at these separated values. This proves nonscalarity on the physical space, not merely on a gauge-variant configuration chart. The chain rule proves the last displayed identity. $\square$

The full-Wilson estimate is finite-volume. A complete fixed plaquette collar can instead define a volume-independent local preparation denominator; that is another allowed J , not the same full-Wilson preparation. Its response to the full W is $-\text{Cov}(W, W_{\text{collar}})$, not automatically $-\text{Var}(W_{\text{collar}})$. None of these formulas sets $J^* W_f J = W_c/2$ or removes the actual electric derivative cost.

21.5 Physical form stability and positive complementary feedback

Lemma 21.7 (The projection is stable in energy). *Let $\mathcal{E}_i(u) = \|u\|^2 + E_{G,i}(u) + E_{H,i}(u) + E_{4,i}(u)$ be the complete kinetic-plus-quartic reference form. At each fixed dressed pair there are finite explicit constants C_J, C_* such that*

$$\mathcal{E}_f(Ju) \leq C_J \mathcal{E}_c(u), \quad \mathcal{E}_c(J^*v) \leq C_* \mathcal{E}_f(v).$$

If $\lambda_- \mathcal{E}_f \leq q_A \leq \lambda_+ \mathcal{E}_f$, then

$$q_A(Pv) \leq C_P q_A(v), \quad q_A(Qv) \leq C_Q q_A(v), \quad C_P = (\lambda_+/\lambda_-) C_J C_*, \quad C_Q = (1 + \sqrt{C_P})^2.$$

The pullback form domain of J is exactly V_c . No bounded QAJ or operator-domain preservation by P is needed.

Proof. The only delicate derivative is that of a tree link, which rotates scalar coordinates as well as the finite Fock frame. Write $m = 12$, $c_H = \sup_{|t|=1} \|\rho_{H*}(t)\|$, and let c_F bound the local Fock charge. On the physical core Gauss' identity gives

$$\sum_{x,\alpha} \|L_{x,\alpha} u\|^2 \leq 14E_{G,c}(u) + 7mNc_F^2 \|u\|^2,$$

since each vertex has six incident electric generators and one Fock charge. For a tree derivative, the orthogonal scalar transform has a coarse angular block, off-diagonal blocks of norms at most $c_H/\sqrt{2}, \sqrt{2}c_H$, and a skew relative block. The latter annihilates the radial Gaussian. Its two off-diagonal terms have Gaussian squared bounds $(\sigma^2 c_H^2/4)|\nabla_c u|^2$ and $(c_H^2/\sigma^2)|q_c u|^2$. Include also the derivatives of r , $F(h)$ and the two-edge coarse path. Summing the six contributions gives

$$\begin{aligned} E_{G,f}(Ju) &\leq 600E_{G,c}(u) + \frac{21}{2}mc_H^2\sigma^2 E_{H,c}(u) + 42mc_H^2\sigma^{-2}E_{2,c}(u) \\ &\quad + (6K_0 + 2352mNc_F^2)\|u\|^2, \\ E_{H,f}(Ju) &= \frac{1}{2}E_{H,c}(u) + 14N\sigma^{-2}\|u\|^2, \\ E_{4,f}(Ju) &\leq 4E_{4,c}(u) + 294N\sigma^4\|u\|^2. \end{aligned}$$

Here one may take $K_0 = \sum_D \sup_{y,h} \int |Dr|^2 d\eta$ over the original fine unit link directions, interpreted through the exact coordinate chain rule; it is finite by (344) and compactness. Since $E_{2,c} \leq N\|u\|^2 + E_{4,c}$ these give, for example,

$$\begin{aligned} C_J &= \max\left\{600, \frac{1}{2} + \frac{21}{2}mc_H^2\sigma^2, 4 + 42mc_H^2\sigma^{-2}, \right. \\ &\quad \left. 1 + 14N\sigma^{-2} + 294N\sigma^4 + 6K_0 \right. \\ &\quad \left. + 2352mNc_F^2 + 42mNc_H^2\sigma^{-2}\right\}. \end{aligned}$$

For the adjoint, differentiate the normalized relative integral in a coarse link direction and apply conditional Cauchy–Schwarz. With $K_1 = \sum_{X_y} \sup_y \int |X_y r|^2 d\eta$, this gives $E_{G,c}(J^*v) \leq 2E_{G,f}(v) + 2K_1\|v\|^2$. The scalar transform and conditional Jensen inequality give respectively $E_{H,c}(J^*v) \leq 2E_{H,f}(v)$ and $E_{4,c}(J^*v) \leq 2E_{4,f}(v)$; thus $C_* = \max(2, 1 + 2K_1)$ suffices.

These are physical estimates: the coarse angular bound used Gauss' law, not an unrestricted kinematic inequality. Complete the invariant cores in their form norms to obtain the stated bounds on the original Hilbert maps. Their composition gives the P bound, and the form triangle inequality gives the Q bound. Finally $J^*J = I$ and the two map bounds prove both inclusions of the pullback-domain equality. $\square$

The same coordinates give the exact scalar compression identities

$$J^*T_{H,f}J = \frac{1}{2}T_{H,c} + 14N/\sigma^2, \quad J^*R_{2,f}J = 2R_{2,c} + 14N\sigma^2,$$

$$J^*R_{4,f}J = \frac{1}{2}R_{4,c} + \frac{21}{2}\sigma^2R_{2,c} + \frac{147}{4}N\sigma^4.$$

They follow from the relative covariance $\mathbb{E}(r_v r_w^T) = (\sigma^2/2)(\delta_{vw} - 1/8)I_4$. In particular, a same-block pair has covariance $-\sigma^2 I_4/16$; discarding it changes the induced scalar edge contact. If $M_e = \langle \rho(T_e) \rangle_{\tau,y}$ for an extra edge in a nontrivial original irreducible matter representation, its Haar mean is zero. Writing the conditional density as its uniform lower bound $\delta = e^{-\tau M}$ plus a positive remainder gives

$$\|M_e\| \leq 1 - \delta, \quad I - M_e^* M_e \geq \delta(2 - \delta)I.$$

Thus an averaged edge is not a unitary replacement link. Its quadratic form retains $|q_x - M_e q_z|^2 + q_z^*(I - M_e^* M_e)q_z$, together with the actual relative covariance. Normal-ordered CAR bilinears compress with the two factors $8^{-1/2}$ and no artificial relative-vacuum occupation. These induced matter formulas do not exhaust the fine electric pullback.

Theorem 21.8 (Energy-space inverse and a nonzero feedback witness). *Let A_J and A_Q be the closed restricted forms on the two physical summands, and fix $\lambda \geq 0$. Put $L = A_J + \lambda$, $D_Q = A_Q + \lambda$, $\eta = (C_P + C_Q)^{-1}$ and $k = 1 - \eta < 1$. There is a bounded form coupling $K : \mathcal{H}_c \rightarrow Q\mathcal{H}_f$, $\|K\| \leq k$, such that, with*

$$S(x, y) = JL^{-1/2}x + D_Q^{-1/2}y, \quad M = \begin{pmatrix} I & K^* \\ K & I \end{pmatrix},$$

$$(A + \lambda)^{-1} = SM^{-1}S^*, \quad J^*(A + \lambda)^{-1}J = L^{-1/2}(I - K^*K)^{-1}L^{-1/2}. \quad (345)$$

*The geometric approximants through $(K^*K)^n$ are positive, increase to the compressed inverse, and have error at most $k^{2n+2}/[(1 + \lambda)(1 - k^2)]$.*

For the core seed in (343) with $\kappa_Y > 0$, let $v = QAJ\phi$ and $u = L\phi$. Then the positive difference from L^{-1} is nonzero; quantitatively,

$$\langle u, [J^*(A + \lambda)^{-1}J - L^{-1}]u \rangle \geq \frac{\ell^4}{C_Q \|A^{3/2}J\phi\|^2 + \lambda \|AJ\phi\|^2} > 0, \quad \ell^2 = \frac{\kappa_Y}{a^2}(7S + 56N\sigma^2).$$

Proof. Represent the off-diagonal form by $q_{A+\lambda}(q, Jf) = \langle D_Q^{1/2}q, KL^{1/2}f \rangle$. The form-splitting estimate is $q_{A+\lambda}(Pz) + q_{A+\lambda}(Qz) \leq (C_P + C_Q)q_{A+\lambda}(z)$, hence $M \geq \eta I$. Testing its two components with the opposite phase gives $\|K\| \leq 1 - \eta$. The map S is a bounded bijection of the direct energy coordinates onto V_f , and $q_{A+\lambda}(S\xi, S\zeta) = \langle \xi, M\zeta \rangle$. Solving this variational equation proves the first inverse; bounded Schur elimination proves the second. It has not formed an unbounded block QAJ . The geometric tail gives the displayed norm bound.

On this particular core seed v is a Hilbert vector and, by form stability, lies in $D(D_Q^{1/2})$. The first feedback term evaluated at u is $\langle v, D_Q^{-1}v \rangle$. Cauchy–Schwarz in the spectral measure of D_Q gives $\langle v, D_Q^{-1}v \rangle \geq \|v\|^4/q_{D_Q}(v)$. Number orthogonality gives $\|v\| \geq \ell$, while the form bound on Q gives the denominator stated above. Core preservation justifies the displayed powers of A on $J\phi$. $\square$

For a fixed-baseline form perturbation $q_\varepsilon = q_A + \varepsilon b$, represent $b(S\xi, S\zeta) = \langle \xi, N\zeta \rangle$. All four blocks of N are retained. The derivative of the compressed inverse is

$$-L^{-1/2}[M^{-1}NM^{-1}]_{cc}L^{-1/2}.$$

The remainder is at most $\varepsilon^2\|N\|^2/[(1 + \lambda)\eta^2(\eta - |\varepsilon|\|N\|)]$ when $|\varepsilon|\|N\| < \eta$. This follows directly from the inverse identity for $M + \varepsilon N$. Moving J adds both differentiated ports; it is not this fixed-baseline formula.

21.6 Polar normalization and ordered, nested elimination

Proposition 21.9 (Full polar chart and exact finite-chain memory). *For $s > 0$ define $X_s = e^{-sA/2}J$ and $C_s = X_s^*X_s$. The positive compact operator C_s is injective, and $V_s C_s^{1/2}u = X_s u$ defines an isometry on the full coarse space by density. Its inverse square root need not be bounded. Compression $O \mapsto V_s^* O V_s$ is a unital completely positive physical map.*

Let J_{01}, J_{12} be two actual dressed inclusions and define $J_{02} = J_{12}J_{01}$. On the complete finest source let $T = e^{-sA}$, $K_0 = I - zT$, with $z \geq 0$ and $h = 1 - ze^{-s} > 0$. For any of these splits write

$$D_0 = QK_0Q|_{Q\mathcal{H}}, \quad \Gamma = J - QD_0^{-1}QK_0J, \quad S_0 = J^*K_0J - J^*K_0QD_0^{-1}QK_0J, \quad G = QD_0^{-1}Q.$$

Then $hI \leq S_0 \leq I$ and

$$K_0^{-1} = \Gamma S_0^{-1} \Gamma^* + G.$$

*Successive elimination uses the entire first Schur operator, not $I - zJ_{12}^*TJ_{12}$. With stage labels on the resulting maps,*

$$S_{02} = S_{01}, \quad \Gamma_{02} = \Gamma_{12}\Gamma_{01}, \quad G_{02} = G_{12} + \Gamma_{12}G_{01}\Gamma_{12}^*. \quad (346)$$

These identities retain every ordered bounded insertion and compose for any finite flag of the actual maps.

Proof. Injectivity of $e^{-sA/2}$ and J proves that of C_s . The identity $\|X_s u\| = \|C_s^{1/2}u\|$ constructs the isometry on the dense range; compactness follows from the fine heat operator. This is a polar construction, not a bounded inverse-Gram assumption. Positivity of matrix compressions proves complete positivity. In particular their multiplicative defect is $V_s^* O^* (I - V_s V_s^*) O V_s \geq 0$ for $O^* O$.

For the second statement, $hI \leq K_0 \leq I$. The vector Γu is the unique minimizer of $\langle Ju + q, K_0(Ju + q) \rangle$ over $q \in Q\mathcal{H}$; its minimum is $\langle u, S_0 u \rangle$. Solving the bounded block equation proves the inverse decomposition. The three orthogonal spaces are $J_{02}\mathcal{H}_0$, $J_{12}Q_{01}\mathcal{H}_1$ and $Q_{12}\mathcal{H}_2$. Minimizing first in the last space and then in the middle space is exactly minimizing in their sum. Uniqueness proves the first two identities in (346); substitution of the two inverse decompositions proves the third.

At an intermediate resolvent in an ordered word substitute the full $\Gamma S^{-1} \Gamma^* + G$. For instance,

$$J^* R_0 O_1 R_1 O_2 R_2 J = F_0 \Gamma_0^* O_1 [\Gamma_1 F_1 \Gamma_1^* + G_1] O_2 \Gamma_2 F_2, \quad F_i = S_i^{-1}.$$

The same substitution at each frequency proves the general word statement, with norm bound $\prod_i \|O_i\| \prod_j h_j^{-1}$. It preserves G at each intermediate observation. Differentiating these bounded identities retains all source blocks and port contacts. No identification with the separately printed intermediate Hamiltonian enters. $\square$

Thus finite-step normalization and elimination are coherent, but not Markovian compression by fiat: in general $J^* e^{-2sA} J \neq (J^* e^{-sA} J)^2$. The strict leakage witness and Theorem 21.8 identify an actual interaction which such a replacement would omit.

21.7 Graph calibration and recovery of original insertions

Proposition 21.10 (Closed graphs and source contacts). *For any actually specified closed densely defined response operator B , the bounded pair*

$$b = (I + B^* B)^{-1/2}, \quad d = Bb$$

records its complete graph. In particular,

$$u \in D(B) \iff \sup_{\varepsilon > 0} \|d(b + \varepsilon I)^{-1}u\| < \infty, \quad Bu = \lim_{\varepsilon \downarrow 0} d(b + \varepsilon I)^{-1}u.$$

The bounded encodings of any jointly admitted collection, their adjoints and actual ordered bounded operations define one positive kernel on the original Hilbert space. They are not identified with physical unbounded source derivatives without calibration.

For the common forms q_s of Theorem 21.1, their full resolvent derivatives are calibrated to the actual forms. With one fixed $R = (H + c)^{-1}$,

$$R_i = -Rq_iR, \quad R_{ij} = Rq_iRq_jR + Rq_jRq_iR - Rq_{ij}R. \quad (347)$$

Every product uses the continuous form maps $V \rightarrow V'$ and extended resolvents $V' \rightarrow V$. For $u, v \in D$ and fixed $x = (H + c)u$, $y = (H + c)v$,

$$q_i(u, v) = -\langle x, R_i y \rangle, \quad q_{ij}(u, v) = -\langle x, R_{ij} y \rangle + q_i(u, Rq_j v) + q_j(u, Rq_i v).$$

Hence dense anchored preparation data for Rq_iR determine the continuous source form, not only its value on one chosen state.

Proof. The map $w \mapsto (bw, dw)$ is an isometry onto the graph of B . If the displayed approximants are bounded, their first components $b(b + \varepsilon)^{-1}u$ converge strongly to u . A weakly convergent subsequence of their second components and weak closedness of the graph prove $u \in D(B)$. Spectral calculus for b on such a graph vector then proves strong convergence to Bu . The converse is the same calculus. Every Gram matrix of actual bounded words is positive; retaining the anchors and the joint words fixes their common cyclic representation, not an unseen orthogonal summand.

For the second part write $E = K^{-1/2}$ and represent the form by a bounded coercive matrix L_s on the fixed reference energy space. Then $R_s(z) = E(L_s - zE^2)^{-1}E$. Differentiating this ordinary bounded inverse proves (347), including its contact term. Pairing with x, y proves the reverse formulas. Since D is form dense, the pairings determine the whole continuous form. One selected preparation alone does not determine it: a bounded operator annihilating that preparation already illustrates the missing information. $\square$

Theorem 21.11 (Matrix unfiltering with all finite source jets). *Let $A = H + c \geq I$. Initially allow an original insertion vector $f_i \in V'$ and set $v_i = A^{-1}f_i \in V$. The complete matrix spectral measure $\mu_{ij}(B) = \langle v_i, E_A(B)v_j \rangle$ determines the original local matrix measure $d\nu_{ij}(t) = t^2 d\mu_{ij}(t)$ and the form-resolvent channel*

$$F_f(z) = z^2 F_v(z) + zG + M_1, \quad G_{ij} = \langle v_i, v_j \rangle, \quad (M_1)_{ij} = \langle A^{1/2}v_i, A^{1/2}v_j \rangle. \quad (348)$$

Here $F_v(z) = \int (t - z)^{-1} d\mu(t)$ and $F_f(z) = \int t^2 (t - z)^{-1} d\mu(t)$. The exact Hilbert-vector criterion is

$$f_i \in \mathcal{H} \iff v_i \in D(A) \iff \int t^2 d\mu_{ii}(t) < \infty.$$

It holds for every original native derivative port in (340).

More generally, suppose A_s is a C^k common-form family with $\alpha K \leq A_s \leq \beta K$ and f_s is C^k as an $\mathcal{H}$ vector. Then the rational cutoff satisfies

$$Q_\Lambda(s)f_s \longrightarrow f_s \quad \text{in } C^k \text{ on compact source sets,} \quad Q_\Lambda(s) = \Lambda(A_s + \Lambda)^{-1}.$$

All preparation and mixed coefficient contacts are differentiated. This does not assert operator-norm convergence $Q_\Lambda \rightarrow I$.

Proof. The energy norm gives $\int t d\mu_{ii} < \infty$. Polarization and spectral Cauchy–Schwarz justify every matrix entry and the form-resolvent integral. Polynomial division $t^2/(t - z) = t + z + z^2/(t - z)$ proves (348); the two polynomial terms cannot be dropped. The spectral characterization of $D(A)$ proves the criterion. Theorem 21.1 supplies it for the stated native ports.

For the finite-jet claim put $R_\Lambda = (A_s + \Lambda)^{-1}$. For every nonempty labelled index set I , bounded form-inverse differentiation gives

$$(Q_\Lambda)_I = \Lambda \sum_{(B_1, \dots, B_m) \in \text{OP}(I)} (-1)^m R_\Lambda q_{B_1} R_\Lambda \cdots q_{B_m} R_\Lambda,$$

where $\text{OP}(I)$ consists of ordered partitions into nonempty blocks. Write $C_B = \|q_B\|_{V \rightarrow V'}$. Spectral calculus gives

$$\|R_\Lambda\|_{\mathcal{H} \rightarrow V}, \|R_\Lambda\|_{V' \rightarrow \mathcal{H}} \leq (2\sqrt{\alpha\Lambda})^{-1}, \quad \|R_\Lambda\|_{V' \rightarrow V} \leq \alpha^{-1}.$$

Each displayed word has $\mathcal{H} \rightarrow \mathcal{H}$ bound $\prod_r C_{B_r}/(4\alpha^m)$, uniformly in Λ . On $V \rightarrow \mathcal{H}$ it has the stronger bound $\sqrt{\beta} \prod_r C_{B_r}/(2\alpha^m \sqrt{\Lambda})$, since the innermost resolvent on V is at most $\sqrt{\beta/\alpha}\Lambda^{-1}$. Similarly $(Q_\Lambda - I) : V \rightarrow \mathcal{H}$ is at most $\sqrt{\beta}/(2\sqrt{\Lambda})$. Density proves strong convergence to zero for every nonempty derivative, uniformly on compact sets of input vectors. Each continuous f_J has compact image on a compact parameter set. The labelled product rule $(Q_\Lambda f)_I = \sum_{J \subset I} (Q_\Lambda)_J f_{I \setminus J}$ therefore proves the assertion. It supplies no rate for arbitrary $\mathcal{H}$ -valued jets; uniformly V -bounded jets have the displayed $O(\Lambda^{-1/2})$ rate. $\square$

For a fixed admitted boundary isometry a_D , Hilbert recovery also gives $a_D^* e^{-itH} f_i = \lim_\Lambda a_D^* e^{-itH} f_{i,\Lambda}$, uniformly in t . A form-dual insertion instead needs an energy-regular boundary for that pairing. If the filtered germ already has an actual meromorphic continuation, (348) transports its complete germ. At a nonzero simple pole p its residue is multiplied by p^2 ; at $p = 0$ the zero of this multiplier is an explicit exception. This is an energy resolvent coordinate, not automatically invariant mass squared. A spectral filter with zeros, a removed ground-state component, or unretained local renormalization contacts cannot be recovered by this formula. Likewise none of these static finite-jet results substitutes for the actual propagator-domain hypotheses of a time-dependent unbounded insertion.

21.8 A broken quark current with a nonzero interacting channel

The same full source gives a concrete nonconserved-current boundary, complementary to its scalar channels. Assume at least one quark Yukawa matrix is nonzero. Choose a real diagonal traceless generation matrix T and define a finite-CAR charge

$$Q_T = d\Gamma(A_T), \quad A_T|_Q = -T, \quad A_T|_{uc} = A_T|_{dc} = -T^T, \quad A_T|_L = A_T|_{ec} = 0.$$

Identity factors on spin, color, weak and site labels are implicit. These are the all-left conjugate-Weyl signs, not the signs of an untransformed right-handed field.

Theorem 21.12 (Broken-current boundary above any finite energy). *There is such a T with $z_r = -i\{T, Y_r\}$ nonzero for $r = u$ or d . The bounded physical charge preserves D , and its literal source derivative is the unbounded physical multiplier*

$$D_T = i[H, Q_T] = a^{-1} \sum_x N_x Y_x(z_u, z_d, 0) : D \rightarrow \mathcal{H}. \quad (349)$$

For every finite L the same Hamiltonian has physical eigenvectors ψ_m, ψ_n with

$$L < E_m < E_n, \quad \langle \psi_n, D_T \psi_m \rangle = i(E_n - E_m) \langle \psi_n, Q_T \psi_m \rangle \neq 0.$$

For the selected lower preparation put $K_m = H - E_m$, $E_+ = \mathbf{1}_{(0, \infty)}(K_m)$, $q = E_+ Q_T \psi_m$ and $d = E_+ D_T \psi_m = iK_m q$. The independently charge-generated space

$$\mathcal{H}_D = \overline{\text{span}}\{f(K_m)q : f \text{ bounded Borel}\}$$

reduces the actual dynamics and contains the nonzero breaking vector d . Its full current/breaking spectral matrix is

$$dM(\omega) = \begin{pmatrix} 1 & i\omega \\ -i\omega & \omega^2 \end{pmatrix} d\mu(\omega), \quad d\mu = \langle q, E_{K_m}(d\omega)q \rangle.$$

For $G_{11}(z) = \int (\omega - z)^{-1} d\mu$, the other resolvent entries are

$$G_{12} = i(\mu_0 + zG_{11}), \quad G_{21} = -i(\mu_0 + zG_{11}), \quad G_{22} = \mu_1 + z\mu_0 + z^2G_{11}, \quad \mu_j = \int \omega^j d\mu. \quad (350)$$

These are actual native excitation channels, not identified hadron masses or a massless conserved-QCD boundary.

Proof. The forward CAR convention gives, for $U_\alpha = e^{i\alpha Q_T}$, $U_\alpha^* H(Y) U_\alpha = H(Y(\alpha))$ with $Y_r(\alpha) = e^{-i\alpha T} Y_r e^{-i\alpha T}$. Every non-Yukawa term is generation blind and commutes with Q_T , including the full shift and spin terms. The charge commutes with Gauss, T_0 and R_4 , and a safe bound is $\|Q_T\| \leq 24N \operatorname{tr} |T|$. Differentiation in $\mathcal{L}(D, \mathcal{H})$ proves (349). For a nonzero matrix entry choose a permutation of $\operatorname{diag}(1, -1, 0)/2$ with $t_i + t_j \neq 0$; its anticommutator entry is nonzero. Thus $\kappa_z = 6\|z_u\|_F^2 + 6\|z_d\|_F^2 > 0$.

Take the normalized physical core vector $\psi_R = 1_{\text{links}} \otimes (\pi R^2)^{-N} e^{-|q|^2/(2R^2)} \otimes |0\rangle$. The pair identity gives

$$\|D_T \psi_R\|^2 = \frac{2\kappa_z R^2}{a^2} \sum_x N_x^2.$$

This proves unboundedness on the physical space, not by using colored nonphysical states. Since H has compact resolvent, the projection $P_L = \mathbf{1}_{(-\infty, L]}(H)$ has finite rank and its eigenvectors span a graph core. If all D_T matrix elements above L vanished, then on that core $D_T = P_L D_T + D_T P_L - P_L D_T P_L$. Symmetry identifies the first term with $(D_T P_L)^*$; the right side has a bounded finite-rank extension. Graph density extends this equality to D , contradicting the displayed norm. The commutator excludes equal energies; Hermiticity allows the pair to be ordered. This proves the high-energy transition statement.

The charge preserves D , so $q \in D(K_m)$ and $d = iK_m q \neq 0$. The truncated functions $i\omega \mathbf{1}_{[0, R]}(\omega)$ show that $d \in \mathcal{H}_D$. Bounded spectral multipliers make this space reducing. The vector identity fixes every entry of its positive matrix measure; polynomial division gives (350) with its finite moments. Compact resolvent supplies at least one nonzero native residue, but the whole regular spectral integral remains part of the channel. The relative frequency K_m has not replaced H as the physical generator. $\square$

This also gives an actual constituent-to-current attachment of the type used in the color source construction [5]. Let O_ℓ be the finite list of literal real and imaginary quark Yukawa insertions, with their original Higgs, weak, spin and color contractions. Put $V_C = \bigoplus_\ell \operatorname{End}(\mathbb{C}^3)$, with its Hilbert–Schmidt inner product, and let P_C be the componentwise conjugation-invariant trace projection, $X \mapsto (\operatorname{Tr} X/3)I_3$. The vectors $\tau_\ell = I_3/\sqrt{3}$ in their respective slots form an orthonormal basis of $P_C V_C$. On that invariant subspace define $j_C(\sum c_\ell \tau_\ell) = \sum c_\ell O_\ell \psi_m$. It is bounded by $(\sum \|O_\ell \psi_m\|^2)^{1/2}$ and the actual source direction fixes x_T with $j_C x_T = D_T \psi_m$. If a_D is the inclusion of the charge-generated boundary, then

$$\|a_D^* e^{-itH} j_C P_C x_T\| = \|d\| > 0.$$

The preparation is the declared stationary excited state, not a vacuum renamed for this purpose. No individual colored constituent has been embedded as a physical state.

Corollary 21.13 (Same-history endpoints and local nonzero persistence). *Use this fixed initial preparation in any admitted graph-continuous full source history through its initial static member, with no additional generation-dependent non-Yukawa term. Then $\psi(t) = U(t, t_0)\psi_m$ and $D_T(t)\psi(t)$ are graph and Hilbert continuous, respectively, and*

$$\int_s^t U(t, r)D_T(r)\psi(r) dr = Q_T\psi(t) - U(t, s)Q_T\psi(s).$$

The integral belongs to the independently specified current-history span $\overline{\text{span}}\{U(t, r)Q_T\psi(r) : r \in I\}$. For $t = t_0 + h$, $s = t_0$, it divided by h converges in $\mathcal{H}$ to $D_T(t_0)\psi_m \neq 0$.

Proof. The common-domain propagator permits differentiating $U(t, r)Q_T\psi(r)$ on D . Its derivative is $U(t, r)i[H(r), Q_T]\psi(r)$; graph continuity makes it a Bochner-integrable Hilbert vector. Integration proves the endpoints and continuity proves the last limit. Therefore its norm is at least $h\|D_T(t_0)\psi_m\|/2$ for sufficiently small positive h . For a moving charge one must add $\dot{Q}$; source differentiation likewise retains $i[H_i, Q] + i[H, Q_i]$ and both mixed cross terms. None is inferred from a derivative of an arbitrarily chosen eigenvector. $\square$

Because the initial transition can lie above any finite energy threshold, the same preparation can meet the finite lower-energy requirement of the constraint-matched native Einstein construction when its other admitted geometric and reference hypotheses hold. Its history then has the nonzero current endpoint just proved. This is genuine overlap of a source and preparation, not equality of that native family with the directly occupied continuum packet. The geometric construction still supplies its own all-test stress and constraint comparison.

The results of this section retain interactions in a finite-source reconstruction: canonical physical neutral letters, a signed common fermionic history, normalized dressed inclusions with nonzero leakage, positive complementary feedback, exact nested observations, and calibrated original insertion vectors. They neither require suppressing the Yukawa coupling nor supply an unstated all-scale spatial state limit. A separate coefficient path that softens interactions can illustrate a spectral mechanism, but is not substituted for this retained full-source chain.

22 Canonical comparison of invariant physical composites

The ground-state estimates identify actual response vectors on the regulated physical source. Their comparison with a physical interface should use observations that preserve that source. We now construct the joint kernel of bounded invariant Higgs, gauge and matter composites, identify its canonical cyclic realization, and retain the actual time generator through its source labels and resolvents. This implements the invariant-correlation reconstruction of BCRS–2EW/H on the selected native cluster source [4, Sec. 12].

22.1 Bounded composites on the joint gauge quotient

Fix the coefficient sequence, spatial prescriptions and one common ground cluster state from Sections 13 and 14. At each regulator the following are actual bounded invariant observations:

$$\begin{aligned} b_x &= \frac{|q_x|^2}{1 + |q_x|^2}, & n_x &= \frac{N_{F,x}}{90}, \\ w_\gamma &= \frac{1}{d_\sigma} \text{Tr } \sigma(U_\gamma), \\ c_{xy} &= \frac{q_x^\dagger \rho_H(U_{xy}) q_y}{\sqrt{1 + |q_x|^2} \sqrt{1 + |q_y|^2}}. \end{aligned} \tag{351}$$

The four real coordinates q_x are here recombined as the same complex Higgs doublet. There are 90 local one-particle fermion modes, so $0 \leq n_x \leq 1$. Wilson paths, their representation and any spatial averaging prescription are retained as part of their comparison. Real and imaginary parts, adjoints, finite products and actual neutral CAR contractions may be included with their specified input incidences.

Lemma 22.1 (Bounded mixed physical generators). *The operators in (351) are parity even and gauge invariant, and their norms are at most one. The mixed edge observation uses the full joint source, with no independent color bundle or gauge-fixed Higgs field.*

Proof. The radial multiplier is invariant under the unitary Higgs action, and local fermion number commutes with the internal representation. Unitarity gives $|\text{Tr } \sigma(U_\gamma)| \leq d_\sigma$; endpoint conjugation leaves a loop trace unchanged. For the mixed edge, Cauchy–Schwarz gives

$$|q_x^\dagger \rho_H(U_{xy}) q_y| \leq |q_x| |q_y|,$$

which proves the stated bound. Under a local gauge transformation,

$$q_x \mapsto \rho_H(k_x) q_x, \quad U_{xy} \mapsto k_x U_{xy} k_y^{-1}, \quad q_y \mapsto \rho_H(k_y) q_y.$$

All endpoint factors in the numerator cancel, and the denominators are invariant. The representation σ and the Higgs representation descend to the same $G_0/\mathbb{Z}_6$ by Lemma 12.1. A bare color fundamental is not substituted for a representation of this quotient. $\square$

The use of complete joint representations agrees with the color contraction and joint-descent results of BCRS–2C [5, Sec. 13]. Those results distinguish an actual invariant operation with its spectator inputs from a merely formal color contraction. Here the displayed operations are already represented on the common physical Hilbert space.

For a dynamical interface first apply the common time-smearing operation of Proposition 13.4 to the selected bounded sequences. Let $\mathfrak{A}$ be the unital norm-closed algebra generated by their quotient classes and all real-time translates. It is a subalgebra of the uniformly time-continuous physical quotient. Retain one family of source-word labels $\{A_\lambda\}_{\lambda \in \Lambda}$, containing the identity, products, adjoints and these time translates, whose linear span is norm dense in $\mathfrak{A}$. The word labels include the prescribed spatial comparisons and repeated uses of the same input. No countability or fixed maximum word length is assumed.

22.2 The joint physical kernel

Let $(\pi, \mathcal{H}_{\text{as}}, \Omega)$ be the physical GNS representation of the selected common cluster state ω . Define

$$\begin{aligned} K(\lambda, \mu) &= \omega(A_\lambda^* A_\mu), \\ \mathcal{H}_{\mathfrak{A}} &= \overline{\text{span}\{\pi(A_\lambda)\Omega : \lambda \in \Lambda\}}. \end{aligned} \tag{352}$$

The compared correlation family consists of these actual invariant word observations. In particular, a Higgs–gauge–matter cross entry is evaluated in the same state as every marginal entry.

Theorem 22.2 (Canonical invariant-composite reconstruction). *The kernel (352) constructs a Hilbert space and bounded source-word generators canonically unitarily equivalent to their cyclic realization on $\mathcal{H}_{\mathfrak{A}}$. Any second cyclic realization with the same full labelled kernel and source operation incidences has a unique intertwining unitary. This unitary preserves all admitted mixed word correlations.*

Proof. For every finite coefficient family and $F = \sum_{\lambda} c_{\lambda} A_{\lambda}$,

$$\sum_{\lambda, \mu} \overline{c_{\lambda}} c_{\mu} K(\lambda, \mu) = \omega(F^* F) \geq 0.$$

Thus K is positive because ω is the already constructed common state. Form the finite formal word space with this sesquilinear form, divide by its null space and complete it to $\mathcal{H}_K$. Every actual linear source relation has zero norm. For a retained generator a with $\|a\| \leq M_a$, the C^* -order inequality gives

$$\omega(F^* a^* a F) \leq M_a^2 \omega(F^* F). \quad (353)$$

Left concatenation therefore preserves the null space and extends to a bounded operator on $\mathcal{H}_K$. This verifies the source generator bound without supplying a second state or an independent positivity assumption.

The map on formal words

$$e_{\lambda} \mapsto \pi(A_{\lambda})\Omega$$

has exactly the inner products $K(\lambda, \mu)$. It descends to an isometry with dense range in $\mathcal{H}_{\mathfrak{A}}$, hence to a unitary. Concatenation proves intertwining. The same construction for a second realization gives its unique labelled unitary; uniqueness follows on the dense word span. All mixed inner products are entries of this one kernel, proving the final assertion. $\square$

This is precisely the positive-kernel and invariant-correlation reconstruction mechanism of BCRS–2EW/H [4, Sec. 12, positive-kernel realization and invariant-correlation descent]. Bounded invariant words have the whole Hilbert space as their operator domain. An undressed gauge-variant Higgs field need not act on that space. Additional current, charged or asymptotic ports enter a larger interface only with their actual cross-kernels, domains and boundary data; they are not recovered from an omitted label by setting its value to zero.

22.3 Time evolution and full generator domains

Theorem 22.3 (Dynamical and resolvent attachment). *The space $\mathcal{H}_{\mathfrak{A}}$ reduces the positive-energy implementing group of Theorem 13.5. Its restriction has a self-adjoint nonnegative generator $L_{\mathfrak{A}}$. A comparison preserving the full kernel and the real-time source labels intertwines this generator and its complete domain.*

Alternatively, for two self-adjoint physical source generators, a labelled unitary intertwining one actual nonreal resolvent intertwines their full domains and generators. Such a resolvent may be retained as a bounded label with its actual joint source kernel; a formal Hamiltonian expression is not a substitute.

Proof. Since the vacuum is fixed,

$$U(t)\pi(A)\Omega = \pi(\alpha_t(A))\Omega.$$

The algebra contains all real-time translates. Thus $\mathcal{H}_{\mathfrak{A}}$ is invariant under both $U(t)$ and $U(-t)$, so it is reducing. Strong continuity and nonnegativity restrict to this space. On the dense word span a label-preserving kernel unitary V obeys $VU_1(t) = U_2(t)V$ for every real t . The characterization of the generator by strong differentiation then gives $VD(L_1) = D(L_2)$ and $VL_1 = L_2V$ there.

For the alternative, let $z \notin \mathbb{R}$ and suppose

$$V(L_1 - z)^{-1} = (L_2 - z)^{-1}V.$$

The range of a nonreal resolvent is exactly the full operator domain. Hence $VD(L_1) = D(L_2)$; inversion on that range gives the generator equality. The joint-kernel theorem constructs such

a unitary when the actual resolvent and its word incidences belong to the compared bounded interface. These are the closed-domain data used in the source reconstruction theorem [4, Sec. 12, closed-operator reconstruction]. $\square$

An implementing unitary or resolvent need not be an element of the original observable algebra. When required as a comparison label, it is an actual bounded operator in the represented source with its joint matrix elements retained. This enlarges the compared interface, not the physical state or its Hilbert representation. Static word values with the time labels and their action forgotten do not by themselves identify an external Hamiltonian.

22.4 The analytic channel of the native cluster source

For $A \in \mathfrak{A}$, put

$$\psi_A = (\pi(A) - \omega(A)I)\Omega, \quad G_A(z) = \langle \psi_A, (L_{\mathfrak{A}} - z)^{-1} \psi_A \rangle. \quad (354)$$

Proposition 22.4 (Positive physical-sheet response). *The channel (354) is the Stieltjes transform of the actual positive measure $\mu_A(\Delta) = \|\mathbf{1}_{\Delta}(L_{\mathfrak{A}})\psi_A\|^2$ on $[0, \infty)$. It is holomorphic on $\mathbb{C} \setminus [0, \infty)$ and satisfies*

$$|G_A(z)| \leq \frac{\|\psi_A\|^2}{\text{dist}(z, [0, \infty))}. \quad (355)$$

It is identically zero exactly when $\psi_A = 0$. The comparisons in Theorem 22.3 preserve this measure and channel for every matched preparation.

Proof. The spectral theorem gives $G_A(z) = \int (E - z)^{-1} d\mu_A(E)$ and the bound, with local dominated differentiation proving holomorphy. If $\psi_A \neq 0$, then for every $s > 0$

$$G_A(-s) = \int \frac{d\mu_A(E)}{E + s} > 0.$$

The converse is immediate. A unitary intertwining the generator intertwines all its spectral projections, and hence the measures and channels of matched vectors. $\square$

Centering removes the component along the chosen vacuum, not necessarily every zero-energy vector. The proposition therefore asserts support in $[0, \infty)$, not automatically in $(0, \infty)$. At every finite regulator the full positive-energy projection and pole are instead supplied by Proposition 14.4.

A time filter with Fourier transform nonzero on the finite positive spectral support, such as a Gaussian, preserves a nonzero response at each regulator. Equations (229) and (230) show why it need not preserve a nonzero finite-energy response after refinement. Once a continuum channel is identified, a nonphysical-sheet continuation and its current residue are further data; reconstruction preserves them when acquired, but does not create their poles [4, Secs. 6 and 12].

22.5 A source-relative criterion for causal comparison

The same joint kernel gives a precise physical locality test. Let π_{cyc} be the representation on the cyclic Hilbert space of the entire compared bounded physical algebra. In the criterion below, the admitted words range over a norm-dense word span of that entire algebra. Thus, if the comparison is restricted to $\mathfrak{A}$, both its operators and the assertion of vanishing are restricted to $\mathcal{H}_{\mathfrak{A}}$; no conclusion on a larger ambient Hilbert space is inferred from those words alone. Let A_n, B_n be actual uniformly bounded even physical comparison sequences, with the time-evolved

sequence admitted by the source, and write $K_n = [\alpha_t^n(A_n), B_n]$. Let D range over a dense cyclic word family for the same represented physical algebra containing $[K_n]$. For a fixed cluster state,

$$\pi_{\text{cyc}}([K_n]) = 0 \iff \lim_{\mathcal{U}} \omega_n(D_n^* K_n^* K_n D_n) = 0 \quad (356)$$

for every such word D .

Indeed each quantity on the right is $\|\pi_{\text{cyc}}([K_n])\pi_{\text{cyc}}(D)\Omega\|^2$. The word vectors are dense in this cyclic Hilbert space, and $[K_n]$ is bounded, which proves both implications. A vacuum-only test does not suffice. When the source comparisons identify the two supports as spacelike, (356) is exactly the commutation assertion on that physical source. It requires neither an operator-norm propagation estimate on every unrelated state nor a raw charged-field representation. The ground moment bounds alone do not control its arbitrary cyclic sibling words.

The resulting interface contains actual bounded invariant composites, one joint state, their mixed correlations, and the acquired positive-energy dynamics. It is canonically reconstructed on that source. Identifying this native cluster realization with the original physical target requires the corresponding current and charged-sector attachments, confined or asymptotic color comparisons, and the nonzero physical-frequency estimates. These comparisons must retain the same source and domains, as in the simultaneous projection theorem of BCRS-3 [3, Sec. 8]; separate sector realizations do not provide their common intersection. The original-source realization in Theorem 11.5 is independently established; the present criterion concerns its comparison with this native cluster.

Part III

Metric response and coupled continuation

The same native quantum source supplies all-component metric response and a source-attached Einstein evolution. The final homogeneous example exhibits the coupling in a simpler smoothed model; its approximation is not substituted for the literal-Hamiltonian construction.

23 Metric response from the common quantum source

Metric response is obtained by varying the same Hamiltonian that generates the quantum evolution. The all-ray construction below identifies the differentiability and domain structure needed to extend bounded source reconstruction to arbitrary metric tests. Its object is a differentiable family of response forms, including its comparison maps, rather than an independently assigned stress tensor.

The operational metric-source construction [25, Definitions 3.7 and 4.4, Proposition 4.5] identifies a derivative with a source only when its graph is the source's actual metric-incidence relation. The normalized-ray reconstruction [26, Definitions 4.1–4.5 and Theorems 4.6 and 4.11] then polarizes the common response into a form-valued distribution. We first establish the required quadratic structure directly in the occupied regulator.

23.1 All-ray response of the full regulated Hamiltonian

At a fixed finite graph, denote the complete physical Hamiltonian by H_θ . Its independent real coefficient entries include the electric, scalar quadratic and quartic, plaquette, scalar hopping, fermion hopping and Yukawa coefficients. Complex coefficients are resolved into their real and imaginary entries before differentiating. The representations, contractions and physical gauge

action are fixed. Restrict to an open coefficient neighborhood on which the quartic coefficient is at least $\lambda_0 > 0$, the electric coefficients are bounded away from zero, and all coefficients are locally bounded. Let $\mathcal{V}$ be the physical form domain with reference norm

$$\|u\|_{\mathcal{V}}^2 = \left\langle u, (T_{G,\text{ref}} + T_H + \lambda_0 \sum_x |q_x|^4 + 1)u \right\rangle, \quad (357)$$

where the right side is understood as a closed quadratic form. The coordinate $q_x \in \mathbb{R}^4$ is the scalar coordinate at that fixed regulator; this statement involves no change of the physical clock.

Proposition 23.1 (Common coefficient-response form). *The forms $\mathbf{q}_\theta$ of H_θ have the common domain $\mathcal{V}$, with locally equivalent shifted form norms. As continuous sesquilinear forms on $\mathcal{V}$, they depend affinely and continuously on the independent coefficient entries. For every nonzero $\psi \in \mathcal{V}$, the readout*

$$R_{[\psi]}(\theta) = \frac{\mathbf{q}_\theta(\psi, \psi)}{\|\psi\|^2}$$

has response

$$W_{[\psi]}(v) = \frac{D\mathbf{q}_\theta[v](\psi, \psi)}{\|\psi\|^2}.$$

Its norm-squared homogeneous extension satisfies the parallelogram identity. Its unique polarization is the differentiated form $D\mathbf{q}_\theta[v]$.

Proof. The quartic coefficient derivative is bounded by $\lambda_0^{-1} \|u\|_{\mathcal{V}}^2$. With n sites,

$$\sum_x |q_x|^2 \leq \frac{1}{2} \sum_x |q_x|^4 + \frac{n}{2}, \quad \sum_{\langle xy \rangle} |q_x - \rho_H(U_{xy})q_y|^2 \leq 12 \sum_x |q_x|^2$$

on the cubic graph. The latter estimate follows from unitarity and the six incidences at each site. Plaquette and fermion hopping terms are bounded at the fixed graph. Each real Yukawa-coefficient derivative is matrix-valued multiplication bounded by a constant times $\sum_x |q_x|$, which is controlled by the same form norm. The electric forms are controlled by their positive reference coefficients. Polarization, or Cauchy–Schwarz for the corresponding positive majorants, gives the mixed form bounds. The lower-order terms are infinitesimally controlled by the positive quartic form; the strictly positive kinetic coefficients give the asserted local equivalence of shifted norms. Consequently the finite sum defining $\mathbf{q}_\theta$ is an affine continuous map into the forms on $\mathcal{V}$. Differentiation at fixed ψ gives the stated response. Sesquilinearity gives phase independence, degree-two homogeneity, the parallelogram identity and polarization. No derivative of a chosen ground vector is used. $\square$

For the fixed local gauge action, $\mathbf{q}_\theta(Uu, Uv) = \mathbf{q}_\theta(u, v)$ at every coefficient point. Differentiating proves gauge covariance of the response forms. They are available on every admitted ray of any physical word interface preserving $\mathcal{V}$; arbitrary bounded operators are not presumed to preserve this domain. These conclusions concern actual coefficient deformations. A general metric tangent cannot be substituted for a coefficient tangent without constructing its action on the same scalar, gauge, spin and current data.

23.2 Differentiable metric families and operator sources

Write $\mathcal{T}_K$ for the real smooth symmetric metric tests supported in a fixed compact set K . Their usual smooth topologies, compatible as K increases, give the compact-support test topology. A metric family is compared on one dense locally convex domain $\mathcal{D}$; its continuous anti-dual is denoted by $\mathcal{D}'$. Inner products and forms are conjugate linear in the first entry.

Theorem 23.2 (All-ray construction from an actual metric form). *Suppose a source owns finite metric-deformation readouts*

$$R_{[\psi]}(g) = \frac{\mathbf{q}_g(\psi, \psi)}{\|\psi\|^2}, \quad 0 \neq \psi \in \mathcal{D},$$

where $\mathbf{q}_g$ is Hermitian and differentiable on each compact test step, compatibly across supports. Let its derivative $\mathbf{a}_g(h)(\varphi, \psi)$ be jointly continuous in the declared domain and test topologies. These hypotheses construct one continuous quadratic response on all normalized rays.

For its attachment as a metric source, retain the exact coverage of the source's metric-response protocols by these deformations, one boundary, normalization, improvement and state-transport convention, and the already established upstream response role. Retain also the lawful transport of metrics, readouts and incidences and the invariance of $\mathcal{D}$ under the symmetry and field operations admitted by this target. Then the operational response and its unique polarized distribution are

$$W_{[\psi]}(h) = \frac{\mathbf{a}_g(h)(\psi, \psi)}{\|\psi\|^2}, \quad \langle \varphi, \hat{T}_g(h)\psi \rangle = 2\mathbf{a}_g(h)(\varphi, \psi), \quad \hat{T}_g(h) : \mathcal{D} \longrightarrow \mathcal{D}'. \quad (358)$$

The support, covariance and Ward or exchange identities of the actual diagonal response lift to all matrix elements. The mass-bearing subtype uses the same upstream mass role; differentiability alone does not assign that role.

Proof. Differentiation gives a real-linear continuous response to every admitted metric test. All rays use the same prequotient form and the same representative conventions. Exact protocol coverage identifies the derivative graph with the source's actual metric incidence graph; thus the operational attachment theorem of [25] applies, rather than merely identifying two formulas. Its convention is $W(h) = \frac{1}{2}\langle T, h \rangle$. The homogeneous diagonal tensor response is therefore $2\mathbf{a}_g(h)(\psi, \psi)$. Sesquilinearity proves its parallelogram identity and identifies its polarization uniquely with $2\mathbf{a}_g(h)$. For fixed ψ , this form is a continuous anti-linear functional of φ , and hence defines $\hat{T}_g(h)\psi \in \mathcal{D}'$, linearly in ψ . Joint continuity gives the form-valued distribution in (358). Polarizing each actual diagonal identity proves its matrix-element version, as in [26]. This constructs no Hilbert-bounded stress operator: continuity on a graph domain is the asserted regularity. $\square$

The distinction between the theorem's two conclusions is important. A differentiable form constructs the all-ray algebra automatically. Its full source attachment additionally uses the actual protocol, role and transport records. None of these records may be replaced by a name assigned to a derivative.

The homogeneous family (415) supplies a concrete application on $\mathcal{V} = D(H_0^{1/2})$:

$$\begin{aligned} \partial_a \mathbf{q}_{a,\phi}(u, v) &= h'(a) \mathbf{q}_{H_0}(u, v) + g \partial_a f(a, \phi) \langle u, Bv \rangle, \\ \partial_\phi \mathbf{q}_{a,\phi}(u, v) &= g \partial_\phi f(a, \phi) \langle u, Bv \rangle. \end{aligned} \quad (359)$$

The lapse and scale-factor source forms are consequently $H_q/(V_\Sigma a^3)$ and $-\partial_a H_q/(3V_\Sigma a^2)$. Their all-ray lift requires no independently postulated stress tensor. Their exchange is already proved in (422)–(423). This metric family is homogeneous; it does not cover all inhomogeneous spacetime tests.

23.3 The derivative information carried by source comparison

Proposition 23.3 (Recovery from a resolvent first jet). *For a differentiable common closed semibounded form family, let $R_g(z) = (H_g - z)^{-1}$, with $z \notin \mathbb{R}$. On its form-domain/anti-dual scale,*

$$DR_g(z)[h] = -R_g(z) \mathbf{a}_g(h) R_g(z). \quad (360)$$

For $u, v \in D(H_g)$,

$$\mathbf{a}_g(h)(u, v) = -\langle (H_g - \bar{z})u, DR_g(z)[h](H_g - z)v \rangle. \quad (361)$$

Thus the source resolvent family and its metric first jet, with the common form-domain comparison, determine the response form.

Proof. The form of $H_g - z$ is an invertible map from its form domain to the anti-dual. Differentiating its inverse gives (360). Multiplying on the dense operator-domain images gives (361); the conjugate on z comes from the first inner-product slot. The operator domain is a form core, so form continuity extends this identity to the common form domain. $\square$

The domain comparison in this proposition is not inferred from norm differentiability of one bounded resolvent. Nor do fixed-metric data determine the derivative. Indeed, the constant family H and $H + sB$, with any existing nonzero bounded invariant self-adjoint B , have identical kernels and resolvents at $s = 0$, but derivatives zero and B . This is an insufficiency test for fixed-metric information, not an assertion that both families are lawful metric sources.

Even a unitary comparison at every metric requires its derivative. For bounded representatives with $F'_g = U_g F_g U_g^*$, put $A_h = (D_h U_g) U_g^*$. Direct differentiation yields

$$D_h F'_g = U_g (D_h F_g) U_g^* + [A_h, F'_g]. \quad (362)$$

The covariant derivative subtracts this comparison commutator. For unbounded forms the corresponding graph or form transport must also be retained. The joint-kernel comparison of [4] therefore transports response derivatives when its labels and ports include the actual deformation family. Unrelated identifications of its individual fibers do not suffice. This is also the fixed-sibling metric-variation requirement of the color response construction [5].

23.4 Uniform first jets in the asymptotic source

Proposition 23.4 (Derivative through the bounded source quotient). *Let $F_n(g)$ be actual bounded self-adjoint representatives of one metric-deformation protocol family, defining bounded sequences throughout the compared neighborhood. On each compact test step, suppose real-linear derivatives $D_n(h)$ satisfy*

$$\begin{aligned} \sup_n \|D_n(h)\| &\leq C_K p_K(h), \\ \sup_n \|F_n(g+v) - F_n(g) - D_n(v)\| &= o(p_K(v)), \end{aligned} \quad (363)$$

in the stated differentiability topology, for a fixed continuous seminorm p_K . Retain the same comparisons and source relations for the representatives, derivatives and their words. Then the bounded asymptotic source family is differentiable with derivative $[D_n(h)]$. In every representation its polarized all-ray response is $2\pi([D_n(h)])$.

Proof. The quotient norm is at most the supremum norm. Thus the uniform remainder in (363) proves differentiability directly in the bounded sequence quotient. Its derivative bound passes to the quotient and to every representation. Evaluation on a normalized ray is continuous and quadratic; polarization gives the asserted operator. An additional source identity passes when it holds as a relation of this same quotient family. $\square$

For the positive-energy, uniformly time-continuous source algebra, the family must belong to that same closed sequence subalgebra throughout the neighborhood. Its uniform norm derivative then belongs to the subalgebra by closure. Without this hypothesis the proposition concerns the full bounded quotient only. For unbounded response forms, the analogous passage requires a

common source-word graph domain and estimates in its all-word graph seminorms; a bound on ground-state expectations alone does not give that passage.

The complete classical gauge–Higgs–Weyl–Yukawa action in [27, Theorems W4B.2 and W4B.4] provides the torsion-free spin lift, derived-connection variation, symmetric source and internal-exchange cancellation that a full metric family must retain. The constructions above give its quantum all-ray and comparison mechanisms. Sections 24 and 25 construct the finite source’s spatial and temporal metric families; Section 26 establishes the stronger native response and coupled evolution. Any limiting comparison retains its corresponding uniform estimates. These constructions do not identify an arbitrary coefficient variation, or a copied classical tensor, with the full quantum metric response.

24 Spatial geometry and momentum response of the interacting source

A metric response must vary the geometry seen by the same scalar, gauge and spin variables. We construct such a family on the physical space of Section 12. It includes arbitrary spatial shear, not only a scale factor, and retains all Yukawa contractions. The canonical matter action of [27, Theorems W4B.2 and W4B.4] fixes the coframe, half-density and derived-connection ancestry of these terms. Its spacetime Ward identities are distinguished below from the identities of this finite spatial representation.

24.1 Anisotropic electric, scalar and magnetic contractions

Fix a geometrically matched cubical member of the original family, with reference spatial metric $h_0 = \delta$ on its coordinate patch, and fix its graph placement, edge paths, plaquette anchors, spin structure and quadrature. Local metric variations extend as the identity outside the patch. The same formulas apply on complete interior stars or the corresponding periodic member. An absent edge at a boundary is not replaced by a new variable. All reference couplings and the faithful representation σ of the joint quotient group remain fixed.

Let h be positive definite and $N > 0$, and put $v = \sqrt{\det h}$. For a positive matrix define

$$\text{Corr}(M)_{ij} = \frac{M_{ij}}{\sqrt{M_{ii}M_{jj}}}.$$

This positive matrix has unit diagonal. For two-form indices $I = (i < j)$, $J = (k < l)$, put

$$B(h)_{IJ} = h^{ik}h^{jl} - h^{il}h^{jk}.$$

It is the positive Gram matrix of h^{-1} on two-forms. An overline denotes the fixed endpoint average on an edge or the average over the four distinct vertices of an elementary plaquette. Each averaged object includes its star anchor x . Define

$$\begin{aligned} m_i^E &= \overline{h_{ii}}/v_{e_i}, & \Gamma^E &= \text{Corr}(h(x)), \\ m_i^H &= \overline{vh^{ii}}_{e_i}, & \Gamma^H &= \text{Corr}(h^{-1}(x)), \\ m_I^B &= \overline{vB(h)_{II}}_{p_I}, & \Gamma^B &= \text{Corr}(B(h(x))). \end{aligned}$$

For $X = E, H, B$, put $M^X = \text{diag}(\sqrt{m^X})$, $\theta_E = \theta_H = 1/2$, and $\theta_B = 1/4$. Write $\bar{N}_i^X$ for the corresponding edge or plaquette lapse average. On the open geometric branch

$$\lambda_{\min}(\Gamma^X) > 1 - \theta_X \quad (X = E, H, B), \quad (364)$$

define the lapse-affine matrices

$$C^X = M^X \left[N_x (\Gamma^X - (1 - \theta_X) I) + \text{diag}(\bar{N}_i^X - \theta_X N_x) \right] M^X. \quad (365)$$

The first matrix in brackets is positive definite, while $\bar{N}_i^X - \theta_X N_x$ is the positive sum of the other averaged lapse samples. Thus C^X is positive definite. Its diagonal is exactly $m_i^X \bar{N}_i^X$; its off-diagonal entries are $N_x \sqrt{m_i^X m_j^X} \Gamma_{ij}^X$. Every conformal point has $\Gamma^X = I$, so (364) contains an open neighborhood of that point and admits every spatial tangent for sufficiently small deformation. These are explicit geometric sampling rules, not a uniqueness theorem for discretization. Exact lapse affinity does not by itself establish the spacetime constraint algebra.

Write $E_{x,i}^a$ for the outgoing self-adjoint electric derivatives, including the square roots of their original positive coefficients. For each internal factor their Lie indices transform in the adjoint at the common initial vertex. Define

$$D_{x,i} = \sqrt{\kappa_{e_i}}(q_x - \rho_H(U_{e_i})q_y), \quad Z_{x,I} = \sqrt{\beta_{p_I}/2}(\sigma(U_{p_I}) - I).$$

The plaquettes are based at that same vertex. The three geometric terms are the form and multipliers

$$\begin{aligned} q_E(u, u) &= \sum_{x,a,i,j} C_{ij}^E(x) \langle E_{x,i}^a u, E_{x,j}^a u \rangle, \\ W_H &= \sum_{x,i,j} C_{ij}^H(x) D_{x,i}^\dagger D_{x,j}, \\ W_B &= \sum_{x,I,J} C_{IJ}^B(x) \text{Tr}(Z_{x,I}^\dagger Z_{x,J}). \end{aligned} \quad (366)$$

Positive Gram matrices make all three nonnegative. The electric operator is the symmetric operator of the displayed form; invariant derivatives on a single non-Abelian link are not presumed to commute. At $N = 1, h = \delta$ the matrices are identity. Since

$$\|\sigma(U) - I\|_{\text{HS}}^2 = 2(\dim \sigma - \text{Re Tr } \sigma(U)),$$

the magnetic term is exactly the original Wilson potential, not a logarithmic replacement. On a conformal metric $h = e^{2s}\delta$, the same prescription gives precisely the diagonal weights

$$\bar{N}_e \overline{e^{-s}}_e, \quad \bar{N}_e \overline{e^s}_e, \quad \bar{N}_p \overline{e^{-s}}_p \quad (367)$$

for electric, Higgs-edge and Wilson terms respectively. For smooth classical geometries and matched refining quadrature, the matrices approach Nh/v , Nvh^{-1} and $Nv\wedge^2 h^{-1}$. This coefficient consistency is not an estimate on rough quantum link configurations.

24.2 The derived spin connection

Choose a smooth spatial coframe section $e(h)$ in the fixed reference frame. Every symmetric tangent k has a lift b satisfying

$$k_{ij} = b_i^a e_j^a + e_i^a b_j^a; \quad b_i^a = \frac{1}{2} e^{aj} k_{ji}$$

is one such lift. The derivative of the selected coframe section differs, if necessary, by its determined infinitesimal frame rotation. In particular all six spatial tensor directions are covered. The intrinsic connection and its spin lift are derived, not varied independently:

$$\begin{aligned} \omega_i^a{}_b &= e_j^a (\partial_i e_b^j + \Gamma_{i\ell}^j(h) e_b^\ell), & \Omega_i &= \frac{1}{4} \omega_{iab} \gamma^{ab}, \\ \delta \Gamma_{i\ell}^j &= \frac{1}{2} h^{jm} (\nabla_i k_{\ell m} + \nabla_\ell k_{im} - \nabla_m k_{i\ell}). \end{aligned} \quad (368)$$

Here $\delta e^{-1} = -e^{-1}(\delta e)e^{-1}$, $c^i = c_a^i \sigma^a$, and $\delta v = \frac{1}{2} v h^{ij} k_{ij}$. For the fixed path from y to x , spin transport satisfies

$$\delta V_{xy} = - \int_0^1 V(1, t) \delta \Omega_i(\gamma(t)) \dot{\gamma}^i(t) V(t, 0) dt. \quad (369)$$

Our orientation gives $V_{xy} = I + a_i \Omega_i(x) + O(a_i^2)$ when $y = x + a_i e_i$. The normal spin hopping coefficient is

$$S_{xy}(h) = \frac{-i\kappa_R}{4a_i} \left(\sqrt{v_x/v_y} c_x^i V_{xy} + \sqrt{v_y/v_x} V_{xy} c_y^i \right). \quad (370)$$

It is paired with the reverse adjoint and the original internal link in the finite CAR hopping. Multiplication by $\bar{N}_e$ is the discrete lapse anticommutator. Thus the scalar density, endpoint Clifford matrices and spin path all participate in the metric derivative. A simultaneous frame change transports these objects by second quantization and preserves the spin epsilon in every Yukawa contraction.

The complete normal source is now

$$H[N, h] = \sum_x N_x (v_x^{-1} |p_x|^2 + v_x V_x + Y_x) + T_E[C^E] + W_H[C^H] + W_B[C^B] + F_{\text{hop}}[N, h], \quad (371)$$

where $V_x = \lambda |q_x|^4 + \mu |q_x|^2$. The scalar coordinates and fermion half-density identification are fixed in this static family. Its Yukawa density consequently has lapse weight N_x and no extra spatial volume factor. Zero explicit shear derivative of that term does not remove its interaction from the source or state.

Theorem 24.1 (Occupied full-spatial response). *On a fixed finite graph, in a bounded nondegenerate geometric neighborhood with a uniform positive margin in (364), (371) has the same nonzero physical Hilbert space and the common form domain*

$$\mathcal{V} = D(T_{\text{ref}}^{1/2}) \cap L^2((1 + \sum_x |q_x|^4) dU dq; \mathcal{F}).$$

Its shifted form norms are locally uniformly equivalent. The forms are affine in the lapse, continuously differentiable in the spatial geometry and, on each compact test support,

$$|Dq_{N,h}[n, k](u, w)| \leq C_{\mathcal{G}, K} (\|n\|_{C^0} + \|k\|_{C^1}) \|u\|_{\mathcal{V}} \|w\|_{\mathcal{V}}. \quad (372)$$

They define self-adjoint, semibounded physical sources with compact resolvent and finite heat trace at every positive inverse temperature. Every compactly supported spatial symmetric-metric test is incident to an actual sufficiently small deformation within this open branch.

Proof. The geometric neighborhood gives positive lower and finite upper bounds for the three Gram matrices, N_x/v_x , and all quartic coefficients. The electric and scalar kinetic forms and positive quartic multiplication therefore control exactly the reference domain. The Higgs term is a nonnegative quadratic multiplier, the magnetic and CAR terms are bounded, and the Yukawa multiplier is linear. Young's inequality gives a locally uniform coercive lower bound and infinitesimal control of the lower-order terms. These prove closedness and equivalence of shifted form norms. All contractions in (366) are at one gauge vertex; the original CAR and Yukawa incidences are unchanged. The same physical projection consequently reduces the source.

Positive-matrix operations and their finite samples are smooth. The derived connection uses the first spatial metric jet, and (369) bounds its transport variation by the corresponding path integral. Taylor remainders in a compact geometric neighborhood prove form-norm differentiation and (372). Reference-form equivalence preserves compact embedding and the harmonic-oscillator/compact-link heat comparison of the original source. Its nonzero physical Gaussian and Fock-vacuum vector still exists. Finally $h + \epsilon k$ is positive for sufficiently small ϵ , and its coframe lift above supplies the claimed deformation for every test; the uniform correlation margin also persists for sufficiently small ϵ . All estimates here are at a fixed graph. $\square$

For fixed spatial geometry, the lapse coefficients give actual local forms $\varepsilon_z[h] = \partial_{N_z} H[N, h]$ with $H[N, h] = \sum_z N_z \varepsilon_z[h]$. They include all terms of the source rather than a separately supplied stress expression. On the common smooth/Schwartz physical core let $J_{zw} = i[\varepsilon_z, \varepsilon_w] = -J_{wz}$. The exact initial form derivative under the static normal dynamics is

$$\left. \frac{d}{dt} \right|_{t=0} \langle e^{-itH} u, \varepsilon_z e^{-itH} u \rangle = - \sum_w N_w \langle u, J_{zw} u \rangle.$$

The displayed differential expressions preserve that core, which lies in every operator-power domain. Thus the derivative is legitimate in the common form envelope. Weighting by N_z and summing cancels every antisymmetric transfer exactly.

Proposition 24.2 (Nonzero physical shear response). *At the flat reference and on a complete star with two nonzero Higgs edge coefficients, a compactly supported pure off-diagonal tangent has a nonscalar physical response form.*

Proof. Choose $k_{12} = k_{21} = s$, with zero diagonal and a nonzero sample at one star anchor, and zero samples at other anchors. A smooth compact test with these values exists. Its volume and diagonal coefficient derivatives vanish, whereas $\delta \text{Corr}(h)_{12} = s$ and $\delta \text{Corr}(h^{-1})_{12} = -s$. The Higgs response contains $-2s \text{Re}(D_{x,1}^\dagger D_{x,2})$. Use physical product vectors constant in the links, radial in each Higgs variable and in the Fock vacuum. Radial symmetry gives

$$\langle \text{Re}(D_{x,1}^\dagger D_{x,2}) \rangle = \sqrt{\kappa_{e_1} \kappa_{e_2}} \langle |q_x|^2 \rangle.$$

Normalized smooth radial annular vectors make the last moment arbitrarily large while retaining bounded other-site moments. The electric term vanishes on link constants, the magnetic response is bounded, and hopping has zero vacuum expectation. Thus the response has unbounded nonconstant diagonal values on physical vectors. It is neither zero nor a scalar form. $\square$

The protocol readouts are $q_{N,h}(u, u)/\|u\|^2$. Their linearization is the derivative just constructed, including the fixed state identification and repeated protocol uses. Theorem 23.2 therefore polarizes their spatial response. Coordinate or frame comparison transports the graph, paths, quadrature and coframe section together with the metric. This is full spatial tensor-type coverage, not injectivity of a finite measurement. A test whose relevant jets vanish along every sampled path can have zero response. The resulting form-valued distribution may be supported on vertices and paths; a continuum smooth stress tensor is not inferred.

24.3 Canonical momentum probes on the same form domain

The common action distinguishes raw covariant momentum from its gauge- and Lorentz-improved spatial generator [27, equations (83)–(92)]. Both this distinction and the fixed real/complex canonical normalization are retained here. Let $P_{x,j}^a$ be the original unrescaled outgoing electric generators. Define a bounded adjoint curvature observation by

$$\mathcal{F}_{x,ij} = \sigma_*^{-1} \text{pr}_{\sigma_*(\mathfrak{g})} \left(\frac{\sigma(U_{p_{ij}}) - \sigma(U_{p_{ij}})^\dagger}{2a_i a_j} \right). \quad (373)$$

The projection uses the fixed invariant inner product; the inverse is only on the faithful Lie-algebra image. Opposite two-form orientation is defined by antisymmetry. This observation is adjoint-covariant and does not replace the Wilson Hamiltonian. Put $\mathcal{D}_i q_x = (\rho_H(U_{x,x+a_i e_i}) q_{x+a_i e_i} - q_x)/a_i$. For a real spatial descriptor ξ , symmetric ordering gives

$$\begin{aligned} P_G[\xi] &= \frac{1}{2} \sum_{x,i,j,a} \xi_x^i (\mathcal{F}_{x,ij}^a P_{x,j}^a + P_{x,j}^a \mathcal{F}_{x,ij}^a), \\ P_H[\xi] &= \frac{1}{2} \sum_{x,i,A} \xi_x^i (p_{x,A} (\mathcal{D}_i q_x)_A + (\mathcal{D}_i q_x)_A p_{x,A}). \end{aligned} \quad (374)$$

Explicitly, with $H = (q_1 + iq_2, q_3 + iq_4)^\top$, use the canonical complex momentum row $\Pi = (p_1 - ip_2, p_3 - ip_4)/2$. Its classical one-form is $\Pi dH + (dH)^\dagger \Pi^\dagger = \sum_A p_A dq_A$. For the displayed forward link difference the same-site real Jacobian is $-I/a_i$; the cross-imaginary ordering commutators therefore vanish. Thus $\Pi \mathcal{D}_i H + (\mathcal{D}_i H)^\dagger \Pi^\dagger$ is precisely the real symmetrized scalar expression above. The CAR momentum uses the original internal link, the reverse adjoint, and spin coefficient

$$S_{xy}^{\text{mom}}[\xi, h] = \frac{-i\bar{\xi}_e^i}{4a_i} \left(\sqrt{v_x/v_y} + \sqrt{v_y/v_x} \right) V_{xy}(h).$$

Summing over the original species and generations defines the bounded operator $P_F[\xi, h]$. Its smooth-profile leading expression is $-i(\xi^i(\partial_i + \Omega_i + \rho_R(A_i)) + \frac{1}{2}\partial_i \xi^i)$ in the fixed spin half-density coordinates. The full family is

$$H[N, h, \xi] = H[N, h] + P_G[\xi] + P_H[\xi] + P_F[\xi, h]. \quad (375)$$

Every normal Yukawa term is retained. Its local scalar density has no separate raw derivative-momentum contribution.

Theorem 24.3 (Occupied normal and momentum family). *For bounded finite descriptor families, (375) defines semibounded closed physical forms on the same $\mathcal{V}$, with compact resolvent, finite heat trace, nonzero physical ground space and self-adjoint unitary dynamics. Its descriptor derivative is exactly $P_G + P_H + P_F$; its joint geometric derivative is a continuous form on that domain.*

Proof. At a fixed graph the curvature multipliers and their group derivatives are bounded. The real-part interpretation of the symmetric gauge form gives

$$|P_G[\xi](u, u)| \leq \epsilon q_E(u, u) + C_{\epsilon, \xi} \|u\|^2.$$

Scalar differences are linear in the finite Higgs coordinates, with unitary bounded link coefficients. Consequently

$$\begin{aligned} |P_H[\xi](u, u)| &\leq C_\xi \|\nabla_q u\| (\|q|u\| + \|u\|) \\ &\leq \epsilon \|\nabla_q u\|^2 + \epsilon \|q|^2 u\|^2 + C_{\epsilon, \xi} \|u\|^2. \end{aligned}$$

The last step uses Young's inequality and $|q|^2 \leq \delta |q|^4 + C_\delta$. The finite CAR term is bounded. Thus the shift form is infinitesimally controlled by the normal coercive form, even for a fixed large finite descriptor. Closedness, compactness and heat comparison follow as above. All terms are parity even and invariant under the same joint gauge action. Differentiation uses the same estimates and derived spin transport, proving the remaining assertions. $\square$

24.4 The exact spatial-deformation discrepancy

Proposition 24.4 (Two-edge discrepancy of the finite momentum). *On a coordinate cycle with at least four distinct vertices, the momentum prescription (374) does not obey the continuum spatial-descriptor bracket as an identity on its common core. Its commutator contains a two-edge scalar principal term absent from every one-edge descriptor of the same prescription.*

Proof. At trivial links, the underlying vector field on one real Higgs component is $(v_\xi q)_x = \xi_x(q_{x+1} - q_x)/a$. The coefficient of q_{x+2} in its bracket is

$$[v_\xi, v_\eta]q_x|_{q_{x+2}} = \frac{\eta_x \xi_{x+1} - \xi_x \eta_{x+1}}{a^2} q_{x+2}.$$

Smooth descriptors can realize samples with nonzero coefficient. Symmetric ordering changes only the zero-order divergence term. The plaquette curvature vanishes at trivial links, so the

gauge momentum's link vector field is zero there. CAR momentum has no scalar-coordinate vector component.

For an invariant physical-core witness, use only the first coordinate direction and choose $\xi_x = 1, \xi_{x+1} = 0, \eta_x = 0, \eta_{x+1} = 1$. Set every outgoing nearest neighbor of x equal to $q_x \neq 0$, but choose q_{x+2e_1} so that $q_x \cdot (q_{x+2e_1} - q_x) \neq 0$. For $r_x = |q_x|^2$,

$$[v_\xi, v_\eta]r_x = -\frac{2}{q^2}q_x \cdot (q_{x+2e_1} - q_x) \neq 0, \quad v_\zeta r_x = 0$$

for every full three-component one-edge descriptor ζ . Local gauge action annihilates r_x . Take a link-constant core vector $F(r_x)$, times compact radial cutoffs at all vertices and the Fock vacuum, with $F(r_0) = 0, F'(r_0) \neq 0$, and the cutoffs constant near the tested configuration. This vector is gauge invariant. Its vanishing value eliminates zero-order ordering terms at that configuration; the other cutoffs contribute no derivative there. Number-preserving CAR momentum annihilates the vacuum. The operator discrepancy on this physical vector consequently has the displayed nonzero value. Its output is smooth and hence nonzero on an open neighborhood, so it is a nonzero physical Hilbert vector, not a measure-zero point test. This disproves the proposed descriptor identity on the physical common core. $\square$

This discrepancy concerns the displayed finite prescription. It does not exclude its controlled refinement, a different action, or a transported graph construction. It gives an explicit term that a continuum comparison must estimate on the actual quantum source, beyond a Taylor expansion on smooth classical profiles.

The distinction matters for spacetime attachment. The common tetrad action in [27, equations (65), (67) and (91)] derives its temporal spin connection and shifts the coframe momentum under the fermion half-density transformation. It also distinguishes raw $V[\xi]$ from $D[\xi] = V[\xi] - G[\iota_\xi A] - J[\iota_\xi \omega]$. An external metric history must retain the corresponding temporal spin/frame comparison. Spatial transport alone is not that comparison. The three momentum responses above therefore are not renamed a completed T^{0i} source.

Finite ordered products of these self-adjoint unitary groups do give actual piecewise constant operational histories on the common physical space. Each factor preserves its form domain, the same set $\mathcal{V}$. This does not identify metric jumps with the covariant action of an arbitrary smooth spacetime history. For a transported finite graph, geometric naturality also retains the graph variation:

$$\delta_g I[\mathcal{L}_X g] + \delta_{\text{fields}} I[\mathcal{L}_X \text{fields}] + \delta_{\text{graph}} I[\mathcal{L}_X \text{graph}] = 0.$$

Dropping its third term while holding the graph fixed requires a proof. Polarization transports an established Ward identity; it cannot remove this actual exchange. This spatial construction supplies lapse response, every spatial tensor test and all raw momentum probes. The action-derived temporal comparison is developed in Section 25; the coupled native Einstein realization is proved in Section 26. Neither construction equates the finite one-edge momentum algebra with its continuum counterpart.

25 Temporal frames and occupied metric histories

The spatial metric family of Section 24 extends to an actual quantum history when its temporal frame comparison is retained. We derive that comparison from the symmetric Weyl action, then construct evolution for the full interacting source. The result covers all components of an external spacetime metric on the declared patch. It is distinct from imposing gravitational field equations on that history.

We use inner products antilinear in the first variable and the Schrödinger convention $i\partial_t u = H(t)u$. The spatial target is closed, or carries the inherited boundary domain on which the

integrations by parts below are licensed. The tetrad and spin connection have the common-action ancestry of [27, Theorems W4B.2 and W4B.4, equations (65), (67), (70)–(79)]. In particular the connection is the Levi–Civita lift of the same tetrad, not an independently varied Palatini variable.

25.1 The canonical half-density identity

For one chiral representation, write the current part of the local action as

$$L_W = \frac{i}{2} \{ \psi^\dagger J^\mu \partial_\mu \psi - (\partial_\mu \psi^\dagger) J^\mu \psi + \psi^\dagger (J^\mu \Omega_\mu - \Omega_\mu^\dagger J^\mu) \psi \}. \quad (376)$$

The current-density matrices J^μ are Hermitian. Choose the time orientation and chiral identification so that $B = J^0 > 0$. The connection includes the derived spin connection and any retained anti-Hermitian internal connection. Yukawa contractions remain separate algebraic terms of the same action. Put

$$K = B^{-1/2}, \quad \psi = K\chi, \quad a^\mu = KJ^\mu K, \quad \Gamma_\mu = K^{-1}(\partial_\mu K + \Omega_\mu K). \quad (377)$$

Lemma 25.1 (Exact canonical pullback). *One has $a^0 = I$, and the action density becomes*

$$L_W = \frac{i}{2} \chi^\dagger \overleftrightarrow{\partial}_t \chi + \frac{i}{2} \chi^\dagger a^i \overleftrightarrow{\partial}_i \chi + \frac{i}{2} \chi^\dagger \sum_{\mu=0}^3 (a^\mu \Gamma_\mu - \Gamma_\mu^\dagger a^\mu) \chi. \quad (378)$$

Here $\chi^\dagger \overleftrightarrow{D} \chi = \chi^\dagger D\chi - (D\chi)^\dagger \chi$, with the intervening matrix retained in each term when present. The corresponding canonical one-particle differential expression is

$$h_W = -i \left(a^i \partial_i + \frac{1}{2} \partial_i a^i \right) - \frac{i}{2} \sum_{\mu=0}^3 (a^\mu \Gamma_\mu - \Gamma_\mu^\dagger a^\mu). \quad (379)$$

Proof. Substitution of $\psi = K\chi$, with $K = K^\dagger$, gives $KJ^\mu K$ as the coefficient of each differentiated χ . The remaining coefficient is

$$\begin{aligned} KJ^\mu \partial_\mu K - (\partial_\mu K)J^\mu K + KJ^\mu \Omega_\mu K - K\Omega_\mu^\dagger J^\mu K \\ = a^\mu \Gamma_\mu - \Gamma_\mu^\dagger a^\mu. \end{aligned}$$

This proves (378), including the derivatives of the field identification. Integration by parts on the specified spatial domain gives (379). The last coefficient is anti-Hermitian, so its contribution to the Hamiltonian expression is Hermitian. $\square$

This is an exact action identity, not an essential-self-adjointness claim for an arbitrary continuum differential expression. The quantum realization constructed below is the full finite source already defined in Section 12. Its physical comparisons retain K , the temporal connection and the state ports, rather than differentiating only the spatial current matrices.

25.2 The derived temporal spin term

Use time gauge on the inherited local slab:

$$\begin{aligned} \theta^0 &= N dt, & \theta^a &= e_i^a (dx^i + \beta^i dt), & h_{ij} &= e_i^a e_j^a, \\ g &= -N^2 dt^2 + h_{ij} (dx^i + \beta^i dt)(dx^j + \beta^j dt), & v &= \sqrt{\det h}, & N &> 0. \end{aligned} \quad (380)$$

The selected coframe section $e = e(h)$ and its residual frame comparisons are retained. The dual normal is $N^{-1}(\partial_t - \beta^i \partial_i)$. For the actual chiral normal symbol c^i , equal to $e_a^i \sigma^a$ for one selected Weyl sign, (377) reads

$$J^0 = vI, \quad K = v^{-1/2}I, \quad a^i = Nc^i - \beta^i I, \quad \Gamma_\mu = \Omega_\mu - \frac{1}{2}\partial_\mu(\log v)I. \quad (381)$$

Thus $\chi = v^{1/2}\psi = h^{1/4}\psi$. Every real scalar density term in Γ_μ cancels in the antisymmetrized connection coefficient. This does not remove the divergence term $\partial_i a^i/2$ in the canonical expression.

Let skew $M = (M - M^\top)/2$, and lower the shift by $\beta_i = h_{ij}\beta^j$. Denote the intrinsic rotational spin connection by Ω_i^{intr} .

Lemma 25.2 (Temporal Levi–Civita rotation). *The rotational temporal connection of (380) is*

$$\omega_{0ab}^{\text{rot}} = -\text{skew}(\dot{e}e^{-1})_{ab} + \frac{1}{2}e_a^\ell e_b^k (\partial_k \beta_\ell - \partial_\ell \beta_k), \quad \Omega_0^{\text{rot}} = \frac{1}{4}\omega_{0ab}^{\text{rot}}\gamma^{ab}|_{\text{chiral}}. \quad (382)$$

In temporal internal gauge the complete canonical expression is

$$\begin{aligned} h_F^{\text{geom}} &= \frac{1}{2}\{N, D_h^{\text{cov}}\} - K_\beta^{\text{cov}} - i\Omega_0^{\text{rot}}, \\ D_h^{\text{cov}} &= -i\left(c^i \partial_i + \frac{1}{2}\partial_i c^i + \frac{1}{2}\{c^i, \Omega_i^{\text{intr}}\} + c^i A_i\right), \\ K_\beta^{\text{cov}} &= -i\left(\beta^i (\partial_i + \Omega_i^{\text{intr}} + A_i) + \frac{1}{2}\partial_i \beta^i\right), \end{aligned} \quad (383)$$

where A_i acts in the retained internal representation. There is no additional volume-dilation or trace-extrinsic-curvature potential in these symmetrized canonical half-density coordinates.

Proof. For the inverse metric in (380), $g^{j0} + \beta^j g^{00} = 0$ and $g^{j\ell} + \beta^j g^{0\ell} = h^{j\ell}$. Hence

$$\Gamma_{0k}^j + \beta^j \Gamma_{0k}^0 = \frac{1}{2}h^{j\ell}(\dot{h}_{\ell k} + \partial_k \beta_\ell - \partial_\ell \beta_k). \quad (384)$$

Substitute this identity into the tetrad-postulate formula of [27, equation (65)]. The inverse-frame derivative contributes $-M$, where $M = \dot{e}e^{-1}$, and the $\dot{h}$ term contributes $(M + M^\top)/2$. The remaining antisymmetric shift term is exactly (382). The analogous spatial Christoffel identity gives the intrinsic rotational connection already used in Section 24.

In the positive chiral time-current inner product, the rotational spin part is anti-Hermitian and the boost part is Hermitian. The spatial boost coefficients contract the symmetric second fundamental form

$$\mathbf{K}_{ij} = \frac{1}{2N}(\dot{h}_{ij} - \mathcal{L}_\beta h_{ij}).$$

In frame indices their contraction with $\sum_i [c^i, \Omega_i^{\text{boost}}]$ is therefore zero, because the Pauli commutator is antisymmetric. The scalar shift part $-\beta^i I$ adds no boost commutator. The temporal boost cancels in $\Omega_0 - \Omega_0^\dagger$, while the real density terms cancel by (381). Expanding the remaining rotational and internal terms in (379) gives (383). $\square$

For zero shift the temporal coefficient is $-\text{skew}(\dot{e}e^{-1})$, and need not vanish even in the symmetric coframe section. For example, let $e(t) = R(\theta(t)) \text{diag}(a, b, c) R(\theta(t))^\top$, where $a, b, c > 0$, $a \neq b$, and R is the standard rotation in the first two coordinates. At $\theta = 0$,

$$\omega_{012}^{\text{rot}} = -\frac{\dot{\theta}(a-b)^2}{2ab}. \quad (385)$$

Indeed $\dot{e}_{12} = \dot{\theta}(a-b)$, so $M_{12} - M_{21} = \dot{\theta}(a-b)^2/(ab)$. A smooth spatially varying angular history supported in the patch supplies an inhomogeneous example. Taking its parameters and deformation in the declared neighborhood preserves the positive correlation margins.

As a separate local sign check, at $h = I, N = 1, e = I$ take $\beta = \varpi(-y, x, 0)$ in a coordinate patch. Then $\omega_{012}^{\text{rot}} = -\varpi$, the spin term is $-\varpi S_z$, and the principal shift term is $+i\beta \cdot \partial = -\varpi L_z$. Their sum is the rotating-coordinate generator $-\varpi(L_z + S_z)$. A cutoff may be placed outside the tested patch; no global rotating-boundary claim is used.

The coframe-momentum shift in the full primary gravitational phase space is the cotangent counterpart of this field identification [27, equation (67)]. For an external metric history its time-dependent state connection must still be retained. It is not a second independent connection variable. In a different temporal internal gauge, the full canonical Gauss multiplier and simultaneous gauge-unitary comparison must be transported together.

25.3 Geometric shift and the explicit CAR convention

The previously defined P_H, P_G, P_F remain the operators of (374)–(375). In particular $P_F[\xi, h]$ has the explicit smooth-profile coefficient $-i(\xi^i D_i + \partial_i \xi^i/2)$. Its relation to a geometric shift must be determined from the action, not from its name. Write $\mathcal{D}_\xi = \xi^i \partial_i + \partial_i \xi^i/2$.

Lemma 25.3 (Relative shift sign). *With the positive time-current action and the annihilation-column convention of the full regulator, the geometric shift contributes $+P_H[\beta] + P_G[\beta] - P_F[\beta, h]$, together with the temporal spin term. A change of CAR polarization or charge-conjugate field coordinates alone does not reverse this relative sign.*

Proof. The reduced odd bracket in [27, equation (79)] is $\{\Psi, \Psi^\dagger\}_D = -i\delta$. The even graded Leibniz rule gives, for $Q(h) = \int \Psi^\dagger h \Psi$,

$$\{\Psi, Q(h)\}_D = -ih\Psi.$$

Thus the conventional-arrow expression $-\frac{i}{2} \int \xi^i \Psi^\dagger \overleftrightarrow{\partial}_i \Psi = Q(-i\mathcal{D}_\xi)$ acts by $-\mathcal{D}_\xi \Psi$. In contrast the scalar canonical momentum $\int p \xi^i \partial_i q$ acts by $+\xi^i \partial_i q$. Right differentiation is already accounted for in the reduced bracket and does not supply a second sign.

The direct Legendre calculation fixes the geometric sign. At flat spatial metric, the Weyl Lagrangian has spatial coefficient $c^i - \beta^i I$, by the dual normal in (380). Subtracting the Lagrangian from the right-odd kinetic one-form therefore gives

$$H_{F,\beta}^{\text{action}} = +\frac{i}{2} \int \beta^i \Psi^\dagger \overleftrightarrow{\partial}_i \Psi = Q(+i\mathcal{D}_\beta).$$

Its field transport is $+\mathcal{D}_\beta \Psi$. The scalar and gauge Legendre maps contain respectively $D_0 H - \beta^i D_i H$ and $F_{0j} - \beta^i F_{ij}$, and give the plus scalar and gauge raw terms [27, equations (83)–(92)]. The potential and Yukawa contractions have no separate derivative-shift momentum.

For finite CAR generators a_i ,

$$[a, a^\dagger h a] = h a, \quad a^\top h a^{\dagger \text{col}} = \text{Tr } h - a^\dagger h^\top a.$$

The first identity reproduces $\dot{a} = -iha$. The second shows that exchanging exterior multiplication and contraction changes both the field identification and the coefficient to $-h^\top$. For the flat centered shift matrix, $\mathcal{D}_\xi$ is real antisymmetric; thus $h = -i\mathcal{D}_\xi$ has $h^\top = -h$ and zero trace, and the coefficient is unchanged. For charge conjugation $b = S a^{\dagger \text{col}}$, the coefficient becomes $-S h^\top S^\dagger$, with its separately recorded scalar trace. The shift has spin symbol I , so either shift sign is again preserved. In contrast the normal Pauli symbol uses $S \sigma^{a^\top} S^\dagger = -\sigma^a$ and transforms with the chiral representation. Transporting that representation also transports its internal and Yukawa incidences; it cannot change only the shift sign. No infinite-dimensional equivalence of Fock polarizations is used. $\square$

The primary holomorphic seed uses exterior multiplication for Ψ and contraction for $\Psi^\dagger$, and records the opposite polarization separately [11, canonical seed and Dirac–Yukawa operator construction]. The finite calculation above makes the comparison independent of that naming convention. For comparison with the compact notation of the source, equations (87)–(88) of [27] do not separately define their arrow or α^i relative to the expanded positive-current action. The possible dictionary

$$\overleftrightarrow{D}_{\text{WDW}} = -\overleftrightarrow{D}_{\text{here}}, \quad \alpha_{\text{WDW}}^i = -c^i$$

reconciles both the normal and raw shorthand displays with that action. It is a possible convention completion, not a convention verified from those displays. We use the expanded action to fix the convention. The present P_F , in contrast, is defined by explicit finite matrices; its geometric comparison in Lemma 25.3 is unambiguous.

25.4 The full physical history on one form domain

Keep the graph, paths, quadrature, spin structure, internal quotient, three generation copies, boundary and all original couplings fixed. Let $H[N, h]$ be precisely the lapse-affine normal family (371). Define a new family through its same reference source by

$$\begin{aligned} H_g(t) &= H[N(t), h(t)] + P_H[\beta(t)] + P_G[\beta(t)] - P_F[\beta(t), h(t)] + B_0[e(t), \beta(t)], \\ B_0[e, \beta] &= \sum_{x, R, r} d\Gamma_{x, R, r}(-i\Omega_0^{\text{rot}}(e, \beta; x)). \end{aligned} \quad (386)$$

The last coefficient is (382) in each actual species representation. Every electric, scalar, magnetic, Higgs-edge, hopping and Yukawa term remains in the source. At $N = 1, h = \delta, \beta = 0, \dot{e} = 0$ the original full regulator is recovered exactly. For $\beta = 0$ this is the temporal-frame extension of the normal family. It does not replace the distinct raw family (375).

The normal discretization retains its matched-smooth-profile and coefficient scope from Section 24. The new canonical signs and temporal coefficients assert neither equality with a continuum action on rough quantum configurations nor a calibration of arbitrary reference propagation coefficients.

Theorem 25.4 (Occupied external-metric history). *On a fixed finite graph and compact time interval I , let $g(t)$ be a smooth history (380) in a bounded nondegenerate geometric neighborhood with uniform positive margins in (364). Keep the coframe section and the source records just specified. Assume the required history jets are bounded so that the coefficients of (386) are continuously differentiable in time; smooth histories bounded through the relevant second jets suffice. The forms q_t of (386) have the common physical domain*

$$\mathcal{V} = D(T_{\text{ref}}^{1/2}) \cap L^2((1 + \sum_x |q_x|^4) dU dq; \mathcal{F}) \cap \mathcal{H}_{\text{phys}}. \quad (387)$$

Their shifted form norms are uniformly equivalent on the specified history neighborhood, and q_t is C^1 into $\mathcal{L}(\mathcal{V}, \mathcal{V}')$.

There is a strongly continuous unitary propagator $U_g(t, s)$ on $\mathcal{H}_{\text{phys}}$, preserving $\mathcal{V}$ with a locally uniform form-norm bound. Initial vectors in $\mathcal{V}$ solve $i\dot{u} = q_t u$ in $\mathcal{V}'$. Every nonzero physical initial vector remains nonzero.

Proof. Theorem 24.1 supplies the common normal coercive form. The scalar momentum is infinitesimally bounded by the scalar kinetic and quartic form, and the gauge momentum is a bounded smooth curvature multiplier times a first link derivative; the estimates in Theorem 24.3 apply unchanged. The finite CAR momentum is bounded. On a finite exterior algebra over a d -dimensional one-particle space, $\|d\Gamma(b)\| \leq d\|b\|$; hence B_0 and its time derivative are bounded on the stated history neighborhood. Every term commutes with the same internal local gauge

action and is parity even. In particular the temporal spin term acts on the existing spin factors and preserves every Yukawa epsilon tensor. The same Haar projection reduces the forms, and the nonzero physical Gaussian–Fock-vacuum preparation remains an admitted initial vector, not an asserted stationary interacting solution.

Young’s inequality and the uniform normal coercivity give a constant c for which $q_t(u, u) + c \|u\|^2$ is positive and uniformly equivalent to $\|u\|_{\mathcal{V}}^2$. Smooth geometric coefficients, the derived spatial transport and (382) give

$$|\dot{q}_t(u, w)| \leq C \|u\|_{\mathcal{V}} \|w\|_{\mathcal{V}}. \quad (388)$$

The compact embedding $\mathcal{V} \hookrightarrow \mathcal{H}_{\text{phys}}$ is the unchanged finite-graph kinetic/quartic compactness.

To construct evolution, choose the eigenbasis of a positive reference compact-resolvent form operator on the physical space. Its spectral projections P_m are uniformly bounded on $\mathcal{V}$ and converge there to the identity. Solve the finite-dimensional Hermitian Schrödinger equation on $P_m \mathcal{H}_{\text{phys}}$ with initial data $P_m u_s$, $u_s \in \mathcal{V}$. Its norm is conserved. Hermiticity also cancels the dynamical contributions to the form energy, giving

$$\frac{d}{dt}(q_t(u_m, u_m) + c \|u_m\|^2) = \dot{q}_t(u_m, u_m). \quad (389)$$

Gronwall and (388) give a uniform $L^\infty(I; \mathcal{V})$ bound. The projected equation gives a uniform $L^\infty(I; \mathcal{V}')$ bound on $\dot{u}_m$. The Gelfand-triple inequality then yields

$$\|u_m(t) - u_m(r)\|_{\mathcal{H}}^2 \leq \|u_m(t) - u_m(r)\|_{\mathcal{V}} \|u_m(t) - u_m(r)\|_{\mathcal{V}'} \leq C|t - r|. \quad (390)$$

Compact embedding and this equicontinuity give a subsequence converging strongly in $C(I; \mathcal{H}_{\text{phys}})$ and weakly-* in $L^\infty(I; \mathcal{V})$. Testing the integral equation against $P_m w \rightarrow w$ in $\mathcal{V}$ identifies the limit as $i\dot{u} = q_t u$ in $\mathcal{V}'$, with the required initial value.

The Hilbert-triple norm chain rule gives $d\|u\|^2/dt = 2 \operatorname{Re}\langle \dot{u}, u \rangle = 0$ in its dual-pairing notation. It follows also by time convolution and passage to the limit, and applies to the difference of two solutions. Thus the solution is unique and the entire Galerkin sequence converges. At each time weak lower semicontinuity gives the representative in $\mathcal{V}$ and its form bound. Repeating the construction backward in time gives the inverse evolution. Norm conservation extends it from the dense domain $\mathcal{V}$ to a unitary propagator on the physical Hilbert space, with strong time continuity and the composition law by uniqueness. No common operator domain, propagator-invariant Schwartz core or occupied quantum constraint kernel was assumed. $\square$

25.5 All metric tangents and the fixed-port response

The coordinate components of an external metric are actual geometric directions, not separately named coefficient parameters.

Proposition 25.5 (Complete local metric-test coverage). *Every smooth compactly supported symmetric spacetime-metric tangent $k_{\mu\nu}$ at a history in Theorem 25.4 is realized by a sufficiently small actual history in the same branch. The corresponding form derivative includes the derivative of (382), including its temporal coframe and lowered-shift derivatives.*

Proof. Differentiation of $g_{0i} = h_{ij}\beta^j$ and $g_{00} = -N^2 + h_{ij}\beta^i\beta^j$ gives the inverse ADM differential

$$\begin{aligned} \delta h_{ij} &= k_{ij}, \\ \delta \beta^i &= h^{ij}(k_{0j} - k_{j\ell}\beta^\ell), \\ \delta N &= \frac{2\beta^j k_{0j} - \beta^i \beta^j k_{ij} - k_{00}}{2N}. \end{aligned} \quad (391)$$

It realizes all ten tensor components. Small deformations preserve lapse positivity, the coframe branch and the strict correlation margins. The selected coframe section and the derived connection give their actual lifts. Smooth finite coefficient operations and the common-form estimates imply, on each compact test step,

$$q_r^\epsilon = q_r + \epsilon a_r[k] + o(\epsilon) \quad \text{in } C(I; \mathcal{L}(\mathcal{V}, \mathcal{V}')). \quad (392)$$

The derivative is bounded by a fixed finite metric-jet seminorm, for example a C^2 seminorm on the declared smooth-history neighborhood. Differentiating only the static spatial coefficients would omit the $\dot{k}$ and shift terms and would not give this derivative. $\square$

Here ten counts external tensor components, not ten propagating or structural gravitational degrees of freedom. Finite sampling need not be injective on the test space, and coverage is not a Ward identity.

Theorem 25.6 (Actual transition-readout derivative). *Fix a base history, its initial and final times, and source ports $\phi, \psi \in \mathcal{V}$. For the actual metric deformation in (392),*

$$D \langle \phi, U_g(t, s) \psi \rangle [k] = -i \int_s^t a_r[k] (U_g(r, t) \phi, U_g(r, s) \psi) dr. \quad (393)$$

At the fixed base and ports this is a Hadamard derivative of the scalar readout, continuous jointly in the two $\mathcal{V}$ vectors and the compact-step metric tangent. It is not an assertion of operator-norm differentiation of the propagator on $\mathcal{H}$, of such differentiation into $\mathcal{L}(\mathcal{V}, \mathcal{V}')$, or of continuity of the derivative as the base history varies.

Proof. The weak cross-pairing identity in the Gelfand triple gives

$$\begin{aligned} & \langle \phi, (U_\epsilon(t, s) - U_0(t, s)) \psi \rangle \\ &= -i \int_s^t (q_r^\epsilon - q_r) (U_0(r, t) \phi, U_\epsilon(r, s) \psi) dr. \end{aligned} \quad (394)$$

Both trajectories have the regularity proved in Theorem 25.4. The deformed solutions are uniformly bounded in $L^\infty(I; \mathcal{V})$. Compactness and uniqueness, by the same Galerkin-limit argument, give strong $C(I; \mathcal{H})$ and weak- $*$ $L^\infty(I; \mathcal{V})$ convergence to the undeformed trajectory. Divide (394) by ϵ . The uniform form remainder tends to zero. The fixed factor $a_r[k](U_0(r, t) \phi, \cdot)$ is an $L^1(I; \mathcal{V}')$ test, so weak- $*$ convergence suffices for the remaining integral. This proves (393). The form derivative and the two propagation bounds give its joint continuity estimate. Convergent tangent sequences have the same uniform remainder and test convergence, proving the stated fixed-base Hadamard assertion. Strong convergence in $\mathcal{V}$ has not been substituted for the proved weak convergence. $\square$

For a Hermitian ray readout take

$$R_{[\psi]}(g) = \frac{\operatorname{Re} \langle \psi, U_g(t, s) \psi \rangle}{\|\psi\|^2}, \quad 0 \neq \psi \in \mathcal{V}. \quad (395)$$

At each history it is the diagonal readout of $(U_g + U_g^\dagger)/2$. The Hermitian part of (393) is a continuous Hermitian form on $\mathcal{V}$; phase invariance, norm-squared extension and the parallelogram identity follow. This supplies an actual differentiable-carrier and all-ray response for the registered history protocol of [25, 26]. It does not assign a mass role merely from quadratic dependence, and does not construct a complete spacetime stress tensor or inherit a Ward identity that the source has not established. Nor does the proof differentiate arbitrary bounded inserted observables without checking preservation of the required form domains.

25.6 Frame covariance and the limit of geometric relabeling

Let a smooth residual time-gauge $\text{Spin}(3)$ rotation have one-particle unitary $S(t)$ and its ordinary finite second quantization $\mathbb{S}(t)$. Spatial coframes, Clifford symbols, internal/spin path attachments and Yukawa tensors are transported together. The derived temporal connection transforms as $\Omega'_0 = S\Omega_0 S^{-1} - \dot{S}S^{-1}$. Consequently

$$\begin{aligned} H'_g &= \mathbb{S}H_g\mathbb{S}^{-1} + i\dot{\mathbb{S}}\mathbb{S}^{-1}, \\ U'_g(t, s) &= \mathbb{S}(t)U_g(t, s)\mathbb{S}(s)^{-1}. \end{aligned} \quad (396)$$

The first identity follows from the complete transformed connection, and differentiation of the second in $\mathcal{V}'$ and uniqueness prove the propagator law. The finite Fock implementation fixes the vacuum phase; no arbitrary scalar phase connection is appended. This preserves the physical space and form domain. It does not assert a finite-dimensional unitary representation of arbitrary Lorentz boosts.

A spatial diffeomorphism may also transport graph placements, edge paths, quadrature and coframes while preserving finite quantum labels. The geometric lift on placements is

$$T_\xi = \sum_v \xi^i(X_v) \frac{\partial}{\partial X_v^i}, \quad [T_\xi, T_\eta] = T_{[\xi, \eta]},$$

with the corresponding path pushforward. This geometric relabeling does not identify its horizontal action with the one-edge scalar momentum inside a fixed quantum fibre.

Proposition 25.7 (Fixed-incidence connection retains the discrepancy). *Freeze graph incidence and identify quantum variables by vertex and edge labels. Suppose the geometric connection differentiates placements, paths, coefficients and spin/internal frames, but introduces neither new scalar jets nor an additional scalar horizontal action beyond the one-edge prescription. Adding that connection to the raw momentum comparison does not restore the continuum descriptor bracket on the physical source. The temporal spin connection in (382) cannot cancel its two-edge scalar discrepancy.*

Proof. At trivial links on a coordinate cycle, $v_\xi q_x = \xi_x(q_{x+1} - q_x)/a$. Its bracket contains

$$[v_\xi, v_\eta]q_x|_{q_{x+2}} = \frac{\eta_x \xi_{x+1} - \xi_x \eta_{x+1}}{a^2} q_{x+2}. \quad (397)$$

Horizontal differentiation of one-edge coefficients changes descriptors, lengths, frames and transports, but not their scalar support (q_x, q_{x+1}) . Thus the terms $T_\xi v_\eta - T_\eta v_\xi - v_{[\xi, \eta]}$ do not cancel the independent second-neighbor coefficient. Spin matrices act on the finite fermion factors, and internal gauge action is local and annihilates $r_x = |q_x|^2$. A coefficient-dependent link derivative acting on a one-edge scalar coefficient likewise does not introduce the second-neighbor scalar variable.

Use the physical witness of Proposition 24.4: choose only the first direction, with $\xi_x = 1, \xi_{x+1} = 0, \eta_x = 0, \eta_{x+1} = 1$, and set all outgoing nearest neighbors equal to $q_x \neq 0$. Vary q_{x+2e_1} . Then

$$[v_\xi, v_\eta]r_x = -2q_x \cdot (q_{x+2e_1} - q_x)/a^2, \quad v_\xi r_x = 0$$

for every full one-edge descriptor ζ . Any extra one-edge term from geometric coefficient differentiation is independent of the varied second neighbor, so cannot cancel this identity for every choice. Take a smooth invariant core function $F(r_x)$, with $F(r_0) = 0, F'(r_0) \neq 0$, times radial cutoffs constant near the tested coordinates, link constants and the Fock vacuum. Its zero value removes ordering multipliers. A choice of the second neighbor gives nonzero output, and smoothness makes that output nonzero on an open neighborhood. This is a nonzero physical Hilbert vector,

not a gauge-variant coordinate or measure-zero test. The unchanged scalar term gives the same obstruction for the new geometric family; its changed fermionic sign and temporal spin matrix do not change scalar principal support. $\square$

For an actual differentiable unitary source trivialization, connection curvature transforms by conjugation on the transported common core: $F'_{\xi,\eta} = \mathbb{U} F_{\xi,\eta} \mathbb{U}^{-1}$. It cannot turn a nonzero physical discrepancy into zero. The result does not exclude a genuinely new horizontal scalar action, nonlocal interpolation, changed admitted incidence or a controlled refinement limit.

The construction supplies an occupied full mixed quantum history, its action-derived temporal and shift coefficients, and a fixed-port response for every external metric component. The distinct raw momentum retains the definition and bracket property established in Section 24. External test coverage alone does not establish a complete spacetime Ward identity, a finite hypersurface-deformation algebra or self-consistent Einstein dynamics. Section 26 constructs the same-source tensor and initial-data comparison needed for an occupied native Einstein realization [10, 2]. A passage beyond the finite graph retains the corresponding history and first-jet estimates on actual source domains. Complete quantum-WDW occupation is a separate gravitational realization method, not an additional premise of the common physical result in Part I.

26 Metric response and coupled continuation of the full native source

The external geometry of Section 25 and a self-consistent Einstein development are different objects. The former defines how the complete interacting Hamiltonian responds to a prescribed metric; the latter must evolve that metric with the source evaluated on the same quantum history. We construct both, retaining their common physical clock, source coefficients and comparison maps. The construction uses the native finite source, not an identification of its spectral levels with the physical Higgs or confined-particle channels.

The nonlinear Einstein law and its metric-source interpretation are downstream realizations of the common premetric architecture [9, 39, 10, 2]. They are not new foundations joined to the quantum source. Here we establish the actual source attachment and coupled evolution. The smoothed interaction of Section 27 remains a valid explicit realization; the family below instead retains the literal unsmoothed native Hamiltonian and its full metric response.

26.1 The common graph domain and its propagation

Fix the physical lattice length ℓ , the graph, all original couplings, the faithful internal quotient, three generation copies, spin structure, paths, anchors and boundary data. Every Hamiltonian is the same physical generator, including its original factor ℓ^{-1} ; no sector receives an independent clock. Let $\mathcal{H}$ be the physical Gauss space and set

$$\mathcal{D} = D(T_0) \cap D(R_4), \quad R_4 = \sum_x |q_x|^4, \quad \|u\|_{\mathcal{D}} = \|u\| + \|T_0 u\| + \|R_4 u\|.$$

Here T_0 is the reference scalar and compact-link kinetic operator. The full history operator is $H[N, h] + P_H + P_G - P_F + B_0$, with precisely the action-derived coefficients of (386).

On a compact strict geometric and quartic envelope this family has common operator domain $\mathcal{D}$. Its graph norms are uniformly equivalent there. In particular the complete electric, scalar-kinetic, quartic, ordered momentum, Wilson, CAR and Yukawa derivative alphabet is bounded from $\mathcal{D}$ to $\mathcal{H}$. These statements are stronger than boundedness as forms on $\mathcal{V}$, but do not assert that every metric insertion maps $\mathcal{V}$ to $\mathcal{H}$. The scalar estimate controlling $q\nabla_q$, the compact-link Casimir estimate controlling products of invariant derivatives, and the quartic graph bound are

all needed. The ordered gauge momentum retains the multiplier derivative in $FP - (i/2)(XF)$; it is not replaced by FP .

Lemma 26.1 (The full metric operator domain). *On the stated finite source envelope, the self-adjoint realization of the complete history has domain $\mathcal{D}$, with uniform two-sided graph comparison there and the invariant smooth/Schwartz physical operator core.*

Proof. Let T_g be the positive normal kinetic term. Its scalar coefficients are external, and the total compact-link Casimir commutes with each invariant generator and its positive Gram contraction. The scalar and link kinetic terms commute. The strict positive coefficients therefore give graph comparison of T_g with T_0 . After a finite analytical shift, its Hermitian normal multiplier satisfies

$$c_1(1 + R_4)I \leq W_g, \quad \|W_g(q, U)\| \leq c_2(1 + R_4).$$

The second scalar and link derivatives of this multiplier are bounded by a constant times $1 + \sum_x |q_x|^2$. Integration by parts in $\|(T_g + W_g)u\|^2$ has a positive leading mixed term: the electric Gram and positive matrix W_g act in separate indices. The differentiated remainder is absorbed into $\epsilon \|W_g u\|^2 + C_\epsilon \|u\|^2$. Consequently

$$\|T_g u\| + \|W_g u\| \leq C(\|(T_g + W_g)u\| + \|u\|).$$

The scalar cutoff, smoothing and weak-equation argument of the original quartic operator-core construction identifies the normal domain as $D(T_0) \cap D(R_4)$, rather than using this inequality alone as a domain proof. Haar averaging preserves that core.

For the scalar momentum, integration by parts gives

$$\| |q_y| \nabla_{q_x} u \|^2 \leq \| |q_y|^2 u \|^2 + C \mathbf{1}_{x=y} \|u\|^2.$$

Quartic absorption and Young's inequality make every such term infinitesimally operator bounded relative to the normal graph norm. For the gauge momentum, F and XF are bounded at the fixed mesh; spectral interpolation makes FP infinitesimally kinetic-operator bounded, and its ordering term is bounded. All actual mesh and electric-weight costs remain. The CAR momentum and temporal spin term are finite-dimensional bounded coefficients. Choosing the total relative bound below one and applying the resolvent Neumann argument gives self-adjointness on the normal domain, the same core, and the received closed form. This identifies the full operator with the already specified physical realization. $\square$

Let s range over a finite-dimensional real neighborhood of original source parameters and let I be a compact physical-time interval. For the following two-response theorem use the actual regularity

$$H_s \in C^2(s; C^1(I; \mathcal{L}(\mathcal{D}, \mathcal{H}))). \quad (398)$$

Smooth native metric and coupling protocols satisfy this condition. Indeed the positive Gram matrices, inverse metric, lapse, lowered shift, determinant and selected coframe are smooth coefficient functions on the strict envelope. Spatial spin transport is differentiated along its actual path, including its endpoint Clifford matrices and ordered connection insertions. Its time derivative uses mixed spacetime metric jets. The temporal spin term differentiates $\dot{e}e^{-1}$ and the curl of the lowered shift, hence uses the retained second history jets. The scalar and gauge shift ordering, real and imaginary Yukawa coordinates, mass and quartic terms remain in the same finite graph-bounded alphabet. Taylor estimates after the full anchor and coordinate pullback prove (398); they do not require a new quantum domain for a differentiated coefficient.

Choose one analytical constant c on this neighborhood such that $A_s(t) = H_s(t) + c \geq I$. Then

$$A_s(t) : \mathcal{D} \longrightarrow \mathcal{H}, \quad A_s(t)^{-1} : \mathcal{H} \longrightarrow \mathcal{D}$$

are uniformly bounded inverse maps and are norm continuous in their displayed spaces. The constant c is used for estimates only; it is absent from physical propagation and gravitational energy.

Theorem 26.2 (Strong graph continuity). *The actual propagator $U_s(t, r)$ is jointly strongly continuous on $\mathcal{D}$ in source parameters and both time endpoints. It has a uniform graph bound on compact envelopes. Neither operator-norm continuity on $\mathcal{H}$ or $\mathcal{D}$, nor graph convergence from a Hilbert limit alone, is asserted.*

Proof. Freeze H on a time partition. The estimate $\|A(t_{j+1})A(t_j)^{-1}\| \leq 1 + C\Delta_j$ and commutation on each frozen interval give a uniform graph bound for the products. Their Hilbert limit is the form propagator of Section 25; weak graph compactness and its closed graph give preservation and the same bound at each time. Reverse products give the inverse bound. This supplies graph boundedness, not yet graph continuity. Its exact cross identity on $u \in \mathcal{D}$ is

$$(U_s(t, r) - U_v(t, r))u = -i \int_r^t U_s(t, \tau)(H_s(\tau) - H_v(\tau))U_v(\tau, r)u \, d\tau.$$

Thus Hilbert parameter continuity follows from the coefficient norm and the existing graph bound, then extends by density to every Hilbert vector. Before graph continuity is known, frozen approximants and their uniform graph bound give weak graph measurability; separability of an equivalent Hilbert graph norm gives the strong measurability needed for this integral. No graph convergence is assumed in this step.

Suppress s , partition $[r, t]$ by t_j , and put

$$E_j = e^{-i(t_{j+1}-t_j)H(t_j)}, \quad J_j = A(t_{j+1})A(t_j)^{-1}.$$

Commutation of E_j with $A(t_j)$ gives the exact bounded factorization

$$A(t)U_\pi(t, r)A(r)^{-1} = (J_{n-1}E_{n-1}) \cdots (J_0E_0), \quad J_j = I + \Delta_j \dot{A}(t_j)A(t_j)^{-1} + \Delta_j o(1), \quad (399)$$

where the remainder is uniform in operator norm. Define $B(t) = \dot{A}(t)A(t)^{-1}$. Expanding the product by its number of bounded B insertions bounds its m -insertion sum by $(\|B\|_\infty |t-r|)^m/m!$. For each fixed m , strong convergence of the intervening frozen propagators and a compact-vector argument give the corresponding ordered integral. The factorial majorant permits summation. Its limit is the bounded Volterra evolution

$$W(t, r) = U(t, r) + \int_r^t U(t, \tau)B(\tau)W(\tau, r) \, d\tau, \quad \|W(t, r)\| \leq e^{\|B\|_\infty |t-r|}.$$

Consequently the left side of (399) converges strongly and

$$U_\pi(t, r)u \longrightarrow A(t)^{-1}W(t, r)A(r)u \quad \text{in } \mathcal{D}.$$

Its known Hilbert limit identifies this vector with $U(t, r)u$. This proves graph convergence rather than inferring it from graph boundedness. Reverse time gives the inverse bound. Finally the same ordered integrals are strongly continuous in s , since U_s is Hilbert-strong continuous and B_s is norm continuous. The uniform factorial majorant and the norm-continuous inverse maps give joint graph continuity. A varying graph-continuous input is handled by the uniform graph bound. $\square$

26.2 Original metric amplitudes and their two responses

Write $H_a = \partial_a H$, $H_{ab} = \partial_a \partial_b H$. They are symmetric on $\mathcal{D}$. A useful domain-safe product rule is

$$\partial_b \langle x, Ty \rangle = \langle x_b, Ty \rangle + \langle x, T_b y \rangle + \langle Tx, y_b \rangle, \quad (400)$$

when x, y are graph continuous and Hilbert C^1 , and T is symmetric and C^1 from $\mathcal{D}$ to $\mathcal{H}$. To prove it, subtract the two pairings as

$$\langle x_h - x, T_h y_h \rangle + \langle x, (T_h - T) y_h \rangle + \langle T x, y_h - y \rangle$$

and take the quotient. Graph continuity makes $T_h y_h \rightarrow T y$ strongly in $\mathcal{H}$. In particular $T y_b$ is never required.

Theorem 26.3 (Strong first and weak mixed second response). *For $u \in \mathcal{D}$, the original propagator has the Hilbert-strong derivative*

$$U_a(t, r)u = -i \int_r^t U(t, \tau) H_a(\tau) U(\tau, r) u \, d\tau. \quad (401)$$

For $\phi, \psi \in \mathcal{D}$, the actual amplitude $\langle \phi, U_s(t, r) \psi \rangle$ is C^2 . With $\Phi(\tau) = U(\tau, t) \phi$ and $\Psi(\tau) = U(\tau, r) \psi$, its mixed derivative is

$$\begin{aligned} \mathcal{B}_{ab}[\phi, \psi] = & -i \int_r^t \langle \Phi(\tau), H_{ab}(\tau) \Psi(\tau) \rangle \, d\tau \\ & - \int_{r < \sigma < \tau < t} \{ \langle H_a(\tau) \Phi(\tau), U(\tau, \sigma) H_b(\sigma) \Psi(\sigma) \rangle + (a \leftrightarrow b) \} \, d\sigma \, d\tau. \end{aligned} \quad (402)$$

If G bounds propagation on $\mathcal{D}$, and L_a, L_{ab} bound the indicated derivative maps, then

$$\|U_a u\| \leq TGL_a \|u\|_{\mathcal{D}}, \quad |\mathcal{B}_{ab}[\phi, \psi]| \leq (TGL_{ab} + T^2 G^2 L_a L_b) \|\phi\|_{\mathcal{D}} \|\psi\|_{\mathcal{D}}.$$

The second assertion is a weak amplitude Hessian. It does not assert that $U_{ab}\psi$ is a Hilbert vector or that $U_b\psi \in \mathcal{D}$.

Proof. Divide the cross identity in the previous proof by a source increment. The coefficient quotient converges in $\mathcal{L}(\mathcal{D}, \mathcal{H})$; unitarity, graph bounds and dominated convergence prove (401). Its continuity as the base source changes uses the strong graph continuity of the right trajectory.

The first scalar derivative is $-i \int \langle \Phi, H_a \Psi \rangle$. Apply (400) to this expression. The contact is H_{ab} . The forward ket derivative contributes one ordered triangle; the reverse-time bra derivative contributes the other. Antilinearity in the bra gives the minus sign in both double terms. Each unbounded insertion acts on a different graph-prepared endpoint, and the middle propagator acts only on $\mathcal{H}$. The two triangles have total area $(t - r)^2$, giving the bound and all interchange of integrals. Graph continuity and the same majorant prove continuity of the second derivatives. $\square$

For C^2 graph-prepared endpoints, all normalization and preparation terms remain:

$$\begin{aligned} M_{ab} = & \langle \phi_{ab}, U \psi \rangle + \langle \phi_a, U_b \psi \rangle + \langle \phi_b, U_a \psi \rangle + \mathcal{B}_{ab}[\phi, \psi] \\ & + \langle \phi_a, U \psi_b \rangle + \langle \phi_b, U \psi_a \rangle + \langle \phi, U_a \psi_b \rangle + \langle \phi, U_b \psi_a \rangle + \langle \phi, U \psi_{ab} \rangle. \end{aligned}$$

For example $(H_s(t_*) + c)^{-1} \Omega_s$, normalized by its actual nonzero norm, is such a preparation when Ω_s is a nonzero C^2 Hilbert seed. Inverse differentiation takes values in $\mathcal{D}$; it does not replace this preparation by a source-independent state.

The coefficient H_{ab} includes all nonlinear ADM, inverse metric, density, coframe, spatial-path and temporal-spin contacts, as well as mixed coupling derivatives. A nonlinear metric protocol adds $DH[k_{ab}]$ to $D^2 H[k_a, k_b]$. The two ordered first insertions in (402) are not counted again inside that coefficient. Original anchor, test and observer maps are differentiated before the formula is applied. Time endpoints in these formulas are fixed physical times.

These conclusions extend to every fixed finite word of the admitted adjoint-closed bounded ports that preserve $\mathcal{D}$, with two source derivatives bounded on both $\mathcal{H}$ and $\mathcal{D}$. Differentiate a

first-insertion term after splitting the word into its two graph-prepared remainders, then use (400). No product $U_a U_b$ on an unproved domain is needed. Actual smooth Wilson, finite-CAR and regularized physical ports satisfy their recorded bounds; arbitrary bounded operators need not. For a domain-preserving frame change,

$$H^S = S H S^{-1} + i \dot{S} S^{-1}, \quad U^S(t, r) = S(t) U(t, r) S(r)^{-1},$$

transporting endpoints and ports gives equality before differentiation. The complete jets consequently retain both endpoint derivatives and the temporal connection. No higher unbounded dynamical jet follows merely from additional coefficient derivatives.

26.3 A reference-resolved source without operator averaging

Choose an admitted periodic cubic member on $\Sigma = \mathbb{T}_L^3$ with periodic spin structure and a nonzero original Yukawa tensor, and put $V_0 = L^3$. Nonzero Yukawa is needed for the strict interaction witness below, not for the domain or response identities. Let $\tilde{C}$ be the 48-element spin lift of the proper cubic rotation group and

$$\Gamma = \mathbb{T}_L^3 \rtimes \tilde{C}, \quad \gamma_{y,r}(x) = R(r)x + y,$$

with normalized Haar measure μ . Another spin structure uses only its preserving subgroup; full cubic isotropization is not assumed for it. In fixed material coordinates retain each complete fiber $H_{g,\gamma} = H_{\gamma^*g}$, and set

$$\hat{\mathcal{H}} = L^2(\Gamma; \mathcal{H}), \quad \hat{\mathcal{V}} = L^2(\Gamma; \mathcal{V}), \quad \hat{q}_g(\Phi, \Psi) = \int_{\Gamma} q_{\gamma^*g}(\Phi_{\gamma}, \Psi_{\gamma}) d\mu. \quad (403)$$

This is a decomposable source, not the single operator $\int H_{\gamma^*g} d\mu$ on $\mathcal{H}$. Uniform coercivity makes its form closed. Uniform graph comparisons identify its operator domain with $\hat{\mathcal{D}} = L^2(\Gamma; \mathcal{D})$ on the source envelopes above. No compact-resolvent assertion is made for the reference lift, whose reference coordinate has no kinetic cost.

Proposition 26.4 (Faithful reference family and normal preparations). *Let $\hat{\mathcal{A}}$ be the algebra of retained uniformly bounded strong-star continuous sections, generated by native ports, actual propagated words, source resolvents and central continuous reference functions. Multiplication on $\hat{\mathcal{H}}$ is faithful. Evaluation at each reference recovers its native algebra, state and encoded operator graphs. Normal reference preparations have the form*

$$\omega(A) = \int_{\Gamma} \text{Tr}(\rho(\gamma)A(\gamma)) d\mu, \quad \rho(\gamma) \geq 0, \quad \int_{\Gamma} \text{Tr} \rho(\gamma) d\mu = 1.$$

Conversely every measurable trace-class density field with this normalization defines such a normal preparation. Point-evaluation states are positive states of $\hat{\mathcal{A}}$, but in general are not normal in its Haar representation and are not bounded point extractions from $L^2(\Gamma; \mathcal{H})$.

Proof. A nonzero continuous section acts nontrivially on a fixed vector throughout an open set of positive Haar measure. A localized reference vector detects it, proving faithfulness. Evaluation preserves products and adjoints; the evaluated resolvents are the native resolvents, hence retain their exact graphs and domains.

A normal state in the concrete representation is a restriction of a positive trace-class spatial vector sum $\sum_j \lambda_j |\Psi_j\rangle\langle\Psi_j|$. Its density field is $\rho(\gamma) = \sum_j \lambda_j |\Psi_j(\gamma)\rangle\langle\Psi_j(\gamma)|$; monotone convergence gives the trace normalization and the displayed formula. Conversely the vectors $\rho(\gamma)^{1/2} e_j$, for a fixed Hilbert basis, give a spatial vector sum of total norm one. This proves the assertion for the normal folium, not for every abstract state. A point measure is singular relative to Haar, already on the central continuous functions, proving the distinction. $\square$

Normal densities concentrated near one reference recover every fixed continuous bounded word and tested form. A norm-Lipschitz section has error at most $C\epsilon$; a form-Lipschitz section has the corresponding $C\epsilon \|\phi\|_{\mathcal{V}} \|\psi\|_{\mathcal{V}}$ bound. Unbounded expectations require their actual graph or form moments. Thus reference completion preserves native evaluations and failures without curing a domain failure or the two-edge discrepancy of Proposition 24.4.

This completion adds classical-reference preparations and central reference observations, not a new Standard Model species. It is a faithful completion of a source *family*, not equivalence of a uniform preparation with a previously fixed point preparation. At a homogeneous metric the constant lift $\iota\psi(\gamma) = \psi$ preserves every original material word exactly. Co-transported material tests $k_\gamma = (\gamma^{-1})^*k_0$ also give identical native responses. A single ambient test applied to every reference is a different protocol. Central reference readouts, and in general its metric responses, distinguish uniform and point preparations.

The common material embedding F is retained through $\hat{S}(F^*g, \Psi)$. Simultaneous transport of g and F leaves this full source unchanged. Independent embeddings for every reference would define additional equations and are not introduced. Time-dependent frame comparisons retain $i\dot{S}S^{-1}$; they cannot be replaced by reference reindexing.

26.4 Literal interacting Einstein evolution on the same state

For $g_b = -dt^2 + b(t)^2\delta_{ij}dx^i dx^j$, $b > 0$, the literal native Hamiltonian is

$$H_b = b^{-3}T_H + b^3(V_4 + V_2) + b^{-1}(T_G + W_G + F_{\text{hop}}) + bW_H + Y. \quad (404)$$

Every summand has its original physical normalization. In particular Y is not heat-smoothed or removed because its explicit scale coefficient is constant. At this metric $\gamma^*g_b = g_b$, so an initially constant-reference state stays constant and evolves by this same H_b .

The reference-resolved action is

$$\hat{S}(g, \Psi) = \int_I \left[\frac{i}{2} (\langle \Psi, \dot{\Psi} \rangle - \langle \dot{\Psi}, \Psi \rangle) - \hat{q}_g(\Psi, \Psi) \right] dt.$$

On an actual prescribed history, holding its evolved state fixed during the metric variation gives the all-test source

$$W_{[\eta]}(k) = -\frac{1}{\|\eta\|^2} \int_I \int_{\Gamma} Dq_{g_b}[\gamma^*k](U_b(t, 0)\eta_\gamma, U_b(t, 0)\eta_\gamma) d\mu dt. \quad (405)$$

It is continuous in the actual finite metric-jet topology. Its degree-two extension polarizes uniquely to

$$\langle u, \hat{T}(k)v \rangle = -2 \int_I \hat{a}_g[k](\hat{U}_b u, \hat{U}_b v) dt, \quad \hat{T}(k) : \hat{\mathcal{V}} \rightarrow \hat{\mathcal{V}}'.$$

The source graph is the derivative graph of this complete action, so the source-side attachment and polarization mechanisms of [25, 26] apply with their actual domain and normalization. The common embedding derivative gives $W(\mathcal{L}_X g) = -J(X)$, and polarization gives $\hat{T}(\mathcal{L}_X g) = -2\hat{J}(X)$. It does not supply the separate mass/Poincare hypotheses of a stronger Module-B subtype.

For $\eta = \iota\psi_0$, reference averaging is performed only after this full derivative. Translation averaging removes nonconstant spatial modes from the expectation; the proper cubic average removes the vector and traceless-tensor mean. All spin, shift and density terms have already been varied. The remaining isotropic tangent has no intrinsic or skew temporal-spin variation. Thus, on *every* compact spacetime metric test, the actual tensor is

$$T^{\mu\nu} = (\rho + P)u^\mu u^\nu + P g_b^{\mu\nu}, \quad \rho = \frac{\langle \psi, H_b \psi \rangle}{V_0 b^3}, \quad P = -\frac{\langle \psi, H'_b \psi \rangle}{3V_0 b^2}, \quad u = \partial_t. \quad (406)$$

This is not the operator identity $D\hat{q}[k] = Dq[\Pi k]$. Source insertions can carry a constant reference state out of its constant subspace.

Theorem 26.5 (Source-attached expanding Einstein family). *Fix $\kappa > 0$, Λ , the native coefficients and the reference prescription. There is a unique local homogeneous coupled solution for each normalized $\psi_0 \in \mathcal{D}$, $b_0 > 0$ and finite canonical momentum, of*

$$\dot{b} = -\frac{\kappa p}{6V_0 b}, \quad \dot{p} = -\frac{\kappa p^2}{12V_0 b^2} - \frac{3\Lambda V_0 b^2}{\kappa} - \langle \psi, H'_b \psi \rangle, \quad i\dot{\psi} = H_b \psi. \quad (407)$$

The constraint

$$\mathcal{C} = -\frac{\kappa p^2}{12V_0 b} + \frac{\Lambda V_0}{\kappa} b^3 + \langle \psi, H_b \psi \rangle$$

is conserved. On $\mathcal{C} = 0$, the C^2 metric and the continuous tensor (406) satisfy all Einstein components and the compact-test Ward identity. The common material embedding force vanishes. There are expanding data with nonzero unsmoothed Yukawa expectation and nonzero reference-resolved metric responses.

Proof. For a candidate Lipschitz scale history in a compact positive interval, the native propagator has a uniform graph bound. Its Duhamel estimate compares two histories in Hilbert norm by $CT \|b - d\|_\infty$. For a symmetric $B : \mathcal{D} \rightarrow \mathcal{H}$,

$$|\langle u, Bu \rangle - \langle v, Bv \rangle| \leq \|u - v\| (\|Bu\| + \|Bv\|).$$

With the graph bounds for H'_b and H''_b , this makes the force a Lipschitz functional of the scale history. The classical integral map is a contraction on a sufficiently short interval. It constructs the coupled solution, not just propagation on an assigned metric.

Writing $E(t) = \langle \psi(t), H_{b(t)} \psi(t) \rangle$, the frozen-step energy identity passes to the actual history as

$$E(t) - E(0) = \int_0^t \dot{b}(s) \langle \psi(s), H'_{b(s)} \psi(s) \rangle ds.$$

No product $H'_b H_b \psi$ is needed. It gives $\dot{\mathcal{C}} = 0$ and $\dot{\rho} + 3(\dot{b}/b)(\rho + P) = 0$. The canonical equations on the constraint become

$$3(\dot{b}/b)^2 = \Lambda + \kappa \rho, \quad 2\ddot{b}/b + (\dot{b}/b)^2 = \Lambda - \kappa P.$$

Because (406) was established on all tests, these, the zero mixed components and isotropy prove the full tensor equation, not merely its homogeneous projection. Conservation gives the common embedding force $W(\mathcal{L}_X g) = 0$.

For explicit data, a sufficiently wide physical radial Gaussian times link constants and the invariant Fock vacuum has arbitrarily large positive quartic energy. For nonzero Yukawa tensor, $a_Y = \|Y\psi_G\|^2 > 0$. Pair-number selection gives

$$\psi_\epsilon = \frac{\psi_G + \epsilon Y \psi_G}{\sqrt{1 + \epsilon^2 a_Y}}, \quad \langle \psi_\epsilon, Y \psi_\epsilon \rangle = \frac{2\epsilon a_Y}{1 + \epsilon^2 a_Y} \neq 0.$$

This holds for complex Yukawa matrices. Choose the width and a small nonzero ϵ so that $M_0 = E_0 + \Lambda V_0 b_0^3 / \kappa > 0$, then set

$$p_0 = -\sqrt{(12V_0/\kappa)b_0 M_0}. \quad (408)$$

This solves the constraint and gives expansion without changing a coupling or subtracting source energy. $\square$

For a precise nonconstant-response witness choose a zero-mean lapse profile with value one at a vertex whose energy expectation is nonzero and zero at the other sampled vertices. Its reference-averaged mean source is zero. Yet, with the original response section $a_k(\gamma)$ and $R_b = (H_b + c)^{-1} : \mathcal{V}' \rightarrow \mathcal{V}$,

$$\int_{\Gamma} \|R_b a_k(\gamma)\psi\|^2 d\mu > 0. \quad (409)$$

At the reference sampling that vertex the insertion has nonzero expectation, so injectivity of R_b and reference continuity give a nonzero vector on an open positive-measure set. Continuity in time also gives a nonzero compact-time integral: localize a nonnegative unit-integral time test where the vector differs from its value at one time by less than half that value's norm. The integral has at least half that norm. This uses the time-dependent filter and actual evolved state, not a filter frozen outside an integral.

Any finite list of nonzero first-response forms can be retained on one expanding branch. Their filtered maps have proper closed kernels in $\mathcal{V}$. The strict energy and Yukawa conditions define a nonempty open set; finite kernel avoidance and density choose a physical core vector in it. At an isotropic base the first temporal rotation variation is the curl of the lowered shift divided by $2b_0^2$; the skew part of $\dot{k}/(2b_0^2) - \dot{b}_0 k/b_0^3$ is zero for symmetric k . Thus these first forms are fixed before the constraint selects $\dot{b}_0$. The finitely many strict filtered diagonals persist on a smaller interval. This is not a claim that every mixed entry, or every higher derivative, is nonzero.

26.5 Differentiable constraint-matched continuation

Initial preparations and geometric data must vary together. Let $\theta \mapsto (b_0, \psi_0)$ be C^1 , with ψ_0 normalized and C^1 in $\mathcal{D}$, bounded graph norm, and a strict positive margin in (408). The same short interval works locally for these preparations.

Theorem 26.6 (Actual first coupled derivative). *The constraint-matched solution map is*

$$\theta \mapsto (b, p, \psi) \in C^1(\theta; C_t^2 \times C_t^1 \times C_t \mathcal{H}),$$

and ψ is continuous with values in $C_t \mathcal{D}$. Writing $\beta = \partial_\theta b$, $\pi = \partial_\theta p$, $\xi = \partial_\theta \psi$, $B = H'_b$, $C = H''_b$, and $F = \langle \psi, B\psi \rangle$, gives

$$\begin{aligned} \xi(t) &= U(t, 0)\eta - i \int_0^t U(t, s)B(s)\psi(s)\beta(s) ds, \\ \delta F &= 2 \operatorname{Re} \langle \xi, B\psi \rangle + \beta \langle \psi, C\psi \rangle, \\ \dot{\beta} &= \frac{\kappa p}{6V_0 b^2} \beta - \frac{\kappa}{6V_0 b} \pi, \\ \dot{\pi} &= \left(\frac{\kappa p^2}{6V_0 b^3} - \frac{6\Lambda V_0 b}{\kappa} \right) \beta - \frac{\kappa p}{6V_0 b^2} \pi - \delta F. \end{aligned} \quad (410)$$

Here $\eta = \partial_\theta \psi_0$, and the initial π is the derivative of the constraint lift. The state derivative is a Hilbert vector; $\xi \in \mathcal{D}$ is not asserted.

Proof. The symmetric force estimate and cross-Duhamel identity give Lipschitz stability of (b, p, ψ) on bounded graph sets in the displayed Hilbert metric. This bounds its difference quotients, but does not prove differentiability. It first gives continuous $b, \dot{b}$ as the initial datum varies, hence norm-continuous $H_b, \dot{H}_b$ from $\mathcal{D}$ to $\mathcal{H}$. The graph-frame proof of Theorem 26.2, which at this step needs only that continuity, proves $\psi_\theta \rightarrow \psi$ in $C_t \mathcal{D}$ without using the derivative sought.

The exact state difference quotient is its propagated initial quotient minus the Duhamel integral of $(H_{b_h} - H_b)\psi/h$. Taylor expansion gives $B\psi\beta_h + o(1)$, uniformly in time. Strong

propagator convergence is uniform on the compact image of $B(t)\psi(t)$, so this quotient satisfies the first equation in (410) with β_h and an error tending to zero. The force quotient is exactly

$$\langle \xi_h, B_h \psi_h \rangle + \langle \psi, [(B_h - B)/h] \psi_h \rangle + \langle B \psi, \xi_h \rangle.$$

Graph continuity, not graph boundedness alone, makes this $2 \operatorname{Re} \langle \xi_h, B \psi \rangle + \beta_h \langle \psi, C \psi \rangle + o(1)$. Eliminating ξ gives a linear Volterra system whose force kernel is

$$2 \operatorname{Re} \langle -iU(t, s)B(s)\psi(s), B(t)\psi(t) \rangle,$$

continuous and bounded on the time triangle. Its ordinary Volterra estimate and Gronwall prove convergence of the actual difference quotients to the unique displayed solution, and continuity as the base datum varies. The classical equations give convergence in C_t^1 for β, π ; differentiating $\dot{b} = -\kappa p/(6V_0 b)$ once in physical time gives the additional C_t^2 conclusion. No derivative of $B\xi$ is used. $\square$

For a nonnormalized seed v_0 ,

$$\eta = \frac{\partial_\theta v_0}{\|v_0\|} - \frac{v_0 \operatorname{Re} \langle v_0, \partial_\theta v_0 \rangle}{\|v_0\|^3}, \quad \delta E_0 = 2 \operatorname{Re} \langle \eta, H_{b_0} \psi_0 \rangle + \beta_0 F_0,$$

and

$$\pi_0 = \frac{p_0}{2} \left[\frac{\beta_0}{b_0} + \frac{\delta E_0 + 3\Lambda V_0 b_0^2 \beta_0 / \kappa}{M_0} \right].$$

These retain normalization and the constraint contact. The identity $\delta \mathcal{C} = 0$ holds along the actual family. Neither a second derivative of the coupled solution nor a graph-valued first state derivative follows from this theorem.

26.6 The action source and its causal two-source kernel

At one of these Einstein histories let ϵ_+, ϵ_- be independent original-source probes on the ket and bra branches. Their complete fiber Hamiltonians are $H_{\gamma^* g_\epsilon, \lambda_\epsilon}$; ambient coupling profiles include their prescribed pullbacks and material profiles keep their material convention. Compactness of Γ makes the graph and two-response estimates uniform. With fixed normalized initial $\Xi = \iota \psi_0$, define

$$Z(\epsilon_+, \epsilon_-) = \left\langle \widehat{U}_{\epsilon_-}(T, 0)\Xi, \widehat{U}_{\epsilon_+}(T, 0)\Xi \right\rangle. \quad (411)$$

This is an actual positive Gram kernel with unit diagonal. Its entries are C^2 by Theorem 26.3; positivity alone would not prove their differentiability.

Let H_i, H_{ij} be the literal source derivatives and $\Psi(t) = \widehat{U}(t, 0)\Xi$. The action response is $W_i = -\int \langle \Psi, H_i \Psi \rangle dt$, a metric source when i is a metric test. At equal branch sources, $\partial_{+i} Z = iW_i$. Put

$$\chi_i(t) = -i \int_0^t \widehat{U}(t, s) H_i(s) \Psi(s) ds, \quad F_{ij}(t, s) = \left\langle H_i(t) \Psi(t), \widehat{U}(t, s) H_j(s) \Psi(s) \right\rangle.$$

Proposition 26.7 (Causal response from the two branches). *The exact derivative of the action-source readout is*

$$\begin{aligned} D_j W_i &= -i (Z_{+j, +i} + Z_{-j, +i}), & Z_{-j, +i} &= \langle \chi_j(T), \chi_i(T) \rangle, \\ D_j W_i &= -\int_0^T \langle \Psi, H_{ij} \Psi \rangle dt - 2 \int_{0 < s < t < T} \operatorname{Im} F_{ij}(t, s) ds dt. \end{aligned} \quad (412)$$

Each double insertion uses one shared reference, not two independently averaged references. In general this causal readout is not symmetric in i, j .

Proof. Differentiating the equal-branch identity varies both branches, proving the first line. The same-branch Hessian is $-i \int \langle \Psi, H_{ij} \Psi \rangle - \int_{s < t} (F_{ij} + F_{ji})$. The cross-branch Gram is the integral over the full time square; its two triangles give $\int_{s < t} (\overline{F_{ij}} + F_{ji})$. Addition and multiplication by $-i$ give the second line, with its contact sign. All pairings are domain safe by the two-response theorem. Equivalently differentiating the source expectation gives $-\int [2 \operatorname{Re} \langle \chi_j, H_i \Psi \rangle + \langle \Psi, H_{ij} \Psi \rangle]$, the same expression without applying H_i to χ_j . $\square$

If the initial source experiment has a varying normalized Ξ_ϵ , the diagonal first derivative also contains $\langle \Xi, \Xi_i \rangle$. Its next derivative contains $\langle \Xi_j, \Xi_i \rangle + \langle \Xi, \Xi_{ij} \rangle$. Normalization does not erase the imaginary preparation connection. These are distinct from changing the preparation parameter θ along the actual coupled family.

For a C^1 compact observer k_θ along that family, let $k_c = \partial_\theta g_\theta$ and, in each reference,

$$A_k = DH[\gamma^* k_\theta], \quad A_{k;\theta} = D^2 H[\gamma^* k_c, \gamma^* k_\theta] + DH[\gamma^* \partial_\theta k_\theta].$$

All actual frame and anchor derivatives are included before these expressions are used. The pairing rule gives the total all-test source derivative

$$\partial_\theta W_\theta(k_\theta) = - \int_I \int_\Gamma [2 \operatorname{Re} \langle \xi, A_k \psi \rangle + \langle \psi, A_{k;\theta} \psi \rangle] d\mu dt. \quad (413)$$

Here ξ includes both its initial preparation term and the metric-feedback integral in (410). This is also the mixed preparation/external first jet obtained by differentiating $\partial_{+i} Z = iW_i$. It requires one coupled derivative, not a second coupled derivative or a three-insertion theorem.

Writing $\mathbb{T} = \sqrt{|g|} T^{\mu\nu}$ and $\mathbb{E} = \sqrt{|g|} (G + \Lambda g)^{\mu\nu}$, the actual C^1 , C_t^2 -valued metric family satisfies

$$DE[g][k_c] = \kappa \delta \mathbb{T}$$

on every compact metric test, with $\delta \mathbb{T}$ given by (413). Density and index variations remain when converting this equation to covariant tensors. The Ward derivative likewise is

$$(\delta W)_{\text{fixed}}(\mathcal{L}_X g) + W(\mathcal{L}_X k_c) = 0,$$

with $W(\mathcal{L}_X g)$ for a moving vector test. A frozen divergence of δT alone would omit this connection contact.

26.7 Exact continuation and its physical scope

Uniqueness gives identity, restart and local composition for the complete state, geometry and source. The restarted history retains its entire quantum preparation and source prescription; neither the bare ray nor (E, P) alone is an autonomous Cauchy datum. For a bounded observer the actual Heisenberg transport preserves every ordered kernel. A closed unbounded observer transports with domain $U^* D(A)$, not an assumed unchanged domain. Compact interior source tests translate with the same history. Endpoint experiments retain their separate canonical boundary terms.

The differentiated restart uses the actual $\xi(s) \in \mathcal{H}$ as its mild initial variation. It does not reapply the coupled theorem to an arbitrary graph-differentiable family at that restart. Reference conditioning also changes the prepared source: the Einstein equation proved for the uniform ensemble is not asserted for every conditioned fiber with its old metric held fixed.

The result is a local, genuinely coupled native Einstein family with full action-source observations, strong first and weak mixed-second external responses, and an actual first coupled derivative. The finite spatial-descriptor failure remains. External ten-component test coverage is not an arbitrary inhomogeneous backreaction theorem or a quantum hypersurface-deformation algebra. The constants retain their mesh and source dependence. Attachment to an occupied physical particle/composite family requires its actual channel, reference, source and continuation comparison; it is not supplied by equality of labels or by the existence of this native family.

27 A coupled regulated matter–gravity realization

The regulator provides one physical source with actual mixed matter response. We now construct a gravitational realization whose source is evaluated on the same evolving quantum state. The metric restriction is a compact constant-curvature spatial quotient with one scale factor; the resulting geometry is locally of FLRW form. Its homogeneous source-response identities do not assert a full local Standard Model stress tensor or the inhomogeneous WDW constraint algebra.

27.1 A smooth physical interaction from the occupied source

Let $H_0 \geq 1$ be the physical regulator Hamiltonian shifted by a fixed target constant. That vacuum/source shift is retained in every subsequent gravitational equation. Let Y be its nonzero Yukawa matrix-multiplication operator on the natural closed physical domain. Quartic coercivity gives, after increasing the fixed shift if necessary,

$$\|Yu\|^2 \leq C \langle u, H_0 u \rangle, \quad YH_0^{-1/2} \in \mathcal{B}(\mathcal{H}).$$

For a fixed $\beta > 0$, define

$$B = e^{-\beta H_0} Y e^{-\beta H_0}. \quad (414)$$

Lemma 27.1 (Nonzero smooth physical interaction). *The operator B is bounded, self-adjoint, compact and nonzero. It preserves the physical gauge space and is not a scalar multiple of the identity there. For all nonnegative integers k, l , $H_0^k B H_0^l$ extends boundedly to $\mathcal{H}$.*

Proof. The factorization

$$H_0^k B H_0^l = (H_0^k e^{-\beta H_0}) (Y H_0^{-1/2}) (H_0^{l+1/2} e^{-\beta H_0})$$

has three bounded factors. Symmetry of the sandwich makes B bounded self-adjoint. Compactness follows from the compact heat factor, and gauge invariance from the physical actions of Y and H_0 . If $B = 0$, injectivity of the left heat factor gives $Y e^{-\beta H_0} = 0$. The closed kernel of Y would contain the dense range of the heat operator, forcing $Y = 0$, contrary to the nonzero physical pair construction. Finally, the physical space is infinite dimensional: independent radial Higgs functions times the invariant Fock vacuum already lie in it. A nonzero compact operator cannot be a nonzero scalar multiple of its identity. $\square$

The heat parameter is a declared regulator coordinate. The operator B is an actual bounded observation from the mixed source; it is not identified with an elementary local Yukawa density without a separate comparison theorem.

27.2 The common Hamiltonian and source

Use units $8\pi G = 1$. Fix a compact hyperbolic reference metric h_k of sectional curvature $k = -1$, and retain its actual fiducial volume $V_\Sigma = \int_\Sigma d\text{vol}_{h_k} > 0$. Its physical volume is $V_\Sigma a^3$; no independent unit-volume normalization is imposed. Let $a > 0$, p_a be a scale factor and its full canonical momentum, and (ϕ, p_ϕ) a scalar clock pair with full canonical momentum. Fix a cosmological constant Λ , and real C^2 functions $V(\phi)$, $h(a) > 0$, $f(a, \phi)$. Fix $g \neq 0$. On the physical space of Section 12 put

$$H_q(a, \phi) = h(a)H_0 + gf(a, \phi)B, \quad D(H_q) = D(H_0). \quad (415)$$

This is the total quantum energy operator, not an energy density; it is self-adjoint and semi-bounded at every retained parameter point. The total Hamiltonian constraint is

$$\begin{aligned} \mathcal{C} = & -\frac{p_a^2}{12V_\Sigma a} - 3V_\Sigma k a + V_\Sigma \Lambda a^3 + \frac{p_\phi^2}{2V_\Sigma a^3} \\ & + V_\Sigma a^3 V(\phi) + \langle \psi, H_q(a, \phi) \psi \rangle = 0, \end{aligned} \quad (416)$$

where $\|\psi\| = 1$. The vacuum shift in H_0 , the functions h, f and the smoothing parameter remain part of this source.

Choose lapse one and the normalized-vector representative of quantum projective space. Hamilton's equations and the quantum symplectic form give

$$\dot{a} = -\frac{p_a}{6V_\Sigma a}, \quad (417)$$

$$\dot{p}_a = -\frac{p_a^2}{12V_\Sigma a^2} + 3V_\Sigma k - 3V_\Sigma \Lambda a^2 + \frac{3p_\phi^2}{2V_\Sigma a^4} - 3V_\Sigma a^2 V(\phi) - \langle \psi, (\partial_a H_q) \psi \rangle, \quad (418)$$

$$\dot{\phi} = \frac{p_\phi}{V_\Sigma a^3}, \quad \dot{p}_\phi = -V_\Sigma a^3 V'(\phi) - \langle \psi, (\partial_\phi H_q) \psi \rangle, \quad (419)$$

$$i\dot{\psi} = H_q(a, \phi)\psi. \quad (420)$$

The partial source derivatives hold the state fixed. Their values along the evolution are subsequently evaluated on the actual changed state.

Theorem 27.2 (Local coupled quantum–gravitational occupation). *For every initial datum with $a_0 > 0$, normalized $\psi_0 \in D(H_0)$ and finite canonical coordinates, (417)–(420) have a unique local solution on an interval where $a > 0$. The state is continuous in $D(H_0)$, strongly differentiable in $\mathcal{H}$, and has constant norm. The total constraint $\mathcal{C}$ is constant. Initial data on $\mathcal{C} = 0$ therefore remain on that surface.*

Proof. Use $X = D(H_0)$ with graph norm. Introduce τ by $\dot{\tau} = h(a)$, $\tau(0) = 0$, and write $\psi = e^{-i\tau H_0} \chi$. Define $B_\tau = e^{i\tau H_0} B e^{-i\tau H_0}$. Lemma 27.1 makes B_τ bounded on X and locally Lipschitz in τ in $\mathcal{B}(X)$. Its derivative is $ie^{i\tau H_0} [H_0, B] e^{-i\tau H_0}$; the commutator and its graph-norm counterpart are bounded by that lemma.

The quantum equation becomes

$$\dot{\chi} = -igf(a, \phi) B_\tau \chi.$$

Every source expression in the canonical equations is a combination of $\langle \chi, H_0 \chi \rangle$ and $\langle \chi, B_\tau \chi \rangle$, with smooth scalar coefficients. These expressions are locally Lipschitz on the product of X and the finite-dimensional variables. The complete interaction-picture system is a locally Lipschitz Banach-space ODE on $a > 0$, so local existence and uniqueness follow. This constructs the nonautonomous Schrödinger evolution on the stated domain; it does not assume its existence independently.

Self-adjointness of B_τ gives $d\|\chi\|^2/dt = 0$. Transforming back proves norm conservation and differentiability of ψ in $\mathcal{H}$. The graph-space quadratic form $\langle \chi, H_0 \chi \rangle$ is differentiable, and the smoothed interaction is differentiable there. Direct differentiation cancels the quantum commutator contribution:

$$\frac{d}{dt} \langle \psi, H_q \psi \rangle = \dot{a} \langle \psi, (\partial_a H_q) \psi \rangle + \dot{\phi} \langle \psi, (\partial_\phi H_q) \psi \rangle. \quad (421)$$

Differentiating the classical part of $\mathcal{C}$ using its Hamilton equations gives the negative of these two terms. Hence $\dot{\mathcal{C}} = 0$. $\square$

The theorem is local. Neither passage through $a = 0$ nor a singularity resolution or unregulated limit is contained in its conclusion.

Proposition 27.3 (Explicit occupied data with a mixed source). *The constraint surface contains normalized smooth quantum data with nonzero mixed force, an expanding initial metric and a locally monotone scalar clock.*

Proof. The nonzero compact self-adjoint B has an eigenvalue $b \neq 0$ and normalized eigenvector ψ_b . Since $\psi_b = b^{-1}B\psi_b$, it belongs to every $D(H_0^m)$. Set

$$a_0 = 1, \quad \phi_0 = 0, \quad p_{\phi,0} = V_\Sigma, \quad \Lambda = 0, \quad V = 0, \quad h(a) = a^{-1}, \quad f(a, \phi) = a + \phi.$$

Choose $0 < |g| \|B\| < 1/2$. Then $H_q(1, 0) \geq 1/2$, so $E_b = \langle \psi_b, H_q(1, 0)\psi_b \rangle > 0$. Define

$$p_{a,0} = -\sqrt{12V_\Sigma[(3 + 1/2)V_\Sigma + E_b]}.$$

Equation (416) holds exactly, while $\dot{a}_0 > 0$ and $\dot{\phi}_0 = 1$. The clock remains monotone on a smaller interval. Both source derivatives contain the nonzero mixed term gb . Because B is nonscalar, its expectation varies between normalized smooth physical states. The source response therefore depends on the quantum state. The corresponding canonical constraint datum is solved for each source energy; it is not held fixed while that energy is changed. $\square$

27.3 Exact homogeneous source exchange

Let

$$E_q = \langle \psi, H_q \psi \rangle, \quad \rho_q = \frac{E_q}{V_\Sigma a^3}, \quad P_q = -\frac{\langle \psi, (\partial_a H_q) \psi \rangle}{3V_\Sigma a^2}.$$

These are the source density and pressure for the homogeneous lapse/scale-factor variation. Equation (421) gives

$$\dot{\rho}_q + 3\frac{\dot{a}}{a}(\rho_q + P_q) = \frac{\dot{\phi}}{V_\Sigma a^3} \langle \psi, (\partial_\phi H_q) \psi \rangle. \quad (422)$$

The scalar quantities $\rho_\phi = \dot{\phi}^2/2 + V$, $P_\phi = \dot{\phi}^2/2 - V$ satisfy

$$\dot{\rho}_\phi + 3\frac{\dot{a}}{a}(\rho_\phi + P_\phi) = -\frac{\dot{\phi}}{V_\Sigma a^3} \langle \psi, (\partial_\phi H_q) \psi \rangle. \quad (423)$$

The two exchange terms cancel exactly. With $p_a = -6V_\Sigma a \dot{a}$, the conserved constraint and the p_a equation become

$$3\left(\frac{\dot{a}}{a}\right)^2 + \frac{3k}{a^2} = \Lambda + \rho_\phi + \rho_q, \quad (424)$$

$$2\frac{\ddot{a}}{a} + \left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} = \Lambda - P_\phi - P_q. \quad (425)$$

Thus $-dt^2 + a(t)^2 h_k$ satisfies the homogeneous Einstein equations with its source computed from the same evolving quantum state and scalar. The quantum source is not frozen during metric evolution.

The construction is a declared regulated coupling of the occupied matter source. Its homogeneous variation does not identify it with the full local continuum Standard Model stress tensor. In particular, (422)–(425) are not substituted for an arbitrary-test local Ward identity or for the complete hypersurface-deformation relations of the WDW source.

Part IV

Further realizations and source comparison

The spectral and quantum-constraint constructions in this part have their own specified sources. They illustrate additional realization methods and exact comparison criteria without identifying distinct physical theories solely from their shared algebraic form.

28 An interacting ground state and gap on the Yang–Mills source

The bounded auxiliary-pair interaction admits a stationary entangled ground state and a quantitative gap. The original Yang–Mills source has a unique vacuum and a positive vacuum-sector gap [18, Corollaries M.2 and M.5]. These supply the starting point for a spectral calculation of the *changed* interacting Hamiltonian; its dynamics are not identified with the uncoupled source.

Let $(\mathcal{H}_Y, \Omega_Y, H_Y)$ be that exact source, with

$$H_Y \geq 0, \quad \ker H_Y = \mathbb{C}\Omega_Y, \quad \text{spec } H_Y \subset \{0\} \cup [\Delta, \infty), \quad \Delta > 0. \quad (426)$$

Corollary M.2 supplies these properties on the reconstructed OS source; Corollary M.5 identifies its time generator with the Minkowski Hamiltonian and transfers the same gap. Retain the bounded self-adjoint gauge-invariant quasilocal observable A constructed in Proposition 12.7, with

$$\langle \Omega_Y, A\Omega_Y \rangle = 0, \quad A\Omega_Y \neq 0, \quad a := \|A\| > 0.$$

Its existence uses cyclicity of the bounded observable algebra and the nontrivial centered source witness [18, Theorem M.6].

On the full two-mode auxiliary *singlet* Fock space

$$\mathcal{F} = \text{span}\{|0\rangle, |1\rangle, |2\rangle, |12\rangle\},$$

put

$$B = a_1^\dagger a_2^\dagger + a_2 a_1, \quad N = a_1^\dagger a_1 + a_2^\dagger a_2, \quad H_g = H_Y \otimes I + I \otimes \omega N + gA \otimes B, \quad (427)$$

where $\omega > 0$ and $g \in \mathbb{R}$. The norm of B is one. It exchanges the vacuum and paired states and vanishes on the two odd one-fermion states. Accordingly the odd restriction of H_g is exactly two copies of $H_Y + \omega$. Those states remain part of the full spectral theorem below; they are not suppressed by considering the even ground state.

Theorem 28.1 (Occupied interacting ground and a fresh gap). *Let*

$$m = \min\{\Delta, \omega\}, \quad \eta = |g|a < m/4.$$

The self-adjoint operator H_g , on $D(H_Y) \otimes \mathcal{F}$, has one simple ground eigenvalue $E_g \in [-\eta, \eta]$. Its remaining spectrum is contained in $[m - \eta, \infty)$. For $g \neq 0$, define

$$\xi = (H_Y + 2\omega)^{-1} A\Omega_Y, \quad b = \|\xi\| > 0, \quad \kappa = \langle A\Omega_Y, \xi \rangle > 0. \quad (428)$$

Then

$$E_g \leq -\frac{g^2 \kappa}{1 + g^2 b^2} < 0. \quad (429)$$

Consequently $H_g - E_g$ has a unique zero-energy ray and a gap at least

$$m - \eta - E_g \geq m - \eta > 3m/4. \quad (430)$$

The ground ray belongs to the even fermion sector.

Proof. The uncoupled operator H_0 has the unique vacuum $\Psi_0 = \Omega_Y \otimes |0\rangle$ and no other spectrum below m . The perturbation is bounded self-adjoint with norm η , so the stated operator domain is unchanged. On the circle $|z| = m/2$,

$$\|(z - H_0)^{-1}\| \leq 2/m, \quad \|g(A \otimes B)(z - H_0)^{-1}\| \leq 2\eta/m < 1/2.$$

The Neumann resolvent formula keeps this circle in the resolvent set along the path sg , $0 \leq s \leq 1$. Its Riesz projection depends continuously on s and retains rank one. The self-adjoint bounded perturbation inclusion places the entire spectrum in

$$[-\eta, \eta] \cup [m - \eta, \infty).$$

Thus the isolated cluster consists of one simple ground eigenvalue.

The inverse in (428) is positive and injective, so $b, \kappa > 0$. Centering and spectral invariance of the vacuum complement give $\xi \perp \Omega_Y$. Evaluate the Rayleigh quotient on $\Psi_0 - g\xi \otimes |12\rangle$. Its squared norm is $1 + g^2 b^2$. Its uncoupled energy is $g^2 \kappa$, the two interaction cross terms contribute $-2g^2 \kappa$, and its pair-pair interaction expectation is zero because B sends the paired state to the vacuum state. The variational principle gives (429). Subtracting E_g from the remaining spectral interval proves (430). Parity commutes with the Riesz projection, which continues the even vacuum projection; the ground ray is therefore even. The unchanged odd block remains bounded below by ω throughout. $\square$

The next result proves entanglement of the *stationary* interacting ground state and provides an explicit nonzero coupling window.

Theorem 28.2 (Quantitative ground-state entanglement). *With a, m, b as above, set*

$$r = \frac{m}{8a}, \quad C = \frac{4}{r^2}, \quad g_{\text{ent}} = \min \left\{ \frac{r}{4}, \frac{b}{4C} \right\} > 0. \quad (431)$$

For every $0 < |g| \leq g_{\text{ent}}$, the normalized ground state of H_g is entangled between $\mathcal{H}_Y$ and the auxiliary pair. Its reduced two-by-two even-Fock density matrix has determinant at least

$$\frac{b^2 g^2}{64} > 0. \quad (432)$$

Proof. Continue the same Riesz projection to complex coupling s with $|s| \leq r$. The Neumann expansion on $|z| = m/2$ gives

$$\|P(s) - P(0)\| \leq \frac{q}{1-q} \leq \frac{1}{3}, \quad q = \frac{2a|s|}{m} \leq \frac{1}{4}.$$

Hence

$$v(s) = \frac{P(s)\Psi_0}{\langle \Psi_0, P(s)\Psi_0 \rangle}$$

is holomorphic and has norm at most two. Its denominator differs from one by at most $1/3$, and its intermediate normalization is fixed. Differentiating the simple-eigenvalue equation at zero gives $E'(0) = 0$ and $v'(0) = -\xi \otimes |12\rangle$. Cauchy's coefficient estimates on the radius- r disc therefore imply

$$\|v(g) - \Psi_0 + g\xi \otimes |12\rangle\| \leq Cg^2 \quad (|g| \leq r/2). \quad (433)$$

Indeed the coefficient of order k has norm at most $2r^{-k}$, and summing from $k = 2$ gives $4g^2/r^2$ in this range.

Parity gives $v(g) = u_0(g) \otimes |0\rangle + u_2(g) \otimes |12\rangle$. Put $R_0 = u_0 - \Omega_Y$ and $R_2 = u_2 + g\xi$; each remainder has norm at most Cg^2 . Since $b \leq a/(\Delta + 2\omega) \leq a/m$, the restriction $|g| \leq r/4$ gives $|g|b \leq 1/32$ and $Cg^2 \leq 1/4$. In the Hilbert exterior square, with the determinant normalization,

$$\|u_0 \wedge u_2 + g\Omega_Y \wedge \xi\| \leq Cg^2(1 + |g|b + Cg^2) \leq 2Cg^2.$$

The further condition $|g| \leq b/(4C)$ gives $\|u_0 \wedge u_2\| \geq b|g|/2$. After normalizing $v(g)$, whose norm is at most two, the reduced-state determinant is

$$\frac{\|u_0 \wedge u_2\|^2}{\|v(g)\|^4} \geq \frac{b^2 g^2}{64}.$$

A pure bipartite state with a mixed reduced state is entangled. The norm-analytic bound (433) controls the remainder; no unbounded formal perturbation series is used. $\square$

This theorem strengthens an actual continuum Yang–Mills source by a new bounded interaction, an occupied mixed ground state and a newly proved spectral estimate. The auxiliary modes remain gauge singlets, not the chiral Standard-Model representation inventory. Coupling to a selected bounded quasilocal A is not asserted to define a local Lorentz-covariant Standard-Model action. The interacting ground marginal is not asserted equal to the original Yang–Mills vacuum marginal. Neither a massless photon sector nor all extended-support confinement and scattering sectors have been added. Thus (430) is a source-specific interacting vacuum-sector result, not an unrestricted Standard-Model total-gap claim.

29 An occupied resonance channel with quantitative pole bounds

An unstable-pole theorem needs an actual channel, an authorized analytic continuation, and a nonzero residue. The response form alone supplies none of these analytic objects. BCRS–2EW/H proves the preservation of an existing simple pole and its physical residue under joint channel transport, together with reconstruction from the full analytic germ [4]. The occupied SYM–8 physical assignment supplies the electroweak channels used directly in Part I [20]. We complement those results with a quantitative existence criterion and an explicitly occupied Friedrichs-type continuum-coupled realization [24].

The construction below is a restricted quantum target. Its exact Hamiltonian, state, channel, continuation, pole and residue are constructed together. A further comparison theorem would be required to identify any of its channels with a physical W , Z or Higgs channel of the Standard Model. In particular, we attach no such particle identity by notation.

29.1 A quantitative analytic acquisition criterion

Lemma 29.1 (Quantitative simple-pole acquisition). *Let $E \in \mathbb{R}$ and let F be holomorphic on a neighborhood of the closed disc $\{z \in \mathbb{C} : |z - E| \leq R\}$, with $|F| \leq M$ there. Fix $0 < r < R$, write $L = M/(R - r)$, and set*

$$D_t(z) = E - z - tF(z), \quad t > 0. \quad (434)$$

If

$$tM < r, \quad tL < 1, \quad \text{Im } F(E) > tLM, \quad (435)$$

then D_t has exactly one zero z_t in $|z - E| < r$, counted with multiplicity. That zero is simple and lies in the lower half-plane. Moreover,

$$|z_t - E| \leq tM, \quad |z_t - E + tF(E)| \leq t^2 LM, \quad (436)$$

and

$$\text{Res}_{z=z_t} \frac{1}{D_t(z)} = -\frac{1}{1 + tF'(z_t)} \neq 0. \quad (437)$$

Proof. On $|z - E| = r$, the perturbation tF has modulus strictly smaller than $|E - z|$. Rouché's theorem therefore gives exactly one zero, counted with multiplicity, in the smaller disc. Cauchy's estimate gives $|F'| \leq L$ on its closure. Consequently $D'_t(z) = -1 - tF'(z)$ cannot vanish there.

At the zero, $z_t - E = -tF(z_t)$, which proves the first bound in (436). The line segment from E to z_t lies in the smaller disc; integration of F' along that segment gives the second bound. Taking imaginary parts yields

$$\operatorname{Im} z_t \leq -t \operatorname{Im} F(E) + t^2 LM < 0.$$

The reciprocal of a holomorphic function at a simple zero has residue $1/D'_t(z_t)$, giving (437). $\square$

No pole or zero is a premise of this lemma. Its application requires an independently constructed channel with the stated continued denominator and analytic bounds. A formal self-energy series is insufficient for that purpose. A physical numerator can also be controlled without assuming a nonzero pole residue in advance.

Corollary 29.2 (Noncancellation by a numerator bound). *Under the hypotheses of Lemma 29.1, let N be holomorphic on the smaller disc, with $N(E) \neq 0$ and*

$$|N(z) - N(E)| < |N(E)| \quad \text{for } |z - E| < r.$$

Then N/D_t has the same simple pole, with nonzero residue $-N(z_t)/(1 + tF'(z_t))$.

Proof. The displayed inequality makes N nonzero throughout the smaller disc. Multiplication of the simple-pole Laurent expansion by N proves the claim. $\square$

29.2 A positive continuum-coupled source

All inner products in this section are conjugate-linear in the first argument. Define

$$\mathcal{H}_1 = \mathbb{C} \oplus L^2((0, \infty), dx), \quad e = (1, 0), \quad v(x) = \sqrt{x} e^{-x/2}. \quad (438)$$

Let M_x be multiplication by x , with $D(M_x) = \{f \in L^2 : xf \in L^2\}$. Since

$$\|v\|^2 = \int_0^\infty x e^{-x} dx = 1,$$

the following is a bounded off-diagonal perturbation of $1 \oplus M_x$:

$$H_\lambda(a, f) = (a + \lambda \langle v, f \rangle, M_x f + \lambda a v), \quad D(H_\lambda) = \mathbb{C} \oplus D(M_x), \quad (439)$$

where λ is real. It is the discrete-continuum coupling parameter of the target.

Proposition 29.3 (Positive occupied source). *For $\lambda^2 < 1$, H_λ is self-adjoint and nonnegative, and $\ker H_\lambda = \{0\}$. On*

$$\mathcal{H} = \mathbb{C}\Omega \oplus \mathcal{H}_1, \quad \|\Omega\| = 1, \quad \widehat{H}_\lambda = 0 \oplus H_\lambda, \quad (440)$$

there is a unique normalized zero-energy vacuum up to phase. The bounded operators

$$O_+ = |e\rangle\langle\Omega|, \quad O_- = O_+^* \quad (441)$$

create the norm-one channel vector $O_+\Omega = e$. For $\lambda \neq 0$, the dynamics couples that vector to the continuum.

Proof. The off-diagonal perturbation in (439) is bounded and self-adjoint, with norm $|\lambda|$. The bounded perturbation theorem therefore gives self-adjointness on the displayed domain. Its associated closed form $\mathfrak{h}_\lambda$, on $\mathbb{C} \oplus D(M_x^{1/2})$, satisfies

$$\mathfrak{h}_\lambda[(a, f)] = \int_0^\infty x \left| f(x) + \lambda a \frac{v(x)}{x} \right|^2 dx + (1 - \lambda^2)|a|^2. \quad (442)$$

Here $\int_0^\infty |v(x)|^2/x dx = 1$. The square is interpreted in the weighted form space: v/x need not belong to ordinary L^2 , while $v/\sqrt{x}$ does. Thus the expression is finite on the form domain. It proves nonnegativity; equality forces $a = 0$ and then $f = 0$ almost everywhere. The vacuum extension has exactly the stated zero eigenspace. The rank-one operators in (441) are bounded, and $H_\lambda e = (1, \lambda v)$ has nonzero continuum component when $\lambda \neq 0$. $\square$

The vector state of Ω is a positive state of the represented operator algebra. Its channel has exact two-point and resolvent responses

$$C_\lambda(\tau) = \langle e, e^{-i\tau H_\lambda} e \rangle, \quad G_\lambda(z) = \langle e, (H_\lambda - z)^{-1} e \rangle. \quad (443)$$

The state, Hamiltonian and source vector belong to one Hilbert realization. They are not assembled from independent marginal states.

29.3 Exact resolvent and its continuation sheet

Lemma 29.4 (Exact scalar channel). *For $\text{Im } z > 0$,*

$$G_\lambda(z) = \frac{1}{1 - z - \lambda^2 F(z)}, \quad F(z) = \int_0^\infty \frac{\rho(x)}{x - z} dx, \quad \rho(x) = x e^{-x}. \quad (444)$$

Proof. Solve $(H_\lambda - z)(a, f) = e$. The continuum equation gives $f = -\lambda a(M_x - z)^{-1}v$. Substitution in the discrete equation gives $(1 - z - \lambda^2 F(z))a = 1$. Its coefficient cannot vanish in the upper half-plane because H_λ is self-adjoint. Taking the discrete matrix element proves the formula. This is an exact block elimination, with numerator one, rather than a truncation of a series. $\square$

Put $a_0 = 1/2$, $b_0 = 3/2$ and $\mathcal{D} = \{z : |z - 1| \leq 1/4\}$. Choose logarithms that are real on the positive real axis. The arguments $b_0 - z$ and $z - a_0$ remain in the disc centered at $1/2$ with radius $1/4$, so these choices are unambiguous throughout a neighborhood of $\mathcal{D}$.

Lemma 29.5 (Constructed continuation and uniform bounds). *The continuation of F from the upper half-plane through (a_0, b_0) is the function*

$$\begin{aligned} F_*(z) &= \int_0^{a_0} \frac{\rho(x)}{x - z} dx + \int_{b_0}^\infty \frac{\rho(x)}{x - z} dx \\ &\quad + \int_{a_0}^{b_0} \frac{\rho(x) - \rho(z)}{x - z} dx \\ &\quad + \rho(z) \{ \log(b_0 - z) - \log(z - a_0) + i\pi \}. \end{aligned} \quad (445)$$

It is holomorphic near $\mathcal{D}$ and satisfies

$$|F_*(z)| < 10 \quad (z \in \mathcal{D}), \quad |F'_*(z)| \leq 80 \quad (|z - 1| \leq 1/8), \quad \text{Im } F_*(1) = \pi/e. \quad (446)$$

Proof. The tail integrals are holomorphic near $\mathcal{D}$, since their denominators stay away from zero and the density decays exponentially. The divided difference in the middle integral extends holomorphically at $x = z$. Direct integration of $1/(x - z)$ over (a_0, b_0) gives (445) when $\text{Im } z > 0$: there $\log(a_0 - z) = \log(z - a_0) - i\pi$. Thus the formula constructs the continuation of the physical

upper-half-plane channel. It does not assume the existence of a continuation sheet. At $z = 1$, the nonlogarithmic terms and the two real logarithms are real, giving the last identity in (446).

For the bound, the tail denominators have modulus at least $1/4$; together those integrals have modulus at most $4 \int_0^\infty \rho(x) dx = 4$. The convex hull of $[a_0, b_0]$ and $\mathcal{D}$ lies in $1/2 \leq \operatorname{Re} w \leq 3/2$, $|\operatorname{Im} w| \leq 1/4$. There

$$|\rho'(w)| = |1 - w|e^{-\operatorname{Re} w} < 0.6.$$

Integration of ρ' along the relevant segment bounds the divided difference by 0.6, and the interval has length one. On $\mathcal{D}$, $|\rho(z)| < 5/8$. Each logarithm has modulus at most $\log 4 + \pi/4 < 2.2$, so the logarithmic bracket has modulus less than 7.6. Its contribution is less than 4.75. The total is therefore less than $4 + 0.6 + 4.75 = 9.35 < 10$. Cauchy's estimate, with distance $1/8$ between the smaller and outer circles, gives the derivative bound. $\square$

29.4 An occupied simple resonance with nonzero residue

Theorem 29.6 (Exact resonance on the continuum-coupled target). *For every real λ with*

$$0 < t = \lambda^2 \leq 1/2000, \quad (447)$$

the exact continued channel

$$G_{\lambda,*}(z) = \frac{1}{1 - z - tF_*(z)}$$

has exactly one simple pole z_λ in $|z - 1| < 1/8$, and $\operatorname{Im} z_\lambda < 0$. Its residue is nonzero, and

$$|z_\lambda - 1| \leq 10t, \quad |z_\lambda - 1 + tF_*(1)| \leq 800t^2, \quad (448)$$

$$\operatorname{Res}_{z=z_\lambda} G_{\lambda,*}(z) = -\frac{1}{1 + tF'_*(z_\lambda)}. \quad (449)$$

Writing $z_\lambda = E_\lambda - i\Gamma_\lambda/2$, its positive width satisfies

$$2t(\pi/e - 800t) \leq \Gamma_\lambda \leq 2t(\pi/e + 800t). \quad (450)$$

The choice $\lambda = 1/50$ is an explicit occupied target.

Proof. Apply Lemma 29.1 with $E = 1$, $R = 1/4$, $r = 1/8$, $M = 10$ and $L = 80$. The independent construction in Lemma 29.5 gives the required holomorphic function and bounds. For (447),

$$tM \leq 1/200 < 1/8, \quad tL \leq 0.04 < 1, \quad tLM \leq 0.4 < \pi/e.$$

All hypotheses therefore hold. The general location and residue formulas give (448) and (449). Taking imaginary parts of the second location bound and using $\operatorname{Im} F_*(1) = \pi/e$ gives (450). Its lower bound is strictly positive. The inverse residue has modulus between $1 - 80t$ and $1 + 80t$, so physical-channel cancellation is impossible. $\square$

Proposition 29.7 (Resonance versus eigenstate). *For nonzero λ in (447), H_λ has no eigenvector at any positive real energy. The continued nonreal pole is not an eigenvalue of the self-adjoint Hamiltonian.*

Proof. At real energy $E > 0$, the continuum equation of an eigenvector would give $f(x) = -\lambda av(x)/(x - E)$. If $a \neq 0$, this is not square integrable, since $v(E) > 0$. If $a = 0$, the continuum equation would require an eigenvector of the multiplication operator M_x , whose point eigenspaces on Lebesgue L^2 are zero. Nonreal eigenvalues are excluded by self-adjointness. $\square$

The theorem constructs a lower-half-plane analytic resonance of an actual positive source channel. It does not assert an exact exponential decay law or a relativistic scattering realization. Neither assertion was used to establish the continued pole. The energy and width bounds are dimensionless properties of the specified comparison target.

29.5 Source identity and the Standard-Model comparison

Proposition 29.8 (Joint source preservation). *Let a unitary map jointly transport $\hat{H}_\lambda$, Ω , e , O_+ and O_- , retaining the marked continuation domain and sheet. It preserves the physical scalar channel, its continuation, the pole z_λ , and the complete Laurent germ, including the residue and regular part.*

Proof. The unitary resolvent identity preserves $\langle e, (H_\lambda - z)^{-1}e \rangle$ on the physical upper half-plane. Uniqueness of analytic continuation on the marked domain preserves its continued function. Equality of meromorphic functions preserves every Laurent coefficient. Source and sheet attachment are included in the transport, so this does not identify independently labelled branches merely because some numerical pole coordinates coincide. $\square$

The full target used here retains the Hilbert space, Hamiltonian, vacuum, source operators, channel vector, density ρ , coupling λ , and marked continuation data. Changing the density, threshold, coupling or source channel generally changes the target. A unitary presentation change does not authorize such a modification.

To apply this separate comparison model to a Standard-Model channel, its operator or physical amplitude must be identified on the intended common target. This is an optional construction route, not a prerequisite for the already attached Higgs channel of Theorem 4.3. One possible comparison is an exact source-preserving reduction of its projected resolvent to an analytic denominator with the contour bounds of Lemma 29.1. The density need not equal the solvable density used above; what matters is that the actual continued channel and bounds have been proved. Mixed neutral channels must retain their full matrix denominator and physical-current numerator, as required by the BCRS-2EW/H transport theorem [4].

An actually identified physical channel can be compared to the response and identity architecture of SYM-8 and the Higgs identity theorem [20, 22]. The comparison must preserve the operator, state, source, gauge and anomaly data, renormalization domain, boundary conditions and continuation. A finite perturbative agreement or a fitted resonance profile does not prove that identification. The present occupied comparison channel consequently supplies a rigorous restricted resonance result. It does not replace the directly received physical assignment or change its established source-normalized Higgs attachment.

30 Finite chiral constraint corrections and the physical-state boundary

The Wheeler–DeWitt sources supply a completed chiral operator family on their declared common source, with the retained renormalization and anomaly construction [11, 12]. WDW-6 also states a raw off-shell Hamiltonian commutator. Its literal full-coframe identification is not received unconditionally here: the torsion-free horizontal spin lift differs from the frozen raw generator by the total Lorentz action, and an admitted degree-zero affine coframe test separates the two off shell. The finite principal-response construction below does not depend on that universal closure claim. CSP-C supplies the matched classical antecedent and its exact quantum-nonpromotion boundary; the quantum operator conclusions used here come separately from WDW-5 and WDW-6 [2]. This section proves a finite extension of the published source-point correction mechanism to chiral coefficients. It also identifies which part of that extension can be made small in the source topology and which part still requires a physical-state construction.

30.1 The received operator and source

Fix the compact, connected, boundaryless spin branch of WDW-5, its admitted trivial gauge bundle, the derived torsion-free spin connection, the declared three-generation minimal chiral representation, one Higgs doublet, and the local scalar-clock source. The topology, global gauge form, numerical couplings, Yukawa matrices, and clock-failure boundary remain fixed source coordinates. In particular, this argument neither adjoins a right-handed neutral field nor identifies a heavy-neutral projection with an additional WDW matter variable.

Use the W5.8-normalized, W4B-symmetrized half-density baseline received by W6R.O18B: left-coefficient ordering, the declared flat configuration-chart connection, zero additional local finite normalization, and the baseline fermionic polarization. Its additional exterior-degree-shift list is empty [11, Secs. 11–12]. Write the completed source as

$$V_B^{\text{ren}} = \bigoplus_{n \geq 0}^{\text{alg}} V_{B,n}^{\text{ren}}, \quad V_{B,n}^{\text{alg}} \subset C^\infty(U_B, \hat{\Lambda}^n \mathfrak{f}). \quad (451)$$

Each degree is completed in the specified graph topology, and the resulting carrier has the locally convex direct-sum topology. Thus every element has finite exterior support; no completion over particle number is introduced. The graph topology retains all registered finite operator words and compact configuration/derivative-test seminorms [12, Sec. 27.2].

For a lapse N , the complete baseline Hamiltonian in degree n is

$$C_N^{(n)} = L_N^{\text{even}} \otimes I + d\Gamma_n(A_{D,N}(q)' + A_{Y,N}(q)'). \quad (452)$$

Here L_N^{even} includes the gravitational, clock, gauge, and Higgs kinetic blocks and the entire multiplication potential. A prime on the one-particle operator denotes its strong transpose. The Dirac and Yukawa terms are not discarded in any result below. In particular, “order zero” for these terms means zero order in differentiation of the *configuration-space section*; the Dirac coefficient still has its spatial differential action and associated Sobolev estimates.

Let $\{C_\alpha\}_{\alpha \in \mathcal{L}_B}$ be the complete received constraint-label family, including all admissible smearings. The relevant published raw-closure formula is the following; below it is a receiving condition only where explicitly imposed:

$$[C_{N_0}, C_{N_1}] = i\hbar (D[s] + G[\iota_s A] + J[\iota_s \omega_{\text{LC}}]). \quad (453)$$

This records the formula of [12, W6R.O18B], not an established identity on every vector of the unchanged full-coframe baseline. Write $V[s] = D[s] + G[\iota_s A] + J[\iota_s \omega_{\text{LC}}]$, with $V[s] = -i\hbar X_V$ on degree-zero coframe cylinders. The distinction can be tested on its actual source. At a smooth configuration q_* , choose lapse jets $N = x^1, M = x^2$ in an adapted normal frame. The raw coframe variation has a nonzero antisymmetric rotation λ_* . Choose a nonnegative smooth bump χ supported where that rotation is nonzero. On the adapted spatial components set $\ell(v) = \int \chi \operatorname{tr}[\lambda_*^T(v e_*^{-1})]$. This fixed smooth compact dual coframe density annihilates every horizontal metric lift and satisfies $\ell(X_V e_*) = \int \chi \|\lambda_*\|^2 > 0$. The centered affine degree-zero cylinder $F(q) = \ell(e - e_*)$ is admitted. Its affine Hessian and degree-zero exterior action vanish identically. For the literal left-coefficient operator $C_N = T_N + V_N + Q_N$, therefore $C_N F = V_N F$. At q_* , the only surviving product Hessian is $dV_M \otimes \ell + \ell \otimes dV_M$. The metric-pullback kinetic tensor has horizontal image, so its contraction with ℓ vanishes. All these are smooth finite-rank contractions. Consequently

$$([C_N, C_M]F)(q_*) = 0, \quad (i\hbar V[s]F)(q_*) = \hbar^2 \int \chi \|\lambda_*\|^2 > 0.$$

The probe is not Lorentz invariant, as appropriate to an off-shell full-source test. It does not refute a reduced identity or remove the independent renormalized source and anomaly results.

A Lorentz-reduced or source-restricted identity requires its own verification; deleting gauge or Lorentz terms does not provide it. Only Corollary 30.5 below invokes the displayed raw-closure equality. The other finite-response and graph estimates use the actual operators and their stated principal coefficients directly.

Lemma 30.1 (Degreewise joint-kernel reduction). *For the baseline family,*

$$Z_B := \bigcap_{\alpha \in \mathcal{L}_B} \ker C_\alpha = \bigoplus_{n \geq 0}^{\text{alg}} Z_{B,n}, \quad Z_{B,n} := \bigcap_{\alpha \in \mathcal{L}_B} \ker(C_\alpha|_{V_{B,n}^{\text{ren}}}). \quad (454)$$

Proof. Every constraint preserves exterior degree on the algebraic source. For the Hamiltonian this follows from (452); for the linear constraints it follows from their configuration derivations and degree-preserving representation actions. The degree projections are continuous in (451). Their commutation with the constraints therefore persists under degreewise graph completion. Applying each projection to a joint solution gives the forward inclusion. Linearity and finite exterior support give the reverse inclusion. $\square$

This reduction does not assert that any $Z_{B,n}$ is nonzero. It does show that occupying one fixed degree does not require estimates uniform as $n \rightarrow \infty$.

Proposition 30.2 (Finite exterior support does not imply nilpotence). *Finite exterior degree alone does not make the fermionic term in (452) nilpotent and does not justify termination of a perturbative inverse series.*

Proof. For one fermionic mode, creation followed by contraction is a number operator N_f satisfying $N_f^2 = N_f \neq 0$. More specifically, on a coupled pair of chiral components a nonzero Yukawa multiplication block has the form

$$Y_H = \begin{pmatrix} 0 & m(H) \\ \frac{0}{m(H)} & 0 \end{pmatrix}, \quad Y_H^2 = |m(H)|^2 I.$$

In degree one its exterior lift is the original one-particle operator. On a region where the fixed Yukawa coupling and Higgs component give $m(H) \neq 0$, every even power of this block is nonzero. This local multiplication calculation requires no assumption about global Dirac eigenfunctions. A perturbative inverse would instead need an independently proved contraction, nilpotent filtration, or convergent inverse estimate. $\square$

30.2 A vector-valued finite neutralizer

The published scalar correction uses four fresh metric probes and the four ordered observations formed from two Hamiltonian rows [13]. Its fourth-jet mechanism extends to Branch B, but its regulated response matrix must not be substituted for the renormalized endpoint response. We construct the required endpoint probes directly.

Let q_δ be the small nonzero gauge–Higgs source germ received from the WDW-4B construction by WDW-5. Its gravity–scalar correction is $O(\delta^2)$ relative to the antecedent source; the two full constraint maps are not identified [11, Sec. 2]. Work on the local configuration chart before quotienting and write $h = h(q_\delta)$. The baseline endpoint has already removed the outer heat regulator: its gravitational principal tensor is the ultralocal tensor $-2\kappa\hbar^2\mathcal{G}(h)$, with

$$\mathcal{G}_{ijkl}(h) = \frac{h_{ik}h_{jl} + h_{il}h_{jk} - h_{ij}h_{kl}}{2\sqrt{\det h}}. \quad (455)$$

These are the accepted momentum and density conventions [11, Secs. 5.2, 11.2]; no heat factor remains in the endpoint principal symbol [12, Part IV, Sec. 9.1].

Choose four pairwise disjoint coordinate balls and nonzero real smooth functions f_j supported compactly inside them, $0 \leq j \leq 3$. The metric covector-densities

$$\ell_j^{ab} = f_j h^{ab} d\mu_h$$

are smooth and therefore belong to the accepted trace-admissible source [12, Part III, Secs. 2.1–2.2]. Let $\varpi : e \mapsto h(e)$ be the local coframe-to-metric map. Pull each covector back at the fixed source point and define the affine coframe cylinders

$$\tilde{\ell}_j = (d\varpi_{e_\delta})^* \ell_j, \quad \eta_j(q) = \tilde{\ell}_j(e - e_\delta).$$

They vanish at q_δ and are independent, since their supports are disjoint and $d\varpi$ is surjective onto metric variations. The canonical identity $P^i_I = 2\pi^{ij} e_{jI}$ identifies their principal contraction with the metric contraction above. This uses only the coframe pullback of the gravitational kinetic term on its Lorentz-reduced sector [11, Sec. 8.2]; it does not invert the Lorentz-null directions or change the baseline ordering. The covectors are fixed at q_δ , so these remain admitted affine cylinders in the original coframe chart.

Take the allowed smooth lapse labels $N_0 = 1$ and N_1 with constant positive value n_j on a neighborhood of $\text{supp } f_j$, where $n_0 \neq n_1$. Such a global smooth N_1 exists by disjoint support and a partition of unity. Let

$$p_a^\delta(z) = \sum_{i,j=0}^3 g_{a,ij}(q_\delta) z_i z_j \quad (456)$$

be the restriction of their actual baseline principal symbols to these four probe covectors. Their responses are explicit:

$$p_0^\delta(z) = \sum_{j=0}^3 a_j z_j^2, \quad p_1^\delta(z) = \sum_{j=0}^3 n_j a_j z_j^2, \quad a_j = 3\kappa\hbar^2 \int_\Sigma f_j^2 d\mu_h > 0. \quad (457)$$

Indeed, the DeWitt contraction of $f_j h^{ij} \sqrt{\det h}$ with itself is $-\frac{3}{2} f_j^2 \sqrt{\det h}$ in three spatial dimensions; the prefactor $-2\kappa\hbar^2$ gives the displayed a_j . All cross contractions vanish by disjoint support and ultralocality. The diagonal coefficient minor from columns 0, 1 is $a_0 a_1 (n_1 - n_0) \neq 0$. Thus p_0^δ, p_1^δ are independent at the actual nonzero-matter source, without assuming continuity of a regulated determinant under heat removal. These calculations retain all other field coordinates at q_δ and require no quotient regularity at zero matter. They establish the rank input of the scalar mechanism anew for the received Branch-B endpoint.

Lemma 30.3 (Top-jet tensorization). *Fix a finite span E_n of accepted algebraic exterior vectors of degree n . Invariance of E_n under the Dirac operator is not required. For $|\alpha| = 4$ and $\xi \in E_n$, set*

$$H_{\alpha,\xi}(q) = \frac{\eta(q)^\alpha}{\alpha!} \xi.$$

Then

$$(C_d C_a H_{\alpha,\xi})(q_\delta) = [z^\alpha] (p_a^\delta(z) p_d^\delta(z)) \xi. \quad (458)$$

All configuration derivatives of $H_{\alpha,\xi}$ through order three vanish at q_δ . Applying any linear constraint and then at most two configuration derivatives also gives zero at that point.

Proof. Two second-order configuration operators can consume at most four derivatives of the section. The section vanishes to order four. If a derivative falls on a coefficient, fewer than four derivatives remain to hit its polynomial factor. A potential or Dirac/Yukawa term likewise consumes too few configuration derivatives. Every such term retains a positive power of η at the source and vanishes there. Gauge, Higgs, and clock derivatives either kill this pure metric jet or differentiate a coefficient, which is the same lower-order case. Thus only the principal–principal

partition survives, with the constant exterior vector unchanged, proving (458). Linear constraints have configuration order at most one, with representation terms of order zero; the last assertion follows from the same derivative count. $\square$

The absence of the Yukawa term from (458) is therefore a proved filtration statement for this response, not a deletion of that term from the quantum theory.

Theorem 30.4 (Finite Branch-B neutralization). *At any source point and matched probe system just described, let A_δ be the four-by-35 matrix*

$$(A_\delta)_{(a,d),\alpha} = [z^\alpha](p_a^\delta p_d^\delta), \quad |\alpha| = 4,$$

with row order 00, 01, 10, 11. Then

$$\text{rank } A_\delta = 3, \quad \ker A_\delta^T = \text{span}\{(0, 1, -1, 0)\}.$$

For target vectors $y_{ad} \in E_n$, a fourth-jet correction with ordered response $-y$ exists if and only if $y_{01} = y_{10}$. When this holds, delete the duplicated row to form $A_{\delta,\text{eff}}$ and set

$$B_\delta = A_{\delta,\text{eff}}^* (A_{\delta,\text{eff}} A_{\delta,\text{eff}}^*)^{-1}, \quad c = -B_\delta y_{\text{eff}}, \quad K = \sum_{|\alpha|=4} \frac{\eta^\alpha}{\alpha!} c_\alpha. \quad (459)$$

This assembled K has exactly the claimed complete ordered response. For real principal coefficients, the adjoints in (459) are ordinary transposes.

Proof. The independent quadratic polynomials p_0^δ, p_1^δ have three independent products $(p_0^\delta)^2, p_0^\delta p_1^\delta, (p_1^\delta)^2$. Otherwise their ratio in the rational function field would satisfy a nonzero scalar quadratic equation and would consequently be constant, contradicting independence. The effective coefficient matrix therefore has row rank three. Commutation of the scalar principal symbols gives the mixed-row duplication, and there are no other row relations. The Gram matrix in (459) is invertible and $A_{\delta,\text{eff}} B_\delta = I$. Apply Lemma 30.3 to the actual finite sum defining K . Its ordered response is $A_\delta c = -y$, including the unreduced mixed row. All target operations are finite scalar linear combinations in E_n ; no exterior component has been replaced by a scalar target. $\square$

For the next corollary only, assume additionally that (453), applied to its chosen input Δ and evaluated at q_δ , equals the actual commutator observation. This is a declared receiving hypothesis, not a discharged property of the full frozen coframe source. Under this hypothesis the stated linear residual implies mixed-row compatibility. Without it, one may instead check $([C_{N_0}, C_{N_1}]\Delta)(q_\delta) = 0$ directly and use Theorem 30.4; vanishing of the printed raw linear residual alone is insufficient.

Corollary 30.5 (Compatibility from the complete commutator). *Let Δ belong to the actual common source and put $y_{ad} = (C_d C_a \Delta)(q_\delta)$. Suppose these four values lie in the chosen finite exterior span and*

$$(D[s] + G[\iota_s A] + J[\iota_s \omega_{\text{LC}}])\Delta(q_\delta) = 0.$$

Then $y_{01} = y_{10}$, and the correction (459) cancels all four observations without changing the value or the first three configuration jets of Δ .

Proof. The difference of the mixed observations is, up to sign, the complete commutator (453) applied to Δ and evaluated at the source. It vanishes under the stated earlier-stage linear condition. Theorem 30.4 supplies the correction, and its fourth-order vanishing proves preservation of the earlier jets. $\square$

These are finite response statements. They neither assume a nonzero joint physical state nor prove that the entire correction is gauge invariant on the open chart. Only the displayed source-point linear residuals are retained. For a concrete source instance, the Branch-B principal coefficients and increment must be evaluated anew; the theorem does not authorize copying the scalar construction's response values.

30.3 Small graph corrections and the limit equation

Finite response coefficients need not satisfy an a priori bound uniform over all subsequent stages. For finitely many earlier graph tests, support shrinking provides a useful stronger statement.

Lemma 30.6 (Finite graph-budget shrinking). *Let $P_N(\eta)\xi$ be a homogeneous fresh-probe polynomial of degree N with an accepted fixed-degree exterior coefficient. Choose a smooth compactly supported cutoff χ equal to one near zero and put*

$$K_\epsilon(q) = \chi(\eta(q)/\epsilon)P_N(\eta(q))\xi.$$

Fix finitely many graph seminorms $p_i(W_i F)$ on compact configuration/derivative-test sets, for registered finite differential words W_i . Include the finitely many proper pullbacks of their test sets when such pullbacks occur. Let R_i bound the number of configuration derivatives falling on the finite-probe factor in that test, and assume $N > R := \max_i R_i$. Then

$$p_i(W_i K_\epsilon) \leq C_i \epsilon^{N-R_i} \quad (460)$$

for sufficiently small positive ϵ . The constants are finite for this fixed family, source record, polynomial, and exterior degree. Every source-point jet of K_ϵ equals that of $P_N \xi$.

Proof. The cutoff is an admitted smooth finite-probe cylinder. On its support $|\eta| = O(\epsilon)$. Differentiating the cutoff a times contributes ϵ^{-a} , while differentiating the homogeneous polynomial b times contributes ϵ^{N-b} , with $a + b \leq R_i$. The finite Leibniz expansion, boundedness of the relevant coefficient derivatives on the selected compact tests, and the registered spatial-contraction and exterior-map seminorm estimates prove (460). Finite proper pullbacks use their compact image sets and the corresponding finite chain rule. No estimate over unbounded word length is asserted. Because χ is identically one near the source, multiplication by it changes no source-point jet. $\square$

If an actual finite-extension construction supplies correction orders N_j eventually exceeding the differential orders of the first j chosen tests, the radii can therefore be chosen after computing the finite coefficients so that stage j costs at most 2^{-j} in those tests. This statement does not assume, or prove, that a chosen countable test family generates the full W6 graph topology. Nor does it supply the required all-label finite-extension construction.

Proposition 30.7 (Point-jet realization does not imply the source equation). *Vanishing of every constraint jet at one source point, even for a normalized smooth section, does not imply vanishing of the constraint on an open configuration chart. Consequently a Borel-type realization of compatible source-point jets is insufficient by itself to populate Z_B .*

Proof. The failure already occurs for a scalar differential operator. Set

$$f(0) = 0, \quad f(x) = e^{-1/x^2} \ (x \neq 0), \quad F = 1 + f, \quad C = \frac{d}{dx}.$$

Then $F(0) = 1$, every derivative of CF vanishes at zero, and CF is nonzero on every punctured neighborhood. The W6 source is a smooth open-chart source with compact derivative-test

seminorms, not a source defined to identify functions having equal jets at the distinguished point. Therefore source-point jet equality cannot replace its actual constraint equation. Even a graph-Cauchy correction sequence gives, by continuity, convergence of its residuals, not that their limit is zero. The latter requires a separate residual-zero argument in the source topology or on a proved determining family. $\square$

There is a second exact distinction in the global coefficient argument. The deterministic selection theorem of the scalar source uses its complete predecessor-ideal reduction and a finite effective-shift interval for the actual serialized operator representation [14]. Finite differential order and smooth new coefficients do not by themselves establish that interval. For example, in $C = e^x d/dx$, the coefficient $F'(0)$ occurs in arbitrarily high differentiated origin rows. Dividing by the invertible e^x gives a valid simpler equation, but doing so is an additional ideal-reduction step. The Branch-B matrix-coefficient representation must satisfy the needed reduction and support-band conditions in its own right. This observation does not change the accepted scalar construction.

30.4 The exact continuation to physical occupation

The physical-state construction in WDW-6 receives an independently constructed normalized joint Branch-A solution and evaluates strong-dual source elements on that solution. Its scale-cylinder comparison transports literally unchanged operators and source topology [15, 12]. The resulting positive rank-one physical form is genuine, but neither it nor the scale identity supplies a changed Branch-B solution or a non-scalar observable algebra.

A complete Branch-B solution by this finite-extension route requires the complete finite-extension property for all retained linear and Hamiltonian labels, their word descendants, and their lower-order coefficient relations. One must then construct a limit in the actual graph source and prove the complete residual zero there, preserving a nonzero normalization. The finite matrix inverse in Theorem 30.4 and the shrinking estimate in Lemma 30.6 supply parts of that construction, but not its all-label compatibility or open-chart residual-zero step.

Closure and principal-symbol matching cannot replace that step even in elementary systems: $C_0 = d^2/dx^2$ and $C_1 = d^2/dx^2 + I$ commute and have the same principal symbol, but their common kernel is zero because $(C_1 - C_0)F = F$. This is not a proposed model of the actual WDW family; it identifies the lower-order left-ideal obstruction that a source-specific extension theorem must exclude.

Finally, a non-scalar physical observable algebra requires more than one formal output label. Suppose actual independent joint solutions $F_1, \dots, F_m$ have been constructed in the exact source. Evaluation gives

$$\Psi(u) = (u(F_1), \dots, u(F_m)), \quad B(u, v) = \langle \Psi(u), \Psi(v) \rangle_{\mathbb{C}^m}.$$

Continuous-dual separation makes this evaluation map surjective, so its positive quotient is $\mathbb{C}^m$. This particular choice of form is not thereby proved unique or target-forced. An actual source observable O has a well-defined action on this quotient only if its transpose respects $\ker \Psi$. A direct sufficient check is $OF_j = \sum_k M_{kj} F_k$, followed by compatibility of the induced matrix action and its adjoint with the chosen form. A non-scalar mixed physical observable requires such a registered O with a non-scalar induced action, not an arbitrarily declared matrix algebra.

Thus the positive result of this section is a chiral finite-response extension with explicit compatibility and graph-budget control. It does not establish nonemptiness of the complete Branch-B physical kernel. That boundary concerns this particular equation and source. It preserves the positive renormalized source/anomaly and structural-necessity results, while keeping the published raw off-shell formula distinct from an actually verified reduced or source-restricted

identity. This optional WDW route is not a prerequisite for the direct physical or native Einstein constructions.

31 Nonpolynomial construction in the full constraint source

Finite jet corrections and nonpolynomial source construction address different parts of the constraint problem. The positive Higgs quartic already prevents a finite Higgs polynomial from being a complete smooth solution at the accepted baseline. Nevertheless, that baseline admits an explicit convergent construction of exponential local functionals in its full graph source. Under an actual Hamilton–Jacobi identity, the construction gives an exact solution rather than an asymptotic WKB substitution.

31.1 Complete word-output representation

Write U_n for the genuine smooth trace core of exterior degree n . For output degree m , let E_m be the Hausdorff completion of the original output source in its inherited locally convex topology. No identification of E_m with an ordinary function space is assumed. Let W range over every registered finite word, including the identity, and let P_m be the exterior projection. The degree-wise graph seminorms are precisely

$$F \mapsto p_m(P_m W F),$$

with p_m an inherited output seminorm [12, Secs. 26–27]. Each fixed word has a finite output-degree band on one input degree.

Lemma 31.1 (Complete-output representation). *The completed degree source V_n is canonically the closure of*

$$J_n(U_n) \subset \prod_{W,m} E_m, \quad (J_n F)_{W,m} = P_m W F.$$

Its extended word-output coordinates jointly separate V_n .

Proof. The pullback of the product topology is exactly the displayed graph topology. The identity-word coordinate makes J_n injective on the Hausdorff core. The product of the complete Hausdorff spaces E_m is complete, and the closure of the image is complete. The universal property of completion gives the identification. Separation is componentwise separation in the product. $\square$

The full source is the locally convex direct sum of the V_n , with finite exterior support. The lemma does not assert that the extended identity coordinate alone is injective. It therefore avoids using point values to identify an arbitrary completed vector.

Proposition 31.2 (Normalized acquisition in the actual source). *Fix a finite set of input degrees and a continuous linear anchor a . Suppose genuine core vectors F_i supported in those degrees are Cauchy in every graph seminorm, $a(F_i) \rightarrow 1$, and, for every constraint C_α , every output degree n , every W, m , and a separating family of $\ell \in E'_m$,*

$$\ell(J_{n;W,m}(P_n C_\alpha F_i)) \longrightarrow 0.$$

Then their limit F belongs to the actual completed source, has $a(F) = 1$, and satisfies $C_\alpha F = 0$ for every retained label. Conversely every such normalized finite-support solution has a core-approximating net with these properties.

Proof. Graph Cauchyness gives a limit in each of the finitely many complete input degrees, hence in their finite direct sum. Continuity preserves the anchor and every scalar output test. Dual separation and Lemma 31.1, applied to each degree of each constraint output, force that output to be zero. Conversely, density of the core in each degree gives an approximating net; continuity gives all the displayed limits. $\square$

The residual tests suffice because graph convergence has already provided their actual vector limit. Weakly vanishing residual tests alone would not construct that limit. All constraint labels remain included: fixed-label continuity does not by itself prove continuity in the label on every completed vector, and no countable cofinal family is assumed here.

31.2 Why a finite Higgs polynomial cannot terminate the construction

The baseline fixes $\lambda > 0$, $v > 0$ and

$$V_H(H) = \lambda(H^\dagger H - v^2/2)^2.$$

Its kinetic coefficients and reference smoothing contain no live Higgs factor. Dirac action is independent of H , Yukawa action is linear in H and $H^\dagger$, and the covariant-gradient potential is quadratic in them [11, Secs. 8, 11]. The renormalized baseline retains this potential and sets only the additional local heat finite profile to zero [12, Part IV, Sec. 9.2].

Theorem 31.3 (Finite-Higgs-degree exclusion). *A nonzero genuine smooth Branch-B source section in the domain of the exact renormalized baseline action, with finite polynomial degree in the real Higgs coordinate, cannot solve the complete Hamiltonian equation for one fixed positive lapse on an open product chart.*

Proof. Let D be the maximal Higgs degree and F_D its nonzero leading homogeneous part. The non-Higgs Hessians preserve degree and the Higgs Hessian lowers it by two. At every fixed positive regulator these statements are literal polynomial identities. Finite-part extraction preserves the degree bound: restrict to a Higgs line in the chart and recover the finitely many polynomial coefficients from $D + 1$ fixed parameter values. The fixed linear interpolation commutes with heat coefficient extraction, subtraction and the retained regular limit. No extra field-dependent finite normalization is introduced.

The Dirac, Yukawa and gradient terms have degree at most D , $D + 1$ and $D + 2$, respectively. The sole degree- $D + 4$ term is

$$\lambda \left[\int_{\Sigma} N \sqrt{h} (H^\dagger H)^2 \right] F_D(H).$$

The bracket is strictly positive for every nonzero smooth Higgs variation. A polynomial identity on a nonempty open affine chart determines its highest homogeneous part: translate the chart centre and restrict to finite-dimensional affine slices. Translation leaves the top quartic and maximal degree unchanged. The displayed product cannot vanish unless $F_D = 0$, a contradiction. The argument uses polynomial continuation, not admissibility of $H = 0$ in the chart. $\square$

The theorem concerns genuine admitted smooth sections, not arbitrary identity-word projections of completed vectors. It permits increasing-degree polynomial approximants to a well-constructed nonpolynomial limit. Positive exterior degree also does not give a small bounded matrix perturbation: the free spatial Dirac principal symbol persists in the classical small-matter limit. Compactly supported spinors of the form ηe^{ikx_1} have a Dirac-to-input Sobolev norm ratio growing linearly with k on any fixed rung. This is consistent with the predecessor's one-derivative-loss estimate and inconsistent with replacing the full Dirac action by a small same-rung bounded operator.

31.3 Local phases and the exact chain rule

Fix the zero-additional-local-finite, left-coefficient baseline r_0 . Write

$$H_N = T_N + V_N, \quad T_N = -\hbar^2 \operatorname{Tr}_{\text{ren}}(\mathbb{K}_N d^2).$$

The exact DeWitt, clock, gauge and Higgs normalizations, including their individual factors of $1/2$, are already in $\mathbb{K}_N$. Let S be a real smeared finite-spatial-jet local functional admitted by the smooth local/separated/smoothing grammar. Put

$$B_N(S, S) = \mathbb{K}_N[dS, dS].$$

The Hessian d^2S is wholly spatial-diagonal and local. Its additional finite coordinate is zero at r_0 , whereas the separated rank-one channel $dS \otimes dS$ has its ordinary retained endpoint. Thus $T_N S = 0$ [12, Part III, Sec. 2.2; Part IV, Sec. 10.1].

For a polynomial p and a genuine fixed-degree core amplitude A , the full channel decomposition of the product Hessian gives

$$\begin{aligned} T_N(p(S)A) &= p(S)T_N A - 2\hbar^2 p'(S)\mathbb{K}_N[dS, dA] \\ &\quad - \hbar^2 p''(S)B_N(S, S)A. \end{aligned} \tag{461}$$

The missing $p'(S)A d^2S$ term is local and has exactly the fixed zero additional finite profile. This is not a normalization selected after choosing S . Potential multiplication and the degreewise Dirac/Yukawa action commute with the scalar factor $p(S)$. The corresponding full-Hamiltonian identity therefore replaces both $T_N(p(S)A)$ and $T_N A$ by $C_N(p(S)A)$ and $C_N A$, respectively.

The registered words also contain the fixed positive-regulator Hessians. For those primitives the chain rule retains the extra term $p'(S)AT_{N,\epsilon}S$, generally nonzero. Their separated contractions are the corresponding regulated ones. At each fixed $\epsilon > 0$ all these coefficients have the inherited smooth compact estimates. No regulator-uniform bound, or assertion $T_{N,\epsilon}S = 0$, is used.

Lemma 31.4 (Finite-word phase estimate). *For every registered finite word W , amplitude A and admitted local S , the polynomial action has a finite expansion*

$$W(p(S)A) = \sum_{\rho, j} A_{W, \rho, j} p^{(j)}(S_\rho),$$

where S_ρ are the finitely many properly transported phases in that word and the coefficients do not depend on p . For every fixed output compact derivative-test seminorm there are finite C, R, J with

$$p_m(P_m W(p(S)A)) \leq C \max_{0 \leq j \leq J} \sup_{|z| \leq R} |p^{(j)}(z)|.$$

Proof. Multiplication and target operations act on the coefficient. A first-order derivative uses the chain rule; the renormalized Hessian uses (461). A fixed regulated Hessian additionally uses its retained smooth local term. Each step raises the derivative order in p by finitely much. Proper pullbacks transport S as well as the coefficient, so retaining a single untransported phase would in general be incorrect. Induction proves the finite expansion. Applying finitely many output derivatives gives another finite chain rule. The coefficient derivatives are bounded on the selected compact tests, and every relevant phase is bounded on the compact image of a proper map. This yields the estimate. Its constants may depend on the fixed word and every fixed regulator label. $\square$

31.4 A convergent nonpolynomial source

For $c \in \mathbb{C}$ define

$$p_k(z) = \sum_{r=0}^k \frac{(cz)^r}{r!}.$$

Every $p_k(S)A$ is a finite trace-core expression. For fixed R, j , its one-variable derivatives converge uniformly to $c^j e^{cz}$; for $k \geq j$ their error is bounded by

$$|c|^j e^{|c|R} \frac{(|c|R)^{k-j+1}}{(k-j+1)!}.$$

Theorem 31.5 (Exponential source acquisition). *The sequence $p_k(S)A$ converges in the actual completed degree source to an element denoted $E_c(S; A)$. It retains finite exterior support, and all its registered word outputs are the limits of the expansions in Lemma 31.4. For $A = 1$ it is nonzero.*

Proof. Apply the lemma to $p_k - p_l$. Uniform convergence of the finitely many one-variable derivatives makes every fixed graph seminorm tend to zero. This holds for every seminorm in the actual family, with no countable cofinality assumption. Degreewise completeness gives the limit and its full output coordinates. For $A = 1$, continuous evaluation at an admitted q_* gives $e^{cS(q_*)} \neq 0$. Multiplication by $e^{-cS(q_*)}$ normalizes that anchor to one. $\square$

This is an explicit convergence theorem, not an assumed exponential calculus for arbitrary continuous operators on a locally convex space. It first concerns the fibre at r_0 . For the orbit carrier, S and A must additionally be actual smooth orbitwise source families whose every finite orbit derivative has the admitted grammar [12, Part IV, Sec. 4.2]. Applying the same estimate after each such derivative then gives the orbitwise extension.

An explicit example is

$$S_H = \int_{\Sigma} \sqrt{h} H^\dagger H.$$

Its natural geometric family transports the fields and density together, and every finite orbit derivative has smooth finite-jet local form. On the closed branch it is spatially, internally and Lorentz invariant. Consequently every $p_k(S_H)$, and then its exponential limit, is annihilated by the complete linear-constraint family. For $c \neq 0$ this is a nonpolynomial source.

The gravitational variation of the volume factor must be retained. Writing $u = H^\dagger H$, its metric momentum is $\frac{1}{2}\sqrt{h} u^{ij}$ and its Higgs momenta are $\sqrt{h} H^\dagger, \sqrt{h} H$. The exact contraction is

$$B_N(S_H, S_H) = \int_{\Sigma} N \sqrt{h} \left(u - \frac{3\kappa}{4} u^2 \right).$$

This does not supply the magnetic, gradient, scalar and curvature terms of the full V_N . The example therefore establishes a genuine linearly constrained source, not a complete Hamiltonian solution.

31.5 Exact Hamilton–Jacobi transfer and chiral amplitudes

Theorem 31.6 (Exact local-phase transfer). *Suppose an actual admitted local S , with its checked orbitwise family when that carrier is used, satisfies on its open chart*

$$\begin{aligned} B_N(S, S) + V_N &= 0 \quad \text{for every retained lapse } N, \\ X_L S &= 0 \quad \text{for every linear-constraint label } L. \end{aligned}$$

For an orbit-carrier conclusion, these identities are required fibrewise at every record of that admitted family; smooth orbitwise membership alone is not a substitute for their transport. Then $E_{i/h}(S; 1)$ is a nonzero joint degree-zero Branch-B constraint solution in the actual graph source.

Proof. The source exists and is nonzero by Theorem 31.5. The linear constraints annihilate each polynomial approximant. For the Hamiltonian the exact core identity is

$$H_N p_k(S) = V_N(\hbar^2 p_k''(S) + p_k(S)).$$

The one-variable residual $\hbar^2 p_k'' + p_k$ and every derivative converge uniformly to zero on bounded sets. Apply Lemma 31.4 with the core coefficient $A = V_N$. Every graph seminorm of the Hamiltonian residual tends to zero. Continuity therefore gives $H_N E_{i/\hbar}(S; 1) = 0$ for each retained N . This proves vanishing in the graph source, not just vanishing of its identity-word pointwise image. $\square$

The local-phase condition is an actual Hamilton–Jacobi equation with the accepted kinetic normalization. The exact transfer works because the local Hessian has the fixed zero additional finite profile and the separated channel is evaluated without approximation. An admitted full-target phase satisfying these equations has not been constructed here. A classical history or a homogeneous principal function does not by itself furnish such a local functional on the full open chart. A nonlocal principal functional can have a retained regular Hessian, for which the vanishing argument no longer applies.

For a genuine degree- n amplitude A , the same core identities give

$$\begin{aligned} C_N E_c(S; A) &= E_c(S; C_N A - 2\hbar^2 c \mathbb{K}_N[dS, dA] \\ &\quad - \hbar^2 c^2 B_N(S, S)A). \end{aligned} \tag{462}$$

The proof passes the polynomial product identity through every graph output, using Theorem 31.5 on its finitely many coefficient terms. The amplitude $C_N A$ retains the full Dirac/Yukawa action. At $c = i/\hbar$ and under the Hamilton–Jacobi identity, its potential cancels against B_N , leaving the genuine kinetic–transport–Dirac–Yukawa equation. It is not a solved amplitude or a finite matrix substitute for the spatial Dirac operator. The linear rows retain $LA + c(X_L S)A$ as well.

The construction gives a nonpolynomial convergence mechanism within the inherited source, including all its declared word outputs. In this Branch-B realization, complete chiral occupation requires an actual phase/amplitude solution, or another construction satisfying Proposition 31.2, followed by positive physical representation and the required mixed observable descent. This criterion concerns the quantum-constraint route, not the completed direct physical realization of Part I.

32 Spatial clock response and the complete constraint equations

The nonpolynomial source of Section 31 makes the choice of initial data consequential. An initial value on a homogeneous restriction need not extend to a solution on the full configuration chart. We prove a more specific statement: at the retained Branch-B baseline, the complete lapse family forces a nonzero smooth initial value to depend on the scalar clock’s spatial shape. An arbitrary shape-dependent initial clock derivative does not remove this restriction.

Throughout this section we use the zero-additional-local-finite, left-coefficient, flat-configuration-connection baseline and its number-preserving holomorphic CAR ordering [12, Secs. 9.2, 10.1 and 11.1]. Thus

$$H_N^B = -\hbar^2 L_N + V_N + Q_N, \quad L_N = \text{FP Contr}_{K_N} d^2. \tag{463}$$

The kinetic tensor contains the DeWitt, real-clock, electric and independent Higgs blocks. The operator Q_N retains the symmetric spatial Dirac map, including its lapse derivatives, and the complete Higgs-dependent Yukawa map. The scalar clock is distinct from the Higgs field.

The theorem concerns actual smooth trace-core sections on a product-open full chart, or a completed sector for which a faithful smooth comparison preserving the stated derivatives has separately been proved. No such comparison is assumed for every element of the graph completion.

32.1 The full lapse family detects transverse clock shape

Let Σ be the retained compact, connected, boundaryless spatial manifold. Fix a positive smooth reference density $d\mu$ and write $d\text{vol}_h = m(z, x)d\mu$, with $m > 0$. The variable z comprises all nonclock even configurations; coefficients may take values in any fixed exterior degree of the inherited source. Choose a fixed smooth $\rho > 0$ with $\int_{\Sigma} \rho d\mu = 1$ and use the affine clock splitting

$$\phi = \phi_* + T\mathbf{1} + \chi, \quad \int_{\Sigma} \rho \chi d\mu = 0. \quad (464)$$

The reference profile and density are configuration independent. For a smooth section F , put

$$u(z, \chi) = F(T_0, z, \chi), \quad v(z, \chi) = \partial_T F(T_0, z, \chi), \quad w(z, \chi) = \partial_T^2 F(T_0, z, \chi).$$

With covectors represented relative to $d\mu$, the scalar kinetic pairing and potential are

$$k_N(\lambda, \eta) = \frac{1}{2} \int_{\Sigma} \frac{N}{m} \lambda \eta d\mu, \quad V_{\phi, N} = \frac{1}{2} \int_{\Sigma} N m (|\nabla \phi|_h^2 + m_{\phi}^2 \phi^2) d\mu. \quad (465)$$

These are the source's original scalar normalizations [11, Secs. 5.1–5.3]. Write R_N for every remaining term of H_N^B , including all nonclock kinetic terms and Q_N . Their coefficients are independent of the real scalar clock.

Theorem 32.1 (Transverse clock response of smooth initial values). *Suppose $H_N^B F = 0$ at $T = T_0$ for every smooth lapse N throughout a product-open full transverse configuration neighborhood. If u is independent of χ , then $u = 0$ throughout that neighborhood. Neither v nor w is assumed to be shape independent.*

Proof. First fix a spatially smooth nonclock configuration z and smooth clock profiles in the chart. Such configurations are dense in the retained Sobolev chart; fixing the clock mean preserves density by subtracting a constant from smooth approximations. The lapse and vector-field smearings used below are therefore smooth admitted smearings. We extend the resulting zero identities to the full chart by continuity only after this calculation. Shape independence of $u = u(z)$ makes its first and second shape derivatives vanish. At the initial slice the scalar Hessian is

$$d_{\phi}^2 F = \rho \otimes \rho w + \rho \otimes d_{\chi} v + d_{\chi} v \otimes \rho, \quad (466)$$

where $d_{\chi} v$ is extended to annihilate the constant direction. These finite-rank separated terms retain the ordinary trace endpoint. For a fixed smooth shape tangent η , with $\int \rho \eta d\mu = 0$, choose the allowed lapse

$$N_{\eta} = \frac{m\eta}{\rho}.$$

It is chosen at the fixed nonclock configuration and is held fixed under the following χ derivatives. Its contraction with the w term vanishes. The two mixed terms in (466), including the factor $1/2$ in (465), give exactly $-\hbar^2 d_{\chi} v[\eta]$. Consequently

$$d_{\chi} v[\eta] = \beta_{\chi}[\eta] = \hbar^{-2} (R_{N_{\eta}} u)(z) + u(z) \int_{\Sigma} \frac{m^2 \eta}{2\hbar^2 \rho} (|\nabla \phi|_h^2 + m_{\phi}^2 \phi^2) d\mu. \quad (467)$$

The first term retains every nonclock response. It is independent of χ , including the terms containing spatial derivatives of N_η in the symmetric Dirac operator.

The exact differential $d_\chi v$ is closed. Differentiating (467) in two constant tangent directions η and ζ , the scalar mass contributions cancel. Thus

$$0 = u(z) \int_{\Sigma} \frac{m^2}{\hbar^2 \rho} (\zeta \nabla \phi \cdot_h \nabla \eta - \eta \nabla \phi \cdot_h \nabla \zeta) d\mu. \quad (468)$$

At a nonconstant profile choose a coordinate ball on which $\nabla \phi \cdot_h \nabla x^1$ has one strict nonzero sign, and a nonzero smooth bump a supported there. Start with $\eta_k = a \cos(kx^1)$ and $\zeta_k = a \sin(kx^1)$, subtracting their respective $\rho d\mu$ means as constant functions to make them tangent. Their means decay faster than every inverse power of k by integration by parts. Before subtraction,

$$\zeta_k \nabla \eta_k - \eta_k \nabla \zeta_k = -ka^2 \nabla x^1.$$

The integral in (468) is therefore

$$-k \int_{\Sigma} \frac{m^2 a^2}{\hbar^2 \rho} \nabla \phi \cdot_h \nabla x^1 d\mu + o(k),$$

which is nonzero for large k . Each fixed direction is an allowed smooth tangent; no frequency-independent neighborhood radius is needed. The nonzero scalar factor in (468) forces $u(z) = 0$. In the Hausdorff exterior coefficient space the same conclusion follows by continuous separating functionals.

Every open shape neighborhood contains a nonconstant profile, even when its central profile is constant. Since u is shape independent, the conclusion holds for the fixed z throughout that neighborhood. The arbitrariness of smooth z , their density in the product-open chart, and continuity of u prove the theorem at every z . $\square$

Corollary 32.2 (Complete constrained Cauchy doublets). *If the complete spatial constraints also vanish on the initial slice under the hypotheses of Theorem 32.1, then $u = v = 0$ there. A nonzero complete initial Cauchy doublet cannot have shape-independent first component.*

Proof. The theorem gives $u = 0$ as a section of every transverse variable, so its transverse configuration derivatives and exterior Lie action vanish. For a spatial vector field ξ , the scalar fundamental field has clock component $\int \rho \xi \cdot \nabla \phi d\mu$. The remaining spatial constraint is exactly

$$-i\hbar v(z, \chi) \int_{\Sigma} \rho \xi \cdot \nabla \phi d\mu = 0.$$

The internal and Lorentz improvements act on the zero initial section. At a spatially smooth nonconstant configuration choose the fixed smearing $\xi = \nabla_h \phi$ evaluated at that configuration. The integral is strictly positive, giving $v = 0$. Smooth nonconstant configurations with the specified mean are dense in the full product-open chart. Continuity therefore extends this result both to constant profiles and to all configurations of its retained Sobolev regularity. $\square$

The result uses the complete lapse and spatial families on the specified full smooth chart. It does not assert that zero initial data have unique subsequent evolution in the complete functional source. Genuinely shape-dependent data and graph-native sources without a faithful smooth comparison remain outside this exclusion.

32.2 The exact nonlocal phase equation

For a scalar real phase S and possibly exterior-valued amplitude A , suppose the functions and their products lie in a common admitted trace core. Require each displayed product-Hessian channel below to have the source's declared finite-part domain. Denote its polarized cross contractions by

$$\mathbb{K}_N[dS, dA] = \text{FP Contr}_{K_N} \text{Sym}(dS \otimes dA), \quad \mathbb{K}_N[dS, dS] = \text{FP Contr}_{K_N}(dS \otimes dS),$$

where Sym includes $1/2$. They agree with ordinary principal-tensor pairings when the separated channel has that retained endpoint. This convention does not multiply arbitrary Sobolev-dual distributions pointwise; smoothness of a functional alone does not establish these contraction domains.

Proposition 32.3 (Full nonlocal phase and amplitude identity). *On the specified common domain,*

$$\begin{aligned} e^{-iS/\hbar} H_N^B(Ae^{iS/\hbar}) &= (\mathbb{K}_N[dS, dS] + V_N)A + Q_N A \\ &\quad - i\hbar(2\mathbb{K}_N[dS, dA] + (L_N S)A) - \hbar^2 L_N A. \end{aligned} \quad (469)$$

Proof. The full product Hessian is

$$\begin{aligned} d^2(Ae^{iS/\hbar}) &= e^{iS/\hbar} (d^2 A + \frac{i}{\hbar} (dS \otimes dA + dA \otimes dS + A d^2 S) \\ &\quad - \hbar^{-2} A dS \otimes dS). \end{aligned}$$

Apply the linear finite-part contraction and the factor $-\hbar^2$. The scalar phase commutes with the exterior fibre operator Q_N ; the spatial Dirac differentiation in that operator acts on its spinor argument, not on S as a new spatial multiplier. These identities give (469) with the full Dirac/Yukawa action retained. $\square$

If the bosonic Hamilton–Jacobi equation $\mathbb{K}_N[dS, dS] + V_N = 0$ holds for every lapse, the exact amplitude equation is still

$$Q_N A - i\hbar(2\mathbb{K}_N[dS, dA] + (L_N S)A) - \hbar^2 L_N A = 0 \quad \text{for every } N.$$

The nonlocal phase Hessian $L_N S$ need not vanish. In exterior degree zero, Q_N vanishes at this ordering, but the second-order amplitude term remains. The linear constraints must also hold on the full product. The local-phase transfer in Theorem 31.6 uses its separately proved local-Hessian identity and graph convergence; neither is inferred for an arbitrary nonlocal phase from the displayed smooth-core formula.

33 Source-preserving extension beyond the regulated targets

The occupied physical source and the constructive native realizations provide different, explicitly identified observations. A direct physical source construction need not pass through the regulator, while a claim made by taking a regulated limit must control that limit and its source comparison. This section gives useful stability results for the latter route and states the complete common-source requirement independently of the chosen method.

33.1 Mixed limits and nontriviality

The mixed-limit theorem of the regulator section constructs a positive state on the increasing union of observable algebras from actual mixed regulated limits and consistent relations. Its argument parallels the finite-cap and overlap construction in the sharp-local Yang–Mills source [17, 18]. The requirement concerns every fixed mixed word with the other observation slots retained. It is stronger than separate gauge, Higgs and matter limits.

Section 13 additionally constructs common bounded cluster states without assuming those ordinary limits. Its sampled continuum CAR and Weyl relations are actual source identities. On the specified physical algebra it also acquires continuous positive-energy dynamics. An identification of this native limit with the physical source would additionally require regular interacting fields, the specified chiral propagation and exact physical source comparison. The regular reference-time family in Theorem 13.6 is explicit, but its local occupation witness approaches the product seed and does not supply the needed uniform mixed lower bound.

Proposition 33.1 (Survival of a mixed response). *Suppose the mixed-limit hypotheses hold, and fixed bounded observation words A, B, AB have the corresponding source limits. If*

$$|\omega_\epsilon(A_\epsilon B_\epsilon) - \omega_\epsilon(A_\epsilon)\omega_\epsilon(B_\epsilon)| \geq \eta > 0$$

throughout the retained regulator family, then the limiting source has

$$|\omega(AB) - \omega(A)\omega(B)| \geq \eta.$$

In particular the selected correlation does not disappear in the limit.

Proof. Each of the three scalar source values converges by hypothesis. Their connected combination and its absolute value therefore converge. The closed lower inequality passes to the limit. $\square$

The nonzero response proved at one regulator is not itself a uniform lower bound on a refining family. The latter is a new source statement. Nor does a single connected correlation establish a complete relativistic scattering theory. Distributional regularity, covariance, locality, operator-domain control and the specified physical anomaly identities remain hypotheses to establish for that continuum reconstruction.

33.2 Analytic continuation must be controlled on its own domain

Theorem 33.2 (Uniformly controlled resonance limits). *Let D_n, N_n be holomorphic on a common neighborhood of a closed disc K , converging uniformly there to D, N . Suppose:*

- (i) $|D_n| \geq \eta > 0$ on ∂K ;
- (ii) each D_n has exactly one zero in K° , counted with multiplicity;
- (iii) $|N_n| \geq \nu > 0$ throughout K ;
- (iv) the zeros satisfy $\text{Im } z_n \leq -\delta < 0$.

Then D has one simple zero $z \in K^\circ$, $z_n \rightarrow z$, $\text{Im } z \leq -\delta$, and N/D has a nonzero residue at z . The residues of N_n/D_n converge to it.

Proof. Uniform convergence makes $|D| \geq \eta$ on the boundary. For sufficiently large n , $|D_n - D| < \eta$ there, so Rouché’s theorem gives the same zero count for D_n and D , namely one with multiplicity. The limiting zero is therefore simple. Every subsequential limit of z_n belongs to the interior and

is a zero of D ; boundary limits are excluded by the uniform boundary lower bound. Uniqueness implies convergence of the whole sequence. The imaginary-part inequality is closed. Uniform convergence also gives $|N| \geq \nu$. Cauchy's formula gives convergence of the derivatives on smaller neighborhoods of the zero, so

$$\frac{N_n(z_n)}{D'_n(z_n)} \longrightarrow \frac{N(z)}{D'(z)} \neq 0.$$

□

The hypothesis concerns actual continued channel functions on a marked common sheet. Real-axis moment convergence alone does not imply it. The compact-resolvent spatial regulator has discrete spectral data; its comparison with a physical unstable channel needs an actual source and continuation map. The direct physical Higgs construction instead uses the occupied channel and transports its entire germ. It does not assert that the regulator spectrum supplies that germ without the additional comparison.

33.3 Color spectral transfer and source identity

The local and nonlocal Yang–Mills results retain their exact observable, vacuum and dynamics identities [16, 19]. Their positive gap must not be transferred to a different total Hamiltonian by retaining only its color label. Section 28 proves a new gap for a specific bounded mixed extension of that source. Its auxiliary singlet-pair target does not include a photon or the complete chiral matter inventory.

Proposition 33.3 (A gap cannot absorb a massless low-energy source). *Let H and h be nonnegative self-adjoint operators and J an isometry intertwining their unitary groups. If $E_H((0, m_*)) = 0$, then $E_h((0, m_*)) = 0$. Consequently no source with nonzero spectral support in every $(0, \varepsilon)$ admits such an embedding into that gapped space.*

Proof. Unitary intertwining implies spectral intertwining, $E_H(B)J = JE_h(B)$, first for the resolvent and then for bounded spectral functions. For $B = (0, m_*)$, the left-hand side is zero. Injectivity of J gives $E_h(B) = 0$. □

This identifies the correct transfer problem: a dynamically meaningful restriction with its actual algebra, state and implemented dynamics, or a new sector-relative spectral proof. The full Standard Model screening quotient also has to be retained. SYM-9 distinguishes that quotient from confinement-phase data [21]; BCRS-2C retains actual source-dependent color continuation [5]. Neither charge-label arithmetic nor a nonzero local interpolating field alone supplies all composite and asymptotic sectors.

33.4 Exact source comparison for alternative realizations

A complete physical comparison must retain all of the following on one target, whether supplied directly or through a controlled constructive extension:

1. actual mixed gauge–Higgs–chiral source observations, with regular physical fields, common domains, interactions and the required renormalization and anomaly structure; when limits are the chosen construction, they must preserve those same mixed observations;
2. a nonzero physical state space and observable algebra satisfying the gauge and other constraints of the chosen realization and carrying those mixed observations;
3. the metric-response source of that same quantum state on the full retained test domain, with compatible coupled dynamics;

4. the acquired electroweak channels and complete color/composite continuations, including their analytic and asymptotic domains;
5. exact source comparisons preserving all sibling inputs and mixed observations.

The direct physical source satisfies these conditions through Theorem 11.5, with the original authority of each observation retained. The criteria also specify what a different realization must preserve to be identified with that source.

The bounded extension and asymptotic-state results establish positive common sources at their declared observation level. The native construction additionally supplies physical domains, exact dressed response and actual nonlinear quantum–Einstein continuation. The direct physical construction supplies represented Higgs/current channels, the complete original color-interface comparison, smooth preparation stress and a source-attached Einstein action and member. Its common-source maps compose these observations explicitly. The corresponding identification of a different regulator limit remains a separate mathematical claim.

On the separate Branch-B quantum-constraint route, Section 31 supplies nonpolynomial graph-domain elements. Full Hamiltonian occupation there still requires the corresponding phase–amplitude solution. That is one method, not an additional universal condition for gravitational attachment. Likewise, a limiting resonance proof must be identified with the physical channel to which it is applied, whereas a directly received physical germ needs no independent comparison resonance. Structural uniqueness applies with the complete fixed source and observation data; it is not a replacement for these identities.

34 Conclusion and the common physical endpoint

A common premetric architecture fixes the source identities of the quantum and Einstein descriptions. The physical comparison developed here preserves that order: source operations are identified before their states, tensor representatives and continuations are compared. The occupied Standard Model assignment is used positively, while new observation maps are proved at the level at which they act.

The direct physical construction begins with the original invariant Higgs mass insertion. Its derivative current and scalar descendant share the actual field domain and positive Hilbert space. A nonzero physical Higgs channel forces a nonzero descendant preparation; bounded spectral inversion then places a nonzero component in an independently generated physical current boundary. The whole continued germ is transported, including regular terms and original source normalization. No unstable pole is interpreted as a Hilbert eigenvector.

The exact Higgs and flavor source transformations preserve the complete gauge and charged-Yukawa interaction inventory. Their Euler, anomaly, nonlinear-path and renormalization contacts identify the actual mixed source combinations with the other inputs fixed. The kinetic coefficient and relative-current constructions cover the entire compact scalar test space. These operations give represented observations where their result is an existing insertion; they do not manufacture separate unbounded operators for every summand of a formal source identity.

Gravity has a directly detectable role in this same physical preparation. The original mass–curvature protocol produces

$$I^{\mu\nu} = 2(\partial^\mu \partial^\nu - \eta^{\mu\nu} \square) O_\rho, \quad \text{tr}_\eta I = -6 \square O_\rho.$$

A fourth source derivative recovers its full physical scalar response with the exact contact subtraction. For a real test f with $\phi = O_\rho(\square f)\Omega \neq 0$, the normalized preparation $(\Omega + s\phi)/\sqrt{1 + s^2 \|\phi\|^2}$ has a strictly nonzero first response to the corresponding local metric contact. For $s \neq 0$, this normalized preparation has finite positive excitation energy; its total stress profile

is smooth throughout the family. The smoothness proof uses reduced Wightman distributions and compact field smearings, not an undefined point-field evaluation.

The physical vacuum stress is fixed on all compact metric tests. Its Einstein matching condition is invariant under the jointly transported homogeneous vacuum convention. More importantly, the metric-dependent quantum action is constructed before the matched member is selected. The original closed-time-path functional supplies the causal stress response, all metric and source contacts, and the diffeomorphism Ward identity. Coupling it to the inherited minimal Einstein action gives the full nonlinear field equation at the original perturbative or retained-order quantum-action authority. The once-selected gravitational constants and the original quantum source then admit an all-test stationary member. This is a member of an actual dynamical theory, not an independently fitted stress tensor. General prepared states are not asserted stationary at the same flat metric.

Color observations retain the same full interacting prescription. The normalized composite insertion and its compact mixed response are constructed before reconstruction; no free Gaussian law replaces the physical source. The actual neutral-boundary color action forces each lawful port to factor through the invariant summand of its complete joint input. Conjugate, baryonic and adjoint-pair singlets are retained, whereas isolated nontrivial-color isometric ports to that interface are excluded. This selection rule is not used as an independent theorem about every physical phase.

Every independently typed localized composite-persistence record has its actual boundary and history witness. The received heavy c/b branch establishes nonemptiness at that physical authority. Finite source processes retain their precise terminal-reader domains, continuation and failures. The reconstruction comparison preserves all distinguishing readouts and the forced non-null persistence history. It neither identifies persistence with a resonance pole nor manufactures an asymptotic state.

The native constructions provide complementary existence and control results. Positive extension on an independently formed bounded observation space has an exact order criterion and a finite positive obstruction. The gauge–Higgs–fermion Hamiltonian has a nonempty physical space with the faithful gauge quotient and nonzero interactions. Its mass and current insertions, dressed physical inclusions, ordered fermionic histories, domain-controlled feedback and asymptotic observations are constructed from that one source. The reference-resolved action then supplies an occupied nonlinear Einstein family driven by the literal interacting Hamiltonian, with a differentiable preparation map and causal response. These are genuine coupled quantum–gravitational results on their declared native target. Their state and continuation identification with the occupied physical Standard Model is not inferred merely from the common interaction grammar.

The full color comparison is established on the independently formed source inputs, not only on a selected list of physical outputs. The source interpretation preserves every original finite constructor, licensed operation and shared operand. Its physical realization preserves the actual reader and the entire operation graph, including intrinsic parameters, domains, source qualifications and failures. Complete spectral, asymptotic and continued objects retain their original authority; neither a finite source jet nor a persistence history is promoted to a stronger physical boundary.

The seven source-completeness conditions consequently hold for the originally declared physical target. Their same-source composition gives Theorem 11.5: the electroweak channels, complete color and composite continuation, interacting quantum source and Einstein response coexist at one occupied point. The whole compatible family is lifted together and has an exact readback. No independent state or parameter choice is made for a different observation.

Physical component	Common-source realization
Electroweak channels	The occupied physical channels are retained. The original Higgs insertion, derivative/current/contact observations and their complete continued germs are attached directly. Complete compact mixed-source operations are received at their original prescription and order. Stronger represented and continued baseline channels remain unchanged.
Color and composite continuation	Every originally formed input and licensed operation has its exact source comparison. The actual neutral ports, normalized compact composite/Higgs responses, persistence processes and invoked whole physical boundaries retain all sibling inputs, source witnesses, domains and continuation authority.
Interacting physical source	The occupied source and represented observations are used jointly. The complete declared mixed-field operations, domains, renormalization and anomaly data, and invoked scattering/asymptotic operations retain that source at their respective physical authority.
Dynamical gravity	The original-source closed-time-path/Einstein action and an occupied stationary member are constructed at the original quantum-action order. The full nonlinear minimal Einstein law is inherited. A universal transverse solution map is not required for this result.
Common physical endpoint	The physical electroweak records, full original color-domain and boundary comparison, and dynamical action coexist at one unchanged quantum source and one fixed joint parameter point. The comparison preserves every originally declared operation and compatible family.

This result is source-relative in the mathematically substantive sense: physical conclusions carry the state, domain, phase and continuation assumptions under which they are proved. The original comparison includes every declared input and use; it neither demands a productive hadronic output from every pre-boundary configuration nor asserts uninvoked all-state asymptotic completeness. The perturbative or retained-order quantum-action construction leaves stronger baseline represented fields and complete physical germs unchanged.

The structural and physical conclusions therefore have a definite order. The premetric architecture fixes admissible identities and their necessary relations; the source constructions supply physical occupancy and operations; exact comparison makes their quantum and Einstein projections jointly realizable. Gravity is present both as an upstream necessity and as the source-dependent metric response of the same physical realization.

A Separation of positivity, constraint occupation and resonances

The positive constructions above distinguish three claims: positivity of specified source data, existence of a physical quantum state, and existence of a resonance in a particular channel. The following exact examples show why an implication between those claims needs its own theorem. They do not contradict a source result whose full hypotheses already construct the stronger object.

Proposition A.1 (Source expectation does not determine state positivity). *Normalization and positivity of a specified source expectation need not give a positive functional on the full observable algebra.*

Proof. On $M_2(\mathbb{C})$, take the source $T = I$ and the normalized Hermitian functional

$$\omega(A) = \frac{3}{2}A_{11} - \frac{1}{2}A_{22}.$$

It satisfies $\omega(I) = \omega(T) = 1$, but the positive projection $P = \text{diag}(0, 1)$ has $\omega(P) = -1/2$. Thus the supplied source expectation does not establish $\omega(A^*A) \geq 0$ for all observable words. $\square$

If a normalized functional has instead been proved positive on the complete declared $\ast$ -algebra, the algebraic GNS construction supplies a cyclic Hilbert realization. Locality, operator closure, spectrum, interacting evolution, source attachment and analytic continuation remain properties to establish on that realization.

Proposition A.2 (Classical occupancy and quantum closure need not give a physical constraint state). *A nonempty regular classical constraint surface, self-adjoint anomaly-free quantum constraint closure and a positive conserved source can coexist with a zero quantum constraint kernel, even on the usual tempered-distribution completion.*

Proof. On the symplectic plane, let

$$C_{\text{cl}}(q, p) = q^2 + p^2 - 2.$$

Its zero set is a nonempty regular circle, and its one-generator Poisson constraint algebra closes. On $L^2(\mathbb{R})$, in units $\hbar = 1$, use

$$\hat{C} = Q^2 + P^2 - 2I$$

with its usual self-adjoint oscillator realization and invariant Schwartz core. Its eigenvalues are $2n - 1$, $n = 0, 1, \dots$, none of which is zero. The source $T = I$ is positive and conserved; the one-generator commutator algebra is anomaly-free. Nevertheless $\ker \hat{C} = \{0\}$ in the Hilbert space.

Hermite expansion identifies $\mathcal{S}(\mathbb{R})$ with rapidly decreasing coefficient sequences and $\mathcal{S}'(\mathbb{R})$ with polynomial-growth sequences. Multiplication by $2n - 1$ and by its reciprocal both preserve these respective sequence spaces continuously. Hence $\hat{C}$ is a continuous automorphism of both $\mathcal{S}(\mathbb{R})$ and $\mathcal{S}'(\mathbb{R})$, and its kernel is also zero in the tempered-distribution completion. $\square$

This example addresses the stated implication, rather than every possible physical-state completion. A Wheeler–DeWitt or GRQM theorem that proves nonempty physical-state realization has stronger antecedents, and those antecedents must be retained when it is imported.

Proposition A.3 (Positive spectral content need not give a pole). *A positive-energy Hilbert realization can have a norm-one source-created vector and a nonzero positive spectral measure while its channel has no isolated resonance pole on its continuation charts.*

Proof. Let $K = M_x$ on $L^2([0, 1], dx)$ and let $v(x) = 1$. Adjoin a zero-energy vacuum and the bounded source creation operator $|v\rangle\langle\Omega|$. The source vector has norm one, the Hamiltonian is nonnegative, and its exact channel is

$$G(z) = \langle v, (K - z)^{-1}v \rangle = \int_0^1 \frac{dx}{x - z} = \log(1 - z) - \log(-z).$$

Its continuations change logarithmic branches. On every ordinary continuation chart away from 0 and 1, the function is holomorphic and has no isolated pole; the endpoint singularities are logarithmic. The positive spectral measure is Lebesgue measure on $[0, 1]$, and is nonzero. $\square$

Thus a nonzero positive spectral measure, even when accompanied by unitary dynamics, cannot be promoted to a resonance without additional analytic channel information. The occupied Friedrichs-type result in Section 29 establishes precisely such information on its declared target. A transfer to a Standard-Model pole needs the corresponding source and channel identification.

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